

GATE

Computer Science & IT

Last 10 Years - Previous Year Question Papers

GATE 2016 to GATE 2025 | every session / shift / set conducted

16

Question papers

975

Questions

532

Pages

2016-2025

Years

What is inside

Every Computer Science paper GATE published in the ten years before GATE 2026 - 2016 through 2025. Where a year ran more than one set, all of them are here: 2025 (2 sets), 2024 (2), 2021 (2), 2017 (2), 2016 (2); 2023, 2022, 2020, 2019 and 2018 were single-session years.

Each of these papers carries 65 questions: Q.1-Q.55 from the CS syllabus (Engineering Mathematics + core subjects) and 10 General Aptitude questions worth 15 marks, so 100 marks in 3 hours.
15 question papers x 65 = 975 questions.

One extra document is included at the end of GATE 2022: the official answer key table for that paper (question type, subject, key and marks for all 65). It is not a question paper, so it is not counted in the total above.

Counts were read off the papers themselves rather than assumed, and the index page numbers were checked against the PDF bookmarks.

Index (1/1)

Year / paper	Session / set	Pg	Q	Starts at page
2025 CS1 master qp	Set 1	64	65	3
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2024 CS1 set5	Set 1 (S5)	36	65	117
2024 CS2 set6	Set 2 (S6)	40	65	154
2023 CS single	Single session	69	65	195
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2019 CS	Single session	16	65	413
2018 CS	Single session	23	65	430
2017 CS set1	Set 1	27	65	454
2017 CS set2	Set 2	25	65	482
2016 CS set1	Set 1	20	65	508
2016 CS set2	Set 2	21	65	529

GATE 2025

Set 1

65 questions | 64 pages

Paper 1 of 16



General Aptitude

Q.1 – Q.5 Carry ONE mark Each

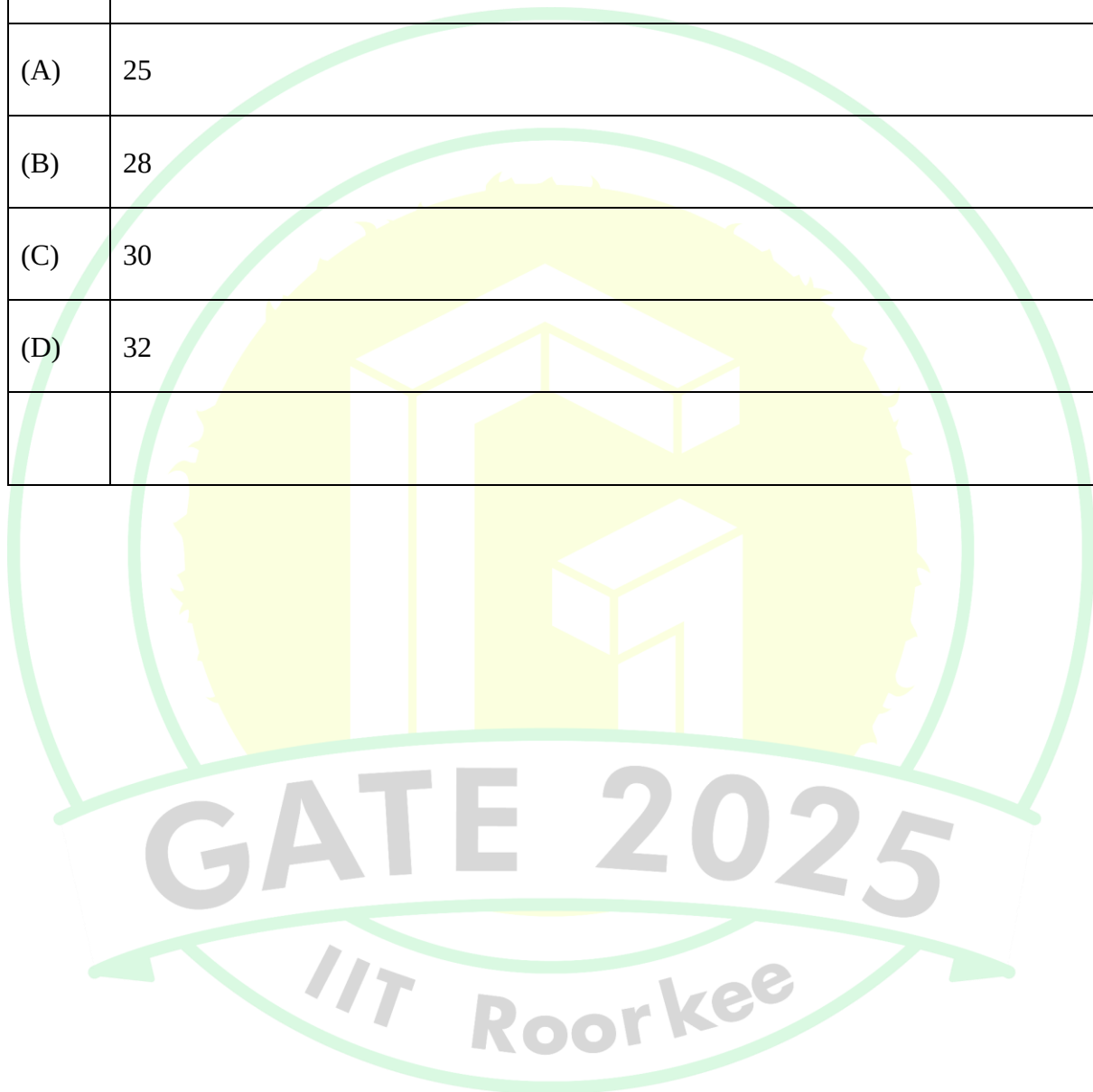
Q.1	Ravi had _____ younger brother who taught at _____ university. He was widely regarded as _____ honorable man. Select the option with the correct sequence of articles to fill in the blanks.
(A)	a; a; an
(B)	the; an; a
(C)	a; an; a
(D)	an; an; a



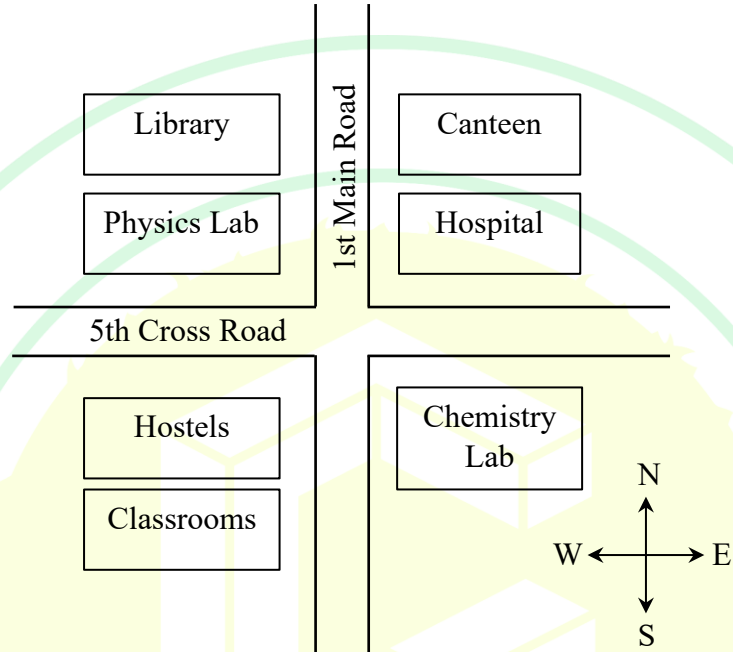
Q.2	The CEO's decision to downsize the workforce was considered <u>myopic</u> because it sacrificed long-term stability to accommodate short-term gains. Select the most appropriate option that can replace the word “myopic” without changing the meaning of the sentence.
(A)	visionary
(B)	shortsighted
(C)	progressive
(D)	innovative



Q.3	The average marks obtained by a class in an examination were calculated as 30.8. However, while checking the marks entered, the teacher found that the marks of one student were entered incorrectly as 24 instead of 42. After correcting the marks, the average becomes 31.4. How many students does the class have?
(A)	25
(B)	28
(C)	30
(D)	32

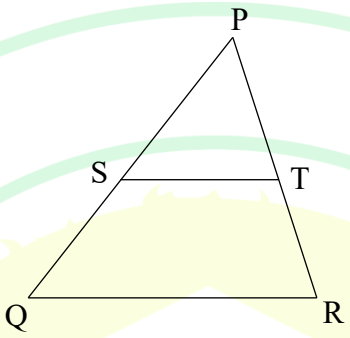


Q.4	Consider the relationships among P, Q, R, S, and T: • P is the brother of Q. • S is the daughter of Q. • T is the sister of S. • R is the mother of Q. The following statements are made based on the relationships given above. (1) R is the grandmother of S. (2) P is the uncle of S and T. (3) R has only one son. (4) Q has only one daughter. Which one of the following options is correct?
(A)	Both (1) and (2) are true.
(B)	Both (1) and (3) are true.
(C)	Only (3) is true.
(D)	Only (4) is true.

Q.5	According to the map shown in the figure, which one of the following statements is correct? Note: The figure shown is representative.
	
(A)	The library is located to the northwest of the canteen.
(B)	The hospital is located to the east of the chemistry lab.
(C)	The chemistry lab is to the southeast of physics lab.
(D)	The classrooms and canteen are next to each other.

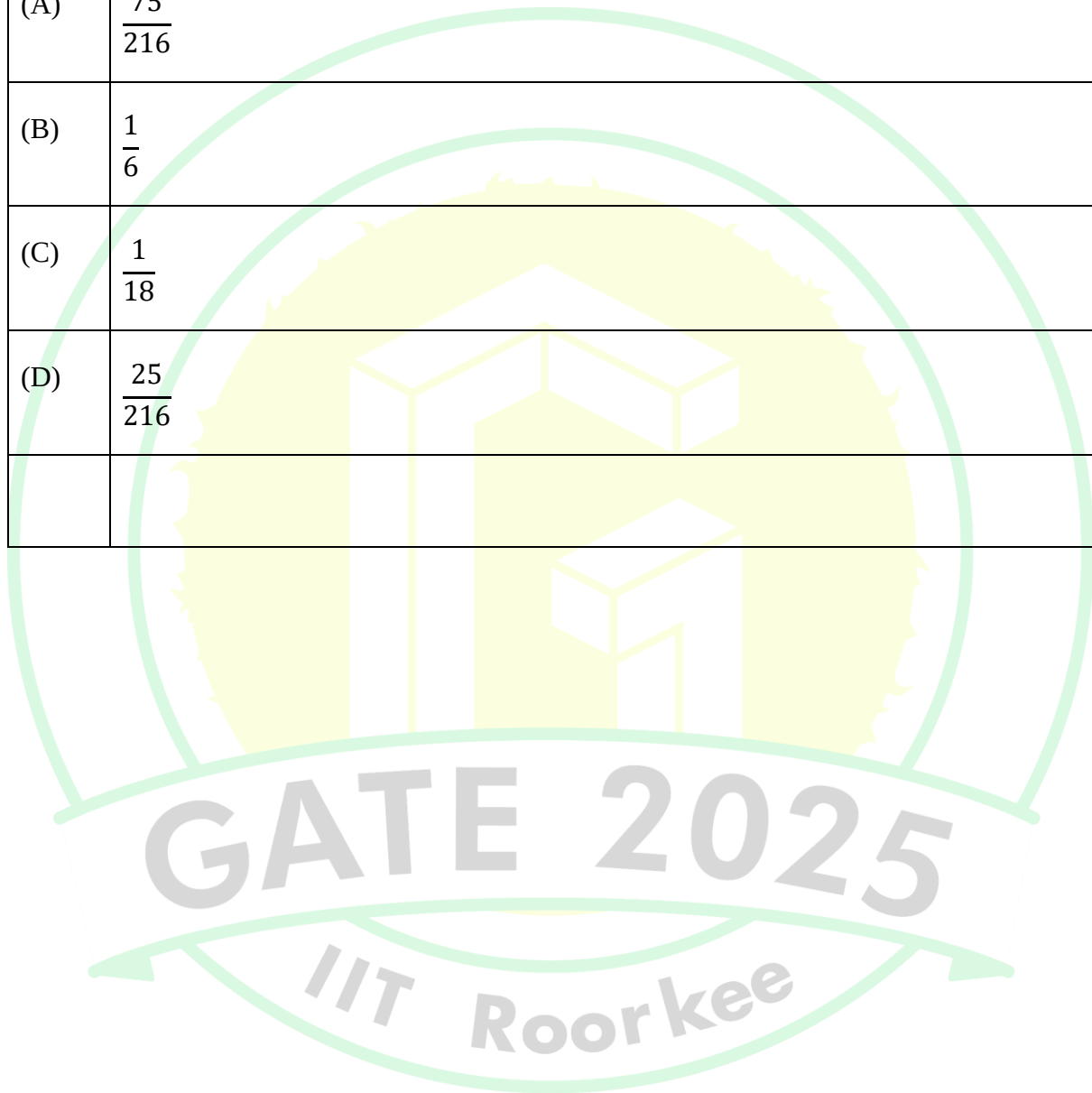
Q.6 – Q.10 Carry TWO marks Each

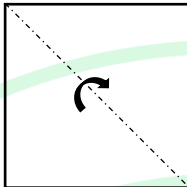
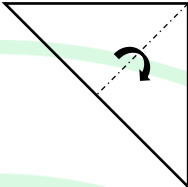
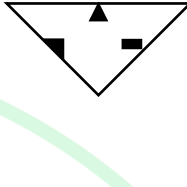
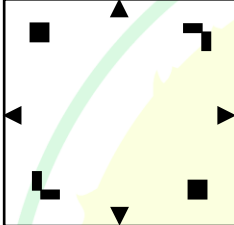
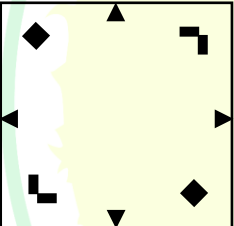
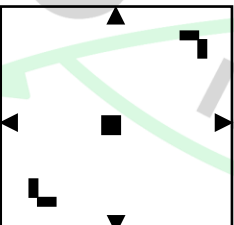
Q.6	“I put the brown paper in my pocket along with the chalks, and possibly other things. I suppose every one must have reflected how primeval and how poetical are the things that one carries in one’s pocket: the pocket-knife, for instance the type of all human tools, the infant of the sword. Once I planned to write a book of poems entirely about the things in my pocket. But I found it would be too long: and the age of the great epics is past.” (From G.K. Chesterton’s “A Piece of Chalk”) Based only on the information provided in the above passage, which one of the following statements is true?
(A)	The author of the passage carries a mirror in his pocket to reflect upon things.
(B)	The author of the passage had decided to write a poem on epics.
(C)	The pocket-knife is described as the infant of the sword.
(D)	Epics are described as too inconvenient to write.

Q.7	In the diagram, the lines QR and ST are parallel to each other. The shortest distance between these two lines is half the shortest distance between the point P and line QR. What is the ratio of the area of the triangle PST to the area of the trapezium SQRT? Note: The figure shown is representative.
	
(A)	$\frac{1}{3}$
(B)	$\frac{1}{4}$
(C)	$\frac{2}{5}$
(D)	$\frac{1}{2}$



Q.8	A fair six-faced dice, with the faces labelled '1', '2', '3', '4', '5', and '6', is rolled thrice. What is the probability of rolling '6' exactly once?
(A)	$\frac{75}{216}$
(B)	$\frac{1}{6}$
(C)	$\frac{1}{18}$
(D)	$\frac{25}{216}$



Q.9	A square paper, shown in figure (I), is folded along the dotted lines as shown in the figures (II) and (III). Then a few cuts are made as shown in figure (IV). Which one of the following patterns will be obtained when the paper is unfolded? Note: The figures shown are representative.
	 (I)  (II)  (III)  (IV)
(A)	
(B)	
(C)	
(D)	

Q.10	A shop has 4 distinct flavors of ice-cream. One can purchase any number of scoops of any flavor. The order in which the scoops are purchased is inconsequential. If one wants to purchase 3 scoops of ice-cream, in how many ways can one make that purchase?
(A)	4
(B)	20
(C)	24
(D)	48

Q.11 – Q.35 Carry ONE mark Each

Q.11	Suppose a program is running on a non-pipelined single processor computer system. The computer is connected to an external device that can interrupt the processor asynchronously. The processor needs to execute the interrupt service routine (ISR) to serve this interrupt. The following steps (not necessarily in order) are taken by the processor when the interrupt arrives: (i) The processor saves the content of the program counter. (ii) The program counter is loaded with the start address of the ISR. (iii) The processor finishes the present instruction. Which ONE of the following is the CORRECT sequence of steps?
(A)	(iii), (i), (ii)
(B)	(i), (iii), (ii)

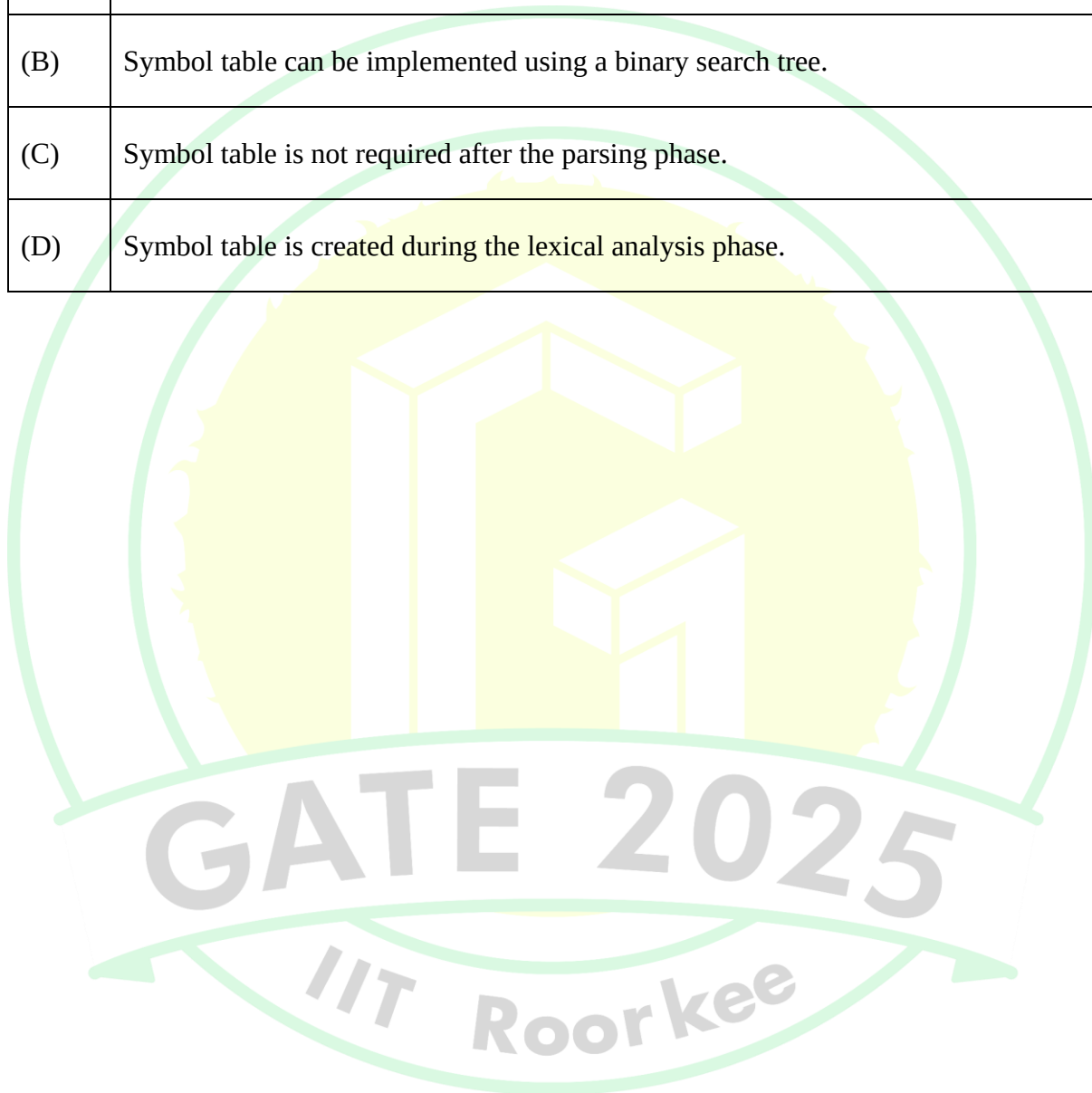


(C)	(i), (ii), (iii)
(D)	(iii), (ii), (i)



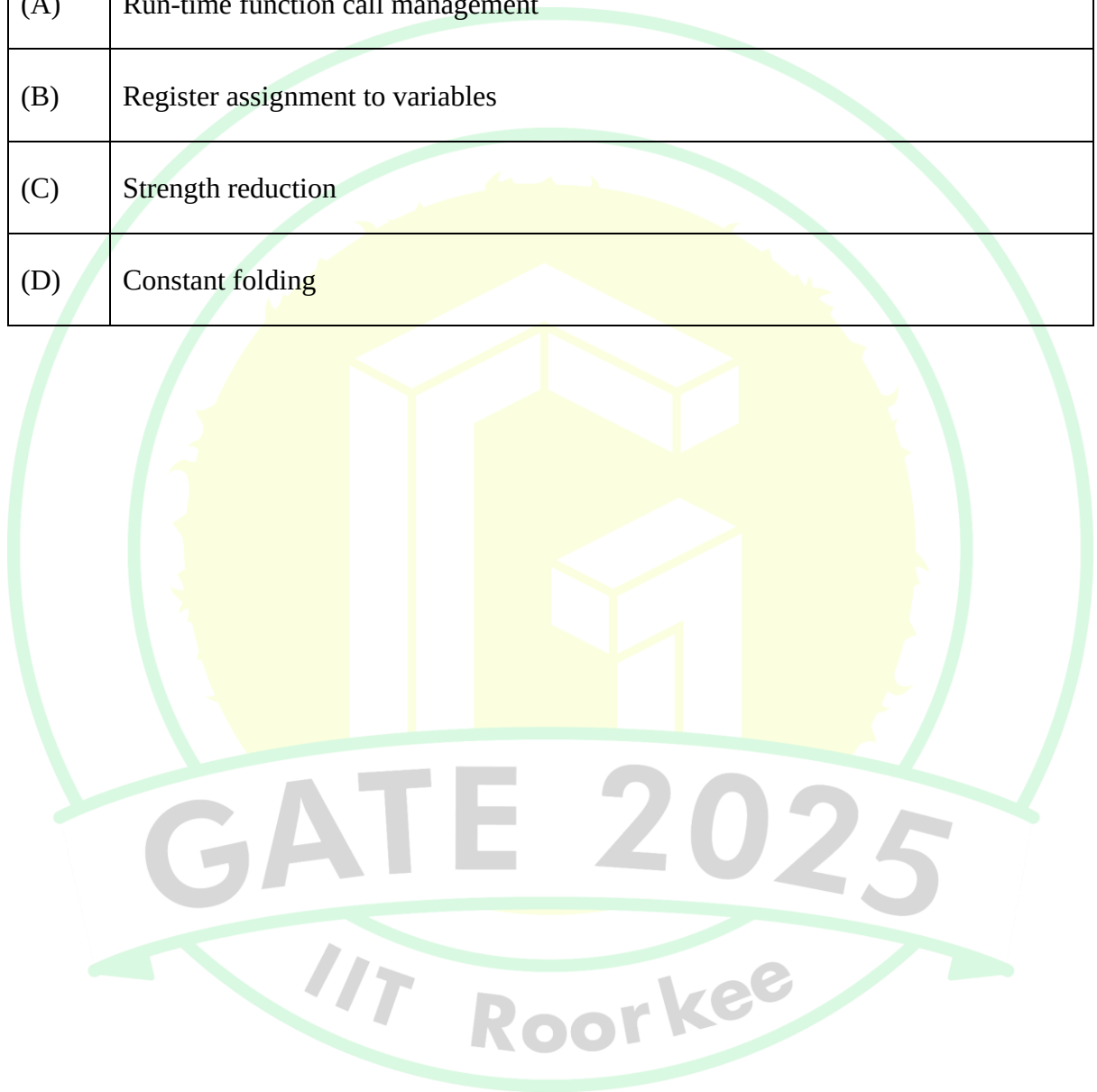


Q.12	Which ONE of the following statements is FALSE regarding the symbol table?
(A)	Symbol table is responsible for keeping track of the scope of variables.
(B)	Symbol table can be implemented using a binary search tree.
(C)	Symbol table is not required after the parsing phase.
(D)	Symbol table is created during the lexical analysis phase.



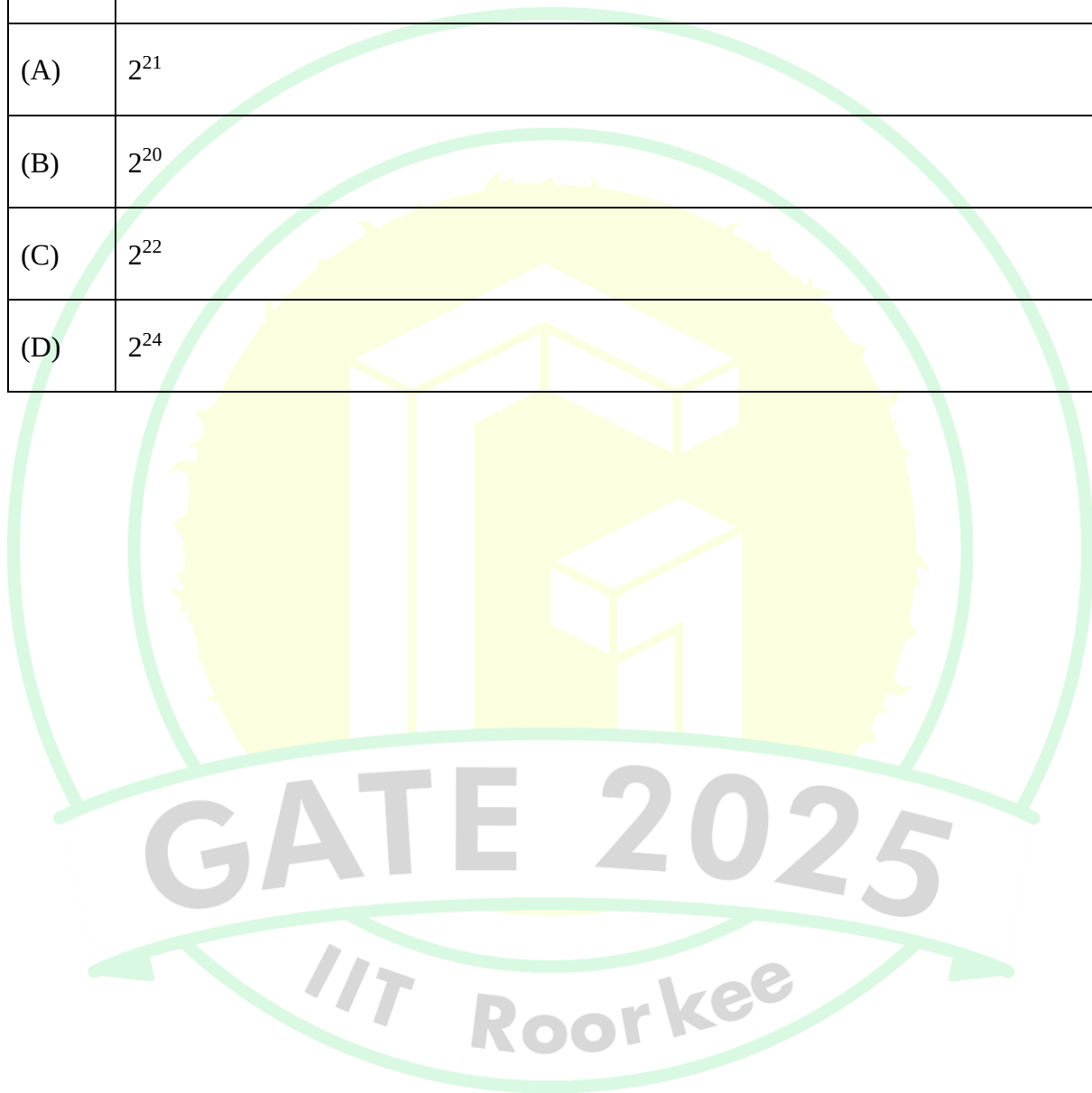


Q.13	Which ONE of the following techniques used in compiler code optimization uses live variable analysis?
(A)	Run-time function call management
(B)	Register assignment to variables
(C)	Strength reduction
(D)	Constant folding





Q.14	Consider a demand paging memory management system with 32-bit logical address, 20-bit physical address, and page size of 2048 bytes. Assuming that the memory is byte addressable, what is the maximum number of entries in the page table?
(A)	2^{21}
(B)	2^{20}
(C)	2^{22}
(D)	2^{24}



Q.15	A schedule of three database transactions T_1 , T_2 , and T_3 is shown. $R_i(A)$ and $W_i(A)$ denote read and write of data item A by transaction $T_i, i = 1,2,3$. The transaction T_1 aborts at the end. Which other transaction(s) will be required to be rolled back?
	$R_1(X) \ W_1(Y) \ R_2(X) \ R_2(Y) \ R_3(Y) \ ABORT(T_1)$
(A)	Only T_2
(B)	Only T_3
(C)	Both T_2 and T_3
(D)	Neither T_2 nor T_3

Q.16	Identify the ONE CORRECT matching between the OSI layers and their corresponding functionalities as shown.								
	<table> <tr> <th><u>OSI Layers</u></th><th><u>Functionalities</u></th></tr> <tr> <td>(a) Network layer</td><td>(I) Packet routing</td></tr> <tr> <td>(b) Transport layer</td><td>(II) Framing and error handling</td></tr> <tr> <td>(c) Datalink layer</td><td>(III) Host to host communication</td></tr> </table>	<u>OSI Layers</u>	<u>Functionalities</u>	(a) Network layer	(I) Packet routing	(b) Transport layer	(II) Framing and error handling	(c) Datalink layer	(III) Host to host communication
<u>OSI Layers</u>	<u>Functionalities</u>								
(a) Network layer	(I) Packet routing								
(b) Transport layer	(II) Framing and error handling								
(c) Datalink layer	(III) Host to host communication								
(A)	(a)-(I), (b)-(II), (c)-(III)								
(B)	(a)-(I), (b)-(III), (c)-(II)								
(C)	(a)-(II), (b)-(I), (c)-(III)								
(D)	(a)-(III), (b)-(II), (c)-(I)								

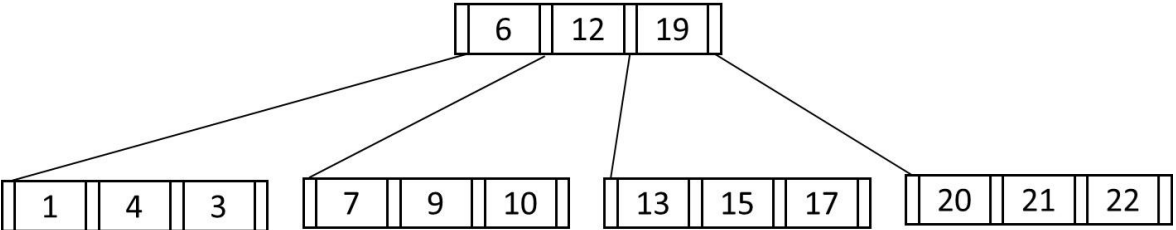


Q.17	$g(.)$ is a function from A to B, $f(.)$ is a function from B to C, and their composition defined as $f(g(.))$ is a mapping from A to C. If $f(.)$ and $f(g(.))$ are onto (surjective) functions, which ONE of the following is TRUE about the function $g(.)$?
(A)	$g(.)$ must be an onto (surjective) function.
(B)	$g(.)$ must be a one-to-one (injective) function.
(C)	$g(.)$ must be a bijective function, that is, both one-to-one and onto.
(D)	$g(.)$ is not required to be a one-to-one or onto function.

Q.18	Let G be any undirected graph with positive edge weights, and T be a minimum spanning tree of G . For any two vertices, u and v , let $d_1(u, v)$ and $d_2(u, v)$ be the shortest distances between u and v in G and T , respectively. Which ONE of the options is CORRECT for all possible G, T, u and v ?
(A)	$d_1(u, v) = d_2(u, v)$
(B)	$d_1(u, v) \leq d_2(u, v)$
(C)	$d_1(u, v) \geq d_2(u, v)$
(D)	$d_1(u, v) \neq d_2(u, v)$

Q.19	Consider the following context-free grammar G, where S, A, and B are the variables (non-terminals), a and b are the terminal symbols, S is the start variable, and the rules of G are described as: $S \rightarrow aaB \mid Abb$ $A \rightarrow a \mid aA$ $B \rightarrow b \mid bB$ Which ONE of the languages $L(G)$ is accepted by G?
(A)	$L(G) = \{a^2b^n \mid n \geq 1\} \cup \{a^nb^2 \mid n \geq 1\}$
(B)	$L(G) = \{a^nb^{2n} \mid n \geq 1\} \cup \{a^{2n}b^n \mid n \geq 1\}$
(C)	$L(G) = \{a^nb^n \mid n \geq 1\}$
(D)	$L(G) = \{a^{2n}b^{2n} \mid n \geq 1\}$

Q.20	Consider the following recurrence relation: $T(n) = 2T(n - 1) + n2^n \text{ for } n > 0, \quad T(0) = 1.$ Which ONE of the following options is CORRECT?
(A)	$T(n) = \Theta(n^2 2^n)$
(B)	$T(n) = \Theta(n 2^n)$
(C)	$T(n) = \Theta((\log n)^2 2^n)$
(D)	$T(n) = \Theta(4^n)$

Q.21	Consider the following B^+ tree with 5 nodes, in which a node can store at most 3 key values. The value 23 is now inserted in the B^+ tree. Which of the following options(s) is/are CORRECT?
	
(A)	None of the nodes will split.
(B)	At least one node will split and redistribute.
(C)	The total number of nodes will remain same.
(D)	The height of the tree will increase.



Q.22	Consider the 3-way handshaking protocol for TCP connection establishment. Let the three packets exchanged during the connection establishment be denoted as P1, P2, and P3, in order. Which of the following option(s) is/are TRUE with respect to TCP header flags that are set in the packets?
(A)	P3: SYN = 1, ACK = 1
(B)	P2: SYN = 1, ACK = 1
(C)	P2: SYN = 0, ACK = 1
(D)	P1: SYN = 1

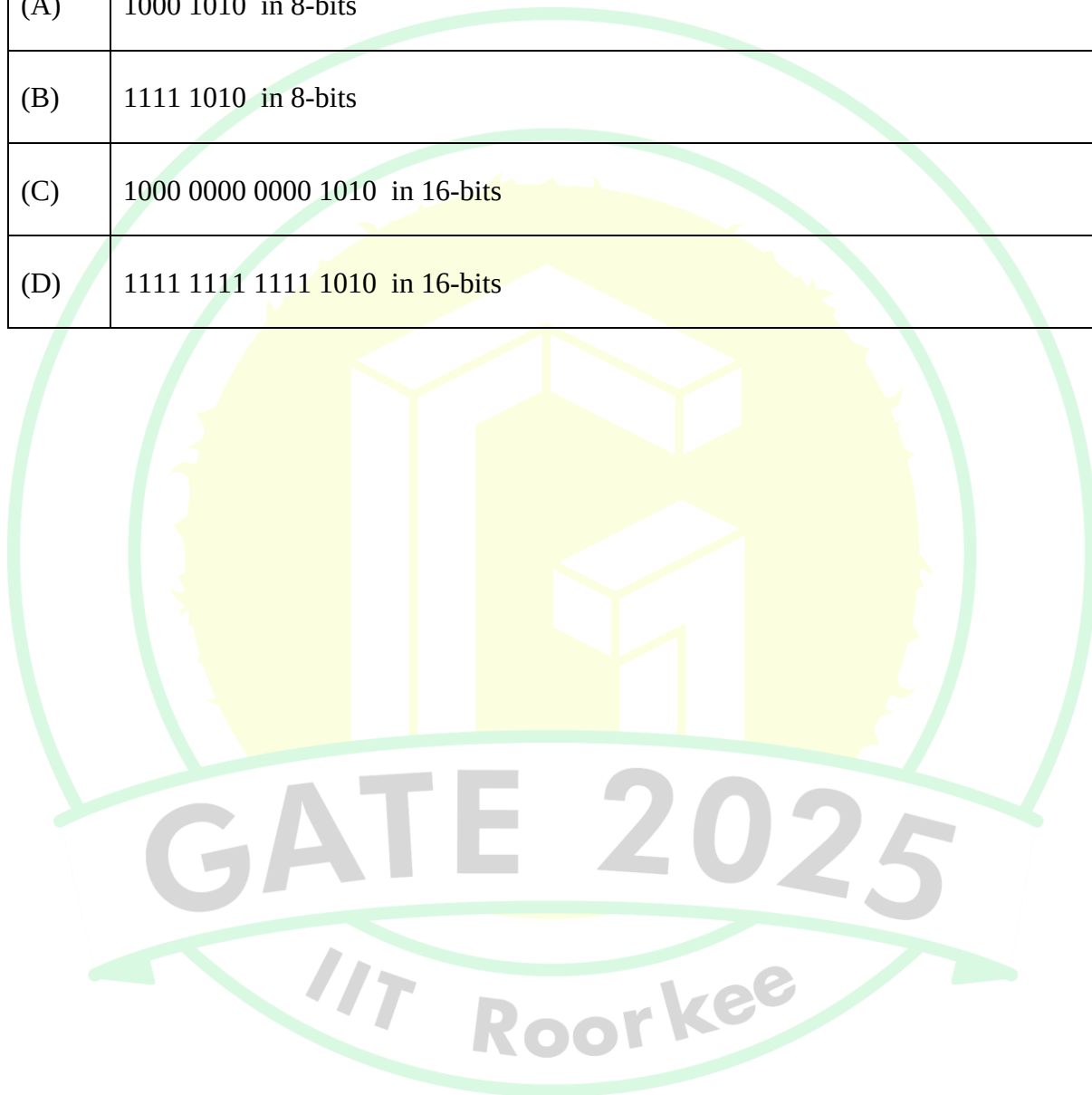


Q.23	Consider the given system of linear equations for variables x and y , where k is a real-valued constant. Which of the following option(s) is/are CORRECT?
	$\begin{aligned}x + ky &= 1 \\ kx + y &= -1\end{aligned}$
(A)	There is exactly one value of k for which the above system of equations has no solution.
(B)	There exist an infinite number of values of k for which the system of equations has no solution.
(C)	There exists exactly one value of k for which the system of equations has exactly one solution.
(D)	There exists exactly one value of k for which the system of equations has an infinite number of solutions.

Q.24	Let X be a 3-variable Boolean function that produces output as '1' when at least two of the input variables are '1'. Which of the following statement(s) is/are CORRECT, where a, b, c, d, e are Boolean variables?
(A)	$X(a, b, X(c, d, e)) = X(X(a, b, c), d, e)$
(B)	$X(a, b, X(a, b, c)) = X(a, b, c)$
(C)	$X(a, b, X(a, c, d)) = (X(a, b, a) \text{ AND } X(c, d, c))$
(D)	$X(a, b, c) = X(a, X(a, b, c), X(a, c, c))$

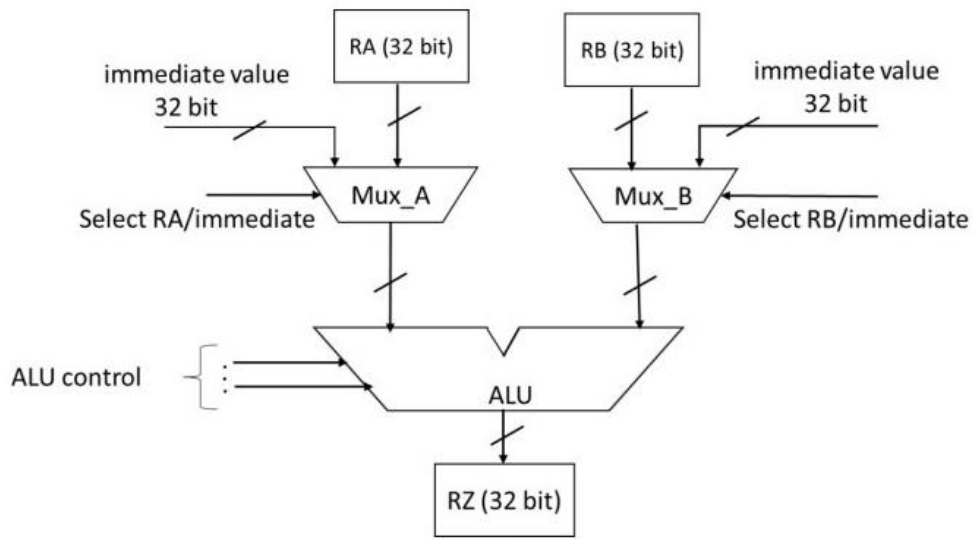


Q.25	The number -6 can be represented as 1010 in 4-bit 2's complement representation. Which of the following is/are CORRECT 2's complement representation(s) of -6 ?
(A)	1000 1010 in 8-bits
(B)	1111 1010 in 8-bits
(C)	1000 0000 0000 1010 in 16-bits
(D)	1111 1111 1111 1010 in 16-bits



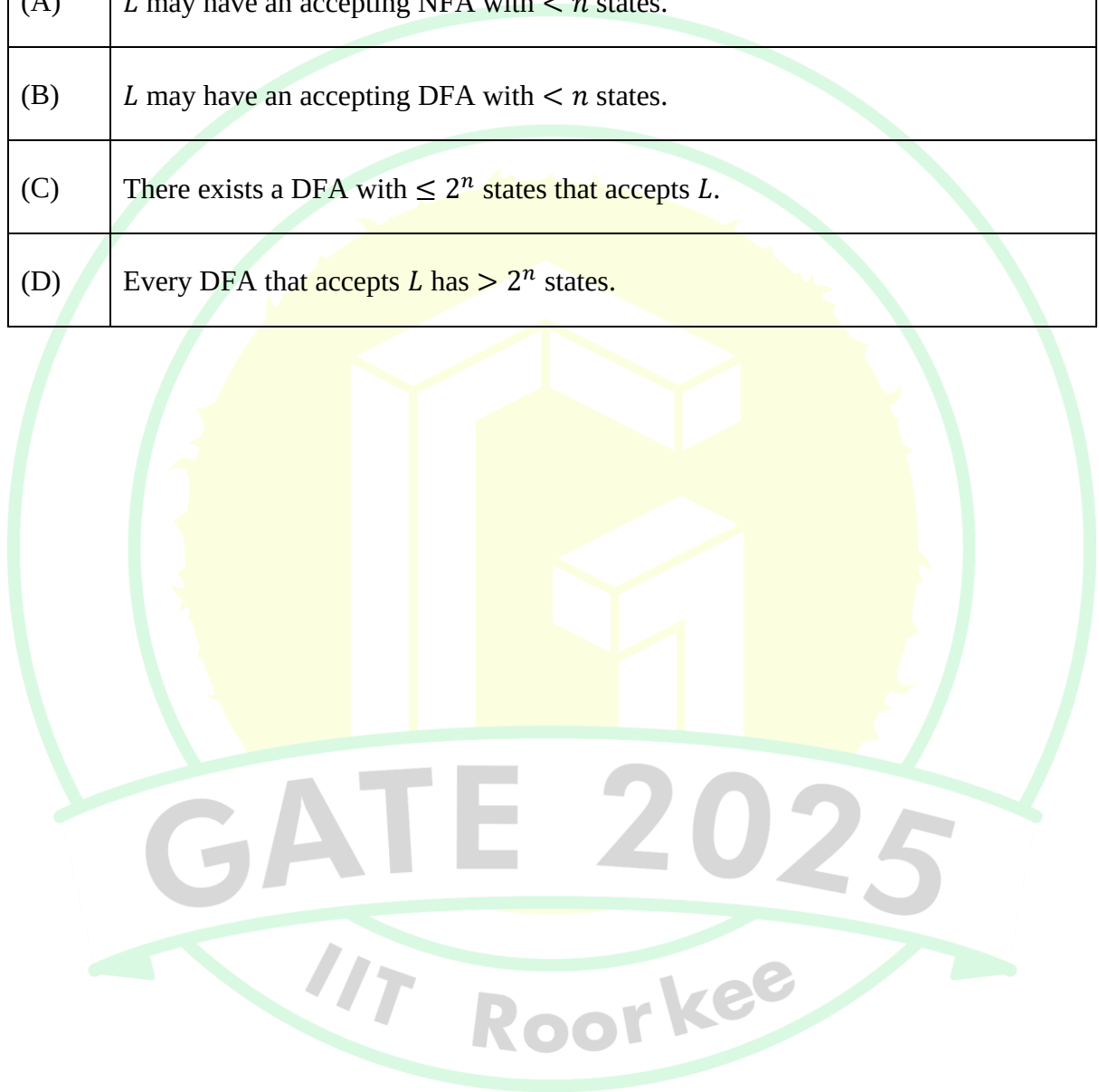


Q.26	Which of the following statement(s) is/are TRUE for any binary search tree (BST) having n distinct integers?
(A)	The maximum length of a path from the root node to any other node is $(n - 1)$.
(B)	An inorder traversal will always produce a sorted sequence of elements.
(C)	Finding an element takes $O(\log_2 n)$ time in the worst case.
(D)	Every BST is also a Min-Heap.

Q.27	A partial data path of a processor is given in the figure, where RA, RB, and RZ are 32-bit registers. Which option(s) is/are CORRECT related to arithmetic operations using the data path as shown?
	
(A)	The data path can implement arithmetic operations involving two registers.
(B)	The data path can implement arithmetic operations involving one register and one immediate value.
(C)	The data path can implement arithmetic operations involving two immediate values.
(D)	The data path can only implement arithmetic operations involving one register and one immediate value.



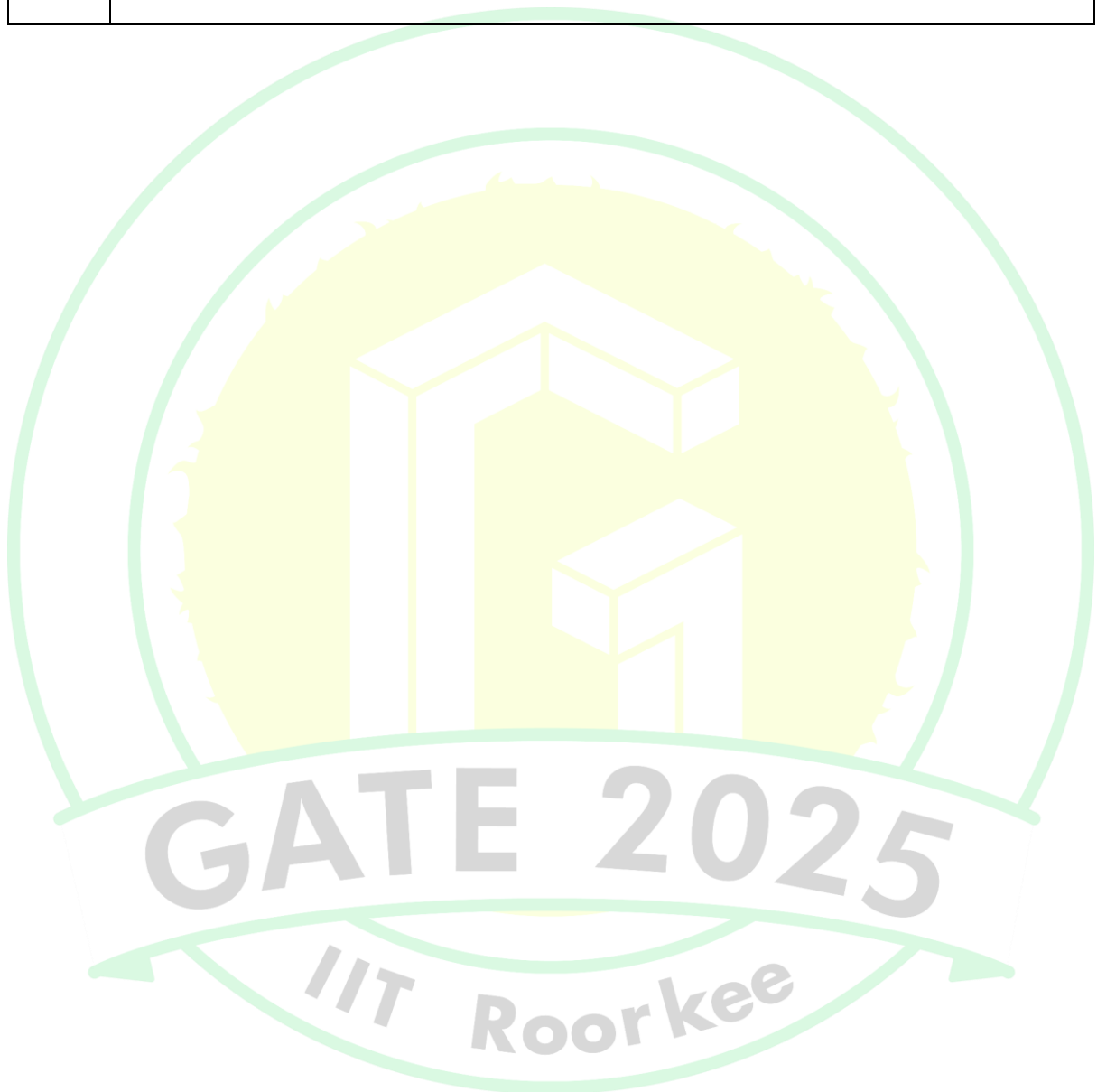
Q.28	A regular language L is accepted by a non-deterministic finite automaton (NFA) with n states. Which of the following statement(s) is/are FALSE?
(A)	L may have an accepting NFA with $< n$ states.
(B)	L may have an accepting DFA with $< n$ states.
(C)	There exists a DFA with $\leq 2^n$ states that accepts L .
(D)	Every DFA that accepts L has $> 2^n$ states.



Q.29	Suppose in a multiprogramming environment, the following C program segment is executed. A process goes into I/O queue whenever an I/O related operation is performed. Assume that there will always be a context switch whenever a process requests for an I/O, and also whenever the process returns from an I/O. The number of times the process will enter the ready queue during its lifetime (not counting the time the process enters the ready queue when it is run initially) is _____. (Answer in integer)
	<pre>int main() { int x=0,i=0; scanf("%d",&x); for(i=0; i<20; i++) { x = x+20; printf("%d\n",x); } return 0; }</pre>

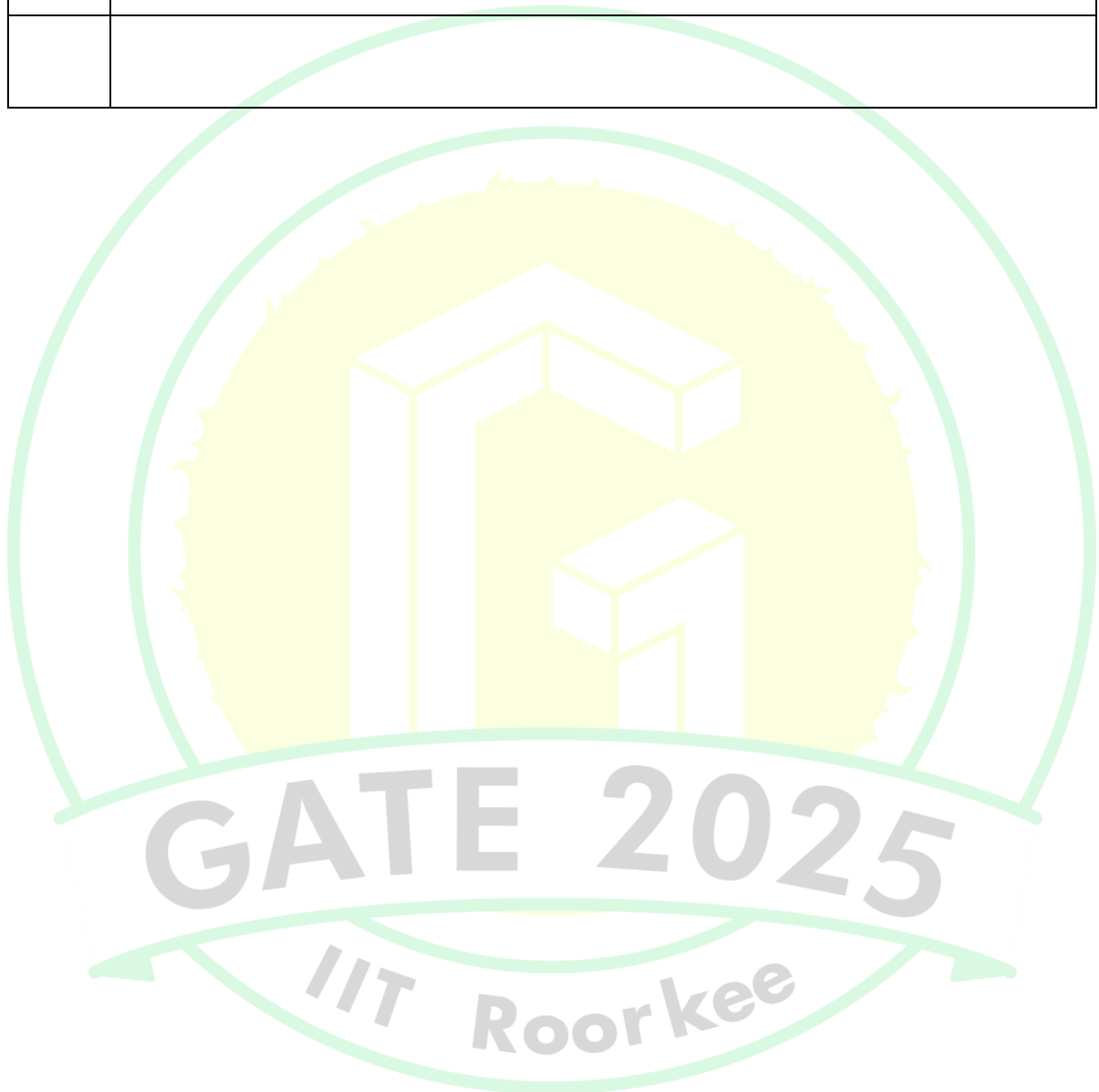


Q.30	Let S be the set of all ternary strings defined over the alphabet $\{a, b, c\}$. Consider all strings in S that contain at least one occurrence of two consecutive symbols, that is, “aa”, “bb” or “cc”. The number of such strings of length 5 that are possible is _____. (Answer in integer)





Q.31	Consider the given function $f(x)$. $f(x) = \begin{cases} ax + b & \text{for } x < 1 \\ x^3 + x^2 + 1 & \text{for } x \geq 1 \end{cases}$ If the function is differentiable everywhere, the value of b must be _____. (rounded off to one decimal place)





Q.32	A box contains 5 coins: 4 regular coins and 1 fake coin. When a regular coin is tossed, the probability $P(head) = 0.5$ and for a fake coin, $P(head) = 1$. You pick a coin at random and toss it twice, and get two heads. The probability that the coin you have chosen is the fake coin is _____. (rounded off to two decimal places)
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Q.33

The pseudocode of a function **fun ()** is given below:

```
fun(int A[0,...,n-1]){  
    for i=0 to n-2  
        for j=0 to n-i-2  
            if (A[j]>A[j+1])  
                then swap A[j] and A[j+1]  
}
```

Let $A[0, \dots, 29]$ be an array storing 30 distinct integers in descending order. The number of swap operations that will be performed, if the function **fun ()** is called with $A[0, \dots, 29]$ as argument, is _____. (Answer in integer)



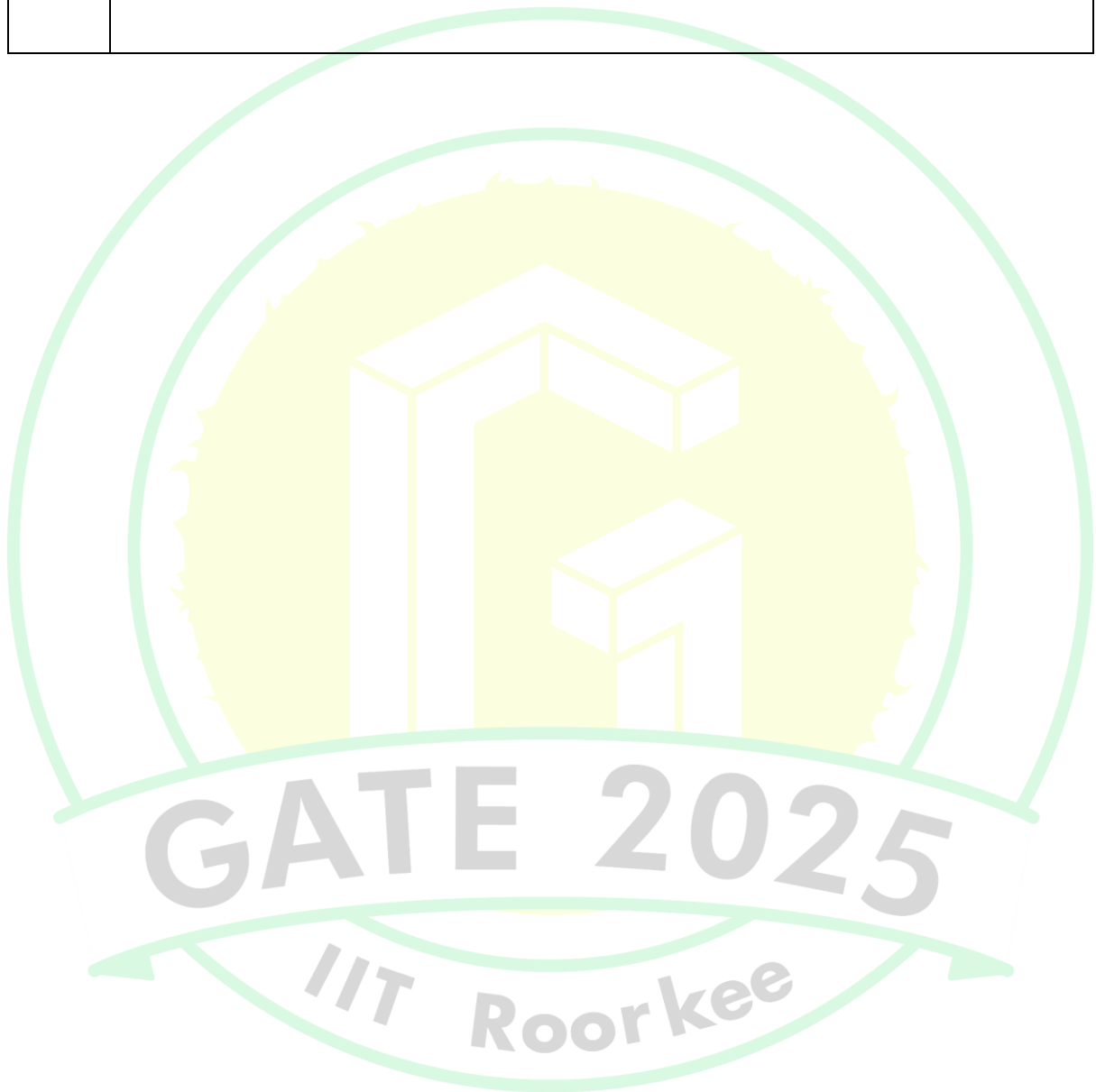
Q.34

```
#include <stdio.h>
void foo(int *p, int x){
    *p=x;
}
int main(){
    int *z;
    int a = 20, b = 25;
    z = &a;
    foo(z,b);
    printf("%d",a);
    return 0;
}
```

The output of the given C program is _____. (Answer in integer)



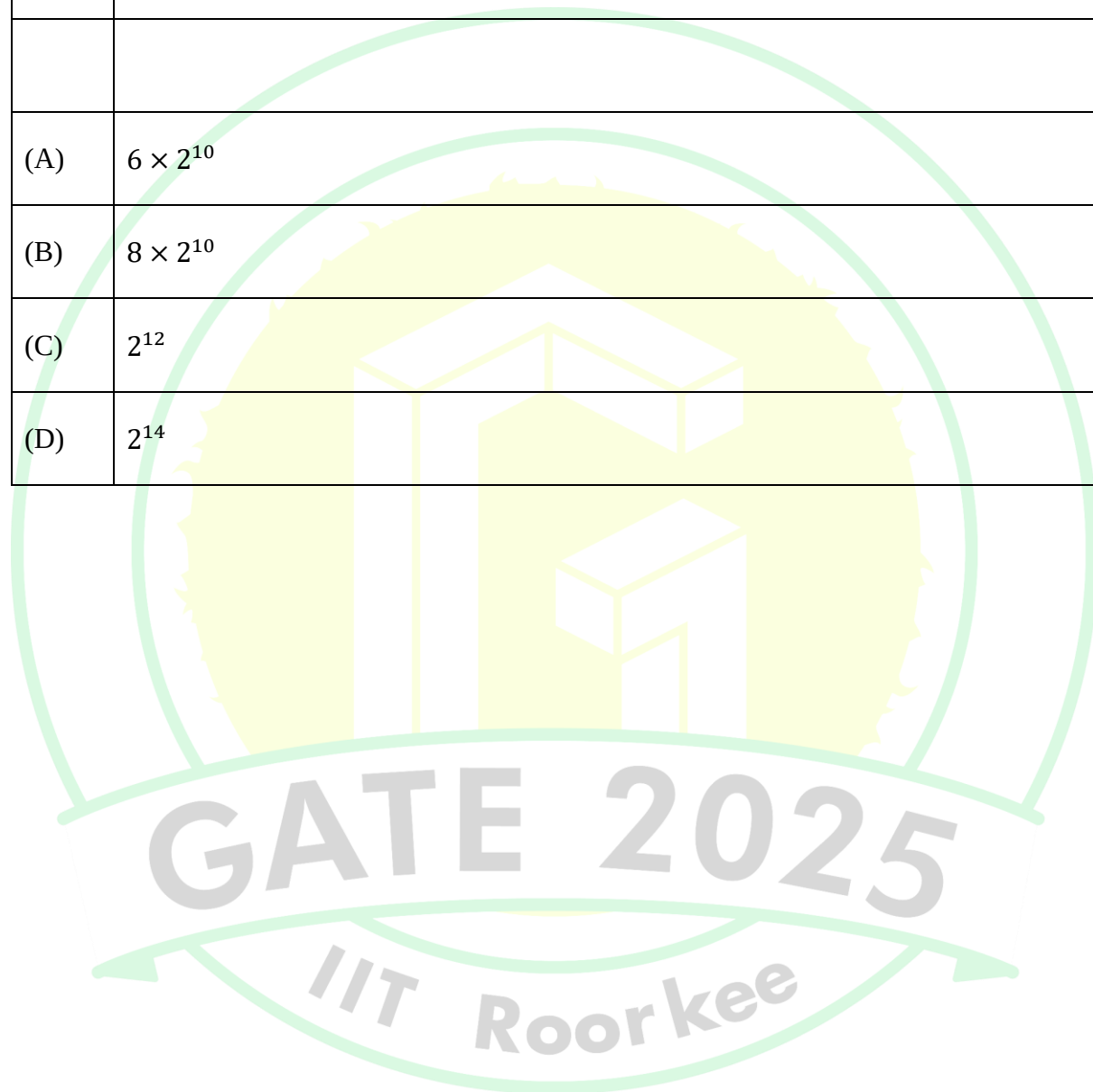
Q.35	The height of any rooted tree is defined as the maximum number of edges in the path from the root node to any leaf node. Suppose a Min-Heap T stores 32 keys. The height of T is _____. (Answer in integer)



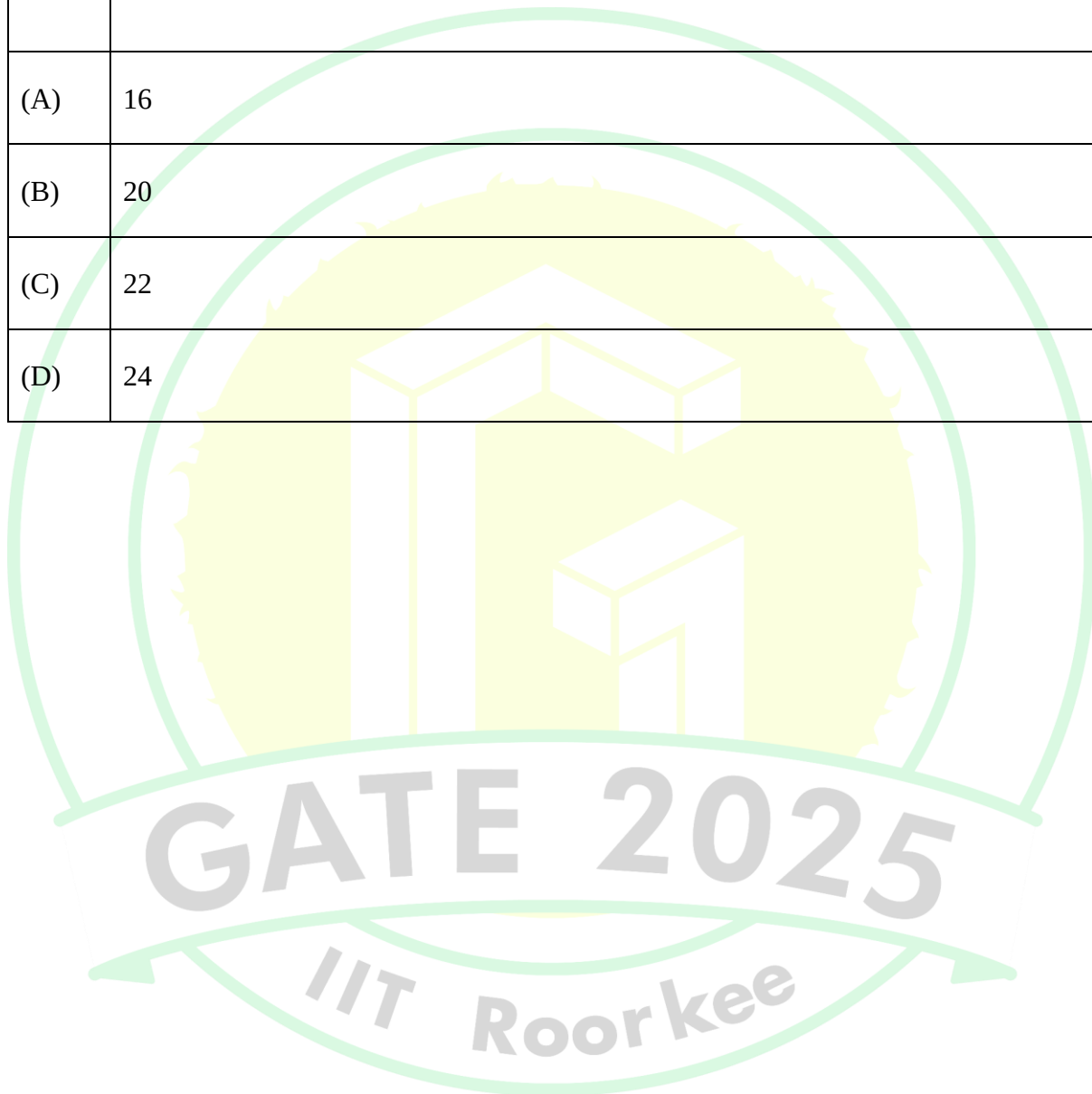


Q.36 – Q.65 Carry TWO marks Each

Q.36	Consider a memory system with 1M bytes of main memory and 16K bytes of cache memory. Assume that the processor generates 20-bit memory address, and the cache block size is 16 bytes. If the cache uses direct mapping, how many bits will be required to store all the <i>tag</i> values? [Assume memory is byte addressable, $1K=2^{10}$, $1M=2^{20}$.]
(A)	6×2^{10}
(B)	8×2^{10}
(C)	2^{12}
(D)	2^{14}



Q.37	A processor has 64 general-purpose registers and 50 distinct instruction types. An instruction is encoded in 32-bits. What is the maximum number of bits that can be used to store the immediate operand for the given instruction? ADD R1, #25 // R1 = R1 + 25
(A)	16
(B)	20
(C)	22
(D)	24



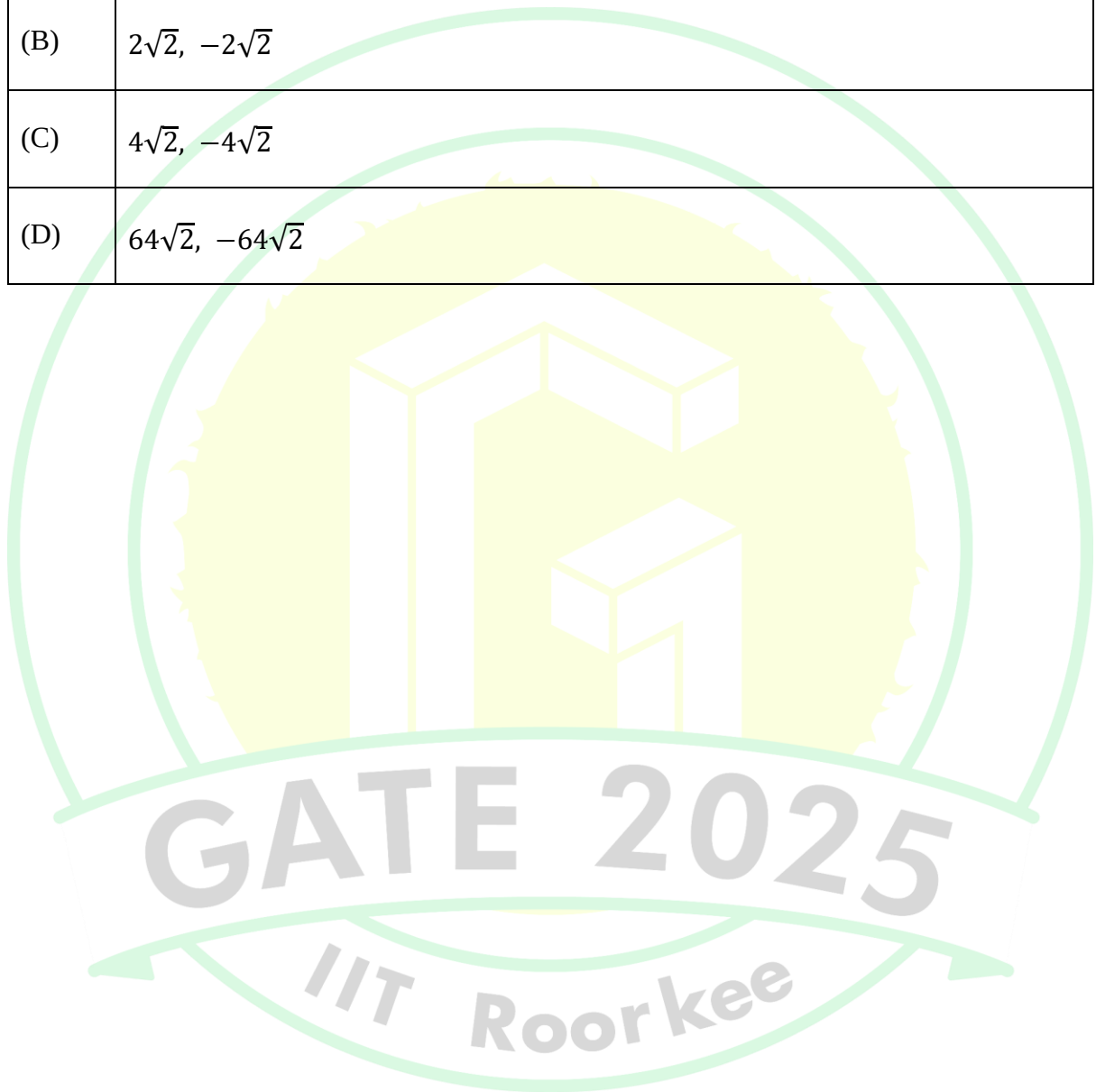
Q.38	A computer has two processors, M_1 and M_2. Four processes P_1, P_2, P_3, P_4 with CPU bursts of 20, 16, 25, and 10 milliseconds, respectively, arrive at the same time and these are the only processes in the system. The scheduler uses non-preemptive priority scheduling, with priorities decided as follows: • M_1 uses priority of execution for the processes as, $P_1 > P_3 > P_2 > P_4$, i.e., P_1 and P_4 have highest and lowest priorities, respectively. • M_2 uses priority of execution for the processes as, $P_2 > P_3 > P_4 > P_1$, i.e., P_2 and P_1 have highest and lowest priorities, respectively. A process P_i is scheduled to a processor M_k, if the processor is free and no other process P_j is waiting with higher priority. At any given point of time, a process can be allocated to any one of the free processors without violating the execution priority rules. Ignore the context switch time. What will be the average waiting time of the processes in milliseconds?
(A)	9.00
(B)	8.75
(C)	6.50
(D)	7.50

Q.39	Consider two relations describing <i>teams</i> and <i>players</i> in a sports league: <i>teams</i>(<i>tid</i>, <i>tname</i>): <i>tid</i>, <i>tname</i> are team-id and team-name, respectively <i>players</i>(<i>pid</i>, <i>pname</i>, <i>tid</i>): <i>pid</i>, <i>pname</i>, and <i>tid</i> denote player-id, player-name and the team-id of the player, respectively Which ONE of the following tuple relational calculus queries returns the name of the players who play for the team having <i>tname</i> as 'MI'?
(A)	$\{ p.pname \mid p \in players \wedge \exists t (t \in teams \wedge p.tid = t.tid \wedge t.tname = 'MI') \}$
(B)	$\{ p.pname \mid p \in teams \wedge \exists t (t \in players \wedge p.tid = t.tid \wedge t.tname = 'MI') \}$
(C)	$\{ p.pname \mid p \in players \wedge \exists t (t \in teams \wedge t.tname = 'MI') \}$
(D)	$\{ p.pname \mid p \in teams \wedge \exists t (t \in players \wedge t.tname = 'MI') \}$

Q.40	A packet with the destination IP address 145.36.109.70 arrives at a router whose routing table is shown. Which interface will the packet be forwarded to?																				
	<table><tr><th>Subnet Address</th><th>Subnet Mask (in CIDR notation)</th><th>Interface</th></tr><tr><td>145.36.0.0</td><td>/16</td><td>E1</td></tr><tr><td>145.36.128.0</td><td>/17</td><td>E2</td></tr><tr><td>145.36.64.0</td><td>/18</td><td>E3</td></tr><tr><td>145.36.255.0</td><td>/24</td><td>E4</td></tr><tr><td>Default</td><td>--</td><td>E5</td></tr></table>			Subnet Address	Subnet Mask (in CIDR notation)	Interface	145.36.0.0	/16	E1	145.36.128.0	/17	E2	145.36.64.0	/18	E3	145.36.255.0	/24	E4	Default	--	E5
Subnet Address	Subnet Mask (in CIDR notation)	Interface																			
145.36.0.0	/16	E1																			
145.36.128.0	/17	E2																			
145.36.64.0	/18	E3																			
145.36.255.0	/24	E4																			
Default	--	E5																			
(A)	E3																				
(B)	E1																				
(C)	E2																				
(D)	E5																				



Q.41	Let A be a 2×2 matrix as given. $A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ What are the eigenvalues of the matrix A^{13} ?
(A)	1, -1
(B)	$2\sqrt{2}$, $-2\sqrt{2}$
(C)	$4\sqrt{2}$, $-4\sqrt{2}$
(D)	$64\sqrt{2}$, $-64\sqrt{2}$



Q.42	Consider the following four variable Boolean function in sum-of-product form $F(b_3, b_2, b_1, b_0) = \sum(0, 2, 4, 8, 10, 11, 12).$ where the value of the function is computed by considering $b_3b_2b_1b_0$ as a 4-bit binary number, where b_3 denotes the most significant bit and b_0 denotes the least significant bit. Note that there are no don't care terms. Which ONE of the following options is the CORRECT minimized Boolean expression for F?
(A)	$\bar{b}_1\bar{b}_0 + \bar{b}_2\bar{b}_0 + b_1\bar{b}_2b_3$
(B)	$\bar{b}_1\bar{b}_0 + \bar{b}_2\bar{b}_0$
(C)	$\bar{b}_2\bar{b}_0 + b_1b_2b_3$
(D)	$\bar{b}_0\bar{b}_2 + \bar{b}_3$

Q.43	Let $G(V, E)$ be an undirected and unweighted graph with 100 vertices. Let $d(u, v)$ denote the number of edges in a shortest path between vertices u and v in V . Let the maximum value of $d(u, v)$, $u, v \in V$ such that $u \neq v$, be 30. Let T be any breadth-first-search tree of G . Which ONE of the given options is CORRECT for every such graph G ?
(A)	The height of T is exactly 15.
(B)	The height of T is exactly 30.
(C)	The height of T is at least 15.
(D)	The height of T is at least 30.

Q.44	Consider the following two languages over the alphabet $\{a, b\}$: $L_1 = \{ \alpha\beta\alpha \mid \alpha \in \{a, b\}^+ \text{ AND } \beta \in \{a, b\}^+ \}$ $L_2 = \{ \alpha\beta\alpha \mid \alpha \in \{a\}^+ \text{ AND } \beta \in \{a, b\}^+ \}$ Which ONE of the following statements is CORRECT?
(A)	Both L_1 and L_2 are regular languages.
(B)	L_1 is a regular language but L_2 is not a regular language.
(C)	L_1 is not a regular language but L_2 is a regular language.
(D)	Neither L_1 nor L_2 is a regular language.

Q.45	Consider the following two languages over the alphabet $\{a, b, c\}$, where m and n are natural numbers. $L_1 = \{a^m b^m c^{m+n} \mid m, n \geq 1\}$ $L_2 = \{a^m b^n c^{m+n} \mid m, n \geq 1\}$ Which ONE of the following statements is CORRECT?
(A)	Both L_1 and L_2 are context-free languages.
(B)	L_1 is a context-free language but L_2 is not a context-free language.
(C)	L_1 is not a context-free language but L_2 is a context-free language.
(D)	Neither L_1 nor L_2 are context-free languages.



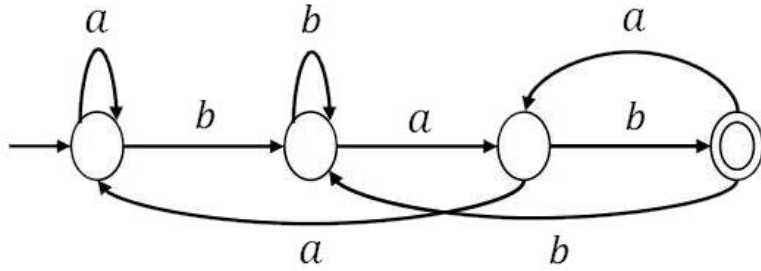
Q.46	Which of the following statement(s) is/are TRUE while computing <i>First</i> and <i>Follow</i> during top down parsing by a compiler?
(A)	For a production $A \rightarrow \epsilon$, ϵ will be added to $First(A)$.
(B)	If there is any input right end marker, it will be added to $First(S)$, where S is the start symbol.
(C)	For a production $A \rightarrow \epsilon$, ϵ will be added to $Follow(A)$.
(D)	If there is any input right end marker, it will be added to $Follow(S)$, where S is the start symbol.



Q.47	Consider a relational schema $team(name, city, owner)$, with functional dependencies $\{name \rightarrow city, name \rightarrow owner\}$. The relation $team$ is decomposed into two relations, $t1(name, city)$ and $t2(name, owner)$. Which of the following statement(s) is/are TRUE?
(A)	The relation $team$ is NOT in BCNF.
(B)	The relations $t1$ and $t2$ are in BCNF.
(C)	The decomposition constitutes a lossless join.
(D)	The relation $team$ is NOT in 3NF.

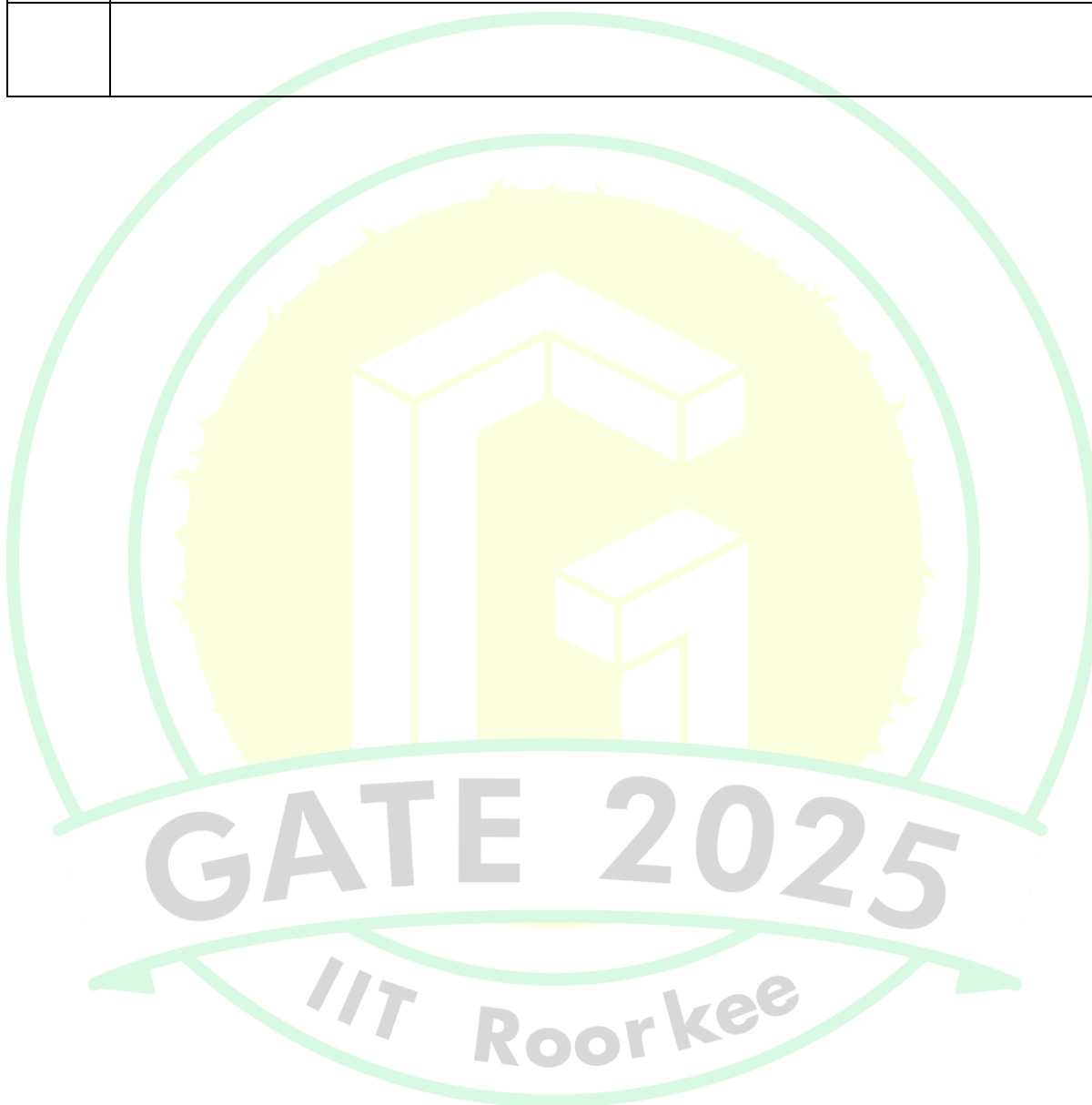
Q.48	Which of the following predicate logic formulae/formula is/are CORRECT representation(s) of the statement: “Everyone has exactly one mother”? The meanings of the predicates used are: • $mother(y, x)$: y is the mother of x • $noteq(x, y)$: x and y are not equal
(A)	$\forall x \exists y \exists z (mother(y, x) \wedge \neg mother(z, x))$
(B)	$\forall x \exists y [mother(y, x) \wedge \forall z (noteq(z, y) \rightarrow \neg mother(z, x))]$
(C)	$\forall x \forall y [mother(y, x) \rightarrow \exists z (mother(z, x) \wedge \neg noteq(z, y))]$
(D)	$\forall x \exists y [mother(y, x) \wedge \neg \exists z (noteq(z, y) \wedge mother(z, x))]$

Q.49	$A = \{0, 1, 2, 3, \dots\}$ is the set of non-negative integers. Let F be the set of functions from A to itself. For any two functions, $f_1, f_2 \in F$, we define $(f_1 \odot f_2)(n) = f_1(n) + f_2(n)$ for every number n in A. Which of the following is/are CORRECT about the mathematical structure $(F, \odot)$?
(A)	$(F, \odot)$ is an Abelian group.
(B)	$(F, \odot)$ is an Abelian monoid.
(C)	$(F, \odot)$ is a non-Abelian group.
(D)	$(F, \odot)$ is a non-Abelian monoid.

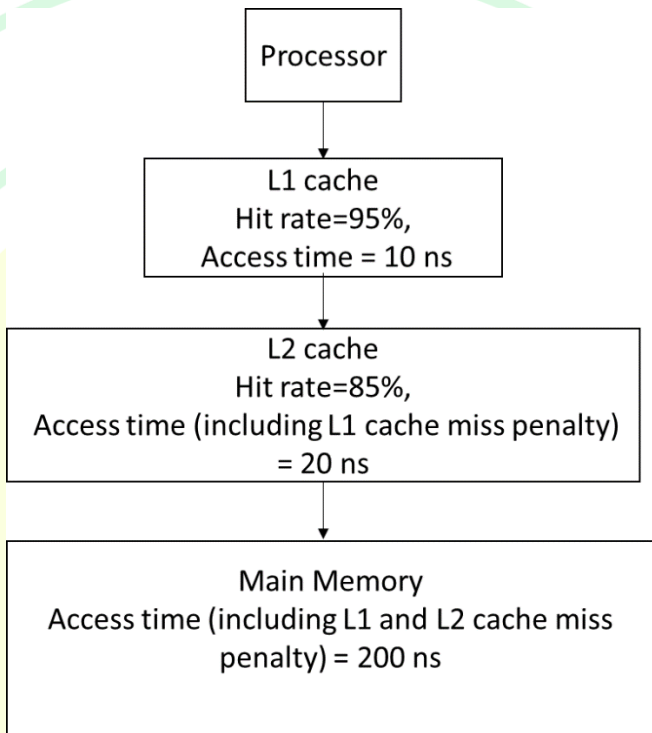
Q.50	Consider the following deterministic finite automaton (DFA) defined over the alphabet, $\Sigma = \{a, b\}$. Identify which of the following language(s) is/are accepted by the given DFA.
	
(A)	The set of all strings containing an even number of b 's.
(B)	The set of all strings containing the pattern bab .
(C)	The set of all strings ending with the pattern bab .
(D)	The set of all strings not containing the pattern aba .



Q.51	A disk of size 512M bytes is divided into blocks of 64K bytes. A file is stored in the disk using linked allocation. In linked allocation, each data block reserves 4 bytes to store the pointer to the next data block. The link part of the last data block contains a <i>NULL</i> pointer (also of 4 bytes). Suppose a file of 1M bytes needs to be stored in the disk. Assume, $1K = 2^{10}$ and $1M = 2^{20}$. The amount of space in bytes that will be wasted due to internal fragmentation is _____. (Answer in integer)



Q.52	Refer to the given 3-address code sequence. This code sequence is split into basic blocks. The number of basic blocks is _____. (Answer in integer)
	<pre> 1001: i = 1 1002: j = 1 1003: t1 = 10*i 1004: t2 = t1+j 1005: t3 = 8*t2 1006: t4 = t3-88 1007: a[t4] = 0.0 1008: j = j+1 1009: if j <= 10 goto 1003 1010: i = i+1 1011: if i <= 10 goto 1002 1012: i = 1 1013: t5 = i-1 1014: t6 = 88*t5 1015: a[t6] = 1.0 1016: i = i+1 1017: if i <= 10 goto 1013 </pre>

Q.53	A computer has a memory hierarchy consisting of two-level cache (L1 and L2) and a main memory. If the processor needs to access data from memory, it first looks into L1 cache. If the data is not found in L1 cache, it goes to L2 cache. If it fails to get the data from L2 cache, it goes to main memory, where the data is definitely available. <i>Hit rates</i> and <i>access times</i> of various memory units are shown in the figure. The <i>average memory access time</i> in nanoseconds (<i>ns</i>) is _____. (rounded off to two decimal places)
	 <pre> graph TD Processor[Processor] --> L1[L1 cache Hit rate=95%, Access time = 10 ns] L1 --> L2[L2 cache Hit rate=85%, Access time (including L1 cache miss penalty) = 20 ns] L2 --> MM[Main Memory Access time (including L1 and L2 cache miss penalty) = 200 ns] </pre>
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Q.54	In optimal page replacement algorithm, information about all future page references is available to the operating system (OS). A modification of the optimal page replacement algorithm is as follows: <i>The OS correctly predicts only up to next 4 page references (including the current page) at the time of allocating a frame to a page.</i> A process accesses the pages in the following order of page numbers: 1, 3, 2, 4, 2, 3, 1, 2, 4, 3, 1, 4. If the system has three memory frames that are initially empty, the number of page faults that will occur during execution of the process is _____. (Answer in integer)

Q.55

Consider the following database tables of a sports league.

player (pid, pname, age)
coach (cid, cname)

team (tid, tname, city, cid)
members (pid, tid)

An instance of the table and an SQL query are given.

player			coach		team				members	
pid	pname	age	cid	cname	tid	tname	city	cid	pid	tid
1	Jaspri	31	101	Ricky	10	MI	Mumbai	102	1	10
2	Atharva	24	102	Mark	20	DC	Delhi	101	2	30
3	Ishan	26	103	Trevor	30	PK	Mohali	103	3	10
4	Axar	30							4	20

```

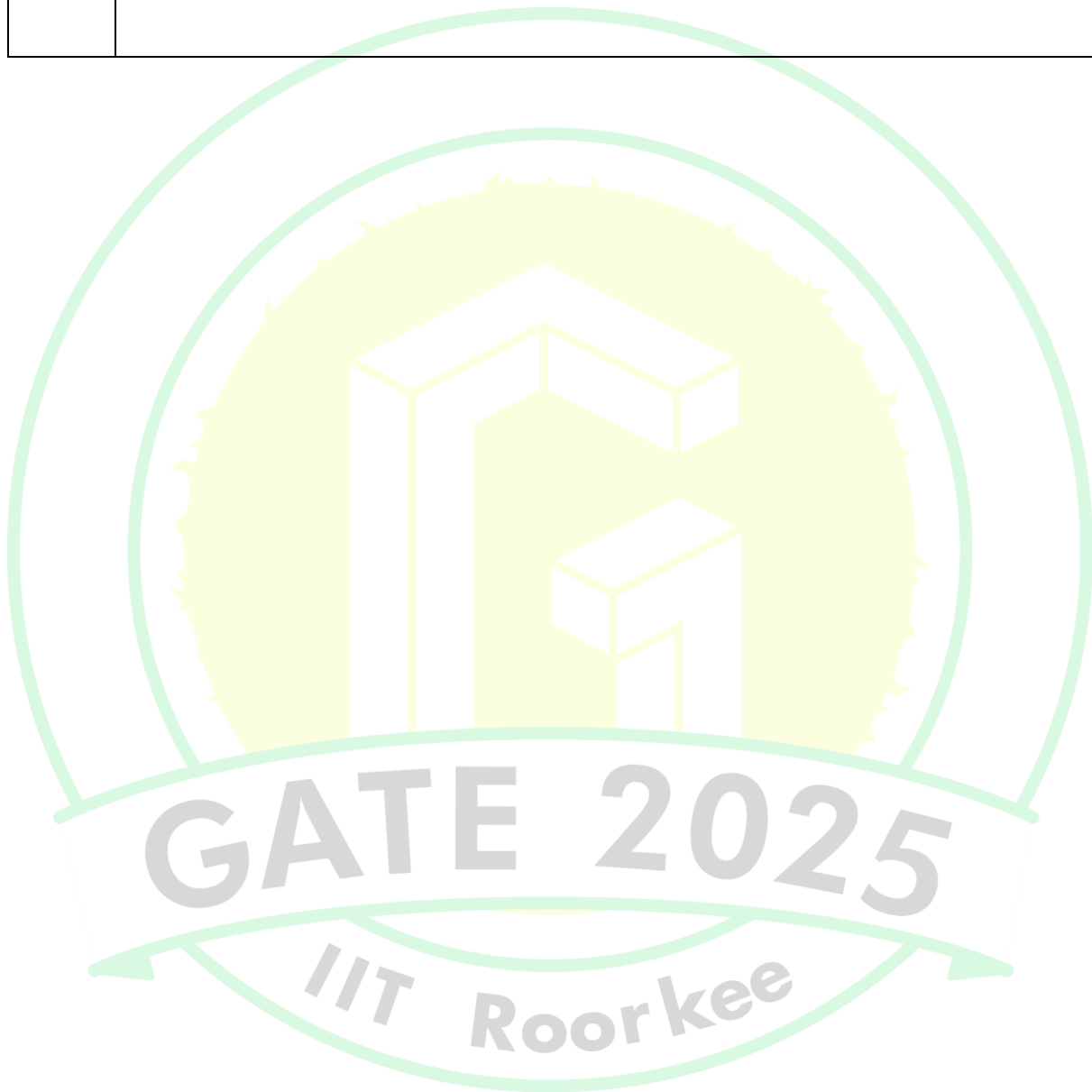
SELECT MIN(P.age)
FROM player P
WHERE P.pid IN (
    SELECT M.pid
    FROM team T, coach C, members M
    WHERE C.cname = 'Mark'
        AND T.cid = C.cid
        AND M.tid = T.tid
)

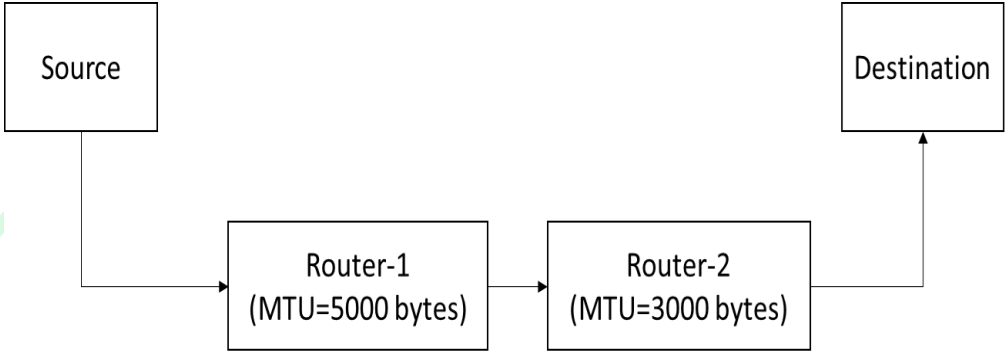
```

The value returned by the given SQL query is _____. (Answer in integer)

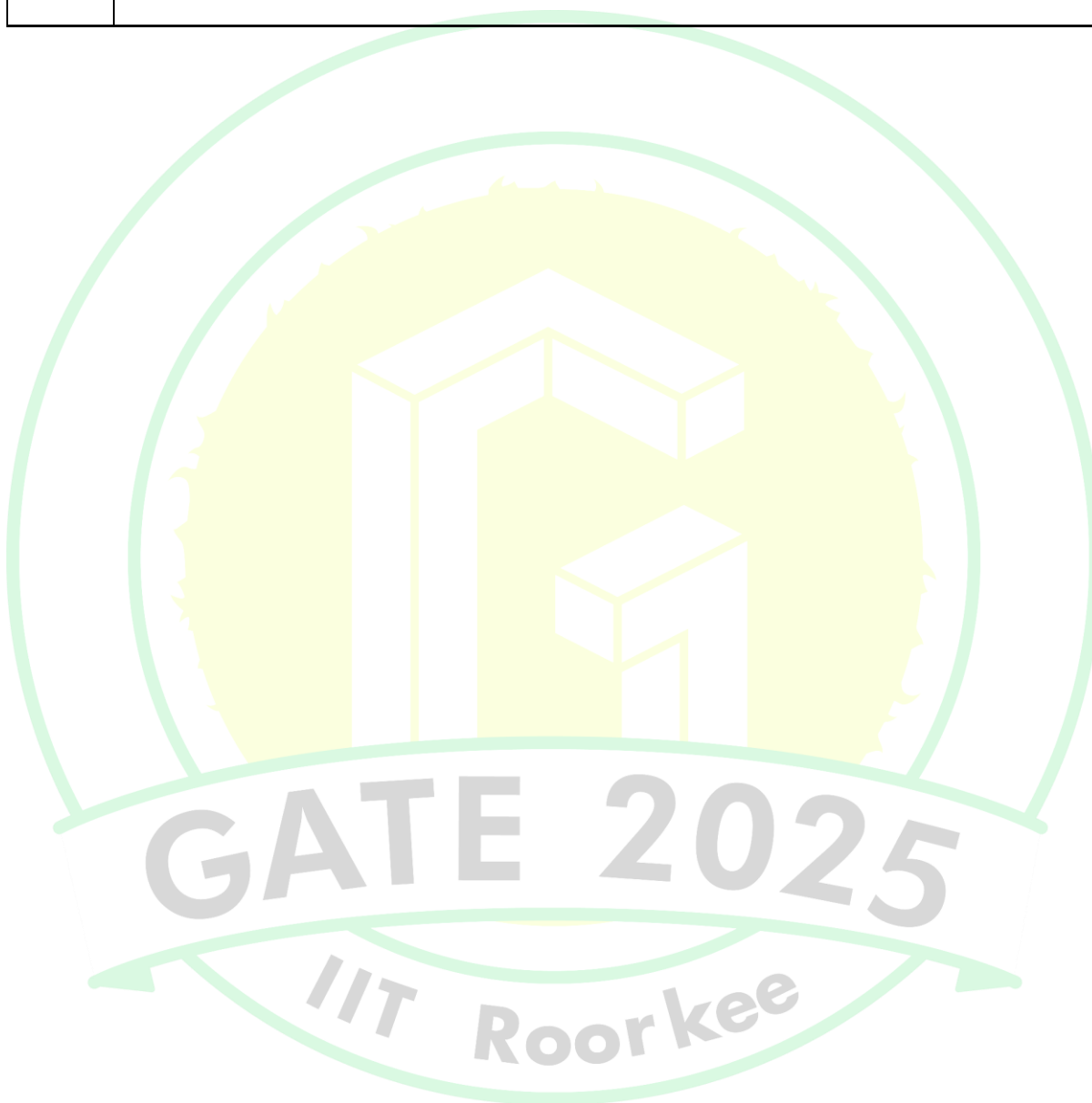


Q.56	Suppose a 5-bit message is transmitted from a source to a destination through a noisy channel. The probability that a bit of the message gets flipped during transmission is 0.01. Flipping of each bit is independent of one another. The probability that the message is delivered error-free to the destination is _____. (rounded off to three decimal places)



Q.57	Suppose a message of size 15000 bytes is transmitted from a source to a destination using IPv4 protocol via two routers as shown in the figure. Each router has a defined maximum transmission unit (MTU) as shown in the figure, including IP header. The number of fragments that will be delivered to the destination is _____. (Answer in integer)
	 <pre> graph LR Source[Source] --> Router1[Router-1 (MTU=5000 bytes)] Router1 --> Router2[Router-2 (MTU=3000 bytes)] Router2 --> Destination[Destination] </pre>

Q.58	Consider a probability distribution given by the density function $P(x)$. $P(x) = \begin{cases} Cx^2, & \text{for } 1 \leq x \leq 4 \\ 0, & \text{for } x < 1 \text{ or } x > 4 \end{cases}$ The probability that x lies between 2 and 3, i.e., $P(2 \leq x \leq 3)$ is _____. (rounded off to three decimal places)
------	---

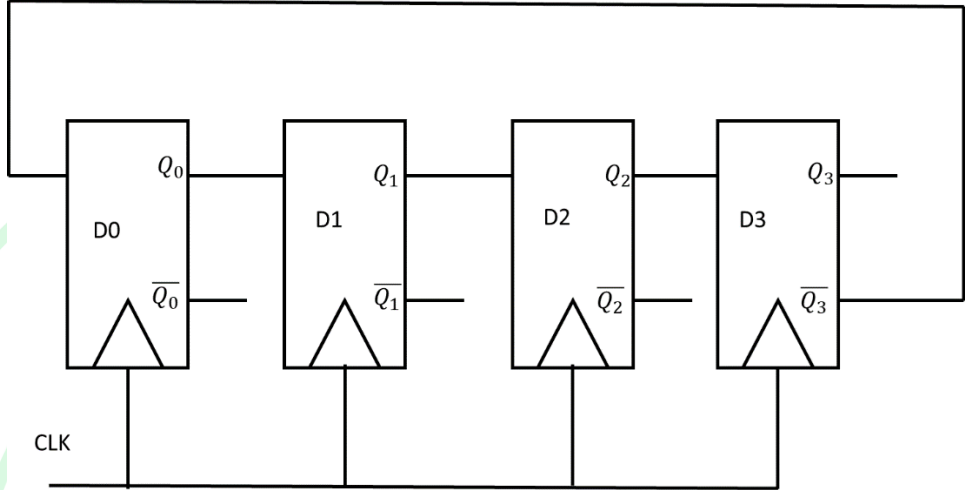


Q.59

Consider a finite state machine (FSM) with one input X and one output f , represented by the given state transition table. The minimum number of states required to realize this FSM is _____. (Answer in integer)

Present state	Next state		Output f	
	$X = 0$	$X = 1$	$X = 0$	$X = 1$
A	F	B	0	0
B	D	C	0	0
C	F	E	0	0
D	G	A	1	0
E	D	C	0	0
F	F	B	1	1
G	G	H	0	1
H	G	A	1	0

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Q.60	Consider the given sequential circuit designed using D-Flip-flops. The circuit is initialized with some value (initial state). The number of distinct states the circuit will go through before returning back to the initial state is _____. (Answer in integer)
	

Q.61

```
#include <stdio.h>
int foo(int S[],int size){
    if(size == 0) return 0;
    if(size == 1) return 1;
    if(S[0] != S[1]) return 1+foo(S+1,size-1);
    return foo(S+1,size-1);
}
int main(){
    int A[]={0,1,2,2,2,0,0,1,1};
    printf("%d",foo(A,9));
    return 0;
}
```

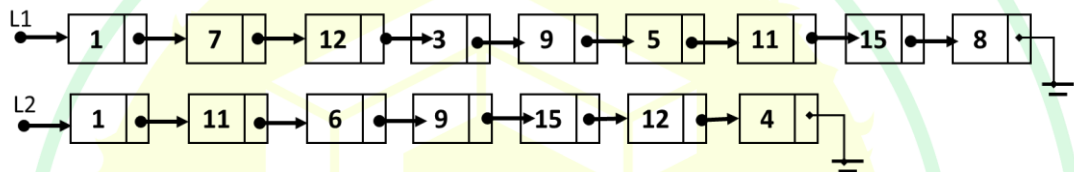
The value printed by the given C program is _____. (Answer in integer)

Q.62 Let **LIST** be a datatype for an implementation of linked list defined as follows:

```
typedef struct list {
    int data;
    struct list *next;
} LIST;
```

Suppose a program has created two linked lists, L1 and L2, whose contents are given in the figure below (code for creating L1 and L2 is not provided here). L1 contains 9 nodes, and L2 contains 7 nodes.

Consider the following C program segment that modifies the list L1. The number of nodes that will be there in L1 after the execution of the code segment is _____. (Answer in integer)



```
int find (int query, LIST *list) {
    while (list != NULL) {
        if(list->data == query) return 1;
        list = list->next;
    }
    return 0;
}

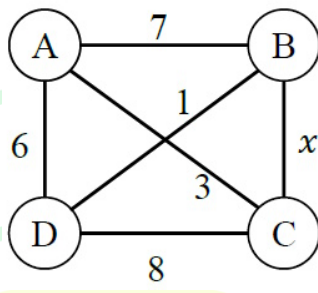
int main () {
    ... ..
    ptr1=L1; ptr2=L2;
    while (ptr1->next != NULL){
        query = ptr1->next->data;
        if (find (query, L2))
            ptr1->next = ptr1->next->next;
        else ptr1 = ptr1->next;
    }
    ... ..
    return 0;
}
```

Q.63

Consider the following C program:

```
#include <stdio.h>
int gate (int n) {
    int d, t, newnum, turn;
    newnum = turn = 0; t=1;
    while (n>=t) t *= 10;
    t /=10;
    while (t>0) {
        d = n/t;
        n = n%t;
        t /= 10;
        if (turn) newnum = 10*newnum + d;
        turn = (turn + 1) % 2;
    }
    return newnum;
}
int main () {
    printf ("%d", gate(14362));
    return 0;
}
```

The value printed by the given C program is _____. (Answer in integer)

Q.64	The maximum value of x such that the edge between the nodes B and C is included in every minimum spanning tree of the given graph is _____. (answer in integer)
	

Q.65	In a double hashing scheme, $h_1(k) = k \bmod 11$ and $h_2(k) = 1 + (k \bmod 7)$ are the auxiliary hash functions. The size m of the hash table is 11. The hash function for the i-th probe in the open address table is $[h_1(k) + i h_2(k)] \bmod m$. The following keys are inserted in the given order: 63, 50, 25, 79, 67, 24. The slot at which key 24 gets stored is _____. (Answer in integer)

GATE 2025

Set 2

65 questions | 48 pages

Paper 2 of 16



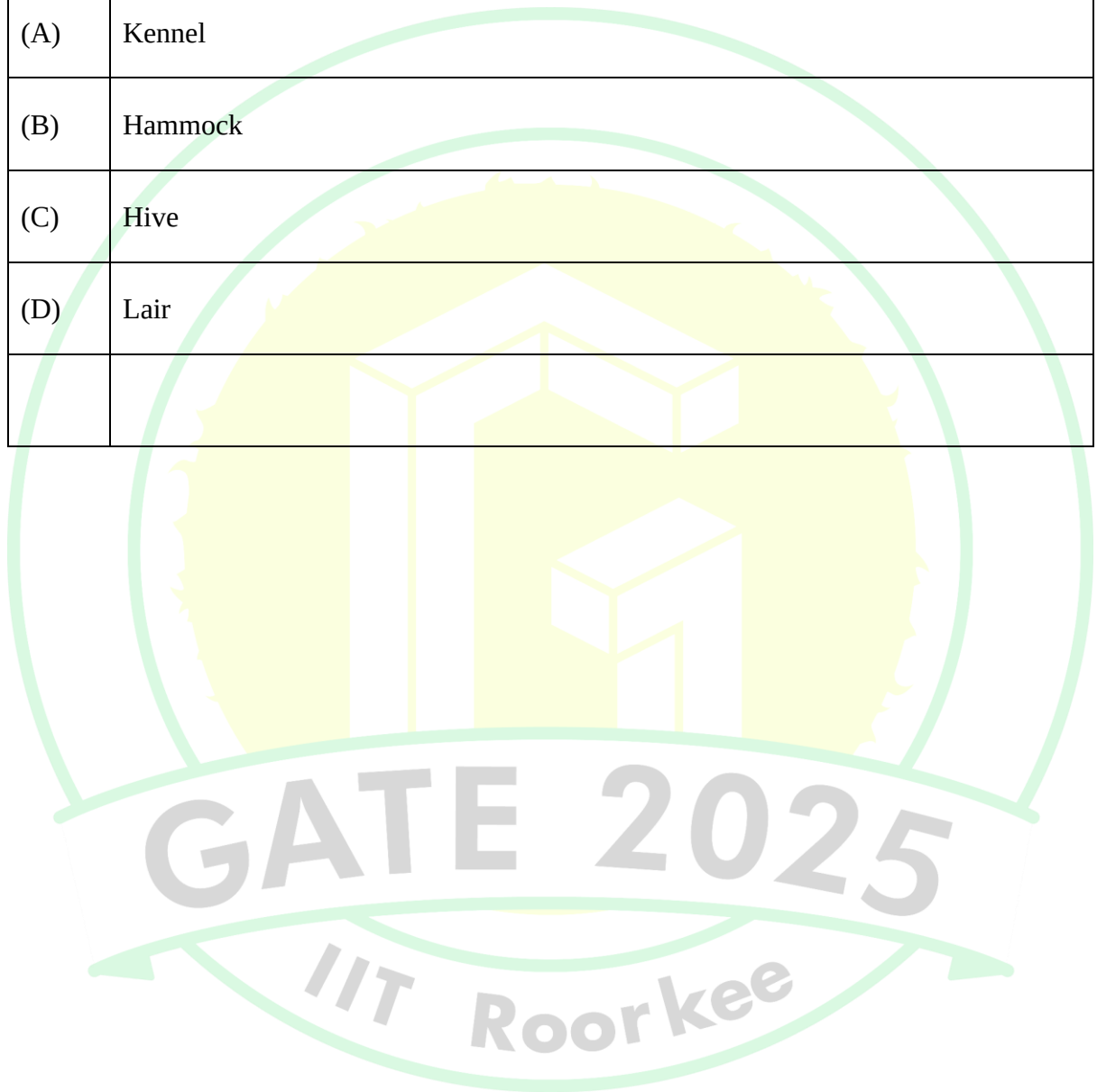
General Aptitude

Q.1 – Q.5 Carry ONE mark Each

Q.1	Despite his initial hesitation, Rehman's _____ to contribute to the success of the project never wavered. Select the most appropriate option to complete the above sentence.
(A)	ambivalence
(B)	satisfaction
(C)	resolve
(D)	revolve

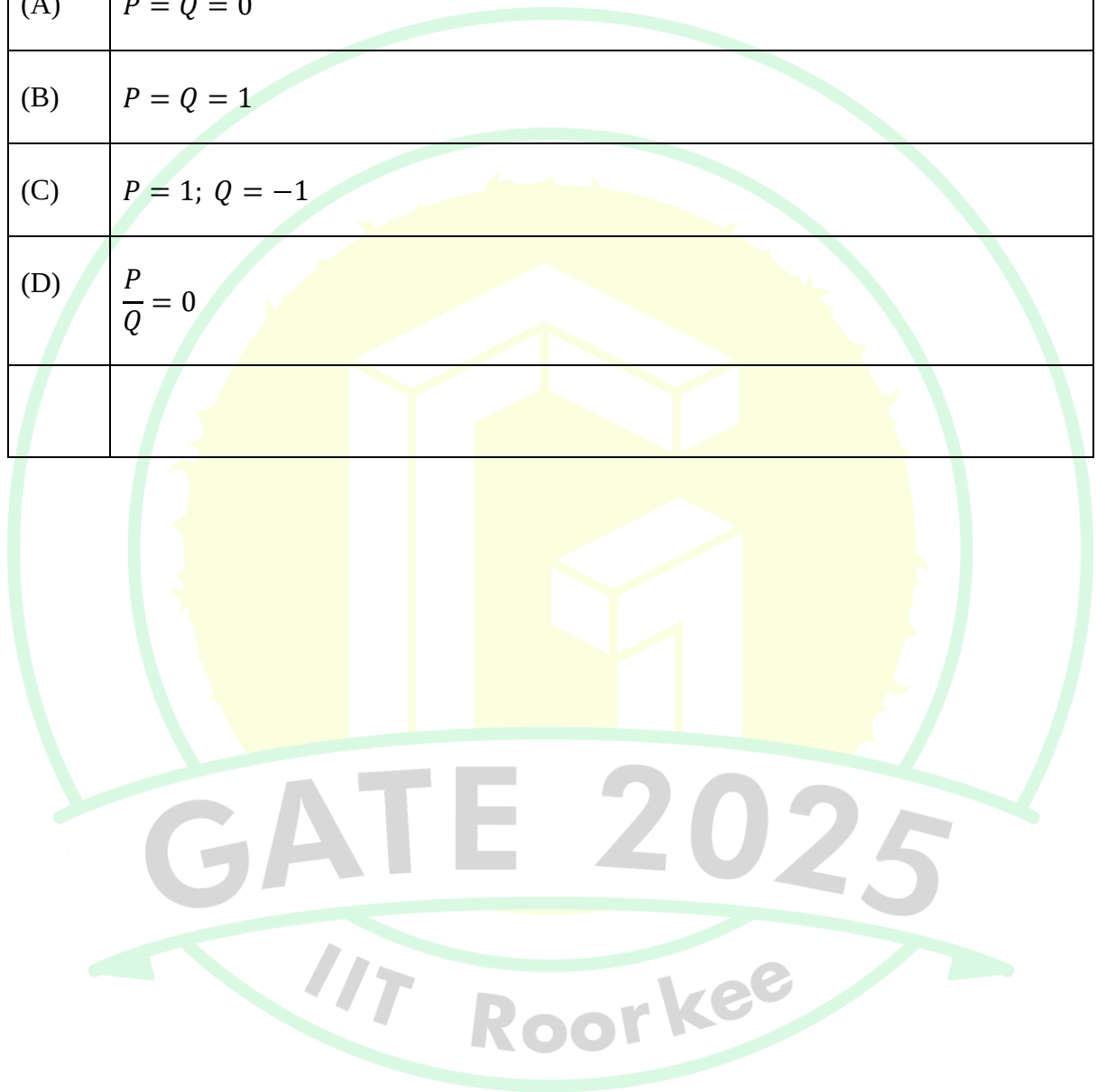


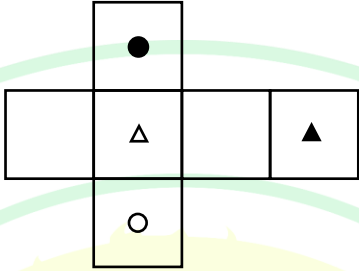
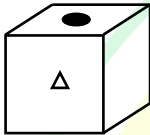
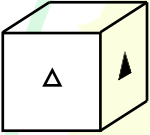
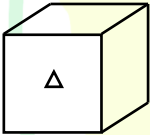
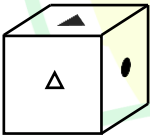
Q.2	Bird : Nest :: Bee : _____ Select the correct option to complete the analogy.
(A)	Kennel
(B)	Hammock
(C)	Hive
(D)	Lair



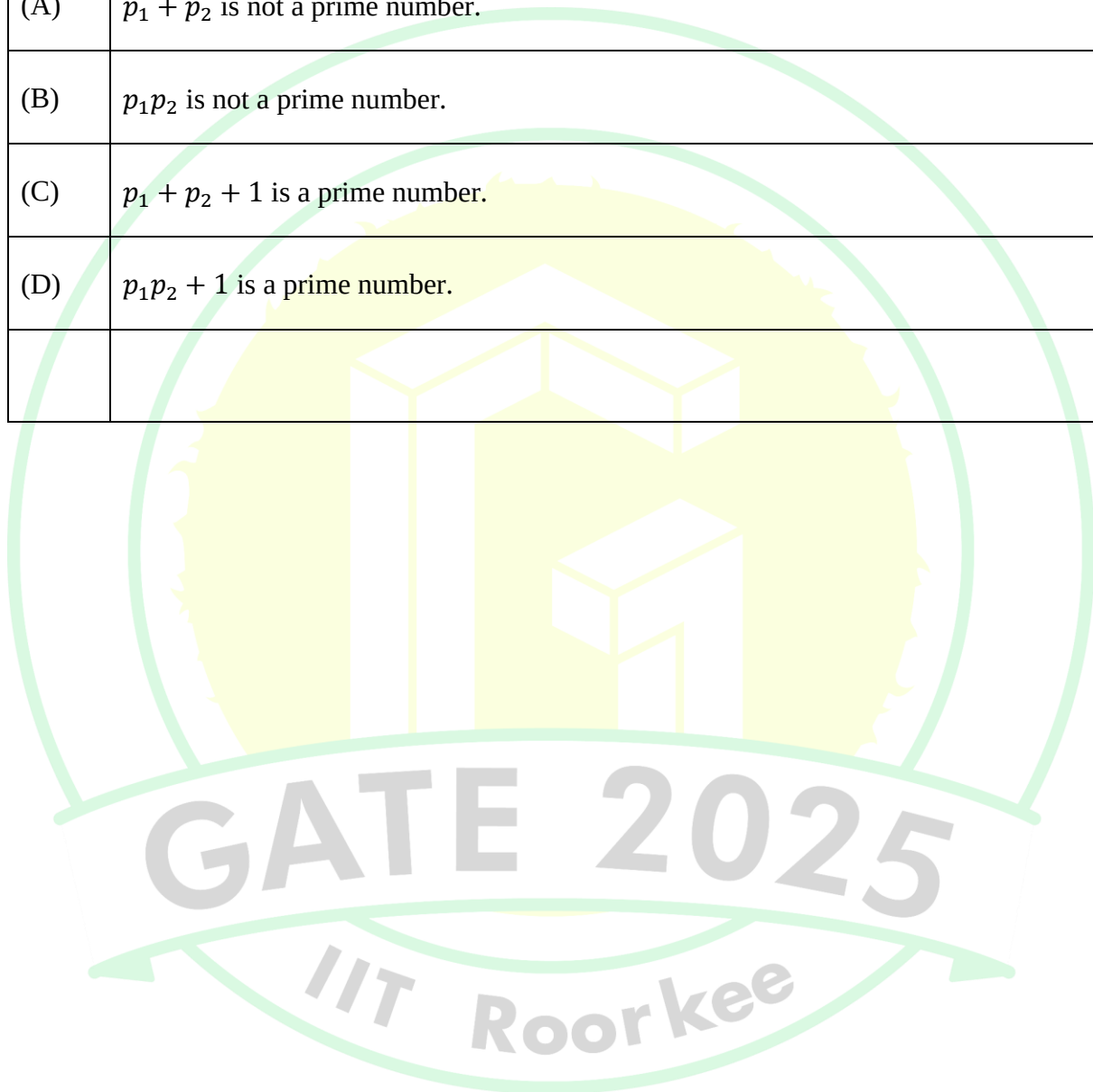


Q.3	If $Pe^x = Qe^{-x}$ for all real values of x , which one of the following statements is true?
(A)	$P = Q = 0$
(B)	$P = Q = 1$
(C)	$P = 1; Q = -1$
(D)	$\frac{P}{Q} = 0$



Q.4	The paper as shown in the figure is folded to make a cube where each square corresponds to a particular face of the cube. Which one of the following options correctly represents the cube? Note: The figures shown are representative.
	
(A)	
(B)	
(C)	
(D)	

Q.5	Let p_1 and p_2 denote two arbitrary prime numbers. Which one of the following statements is correct for all values of p_1 and p_2 ?
(A)	$p_1 + p_2$ is not a prime number.
(B)	$p_1 p_2$ is not a prime number.
(C)	$p_1 + p_2 + 1$ is a prime number.
(D)	$p_1 p_2 + 1$ is a prime number.



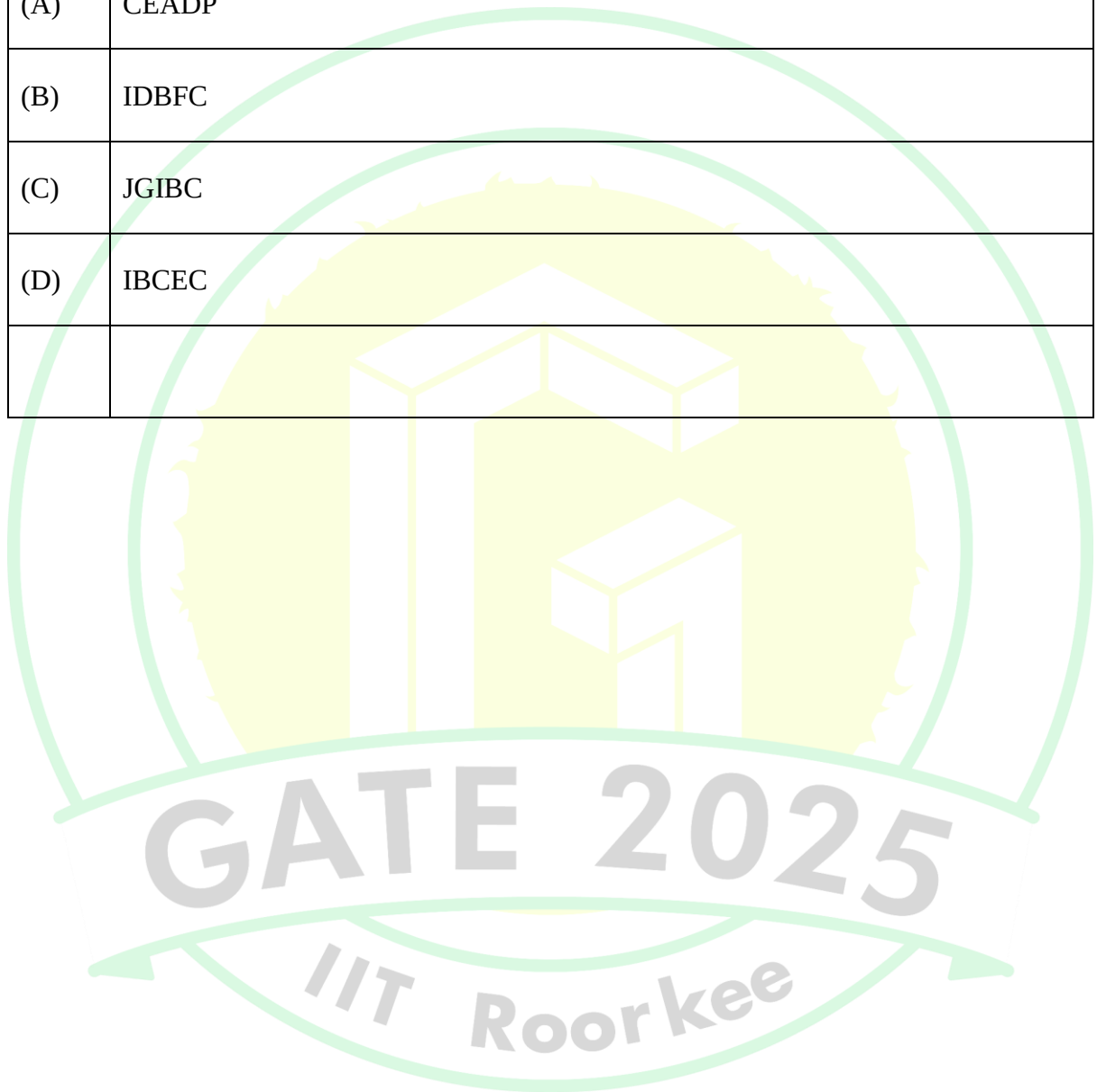


Q.6 – Q.10 Carry TWO marks Each

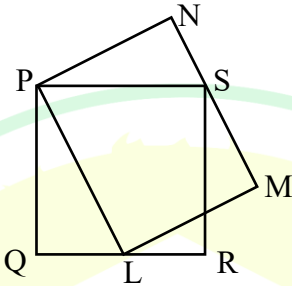
Q.6	Based only on the conversation below, identify the logically correct inference: “Even if I had known that you were in the hospital, I would not have gone there to see you”, Ramya told Josephine.
(A)	Ramya knew that Josephine was in the hospital.
(B)	Ramya did not know that Josephine was in the hospital.
(C)	Ramya and Josephine were once close friends; but now, they are not.
(D)	Josephine was in the hospital due to an injury to her leg.

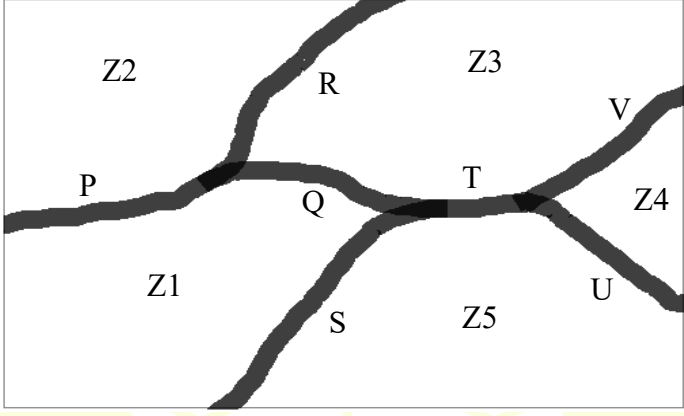


Q.7	If IMAGE and FIELD are coded as FHBNJ and EMFJG respectively then, which one among the given options is the most appropriate code for BEACH ?
(A)	CEADP
(B)	IDBFC
(C)	JGIBC
(D)	IBCEC



Q.8	Which one of the following options is correct for the given data in the table?																				
	<table><tr><td>Iteration (i)</td><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>Input (I)</td><td>20</td><td>-4</td><td>10</td><td>15</td></tr><tr><td>Output (X)</td><td>20</td><td>16</td><td>26</td><td>41</td></tr><tr><td>Output (Y)</td><td>20</td><td>-80</td><td>-800</td><td>-12000</td></tr></table>	Iteration (i)	0	1	2	3	Input (I)	20	-4	10	15	Output (X)	20	16	26	41	Output (Y)	20	-80	-800	-12000
Iteration (i)	0	1	2	3																	
Input (I)	20	-4	10	15																	
Output (X)	20	16	26	41																	
Output (Y)	20	-80	-800	-12000																	
(A)	$X(i) = X(i - 1) + I(i); \quad Y(i) = Y(i - 1)I(i); \quad i > 0$																				
(B)	$X(i) = X(i - 1)I(i); \quad Y(i) = Y(i - 1) + I(i); \quad i > 0$																				
(C)	$X(i) = X(i - 1)I(i); \quad Y(i) = Y(i - 1)I(i); \quad i > 0$																				
(D)	$X(i) = X(i - 1) + I(i); \quad Y(i) = Y(i - 1)I(i - 1); \quad i > 0$																				

Q.9	In the given figure, PQRS is a square of side 2 cm and PLMN is a rectangle. The corner L of the rectangle is on the side QR. Side MN of the rectangle passes through the corner S of the square. What is the area (in cm^2) of the rectangle PLMN? Note: The figure shown is representative.
	
(A)	$2\sqrt{2}$
(B)	2
(C)	8
(D)	4

Q.10	The diagram below shows a river system consisting of 7 segments, marked P, Q, R, S, T, U, and V. It splits the land into 5 zones, marked Z1, Z2, Z3, Z4, and Z5. We need to connect these zones using the least number of bridges. Out of the following options, which one is correct? Note: The figure shown is representative.
	
(A)	Bridges on P, Q, and T
(B)	Bridges on P, Q, S, and T
(C)	Bridges on Q, R, T, and V
(D)	Bridges on P, Q, S, U, and V



Q.11 – Q.35 Carry ONE mark Each

Q.11	If $A = \begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix}$, then which ONE of the following is A^8 ?
(A)	$\begin{pmatrix} 25 & 0 \\ 0 & 25 \end{pmatrix}$
(B)	$\begin{pmatrix} 125 & 0 \\ 0 & 125 \end{pmatrix}$
(C)	$\begin{pmatrix} 625 & 0 \\ 0 & 625 \end{pmatrix}$
(D)	$\begin{pmatrix} 3125 & 0 \\ 0 & 3125 \end{pmatrix}$
Q.12	The value of x such that $x > 1$, satisfying the equation $\int_1^x t \ln t \, dt = \frac{1}{4}$ is
(A)	$\sqrt{e}$
(B)	e
(C)	e^2
(D)	$e - 1$

Q.13	Consider a binary tree T in which every node has either zero or two children. Let $n > 0$ be the number of nodes in T. Which ONE of the following is the number of nodes in T that have exactly two children?
(A)	$\frac{n-2}{2}$
(B)	$\frac{n-1}{2}$
(C)	$\frac{n}{2}$
(D)	$\frac{n+1}{2}$

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Q.14	Let L, M, and N be non-singular matrices of order 3 satisfying the equations $L^2 = L^{-1}$, $M = L^8$ and $N = L^2$. Which ONE of the following is the value of the determinant of $(M - N)$?
(A)	0
(B)	1
(C)	2
(D)	3
Q.15	Let $P(x)$ be an arbitrary predicate over the domain of natural numbers. Which ONE of the following statements is TRUE?
(A)	$(P(0) \wedge (\forall x [P(x) \Rightarrow P(x + 1)])) \Rightarrow (\forall x P(x))$
(B)	$(P(0) \wedge (\forall x [P(x) \Rightarrow P(x - 1)])) \Rightarrow (\forall x P(x))$
(C)	$(P(1000) \wedge (\forall x [P(x) \Rightarrow P(x - 1)])) \Rightarrow (\forall x P(x))$
(D)	$(P(1000) \wedge (\forall x [P(x) \Rightarrow P(x + 1)])) \Rightarrow (\forall x P(x))$

Q.16	Consider the following statements: (i) Address Resolution Protocol (ARP) provides a mapping from an IP address to the corresponding hardware (link-layer) address. (ii) A single TCP segment from a sender S to a receiver R cannot carry both data from S to R and acknowledgement for a segment from R to S. Which ONE of the following is CORRECT?
(A)	Both (i) and (ii) are TRUE
(B)	(i) is TRUE and (ii) is FALSE
(C)	(i) is FALSE and (ii) is TRUE
(D)	Both (i) and (ii) are FALSE

Q.17	Consider the routing protocols given in List I and the names given in List II: <table border="1" data-bbox="320 353 1385 584"> <thead> <tr> <th data-bbox="320 353 852 427">List I</th><th data-bbox="852 353 1385 427">List II</th></tr> </thead> <tbody> <tr> <td data-bbox="320 427 852 501">(i) Distance vector routing</td><td data-bbox="852 427 1385 501">(a) Bellman-Ford</td></tr> <tr> <td data-bbox="320 501 852 584">(ii) Link state routing</td><td data-bbox="852 501 1385 584">(b) Dijkstra</td></tr> </tbody> </table> For matching of items in List I with those in List II, which ONE of the following options is CORRECT?	List I	List II	(i) Distance vector routing	(a) Bellman-Ford	(ii) Link state routing	(b) Dijkstra
List I	List II						
(i) Distance vector routing	(a) Bellman-Ford						
(ii) Link state routing	(b) Dijkstra						
(A)	(i) – (a) and (ii) – (b)						
(B)	(i) – (a) and (ii) – (a)						
(C)	(i) – (b) and (ii) – (a)						
(D)	(i) – (b) and (ii) – (b)						



Q.18	A machine receives an IPv4 datagram. The protocol field of the IPv4 header has the protocol number of a protocol X. Which ONE of the following is NOT a possible candidate for X?
(A)	Internet Control Message Protocol (ICMP)
(B)	Internet Group Management Protocol (IGMP)
(C)	Open Shortest Path First (OSPF)
(D)	Routing Information Protocol (RIP)

Q.19	Consider the following C program: <pre>#include <stdio.h> void strcpy(char *, char *); int main(){ char a[30] = "@#Hello World!"; strcpy(a, a + 2); printf("%s\n", a); return 0; } void strcpy(char *s, char *t) { while(*t) *s++ = *t++; }</pre> Which ONE of the following will be the output of the program?
(A)	@#Hello World!
(B)	Hello World!
(C)	ello World!
(D)	Hello World!d!



Q.20	Consider an unordered list of N distinct integers. What is the minimum number of element comparisons required to find an integer in the list that is NOT the largest in the list?
(A)	1
(B)	$N - 1$
(C)	N
(D)	$2N - 1$
Q.21	Consider the following statements about the use of backpatching in a compiler for intermediate code generation: (I) Backpatching can be used to generate code for Boolean expression in one pass. (II) Backpatching can be used to generate code for flow-of-control statements in one pass. Which ONE of the following options is CORRECT?
(A)	Only (I) is correct.
(B)	Only (II) is correct.
(C)	Both (I) and (II) are correct.
(D)	Neither (I) nor (II) is correct.

Q.22	Given the following syntax directed translation rules: Rule 1: $R \rightarrow AB \{B.i = R.i - 1; A.i = B.i; R.i = A.i + 1;\}$ Rule 2: $P \rightarrow CD \{P.i = C.i + D.i; D.i = C.i + 2;\}$ Rule 3: $Q \rightarrow EF \{Q.i = E.i + F.i;\}$ Which ONE is the CORRECT option among the following?
(A)	Rule 1 is S-attributed and L-attributed; Rule 2 is S-attributed and not L-attributed; Rule 3 is neither S-attributed nor L-attributed
(B)	Rule 1 is neither S-attributed nor L-attributed; Rule 2 is S-attributed and L-attributed; Rule 3 is S-attributed and L-attributed
(C)	Rule 1 is neither S-attributed nor L-attributed; Rule 2 is not S-attributed and is L-attributed; Rule 3 is S-attributed and L-attributed
(D)	Rule 1 is S-attributed and not L-attributed; Rule 2 is not S-attributed and is L-attributed; Rule 3 is S-attributed and L-attributed



Q.23	Consider a network that uses Ethernet and IPv4. Assume that IPv4 headers do not use any options field. Each Ethernet frame can carry a maximum of 1500 bytes in its data field. A UDP segment is transmitted. The payload (data) in the UDP segment is 7488 bytes. Which ONE of the following choices has the CORRECT total number of fragments transmitted and the size of the last fragment including IPv4 header?
(A)	5 fragments, 1488 bytes
(B)	6 fragments, 88 bytes
(C)	6 fragments, 108 bytes
(D)	6 fragments, 116 bytes
Q.24	Which ONE of the following languages is accepted by a deterministic pushdown automaton?
(A)	Any regular language.
(B)	Any context-free language.
(C)	Any language accepted by a non-deterministic pushdown automaton.
(D)	Any decidable language.

Q.25	Let G_1, G_2 be Context Free Grammars (CFGs) and R be a regular expression. For a grammar G, let $L(G)$ denote the language generated by G. Which ONE among the following questions is decidable?
(A)	Is $L(G_1) = L(G_2)$?
(B)	Is $L(G_1) \cap L(G_2) = \emptyset$?
(C)	Is $L(G_1) = L(R)$?
(D)	Is $L(G_1) = \emptyset$?
Q.26	Processes P_1, P_2, P_3, P_4 arrive in that order at times 0, 1, 2, and 8 milliseconds respectively, and have execution times of 10, 13, 6, and 9 milliseconds respectively. Shortest Remaining Time First (SRTF) algorithm is used as the CPU scheduling policy. Ignore context switching times. Which ONE of the following correctly gives the average turnaround time of the four processes in milliseconds?
(A)	22
(B)	15
(C)	37
(D)	19



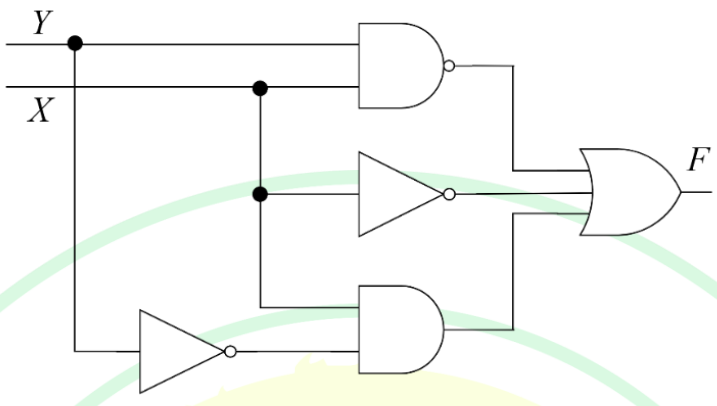
Q.27	An audit of a banking transactions system has found that on an earlier occasion, two joint holders of account <i>A</i> attempted simultaneous transfers of Rs. 10000 each from account <i>A</i> to account <i>B</i>. Both transactions read the same value, Rs. 11000, as the initial balance in <i>A</i> and were allowed to go through. <i>B</i> was credited Rs. 10000 twice. <i>A</i> was debited only once and ended up with a balance of Rs. 1000. Which of the following properties is/are certain to have been violated by the system?
(A)	Atomicity
(B)	Consistency
(C)	Isolation
(D)	Durability
Q.28	Which of the following is/are part of an Instruction Set Architecture of a processor?
(A)	The size of the cache memory
(B)	The clock frequency of the processor
(C)	The number of cache memory levels
(D)	The total number of registers



Q.29	Which of the following statements regarding Breadth First Search (BFS) and Depth First Search (DFS) on an undirected simple graph G is/are TRUE?
(A)	A DFS tree of G is a Shortest Path tree of G .
(B)	Every non-tree edge of G with respect to a DFS tree is a forward/back edge.
(C)	If (u, v) is a non-tree edge of G with respect to a BFS tree, then the distances from the source vertex s to u and v in the BFS tree are within ± 1 of each other.
(D)	Both BFS and DFS can be used to find the connected components of G .

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Q.30	Consider the two lists List I and List II given below: <table border="1" data-bbox="349 344 1358 752"> <thead> <tr> <th data-bbox="349 344 823 443">List I</th><th data-bbox="823 344 1358 443">List II</th></tr> </thead> <tbody> <tr> <td data-bbox="349 443 823 546">(i) Context free languages</td><td data-bbox="823 443 1358 546">(a) Closed under union</td></tr> <tr> <td data-bbox="349 546 823 649">(ii) Recursive languages</td><td data-bbox="823 546 1358 649">(b) Not closed under complementation</td></tr> <tr> <td data-bbox="349 649 823 752">(iii) Regular languages</td><td data-bbox="823 649 1358 752">(c) Closed under intersection</td></tr> </tbody> </table> For matching of items in List I with those in List II, which of the following option(s) is/are CORRECT?	List I	List II	(i) Context free languages	(a) Closed under union	(ii) Recursive languages	(b) Not closed under complementation	(iii) Regular languages	(c) Closed under intersection
List I	List II								
(i) Context free languages	(a) Closed under union								
(ii) Recursive languages	(b) Not closed under complementation								
(iii) Regular languages	(c) Closed under intersection								
(A)	(i) – (a), (ii) – (b), and (iii) – (c)								
(B)	(i) – (b), (ii) – (a), and (iii) – (c)								
(C)	(i) – (b), (ii) – (c), and (iii) – (a)								
(D)	(i) – (a), (ii) – (c), and (iii) – (b)								

Q.31	Consider the following logic circuit diagram.  Which is/are the CORRECT option(s) for the output function F?
(A)	$\bar{X}\bar{Y}$
(B)	$\bar{X} + \bar{Y} + X\bar{Y}$
(C)	$\bar{X}\bar{Y} + \bar{X} + X\bar{Y}$
(D)	$X + \bar{Y}$
Q.32	The following two signed 2's complement numbers (multiplicand M and multiplier Q) are being multiplied using Booth's algorithm: M: 1100 1101 1110 1101 and Q: 1010 0100 1010 1010 The total number of addition and subtraction operations to be performed is _____. (Answer in integer)

Q.33	<pre>int x=126,y=105; do { if(x>y) x=x-y; else y=y-x; } while(x!=y); printf("%d",x);</pre> The output of the given C code segment is _____. (Answer in integer)
Q.34	In a 4-bit ripple counter, if the period of the waveform at the last flip-flop is 64 microseconds, then the frequency of the ripple counter in kHz is _____. (Answer in integer)
Q.35	Suppose the values 10, -4, 15, 30, 20, 5, 60, 19 are inserted in that order into an initially empty binary search tree. Let T be the resulting binary search tree. The number of edges in the path from the node containing 19 to the root node of T is _____. (Answer in integer)



Q.36 – Q.65 Carry TWO marks Each

Q.36	Suppose we are transmitting frames between two nodes using Stop-and-Wait protocol. The frame size is 3000 bits. The transmission rate of the channel is 2000 bps (bits/second) and the propagation delay between the two nodes is 100 milliseconds. Assume that the processing times at the source and destination are negligible. Also, assume that the size of the acknowledgement packet is negligible. Which ONE of the following most accurately gives the channel utilization for the above scenario in percentage?
(A)	88.23
(B)	93.75
(C)	85.44
(D)	66.67

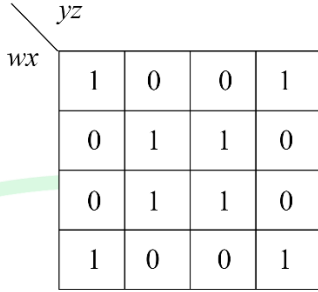
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Q.37	Let G be an edge-weighted undirected graph with positive edge weights. Suppose a positive constant α is added to the weight of every edge. Which ONE of the following statements is TRUE about the minimum spanning trees (MSTs) and shortest paths (SPs) in G before and after the edge weight update?
(A)	Every MST remains an MST, and every SP remains an SP.
(B)	MSTs need not remain MSTs, and every SP remains an SP.
(C)	Every MST remains an MST, and SPs need not remain SPs.
(D)	MSTs need not remain MSTs, and SPs need not remain SPs.

Q.38	A meld operation on two instances of a data structure combines them into one single instance of the same data structure. Consider the following data structures: P: Unsorted doubly linked list with pointers to the head node and tail node of the list. Q: Min-heap implemented using an array. R: Binary Search Tree. Which ONE of the following options gives the worst-case time complexities for meld operation on instances of size n of these data structures?
(A)	P: $\Theta(1)$, Q: $\Theta(n)$, R: $\Theta(n)$
(B)	P: $\Theta(1)$, Q: $\Theta(n \log n)$, R: $\Theta(n)$
(C)	P: $\Theta(n)$, Q: $\Theta(n \log n)$, R: $\Theta(n^2)$
(D)	P: $\Theta(1)$, Q: $\Theta(n)$, R: $\Theta(n \log n)$

Q.39	For a direct-mapped cache, 4 bits are used for the tag field and 12 bits are used to index into a cache block. The size of each cache block is one byte. Assume that there is no other information stored for each cache block. Which ONE of the following is the CORRECT option for the sizes of the main memory and the cache memory in this system (byte addressable), respectively?
(A)	64 KB and 4 KB
(B)	128 KB and 16 KB
(C)	64 KB and 8 KB
(D)	128 KB and 6 KB
Q.40	Given a Context-Free Grammar G as follows: $S \rightarrow Aa \mid bAc \mid dc \mid bda$ $A \rightarrow d$ Which ONE of the following statements is TRUE?
(A)	G is neither LALR(1) nor SLR(1)
(B)	G is CLR(1), not LALR(1)
(C)	G is LALR(1), not SLR(1)
(D)	G is LALR(1), also SLR(1)

Q.41	An array A of length n with distinct elements is said to be bitonic if there is an index $1 \leq i \leq n$ such that $A[1..i]$ is sorted in the non-decreasing order and $A[i+1..n]$ is sorted in the non-increasing order. Which ONE of the following represents the best possible asymptotic bound for the worst-case number of comparisons by an algorithm that searches for an element in a bitonic array A?
(A)	$\Theta(n)$
(B)	$\Theta(1)$
(C)	$\Theta(\log^2 n)$
(D)	$\Theta(\log n)$
Q.42	Let $\mathcal{F}$ be the set of all functions from $\{1, \dots, n\}$ to $\{0,1\}$. Define the binary relation $\preceq$ on $\mathcal{F}$ as follows: $\forall f, g \in \mathcal{F}, f \preceq g$ if and only if $\forall x \in \{1, \dots, n\}, f(x) \leq g(x)$, where $0 \leq 1$. Which of the following statement(s) is/are TRUE?
(A)	$\preceq$ is a symmetric relation
(B)	$(\mathcal{F}, \preceq)$ is a partial order
(C)	$(\mathcal{F}, \preceq)$ is a lattice
(D)	$\preceq$ is an equivalence relation

Q.43	Given the following Karnaugh Map for a Boolean function $F(w, x, y, z)$:  Which one or more of the following Boolean expression(s) represent(s) F?
(A)	$\bar{w}\bar{x}\bar{y}\bar{z} + w\bar{x}\bar{y}\bar{z} + \bar{w}\bar{x}y\bar{z} + w\bar{x}y\bar{z} + xz$
(B)	$\bar{w}\bar{x}\bar{y}\bar{z} + \bar{w}\bar{x}y\bar{z} + w\bar{x}yz + xz$
(C)	$\bar{w}\bar{x}\bar{y}\bar{z} + w\bar{x}\bar{y}\bar{z} + w\bar{x}\bar{y}z + xz$
(D)	$\bar{x}\bar{z} + xz$

Q.44	Consider a system of linear equations $PX = Q$ where $P \in \mathbb{R}^{3 \times 3}$ and $Q \in \mathbb{R}^{3 \times 1}$. Suppose P has an LU decomposition, $P = LU$, where $L = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \text{ and } U = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}.$ Which of the following statement(s) is/are TRUE?
(A)	The system $PX = Q$ can be solved by first solving $LY = Q$ and then $UX = Y$.
(B)	If P is invertible, then both L and U are invertible.
(C)	If P is singular, then at least one of the diagonal elements of U is zero.
(D)	If P is symmetric, then both L and U are symmetric.

Q.45	Consider a stack data structure into which we can PUSH and POP records. Assume that each record pushed in the stack has a positive integer key and that all keys are distinct. We wish to augment the stack data structure with an $O(1)$ time MIN operation that returns a pointer to the record with smallest key present in the stack 1) without deleting the corresponding record, and 2) without increasing the complexities of the standard stack operations. Which one or more of the following approach(es) can achieve it?
(A)	Keep with every record in the stack, a pointer to the record with the smallest key below it.
(B)	Keep a pointer to the record with the smallest key in the stack.
(C)	Keep an auxiliary array in which the key values of the records in the stack are maintained in sorted order.
(D)	Keep a Min-Heap in which the key values of the records in the stack are maintained.

Q.46	Consider the following relational schema along with all the functional dependencies that hold on them. $R1(A, B, C, D, E): \{ D \rightarrow E, EA \rightarrow B, EB \rightarrow C \}$ $R2(A, B, C, D): \{ A \rightarrow D, A \rightarrow B, C \rightarrow A \}$ Which of the following statement(s) is/are TRUE?
(A)	$R1$ is in 3NF
(B)	$R2$ is in 3NF
(C)	$R1$ is NOT in 3NF
(D)	$R2$ is NOT in 3NF

Q.47

Consider a demand paging system with three frames, and the following page reference string: 1 2 3 4 5 4 1 6 4 5 1 3 2. The contents of the frames are as follows initially and after each reference (from left to right):

<i>initially</i>	<i>after</i>												
-	1*	2*	3*	4*	5*	4	1	6*	4	5	1*	3*	2*
-	1	1	1	1	1	1	1	6	6	6	6	6	2
-	-	2	2	4	4	4	4	4	4	4	1	1	1
-	-	-	3	3	5	5	5	5	5	5	5	3	3

The *-marked references cause page replacements.

Which one or more of the following could be the page replacement policy/policies in use?

(A)

Least Recently Used page replacement policy

(B)

Least Frequently Used page replacement policy

(C)

Most Frequently Used page replacement policy

(D)

Optimal page replacement policy

Q.48	$P = \{P_1, P_2, P_3, P_4\}$ consists of all active processes in an operating system. $R = \{R_1, R_2, R_3, R_4\}$ consists of single instances of distinct types of resources in the system. The resource allocation graph has the following assignment and claim edges. Assignment edges: $R_1 \rightarrow P_1, R_2 \rightarrow P_2, R_3 \rightarrow P_3, R_4 \rightarrow P_4$ (the assignment edge $R_1 \rightarrow P_1$ means resource R_1 is assigned to process P_1, and so on for others) Claim edges: $P_1 \rightarrow R_2, P_2 \rightarrow R_3, P_3 \rightarrow R_1, P_2 \rightarrow R_4, P_4 \rightarrow R_2$ (the claim edge $P_1 \rightarrow R_2$ means process P_1 is waiting for resource R_2, and so on for others) Which of the following statement(s) is/are CORRECT?
(A)	Aborting P_1 makes the system deadlock free.
(B)	Aborting P_3 makes the system deadlock free.
(C)	Aborting P_2 makes the system deadlock free.
(D)	Aborting P_1 and P_4 makes the system deadlock free.

Q.49	Three floating point numbers X, Y, and Z are stored in three registers R_X, R_Y, and R_Z, respectively in IEEE 754 single precision format as given below in hexadecimal: $R_X = 0xC1100000, R_Y = 0x40C00000$, and $R_Z = 0x41400000$ Which of the following option(s) is/are CORRECT?
(A)	$4(X + Y) + Z = 0$
(B)	$2Y - Z = 0$
(C)	$4X + 3Z = 0$
(D)	$X + Y + Z = 0$
Q.50	Which of the following Boolean algebraic equation(s) is/are CORRECT?
(A)	$\bar{A} B C + A \bar{B} \bar{C} + \bar{A} \bar{B} \bar{C} + A \bar{B} C + A B C = B C + \bar{B} \bar{C} + \bar{A} \bar{B}$
(B)	$A B + \bar{A} C + B C = A B + \bar{A} C$
(C)	$(A + C) (\bar{A} + B) = A B + \bar{A} C$
(D)	$\overline{(A + \bar{B} + \bar{D}) (C + D) (\bar{A} + C + D) (A + B + \bar{D})} = \bar{A} D + \bar{C} \bar{D}$

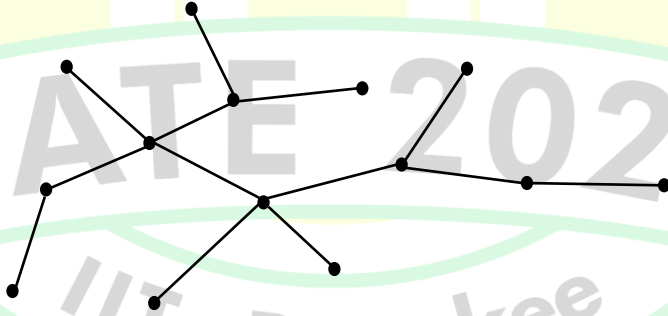
Q.51	Consider two grammars G_1 and G_2 with the production rules given below: $G_1: S \rightarrow \text{if } E \text{ then } S \mid \text{if } E \text{ then } S \text{ else } S \mid a$ $E \rightarrow b$ $G_2: S \rightarrow \text{if } E \text{ then } S \mid M$ $M \rightarrow \text{if } E \text{ then } M \text{ else } S \mid c$ $E \rightarrow b$ where <i>if</i>, <i>then</i>, <i>else</i>, <i>a</i>, <i>b</i>, <i>c</i> are the terminals. Which of the following option(s) is/are CORRECT?
(A)	G_1 is not $LL(1)$ and G_2 is $LL(1)$.
(B)	G_1 is $LL(1)$ and G_2 is not $LL(1)$.
(C)	G_1 and G_2 are not $LL(1)$.
(D)	G_1 and G_2 are ambiguous.

Q.52	Let $\Sigma = \{a, b, c\}$. For $x \in \Sigma^*$, and $\alpha \in \Sigma$, let $\#_{\alpha}(x)$ denote the number of occurrences of α in x. Which one or more of the following option(s) define(s) regular language(s)?
(A)	$\{a^m b^n \mid m, n \geq 0\}$
(B)	$\{a, b\}^* \cap \{a^m b^n c^{m-n} \mid m \geq n \geq 0\}$
(C)	$\{w \mid w \in \{a, b\}^*, \#_a(w) \equiv 2 \pmod{7}, \text{ and } \#_b(w) \equiv 3 \pmod{9}\}$
(D)	$\{w \mid w \in \{a, b\}^*, \#_a(w) \equiv 2 \pmod{7}, \text{ and } \#_a(w) = \#_b(w)\}$

Q.53	Consider the database transactions T1 and T2, and data items X and Y. Which of the schedule(s) is/are conflict serializable?
	Transaction T1 R1(X) W1(Y) R1(X) W1(X) COMMIT(T1) Transaction T2 W2(X) W2(Y) COMMIT(T2)
(A)	R1(X), W2(X), W1(Y), W2(Y), R1(X), W1(X), COMMIT(T2), COMMIT(T1)
(B)	W2(X), R1(X), W2(Y), W1(Y), R1(X), COMMIT(T2), W1(X), COMMIT(T1)
(C)	R1(X), W1(Y), W2(X), W2(Y), R1(X), W1(X), COMMIT(T1), COMMIT(T2)
(D)	W2(X), R1(X), W1(Y), W2(Y), R1(X), COMMIT(T2), W1(X), COMMIT(T1)

Q.54	Consider the following relational schema: Students (<u>rollno</u>: integer, name: string, age: integer, cgpa: real) Courses (<u>courseno</u>: integer, cname: string, credits: integer) Enrolled (<u>rollno</u>: integer, <u>courseno</u>: integer, grade: string) Which of the following options is/are correct SQL query/queries to retrieve the names of the students enrolled in course number (i.e., courseno) 1470?
(A)	<pre>SELECT S.name FROM Students S WHERE EXISTS (SELECT * FROM Enrolled E WHERE E.courseno = 1470 AND E.rollno = S.rollno);</pre>
(B)	<pre>SELECT S.name FROM Students S WHERE SIZEOF (SELECT * FROM Enrolled E WHERE E.courseno = 1470 AND E.rollno = S.rollno) > 0;</pre>
(C)	<pre>SELECT S.name FROM Students S WHERE 0 < (SELECT COUNT(*) FROM Enrolled E WHERE E.courseno = 1470 AND E.rollno = S.rollno);</pre>
(D)	<pre>SELECT S.name FROM Students S NATURAL JOIN Enrolled E WHERE E.courseno = 1470;</pre>

Q.55	Given a computing system with two levels of cache (L1 and L2) and a main memory. The first level (L1) cache access time is 1 nanosecond (ns) and the “hit rate” for L1 cache is 90% while the processor is accessing the data from L1 cache. Whereas, for the second level (L2) cache, the “hit rate” is 80% and the “miss penalty” for transferring data from L2 cache to L1 cache is 10 ns. The “miss penalty” for the data to be transferred from main memory to L2 cache is 100 ns. Then the average memory access time in this system in nanoseconds is _____. (rounded off to one decimal place)
Q.56	A 5-stage instruction pipeline has stage delays of 180, 250, 150, 170, and 250, respectively, in nanoseconds. The delay of an inter-stage latch is 10 nanoseconds. Assume that there are no pipeline stalls due to branches and other hazards. The time taken to process 1000 instructions in microseconds is _____. (rounded off to two decimal places)
Q.57	In a B⁺- tree where each node can hold at most four key values, a root to leaf path consists of the following nodes: $A = (49, 77, 83, -), B = (7, 19, 33, 44), C = (20^*, 22^*, 25^*, 26^*)$ The *-marked keys signify that these are data entries in a leaf. Assume that a pointer between keys k_1 and k_2 points to a subtree containing keys in $[k_1, k_2)$, and that when a leaf is created, the smallest key in it is copied up into its parent. A record with key value 23 is inserted into the B⁺- tree. The smallest key value in the parent of the leaf that contains 25* is _____. (Answer in integer)

Q.58	A computer system supports a logical address space of 2^{32} bytes. It uses two-level hierarchical paging with a page size of 4096 bytes. A logical address is divided into a b-bit index to the outer page table, an offset within the page of the inner page table, and an offset within the desired page. Each entry of the inner page table uses eight bytes. All the pages in the system have the same size. The value of b is _____. (Answer in integer)
Q.59	Consider the following algorithm <code>someAlgo</code> that takes an undirected graph G as input. <pre>someAlgo(G)</pre> 1. Let v be any vertex in G. Run BFS on G starting at v. Let u be a vertex in G at maximum distance from v as given by the BFS. 2. Run BFS on G again with u as the starting vertex. Let z be the vertex at maximum distance from u as given by the BFS. 3. Output the distance between u and z in G. The output of <code>someAlgo(T)</code> for the tree shown in the given figure is _____. (Answer in integer)
	

Q.60	Let $\Sigma = \{1,2,3,4\}$. For $x \in \Sigma^*$, let $prod(x)$ be the product of symbols in x modulo 7. We take $prod(\epsilon) = 1$, where ϵ is the null string. For example, $prod(124) = (1 \times 2 \times 4) \bmod 7 = 1$. Define $L = \{x \in \Sigma^* \mid prod(x) = 2\}$. The number of states in a minimum state DFA for L is _____. (Answer in integer)															
Q.61	An application executes 6.4×10^8 number of instructions in 6.3 seconds. There are four types of instructions, the details of which are given in the table. The duration of a clock cycle in nanoseconds is _____. (rounded off to one decimal place) <table><tr><th>Instruction type</th><th>Clock cycles required per instruction (CPI)</th><th>Number of instructions executed</th></tr><tr><td>Branch</td><td>2</td><td>2.25×10^8</td></tr><tr><td>Load</td><td>5</td><td>1.20×10^8</td></tr><tr><td>Store</td><td>4</td><td>1.65×10^8</td></tr><tr><td>Arithmetic</td><td>3</td><td>1.30×10^8</td></tr></table>	Instruction type	Clock cycles required per instruction (CPI)	Number of instructions executed	Branch	2	2.25×10^8	Load	5	1.20×10^8	Store	4	1.65×10^8	Arithmetic	3	1.30×10^8
Instruction type	Clock cycles required per instruction (CPI)	Number of instructions executed														
Branch	2	2.25×10^8														
Load	5	1.20×10^8														
Store	4	1.65×10^8														
Arithmetic	3	1.30×10^8														

Q.62

Consider the following C program:

```
#include <stdio.h>

int main(){

    int a;

    int arr[5] = {30,50,10};

    int *ptr;

    ptr = &arr[0] + 1;

    a = *ptr;

    (*ptr)++;

    ptr++;

    printf("%d", a + (*ptr) + arr[1]);

    return 0;

}
```

The output of the above program is _____. (Answer in integer)

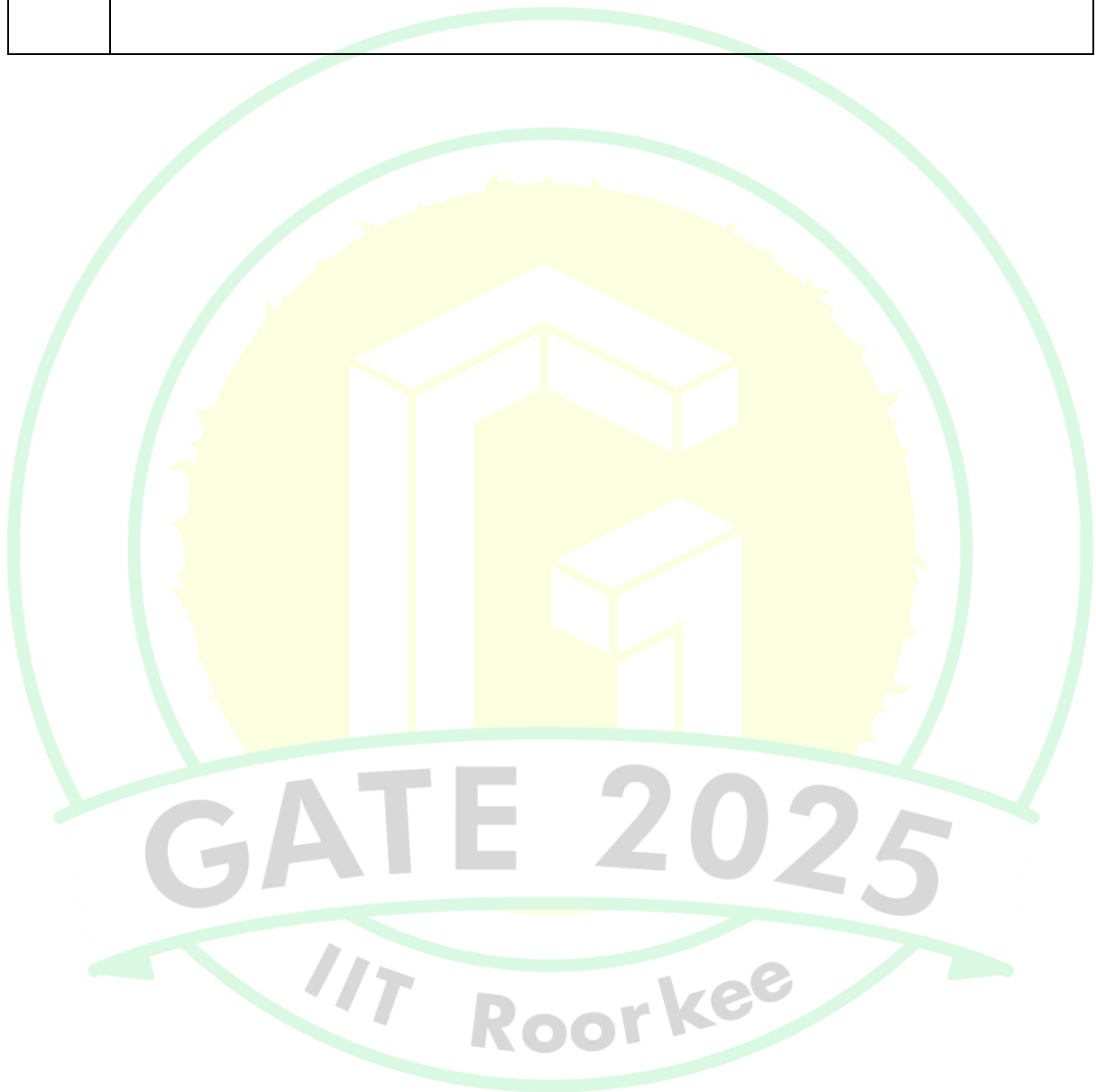
GATE 2025

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Q.63	Consider the following C program: <pre>#include <stdio.h> int g(int n) { return (n+10); } int f(int n) { return g(n*2); } int main() { int sum, n; sum=0; for (n=1; n<3; n++) sum += g(f(n)); printf ("%d", sum); return 0; }</pre> The output of the given C program is _____. (Answer in integer)
Q.64	A quadratic polynomial $(x - \alpha)(x - \beta)$ over complex numbers is said to be square invariant if $(x - \alpha)(x - \beta) = (x - \alpha^2)(x - \beta^2)$. Suppose from the set of all square invariant quadratic polynomials we choose one at random. The probability that the roots of the chosen polynomial are equal is _____. (rounded off to one decimal place)



Q.65	The unit interval $(0,1)$ is divided at a point chosen uniformly distributed over $(0,1)$ in $\mathbb{R}$ into two disjoint subintervals. The expected length of the subinterval that contains 0.4 is _____. (rounded off to two decimal places)



GATE 2024

Set 1 (S5)

65 questions | 36 pages

Paper 3 of 16

General Aptitude (GA)

Q.1 – Q.5 Carry ONE mark Each

Q.1 If '→' denotes increasing order of intensity, then the meaning of the words
[dry → arid → parched] is analogous to [diet → fast → _____].

Which one of the given options is appropriate to fill the blank?

- (A) starve
- (B) reject
- (C) feast
- (D) deny

Q.2 If two distinct non-zero real variables x and y are such that $(x + y)$ is proportional to $(x - y)$ then the value of $\frac{x}{y}$

- (A) depends on xy
- (B) depends only on x and not on y
- (C) depends only on y and not on x
- (D) is a constant

Q.3 Consider the following sample of numbers:

9, 18, 11, 14, 15, 17, 10, 69, 11, 13

The median of the sample is

(A) 13.5

(B) 14

(C) 11

(D) 18.7

Q.4 The number of coins of ₹1, ₹5, and ₹10 denominations that a person has are in the ratio 5:3:13. Of the total amount, the percentage of money in ₹5 coins is

(A) 21%

(B) $14\frac{2}{7}\%$

(C) 10%

(D) 30%

Q.5 For positive non-zero real variables p and q , if

$$\log (p^2 + q^2) = \log p + \log q + 2 \log 3 ,$$

then, the value of $\frac{p^4 + q^4}{p^2 q^2}$ is

- (A) 79
- (B) 81
- (C) 9
- (D) 83

GATE 2024

Q.6 – Q.10 Carry TWO marks Each

Q.6 In the given text, the blanks are numbered (i)–(iv). Select the best match for all the blanks.

Steve was advised to keep his head _____ (i) before heading _____ (ii) to bat; for, while he had a head _____ (iii) batting, he could only do so with a cool head _____ (iv) his shoulders.

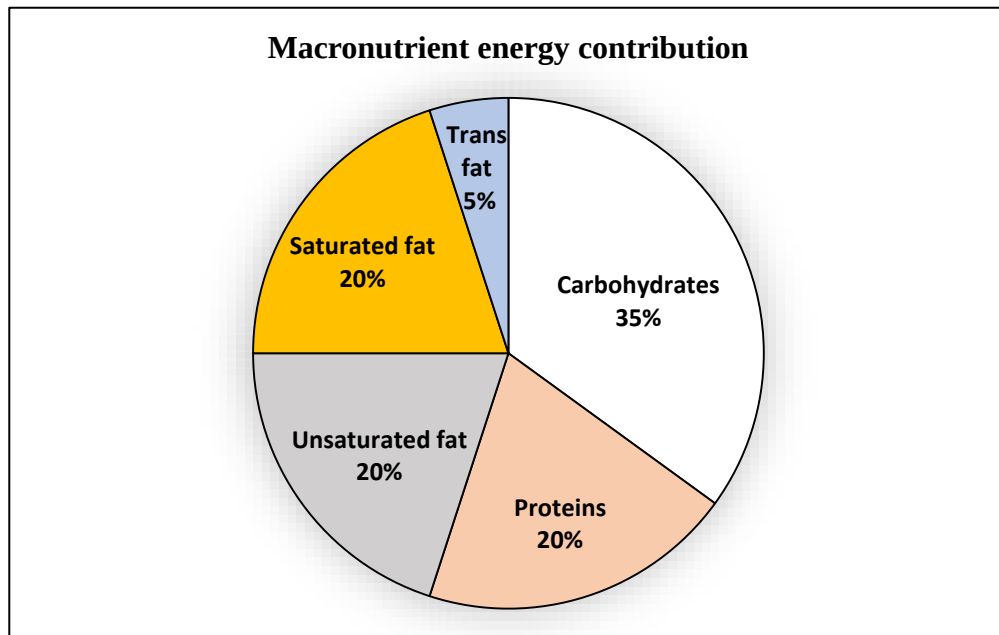
- (A) (i) down (ii) down (iii) on (iv) for
- (B) (i) on (ii) down (iii) for (iv) on
- (C) (i) down (ii) out (iii) for (iv) on
- (D) (i) on (ii) out (iii) on (iv) for

- Q.7 A rectangular paper sheet of dimensions $54 \text{ cm} \times 4 \text{ cm}$ is taken. The two longer edges of the sheet are joined together to create a cylindrical tube. A cube whose surface area is equal to the area of the sheet is also taken.

Then, the ratio of the volume of the cylindrical tube to the volume of the cube is

- (A) $1/\pi$
- (B) $2/\pi$
- (C) $3/\pi$
- (D) $4/\pi$

Q.8 The pie chart presents the percentage contribution of different macronutrients to a typical 2,000 kcal diet of a person.



The typical energy density (kcal/g) of these macronutrients is given in the table.

Macronutrient	Energy density (kcal/g)
Carbohydrates	4
Proteins	4
Unsaturated fat	9
Saturated fat	9
Trans fat	9

The total fat (all three types), in grams, this person consumes is

- (A) 44.4
- (B) 77.8
- (C) 100
- (D) 3,600

Q.11 – Q.35 Carry ONE mark Each

Q.11 Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function such that $f(x) = \max\{x, x^3\}$, $x \in \mathbb{R}$, where $\mathbb{R}$ is the set of all real numbers. The set of all points where $f(x)$ is NOT differentiable is

- (A) $\{-1, 1, 2\}$
- (B) $\{-2, -1, 1\}$
- (C) $\{0, 1\}$
- (D) $\{-1, 0, 1\}$

Q.12 The product of all eigenvalues of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$ is

- (A) -1
- (B) 0
- (C) 1
- (D) 2

Q.13 Consider a system that uses 5 bits for representing signed integers in 2's complement format. In this system, two integers A and B are represented as $A=01010$ and $B=11010$. Which one of the following operations will result in either an arithmetic overflow or an arithmetic underflow?

(A) $A + B$

(B) $A - B$

(C) $B - A$

(D) $2 * B$

Q.14 Consider a permutation sampled uniformly at random from the set of all permutations of $\{1, 2, 3, \dots, n\}$ for some $n \geq 4$. Let X be the event that 1 occurs before 2 in the permutation, and Y the event that 3 occurs before 4. Which one of the following statements is TRUE?

(A) The events X and Y are mutually exclusive

(B) The events X and Y are independent

(C) Either event X or Y must occur

(D) Event X is more likely than event Y

Q.15 Which one of the following statements is FALSE?

- (A) In the cycle stealing mode of DMA, one word of data is transferred between an I/O device and main memory in a stolen cycle
- (B) For bulk data transfer, the burst mode of DMA has a higher throughput than the cycle stealing mode
- (C) Programmed I/O mechanism has a better CPU utilization than the interrupt driven I/O mechanism
- (D) The CPU can start executing an interrupt service routine faster with vectored interrupts than with non-vectored interrupts

Q.16 A user starts browsing a webpage hosted at a remote server. The browser opens a single TCP connection to fetch the entire webpage from the server. The webpage consists of a top-level index page with multiple embedded image objects. Assume that all caches (e.g., DNS cache, browser cache) are all initially empty. The following packets leave the user's computer in some order.

- (i) HTTP GET request for the index page
- (ii) DNS request to resolve the web server's name to its IP address
- (iii) HTTP GET request for an image object
- (iv) TCP SYN to open a connection to the web server

Which one of the following is the CORRECT chronological order (earliest in time to latest) of the packets leaving the computer ?

- (A) (iv), (ii), (iii), (i)
- (B) (ii), (iv), (iii), (i)
- (C) (ii), (iv), (i), (iii)
- (D) (iv), (ii), (i), (iii)

Q.17 Given an integer array of size N , we want to check if the array is sorted (in either ascending or descending order). An algorithm solves this problem by making a single pass through the array and comparing each element of the array only with its adjacent elements. The worst-case time complexity of this algorithm is

- (A) both $O(N)$ and $\Omega(N)$
- (B) $O(N)$ but not $\Omega(N)$
- (C) $\Omega(N)$ but not $O(N)$
- (D) neither $O(N)$ nor $\Omega(N)$

Q.18 Consider the following C program:

```
#include <stdio.h>

int main() {
    int a = 6;
    int b = 0;
    while(a < 10) {
        a = a / 12 + 1;
        a += b; }
    printf("%d", a);
    return 0; }
```

Which one of the following statements is CORRECT?

- (A) The program prints 9 as output
- (B) The program prints 10 as output
- (C) The program gets stuck in an infinite loop
- (D) The program prints 6 as output

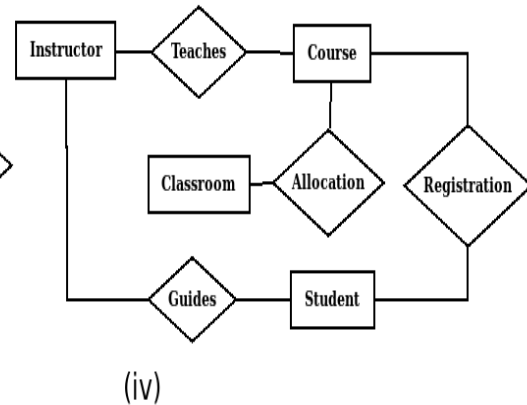
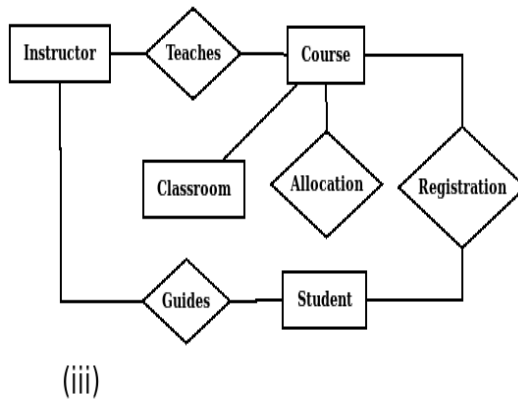
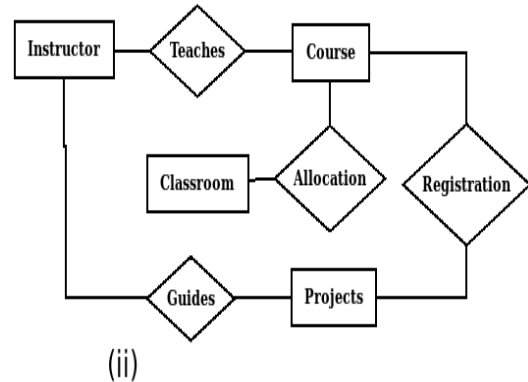
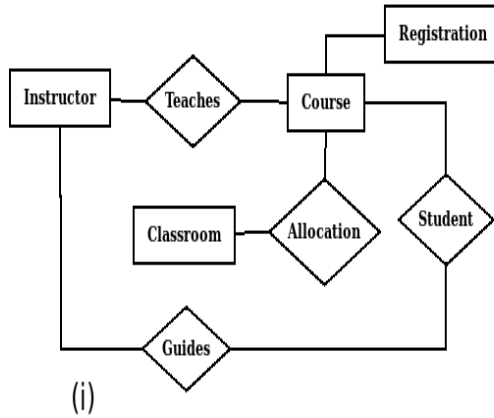
Q.19 Consider the following C program:

<pre>#include <stdio.h> void fX(); int main(){ fX(); return 0;}</pre>	<pre>void fX(){ char a; if((a=getchar()) != '\n') fX(); if(a != '\n') putchar(a);}</pre>
---	---

Assume that the input to the program from the command line is 1234 followed by a newline character. Which one of the following statements is CORRECT?

- (A) The program will not terminate
- (B) The program will terminate with no output
- (C) The program will terminate with 4321 as output
- (D) The program will terminate with 1234 as output

Q.20 Let S be the specification: "Instructors teach courses. Students register for courses. Courses are allocated classrooms. Instructors guide students." Which one of the following ER diagrams CORRECTLY represents S?



- (A) (i)
(B) (ii)
(C) (iii)
(D) (iv)

Q.21 In a B+ tree, the requirement of at least half-full (50%) node occupancy is relaxed for which one of the following cases?

- (A) Only the root node
- (B) All leaf nodes
- (C) All internal nodes
- (D) Only the leftmost leaf node

Q.22 Which of the following statements about a relation **R** in first normal form (1NF) is/are TRUE ?

- (A) **R** can have a multi-attribute key
- (B) **R** cannot have a foreign key
- (C) **R** cannot have a composite attribute
- (D) **R** cannot have more than one candidate key

Q.23 Let L_1, L_2 be two regular languages and L_3 a language which is not regular. Which of the following statements is/are *always* TRUE?

- (A) $L_1 = L_2$ if and only if $L_1 \cap \overline{L_2} = \phi$
- (B) $L_1 \cup L_3$ is not regular
- (C) $\overline{L_3}$ is not regular
- (D) $\overline{L_1} \cup \overline{L_2}$ is regular

Q.24 Which of the following statements about threads is/are TRUE?

- (A) Threads can only be implemented in kernel space
- (B) Each thread has its own file descriptor table for open files
- (C) All the threads belonging to a process share a common stack
- (D) Threads belonging to a process are by default not protected from each other

Q.25 Which of the following process state transitions is/are NOT possible?

- (A) Running to Ready
- (B) Waiting to Running
- (C) Ready to Waiting
- (D) Running to Terminated

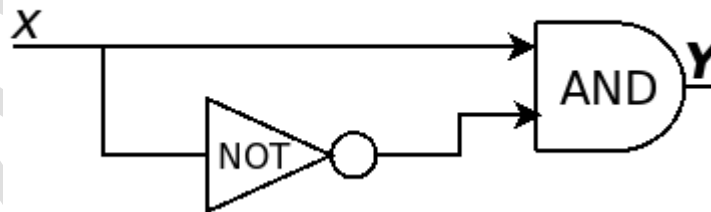
Q.26 Which of the following is/are Bottom-Up Parser(s)?

- (A) Shift-reduce Parser
- (B) Predictive Parser
- (C) LL(1) Parser
- (D) LR Parser

Q.27 Let A and B be two events in a probability space with $P(A) = 0.3$, $P(B) = 0.5$, and $P(A \cap B) = 0.1$. Which of the following statements is/are TRUE?

- (A) The two events A and B are independent
- (B) $P(A \cup B) = 0.7$
- (C) $P(A \cap B^c) = 0.2$, where B^c is the complement of the event B
- (D) $P(A^c \cap B^c) = 0.4$, where A^c and B^c are the complements of the events A and B , respectively

Q.28 Consider the circuit shown below where the gates may have propagation delays. Assume that all signal transitions occur instantaneously and that wires have no delays. Which of the following statements about the circuit is/are CORRECT?



- (A) With no propagation delays, the output Y is always logic Zero
- (B) With no propagation delays, the output Y is always logic One
- (C) With propagation delays, the output Y can have a transient logic One after X transitions from logic Zero to logic One
- (D) With propagation delays, the output Y can have a transient logic Zero after X transitions from logic One to logic Zero

Q.29 TCP client P successfully establishes a connection to TCP server Q. Let N_P denote the sequence number in the SYN sent from P to Q. Let N_Q denote the acknowledgement number in the SYN ACK from Q to P. Which of the following statements is/are CORRECT?

- (A) The sequence number N_P is chosen randomly by P
- (B) The sequence number N_P is always 0 for a new connection
- (C) The acknowledgement number N_Q is equal to N_P
- (D) The acknowledgement number N_Q is equal to $N_P + 1$

Q.30 Consider a 5-stage pipelined processor with Instruction Fetch (IF), Instruction Decode (ID), Execute (EX), Memory Access (MEM), and Register Writeback (WB) stages. Which of the following statements about *forwarding* is/are CORRECT?

- (A) In a pipelined execution, forwarding means the result from a source stage of an earlier instruction is passed on to the destination stage of a later instruction
- (B) In forwarding, data from the output of the MEM stage can be passed on to the input of the EX stage of the next instruction
- (C) Forwarding cannot prevent all pipeline stalls
- (D) Forwarding does not require any extra hardware to retrieve the data from the pipeline stages

Q.31 Which of the following fields is/are modified in the IP header of a packet going out of a network address translation (NAT) device from an internal network to an external network?

- (A) Source IP
- (B) Destination IP
- (C) Header Checksum
- (D) Total Length

Q.32 Let A and B be non-empty finite sets such that there exist one-to-one and onto functions (i) from A to B and (ii) from $A \times A$ to $A \cup B$. The number of possible values of $|A|$ is _____

Q.33 Consider the operator precedence and associativity rules for the *integer* arithmetic operators given in the table below.

Operator	Precedence	Associativity
+	Highest	Left
−	High	Right
*	Medium	Right
/	Low	Right

The value of the expression $3 + 1 + 5 * 2 / 7 + 2 - 4 - 7 - 6 / 2$ as per the above rules is _____

Q.34 The number of spanning trees in a *complete* graph of 4 vertices labelled A, B, C, and D is _____

Q.35 Consider the following two relations, $R(A,B)$ and $S(A,C)$:

R	
A	B
10	20
20	30
30	40
30	50
50	95

S	
A	C
10	90
30	45
40	80

The total number of tuples obtained by evaluating the following expression

$$\sigma_{B < C}(R \bowtie_{R.A=S.A} S)$$

is _____

Q.36 – Q.65 Carry TWO marks Each

Q.36

Consider a network path P—Q—R between nodes P and R via router Q. Node P sends a file of size 10^6 bytes to R via this path by splitting the file into chunks of 10^3 bytes each. Node P sends these chunks one after the other without any wait time between the successive chunk transmissions. Assume that the size of extra headers added to these chunks is negligible, and that the chunk size is less than the MTU.

Each of the links P—Q and Q—R has a bandwidth of 10^6 bits/sec, and negligible propagation latency. Router Q immediately transmits every packet it receives from P to R, with negligible processing and queueing delays. Router Q can simultaneously receive on link P—Q and transmit on link Q—R.

Assume P starts transmitting the chunks at time $t = 0$.

Which one of the following options gives the time (in seconds, rounded off to 3 decimal places) at which R receives all the chunks of the file?

- (A) 8.000
- (B) 8.008
- (C) 15.992
- (D) 16.000

Q.37 Consider the following syntax-directed definition (SDD).

$S \rightarrow DHTU$	$\{ S.val = D.val + H.val + T.val + U.val; \}$
$D \rightarrow "M"D_1$	$\{ D.val = 5 + D_1.val; \}$
$D \rightarrow \epsilon$	$\{ D.val = -5; \}$
$H \rightarrow "L"H_1$	$\{ H.val = 5 * 10 + H_1.val; \}$
$H \rightarrow \epsilon$	$\{ H.val = -10; \}$
$T \rightarrow "C"T_1$	$\{ T.val = 5 * 100 + T_1.val; \}$
$T \rightarrow \epsilon$	$\{ T.val = -5; \}$
$U \rightarrow "K"$	$\{ U.val = 5; \}$

Given "MMLK" as the input, which one of the following options is the CORRECT value computed by the SDD (in the attribute $S.val$)?

- (A) 45
- (B) 50
- (C) 55
- (D) 65

- Q.38 Consider the following grammar G , with S as the start symbol. The grammar G has three incomplete productions denoted by (1), (2), and (3).

$$\begin{aligned} S &\rightarrow daT \mid \underline{\quad (1) \quad} \\ T &\rightarrow aS \mid bT \mid \underline{\quad (2) \quad} \\ R &\rightarrow \underline{\quad (3) \quad} \mid \epsilon \end{aligned}$$

The set of terminals is $\{a, b, c, d, f\}$. The FIRST and FOLLOW sets of the different non-terminals are as follows.

$$\text{FIRST}(S) = \{c, d, f\}, \text{FIRST}(T) = \{a, b, \epsilon\}, \text{FIRST}(R) = \{c, \epsilon\}$$

$$\text{FOLLOW}(S) = \text{FOLLOW}(T) = \{c, f, \$\}, \text{FOLLOW}(R) = \{f\}$$

Which one of the following options CORRECTLY fills in the incomplete productions?

- (A) (1) $S \rightarrow Rf$ (2) $T \rightarrow \epsilon$ (3) $R \rightarrow cTR$
- (B) (1) $S \rightarrow fR$ (2) $T \rightarrow \epsilon$ (3) $R \rightarrow cTR$
- (C) (1) $S \rightarrow fR$ (2) $T \rightarrow cT$ (3) $R \rightarrow cR$
- (D) (1) $S \rightarrow Rf$ (2) $T \rightarrow cT$ (3) $R \rightarrow cR$

Q.39 Consider the following pseudo-code.

```
L1:      t1 = -1
L2:      t2 = 0
L3:      t3 = 0
L4:      t4 = 4 * t3
L5:      t5 = 4 * t2
L6:      t6 = t5 * M
L7:      t7 = t4 + t6
L8:      t8 = a[t7]
L9:      if t8 <= max goto L11
L10:     t1 = t8
L11:     t3 = t3 + 1
L12:     if t3 < M goto L4
L13:     t2 = t2 + 1
L14:     if t2 < N goto L3
L15:     max = t1
```

Which one of the following options CORRECTLY specifies the number of basic blocks and the number of instructions in the largest basic block, respectively ?

- (A) 6 and 6
- (B) 6 and 7
- (C) 7 and 7
- (D) 7 and 6

- Q.40 Consider the following two threads T1 and T2 that update two shared variables a and b. Assume that initially $a = b = 1$. Though context switching between threads can happen at any time, each statement of T1 or T2 is executed atomically without interruption.

T1	T2
$a = a + 1;$	$b = 2 * b;$
$b = b + 1;$	$a = 2 * a;$

Which one of the following options lists all the possible combinations of values of a and b after both T1 and T2 finish execution?

- (A) $(a = 4, b = 4); (a = 3, b = 3); (a = 4, b = 3)$
- (B) $(a = 3, b = 4); (a = 4, b = 3); (a = 3, b = 3)$
- (C) $(a = 4, b = 4); (a = 4, b = 3); (a = 3, b = 4)$
- (D) $(a = 2, b = 2); (a = 2, b = 3); (a = 3, b = 4)$

- Q.41 An array [82, 101, 90, 11, 111, 75, 33, 131, 44, 93] is heapified. Which one of the following options represents the first three elements in the heapified array?

- (A) 82, 90, 101
- (B) 82, 11, 93
- (C) 131, 11, 93
- (D) 131, 111, 90

Q.42 Consider the following recurrence relation:

$$T(n) = \begin{cases} \sqrt{n}T(\sqrt{n}) + n & \text{for } n \geq 1, \\ 1 & \text{for } n = 1. \end{cases}$$

Which one of the following options is CORRECT?

(A) $T(n) = \Theta(n \log \log n)$

(B) $T(n) = \Theta(n \log n)$

(C) $T(n) = \Theta(n^2 \log n)$

(D) $T(n) = \Theta(n^2 \log \log n)$

Q.43 Consider a binary min-heap containing 105 distinct elements. Let k be the index (in the underlying array) of the maximum element stored in the heap. The number of possible values of k is

(A) 53

(B) 52

(C) 27

(D) 1

Q.44 The symbol $\rightarrow$ indicates functional dependency in the context of a relational database. Which of the following options is/are TRUE?

- (A) $(X, Y) \rightarrow (Z, W)$ implies $X \rightarrow (Z, W)$
- (B) $(X, Y) \rightarrow (Z, W)$ implies $(X, Y) \rightarrow Z$
- (C) $((X, Y) \rightarrow Z \text{ and } W \rightarrow Y)$ implies $(X, W) \rightarrow Z$
- (D) $(X \rightarrow Y \text{ and } Y \rightarrow Z)$ implies $X \rightarrow Z$

Q.45 Let G be a directed graph and T a depth first search (DFS) spanning tree in G that is rooted at a vertex v . Suppose T is also a breadth first search (BFS) tree in G , rooted at v . Which of the following statements is/are TRUE for *every* such graph G and tree T ?

- (A) There are no back-edges in G with respect to the tree T
- (B) There are no cross-edges in G with respect to the tree T
- (C) There are no forward-edges in G with respect to the tree T
- (D) The only edges in G are the edges in T

Q.46 Consider the following read-write schedule S over three transactions T_1 , T_2 , and T_3 , where the subscripts in the schedule indicate transaction IDs:

$S: r_1(z); w_1(z); r_2(x); r_3(y); w_3(y); r_2(y); w_2(x); w_2(y);$

Which of the following transaction schedules is/are conflict equivalent to S ?

(A) $T_1T_2T_3$

(B) $T_1T_3T_2$

(C) $T_3T_2T_1$

(D) $T_3T_1T_2$

Q.47 Consider a Boolean expression given by $F(X, Y, Z) = \sum(3, 5, 6, 7)$.

Which of the following statements is/are CORRECT?

(A) $F(X, Y, Z) = \Pi(0, 1, 2, 4)$

(B) $F(X, Y, Z) = XY + YZ + XZ$

(C) $F(X, Y, Z)$ is independent of input Y

(D) $F(X, Y, Z)$ is independent of input X

Q.48 Consider the following C function definition.

```
int f(int x, int y) {  
    for (int i=0; i<y; i++) {  
        x=x+x+y;  
    }  
    return x;  
}
```

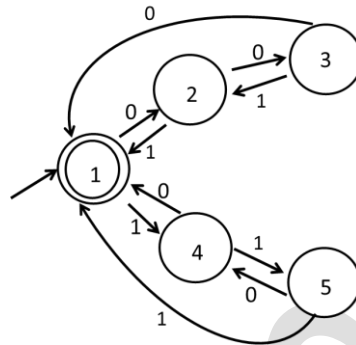
Which of the following statements is/are TRUE about the above function?

- (A) If the inputs are $x=20$, $y=10$, then the return value is greater than 2^{20}
- (B) If the inputs are $x=20$, $y=20$, then the return value is greater than 2^{20}
- (C) If the inputs are $x=20$, $y=10$, then the return value is less than 2^{10}
- (D) If the inputs are $x=10$, $y=20$, then the return value is greater than 2^{20}

Q.49 Let A be any $n \times m$ matrix, where $m > n$. Which of the following statements is/are TRUE about the system of linear equations $Ax = \mathbf{0}$?

- (A) There exist at least $m - n$ linearly independent solutions to this system
- (B) There exist $m - n$ linearly independent vectors such that every solution is a linear combination of these vectors
- (C) There exists a non-zero solution in which at least $m - n$ variables are 0
- (D) There exists a solution in which at least n variables are non-zero

- Q.50 Consider the 5-state DFA M accepting the language $L(M) \subset (0 + 1)^*$ shown below. For any string $w \in (0 + 1)^*$ let $n_0(w)$ be the number of 0's in w and $n_1(w)$ be the number of 1's in w .



Which of the following statements is/are FALSE?

- (A) States 2 and 4 are distinguishable in M
- (B) States 3 and 4 are distinguishable in M
- (C) States 2 and 5 are distinguishable in M
- (D) Any string w with $n_0(w) = n_1(w)$ is in $L(M)$

Q.51 The chromatic number of a graph is the minimum number of colours used in a *proper* colouring of the graph. Let G be any graph with n vertices and chromatic number k . Which of the following statements is/are always TRUE?

- (A) G contains a complete subgraph with k vertices
- (B) G contains an independent set of size at least n/k
- (C) G contains at least $k(k - 1)/2$ edges
- (D) G contains a vertex of degree at least k

Q.52 Consider the operators $\diamond$ and $\square$ defined by $a \diamond b = a + 2b$, $a \square b = ab$, for positive integers. Which of the following statements is/are TRUE?

- (A) Operator $\diamond$ obeys the associative law
- (B) Operator $\square$ obeys the associative law
- (C) Operator $\diamond$ over the operator $\square$ obeys the distributive law
- (D) Operator $\square$ over the operator $\diamond$ obeys the distributive law

Q.53 Consider two set-associative cache memory architectures: **WBC**, which uses the write back policy, and **WTC**, which uses the write through policy. Both of them use the LRU (*Least Recently Used*) block replacement policy. The cache memory is connected to the main memory. Which of the following statements is/are TRUE?

- (A) A read miss in **WBC** never evicts a dirty block
- (B) A read miss in **WTC** never triggers a write back operation of a cache block to main memory
- (C) A write hit in **WBC** can modify the value of the dirty bit of a cache block
- (D) A write miss in **WTC** always writes the victim cache block to main memory before loading the missed block to the cache

Q.54 Consider a 512 GB hard disk with 32 storage surfaces. There are 4096 sectors per track and each sector holds 1024 bytes of data. The number of cylinders in the hard disk is _____

Q.55 The baseline execution time of a program on a 2 GHz single core machine is 100 nanoseconds (*ns*). The code corresponding to 90% of the execution time can be fully parallelized. The overhead for using an additional core is 10 *ns* when running on a multicore system. Assume that all cores in the multicore system run their share of the parallelized code for an equal amount of time.

The number of cores that minimize the execution time of the program is _____

Q.56 A given program has 25% load/store instructions. Suppose the ideal CPI (cycles per instruction) without any memory stalls is 2. The program exhibits 2% miss rate on instruction cache and 8% miss rate on data cache. The miss penalty is 100 cycles. The speedup (*rounded off to two decimal places*) achieved with a perfect cache (i.e., with NO data or instruction cache misses) is _____

- Q.57 Consider the following code snippet using the `fork()` and `wait()` system calls. Assume that the code compiles and runs correctly, and that the system calls run successfully without any errors.

```
int x = 3;
while(x > 0) {
    fork();
    printf("hello");
    wait(NULL);
    x--;
}
```

The total number of times the `printf` statement is executed is _____

- Q.58 Consider the entries shown below in the forwarding table of an IP router. Each entry consists of an IP prefix and the corresponding next hop router for packets whose destination IP address matches the prefix. The notation “/N” in a prefix indicates a subnet mask with the most significant N bits set to 1.

Prefix	Next hop router
10.1.1.0/24	R1
10.1.1.128/25	R2
10.1.1.64/26	R3
10.1.1.192/26	R4

This router forwards 20 packets each to 5 hosts. The IP addresses of the hosts are 10.1.1.16, 10.1.1.72, 10.1.1.132, 10.1.1.191, and 10.1.1.205. The number of packets forwarded via the next hop router R2 is _____

- Q.59 Let $G = (V, \Sigma, S, P)$ be a context-free grammar in Chomsky Normal Form with $\Sigma = \{a, b, c\}$ and V containing 10 variable symbols including the start symbol S . The string $w = a^{30}b^{30}c^{30}$ is derivable from S . The number of steps (application of rules) in the derivation $S \rightarrow^* w$ is _____

Q.60 The number of edges present in the forest generated by the DFS traversal of an undirected graph G with 100 vertices is 40. The number of connected components in G is _____

Q.61 Consider the following two regular expressions over the alphabet $\{0,1\}$:

$$r = 0^* + 1^*$$

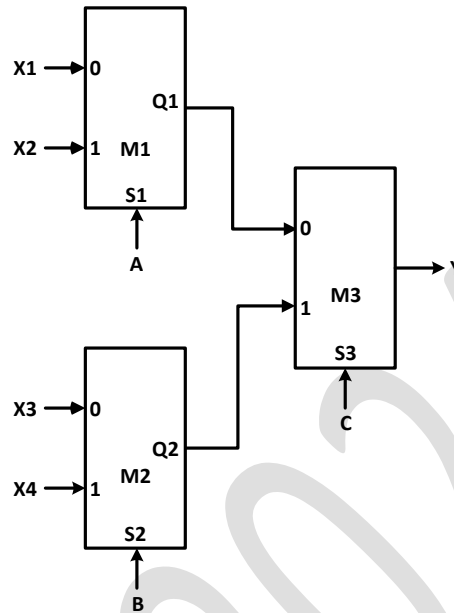
$$s = 01^* + 10^*$$

The total number of strings of length less than or equal to 5, which are neither in r nor in s , is _____

Q.62 Consider a memory management system that uses a page size of 2 KB. Assume that both the physical and virtual addresses start from 0. Assume that the pages 0, 1, 2, and 3 are stored in the page frames 1, 3, 2, and 0, respectively. The physical address (*in decimal format*) corresponding to the virtual address 2500 (*in decimal format*) is _____

Q.63 A bag contains 10 red balls and 15 blue balls. Two balls are drawn randomly without replacement. Given that the first ball drawn is red, the probability (*rounded off to 3 decimal places*) that both balls drawn are red is _____

- Q.64 Consider a digital logic circuit consisting of three 2-to-1 multiplexers M1, M2, and M3 as shown below. X1 and X2 are inputs of M1. X3 and X4 are inputs of M2. A, B, and C are select lines of M1, M2, and M3, respectively.



For an instance of inputs $X1=1$, $X2=1$, $X3=0$, and $X4=0$, the number of combinations of A, B, C that give the output $Y=1$ is _____

- Q.65 Consider sending an IP datagram of size 1420 bytes (including 20 bytes of IP header) from a sender to a receiver over a path of two links with a router between them. The first link (sender to router) has an MTU (Maximum Transmission Unit) size of 542 bytes, while the second link (router to receiver) has an MTU size of 360 bytes. The number of fragments that would be delivered at the receiver is _____

GATE 2024

Set 2 (S6)

65 questions | 40 pages

Paper 4 of 16

General Aptitude (GA)

Q.1 – Q.5 Carry ONE mark Each

Q.1 If '→' denotes increasing order of intensity, then the meaning of the words [walk → jog → sprint] is analogous to [bothered → _____ → daunted]. Which one of the given options is appropriate to fill the blank?

- (A) phased
- (B) phrased
- (C) fazed
- (D) fused

Q.2 Two wizards try to create a spell using all the four elements, *water*, *air*, *fire*, and *earth*. For this, they decide to mix all these elements in all possible orders. They also decide to work independently. After trying all possible combination of elements, they conclude that the spell does not work. How many attempts does each wizard make before coming to this conclusion, independently?

- (A) 24
- (B) 48
- (C) 16
- (D) 12

- Q.3 In an engineering college of 10,000 students, 1,500 like neither their core branches nor other branches. The number of students who like their core branches is $\frac{1}{4}^{\text{th}}$ of the number of students who like other branches. The number of students who like both their core and other branches is 500.

The number of students who like their core branches is

- (A) 1,800
- (B) 3,500
- (C) 1,600
- (D) 1,500

- Q.4 For positive non-zero real variables x and y , if

$$\ln\left(\frac{x+y}{2}\right) = \frac{1}{2}[\ln(x) + \ln(y)]$$

then, the value of $\frac{x}{y} + \frac{y}{x}$ is

- (A) 1
- (B) $\frac{1}{2}$
- (C) 2
- (D) 4

Q.5 In the sequence 6, 9, 14, x , 30, 41, a possible value of x is

(A) 25

(B) 21

(C) 18

(D) 20

GATE 2024

Q.6 – Q.10 Carry TWO marks Each

Q.6 Sequence the following sentences in a coherent passage.

P: This fortuitous geological event generated a colossal amount of energy and heat that resulted in the rocks rising to an average height of 4 km across the contact zone.

Q: Thus, the geophysicists tend to think of the Himalayas as an active geological event rather than as a static geological feature.

R: The natural process of the cooling of this massive edifice absorbed large quantities of atmospheric carbon dioxide, altering the earth's atmosphere and making it better suited for life.

S: Many millennia ago, a breakaway chunk of bedrock from the Antarctic Plate collided with the massive Eurasian Plate.

(A) QPSR

(B) QSPR

(C) SPRQ

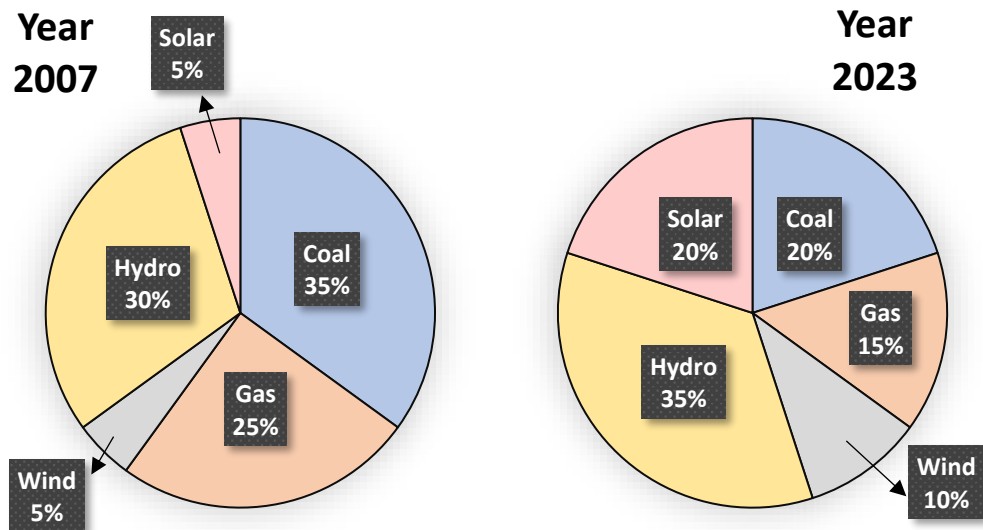
(D) SRPQ

Q.7 A person sold two different items at the same price. He made 10% profit in one item, and 10% loss in the other item. In selling these two items, the person made a total of

- (A) 1% profit
- (B) 2% profit
- (C) 1% loss
- (D) 2% loss

GATE 2024

Q.8 The pie charts depict the shares of various power generation technologies in the total electricity generation of a country for the years 2007 and 2023.



The renewable sources of electricity generation consist of Hydro, Solar and Wind. Assuming that the total electricity generated remains the same from 2007 to 2023, what is the percentage increase in the share of the renewable sources of electricity generation over this period?

- (A) 25%
- (B) 50%
- (C) 77.5%
- (D) 62.5%

Q.9 A cube is to be cut into 8 pieces of equal size and shape. Here, each cut should be straight and it should not stop till it reaches the other end of the cube.

The minimum number of such cuts required is

- (A) 3
- (B) 4
- (C) 7
- (D) 8

GATE 2024

- Q.10 In the 4×4 array shown below, each cell of the first three rows has either a cross (X) or a number.

1	X	4	3
X	5	5	4
3	X	6	X

The number in a cell represents the count of the immediate neighboring cells (left, right, top, bottom, diagonals) NOT having a cross (X). Given that the last row has no crosses (X), the sum of the four numbers to be filled in the last row is

- (A) 11
(B) 10
(C) 12
(D) 9

Q.11 – Q.35 Carry ONE mark Each

Q.11 Consider a computer with a 4 MHz processor. Its DMA controller can transfer 8 bytes in 1 cycle from a device to main memory through cycle stealing at regular intervals. Which one of the following is the data transfer rate (*in bits per second*) of the DMA controller if 1% of the processor cycles are used for DMA?

- (A) 2,56,000
- (B) 3,200
- (C) 25,60,000
- (D) 32,000

Q.12 Let p and q be the following propositions:

p : Fail grade can be given.

q : Student scores more than 50% marks.

Consider the statement: “*Fail grade cannot be given when student scores more than 50% marks.*”

Which one of the following is the CORRECT representation of the above statement in propositional logic?

- (A) $q \rightarrow \neg p$
- (B) $q \rightarrow p$
- (C) $p \rightarrow q$
- (D) $\neg p \rightarrow q$

- Q.13 Consider the following C program. Assume parameters to a function are evaluated from *right to left*.

```
#include <stdio.h>

int g(int p) { printf("%d", p); return p; }
int h(int q) { printf("%d", q); return q; }

void f(int x, int y) {
    g(x);
    h(y);
}

int main() {
    f(g(10), h(20));
}
```

Which one of the following options is the CORRECT output of the above C program?

- (A) 20101020
- (B) 10202010
- (C) 20102010
- (D) 10201020

Q.14 The format of a single-precision floating-point number as per the IEEE 754 standard is:

Sign (1bit)	Exponent (8 bits)	Mantissa (23 bits)
----------------	----------------------	-----------------------

Choose the largest floating-point number among the following options.

(A)

Sign	Exponent	Mantissa
0	0111 1111	1111 1111 1111 1111 1111 111

(B)

Sign	Exponent	Mantissa
0	1111 1110	1111 1111 1111 1111 1111 111

(C)

Sign	Exponent	Mantissa
0	1111 1111	1111 1111 1111 1111 1111 111

(D)

Sign	Exponent	Mantissa
0	0111 1111	0000 0000 0000 0000 0000 000

Q.15 Let $T(n)$ be the recurrence relation defined as follows:

$$T(0) = 1,$$

$$T(1) = 2, \text{ and}$$

$$T(n) = 5T(n-1) - 6T(n-2) \quad \text{for } n \geq 2$$

Which one of the following statements is TRUE?

(A) $T(n) = \Theta(2^n)$

(B) $T(n) = \Theta(n2^n)$

(C) $T(n) = \Theta(3^n)$

(D) $T(n) = \Theta(n3^n)$

Q.16 Let $f(x)$ be a continuous function from $\mathbb{R}$ to $\mathbb{R}$ such that

$$f(x) = 1 - f(2-x)$$

Which one of the following options is the CORRECT value of $\int_0^2 f(x) dx$?

(A) 0

(B) 1

(C) 2

(D) -1

Q.17 Let A be the adjacency matrix of a simple undirected graph G . Suppose A is its own inverse. Which one of the following statements is *always* TRUE?

- (A) G is a cycle
- (B) G is a perfect matching
- (C) G is a complete graph
- (D) There is no such graph G

Q.18 When six unbiased dice are rolled simultaneously, the probability of getting all distinct numbers (*i.e.*, 1, 2, 3, 4, 5, and 6) is

- (A) $\frac{1}{324}$
- (B) $\frac{5}{324}$
- (C) $\frac{7}{324}$
- (D) $\frac{11}{324}$

Q.19 Once the DBMS informs the user that a transaction has been successfully completed, its effect should persist even if the system crashes before all its changes are reflected on disk. This property is called

- (A) durability
- (B) atomicity
- (C) consistency
- (D) isolation

Q.20 In the context of owner and weak entity sets in the ER (Entity-Relationship) data model, which one of the following statements is TRUE?

- (A) The weak entity set **MUST** have total participation in the identifying relationship
- (B) The owner entity set **MUST** have total participation in the identifying relationship
- (C) Both weak and owner entity sets **MUST** have total participation in the identifying relationship
- (D) Neither weak entity set nor owner entity set **MUST** have total participation in the identifying relationship

Q.21 Consider the following two sets:

Set X

- P. Lexical Analyzer
- Q. Syntax Analyzer
- R. Intermediate Code Generator
- S. Code Optimizer

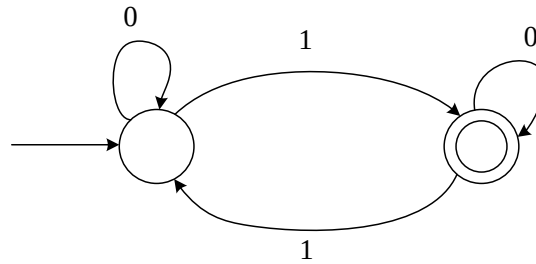
Set Y

- 1. Abstract Syntax Tree
- 2. Token
- 3. Parse Tree
- 4. Constant Folding

Which one of the following options is the CORRECT match from **Set X** to **Set Y**?

- (A) P – 4; Q – 1; R – 3; S – 2
- (B) P – 2; Q – 3; R – 1; S – 4
- (C) P – 2; Q – 1; R – 3; S – 4
- (D) P – 4; Q – 3; R – 2; S – 1

Q.22 Which one of the following regular expressions is equivalent to the language accepted by the DFA given below?



- (A) $0^*1(0 + 10^*1)^*$
- (B) $0^*(10^*11)^*0^*$
- (C) $0^*1(010^*1)^*0^*$
- (D) $0(1 + 0^*10^*1)^*0^*$

Q.23 Node X has a TCP connection open to node Y. The packets from X to Y go through an intermediate IP router R. Ethernet switch S is the first switch on the network path between X and R. Consider a packet sent from X to Y over this connection.

Which of the following statements is/are TRUE about the destination IP and MAC addresses on this packet at the time it leaves X?

- (A) The destination IP address is the IP address of R
- (B) The destination IP address is the IP address of Y
- (C) The destination MAC address is the MAC address of S
- (D) The destination MAC address is the MAC address of Y

Q.24 Which of the following tasks is/are the responsibility/responsibilities of the memory management unit (MMU) in a system with paging-based memory management?

- (A) Allocate a new page table for a newly created process
- (B) Translate a virtual address to a physical address using the page table
- (C) Raise a trap when a virtual address is not found in the page table
- (D) Raise a trap when a process tries to write to a page marked with read-only permission in the page table

Q.25 Consider a process P running on a CPU. Which one or more of the following events will *always* trigger a context switch by the OS that results in process P moving to a non-running state (e.g., ready, blocked)?

- (A) P makes a blocking system call to read a block of data from the disk
- (B) P tries to access a page that is in the swap space, triggering a page fault
- (C) An interrupt is raised by the disk to deliver data requested by some other process
- (D) A timer interrupt is raised by the hardware

Q.26 Which of the following file organizations is/are I/O efficient for the scan operation in DBMS?

- (A) Sorted
- (B) Heap
- (C) Unclustered tree index
- (D) Unclustered hash index

Q.27 Which of the following statements about the Two Phase Locking (2PL) protocol is/are TRUE?

- (A) 2PL permits only serializable schedules
- (B) With 2PL, a transaction always locks the data item being read or written just before every operation and always releases the lock just after the operation
- (C) With 2PL, once a lock is released on any data item inside a transaction, no more locks on any data item can be obtained inside that transaction
- (D) A deadlock is possible with 2PL

Q.28 Which of the following statements about IPv4 fragmentation is/are TRUE?

- (A) The fragmentation of an IP datagram is performed *only* at the source of the datagram
- (B) The fragmentation of an IP datagram is performed at any IP router which finds that the size of the datagram to be transmitted exceeds the MTU
- (C) The reassembly of fragments is performed *only* at the destination of the datagram
- (D) The reassembly of fragments is performed at all intermediate routers along the path from the source to the destination

Q.29 Which of the following statements is/are FALSE?

- (A) An attribute grammar is a syntax-directed definition (SDD) in which the functions in the semantic rules have no side effects
- (B) The attributes in a L-attributed definition cannot always be evaluated in a depth-first order
- (C) Synthesized attributes can be evaluated by a bottom-up parser as the input is parsed
- (D) All L-attributed definitions based on LR(1) grammar can be evaluated using a bottom-up parsing strategy

Q.30 For a Boolean variable x , which of the following statements is/are FALSE?

- (A) $x \cdot 1 = x$
- (B) $x + 1 = x$
- (C) $x \cdot x = 0$
- (D) $x + \bar{x} = 1$

Q.31 An instruction format has the following structure:

Instruction Number: *Opcode destination reg, source reg-1, source reg-2*

Consider the following sequence of instructions to be executed in a pipelined processor:

I1: DIV R3, R1, R2

I2: SUB R5, R3, R4

I3: ADD R3, R5, R6

I4: MUL R7, R3, R8

Which of the following statements is/are TRUE?

- (A) There is a RAW dependency on R3 between I1 and I2
- (B) There is a WAR dependency on R3 between I1 and I3
- (C) There is a RAW dependency on R3 between I2 and I3
- (D) There is a WAW dependency on R3 between I3 and I4

Q.32 Which of the following fields of an IP header is/are *always* modified by any router before it forwards the IP packet?

- (A) Source IP Address
- (B) Protocol
- (C) Time to Live (TTL)
- (D) Header Checksum

Q.33 Consider the following C function definition.

```
int fX(char *a) {  
    char *b = a;  
    while(*b)  
        b++;  
    return b - a; }
```

Which of the following statements is/are TRUE?

- (A) The function call `fX("abcd")` will always return a value
- (B) Assuming a character array `c` is declared as `char c[] = "abcd"` in `main()`, the function call `fX(c)` will always return a value
- (C) The code of the function will not compile
- (D) Assuming a character pointer `c` is declared as `char *c = "abcd"` in `main()`, the function call `fX(c)` will always return a value

Q.34 Let P be the partial order defined on the set $\{1,2,3,4\}$ as follows

$$P = \{(x, x) \mid x \in \{1,2,3,4\}\} \cup \{(1,2), (3,2), (3,4)\}$$

The number of total orders on $\{1,2,3,4\}$ that contain P is _____

Q.35 Let A be an array containing integer values. The distance of A is defined as the minimum number of elements in A that must be replaced with another integer so that the resulting array is sorted in non-decreasing order. The distance of the array $[2, 5, 3, 1, 4, 2, 6]$ is _____

Q.36 – Q.65 Carry TWO marks Each

Q.36 What is the output of the following C program?

```
#include <stdio.h>

int main() {

    double a[2]={20.0, 25.0}, *p, *q;

    p = a;

    q = p + 1;

    printf("%d,%d", (int)(q - p), (int)(*q - *p));

    return 0;}
```

- (A) 4,8
- (B) 1,5
- (C) 8,5
- (D) 1,8

- Q.37 Consider a single processor system with four processes A, B, C, and D, represented as given below, where for each process the first value is its arrival time, and the second value is its CPU burst time.

A (0, 10), B (2, 6), C (4, 3), and D (6, 7).

Which one of the following options gives the average waiting times when preemptive Shortest Remaining Time First (SRTF) and Non-Preemptive Shortest Job First (NP-SJF) CPU scheduling algorithms are applied to the processes?

- (A) SRTF = 6, NP-SJF = 7
- (B) SRTF = 6, NP-SJF = 7.5
- (C) SRTF = 7, NP-SJF = 7.5
- (D) SRTF = 7, NP-SJF = 8.5

- Q.38 Which one of the following CIDR prefixes exactly represents the range of IP addresses 10.12.2.0 to 10.12.3.255?

- (A) 10.12.2.0/23
- (B) 10.12.2.0/24
- (C) 10.12.0.0/22
- (D) 10.12.2.0/22

- Q.39 You are given a set V of distinct integers. A binary search tree T is created by inserting all elements of V one by one, starting with an empty tree. The tree T follows the convention that, at each node, all values stored in the left subtree of the node are smaller than the value stored at the node. You are not aware of the sequence in which these values were inserted into T , and you do not have access to T .

Which one of the following statements is TRUE?

- (A) Inorder traversal of T can be determined from V
- (B) Root node of T can be determined from V
- (C) Preorder traversal of T can be determined from V
- (D) Postorder traversal of T can be determined from V

Q.40 Consider the following context-free grammar where the start symbol is S and the set of terminals is $\{a,b,c,d\}$.

$$S \rightarrow AaAb \mid BbBa$$

$$A \rightarrow cS \mid \epsilon$$

$$B \rightarrow dS \mid \epsilon$$

The following is a partially-filled LL(1) parsing table.

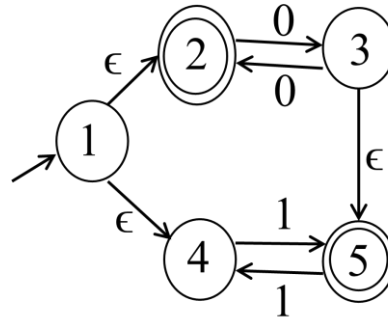
	a	b	c	d	$\$$
S	$S \rightarrow AaAb$	$S \rightarrow BbBa$	(1)	(2)	
A	$A \rightarrow \epsilon$	(3)	$A \rightarrow cS$		
B	(4)	$B \rightarrow \epsilon$		$B \rightarrow dS$	

Which one of the following options represents the CORRECT combination for the numbered cells in the parsing table?

Note: In the options, "blank" denotes that the corresponding cell is empty.

- (A) (1) $S \rightarrow AaAb$ (2) $S \rightarrow BbBa$ (3) $A \rightarrow \epsilon$ (4) $B \rightarrow \epsilon$
- (B) (1) $S \rightarrow BbBa$ (2) $S \rightarrow AaAb$ (3) $A \rightarrow \epsilon$ (4) $B \rightarrow \epsilon$
- (C) (1) $S \rightarrow AaAb$ (2) $S \rightarrow BbBa$ (3) blank (4) blank
- (D) (1) $S \rightarrow BbBa$ (2) $S \rightarrow AaAb$ (3) blank (4) blank

Q.41 Let M be the 5-state NFA with ϵ -transitions shown in the diagram below.



Which one of the following regular expressions represents the language accepted by M ?

- (A) $(00)^* + 1(11)^*$
- (B) $0^* + (1 + 0(00)^*)(11)^*$
- (C) $(00)^* + (1 + (00)^*)(11)^*$
- (D) $0^+ + 1(11)^* + 0(11)^*$

Q.42 Consider an array X that contains n positive integers. A subarray of X is defined to be a sequence of array locations with consecutive indices.

The C code snippet given below has been written to compute the length of the longest subarray of X that contains at most two distinct integers. The code has two missing expressions labelled (P) and (Q) .

```
int first=0, second=0, len1=0, len2=0, maxlen=0;
for (int i=0; i < n; i++) {
    if (X[i] == first) {
        len2++; len1++;
    } else if (X[i] == second) {
        len2++;
        len1 = (P) ;
        second = first;
    } else {
        len2 = (Q) ;
        len1 = 1; second = first;
    }
    if (len2 > maxlen) {
        maxlen = len2;
    }
    first = X[i];
}
```

Which one of the following options gives the CORRECT missing expressions?

(Hint: At the end of the i -th iteration, the value of $len1$ is the length of the longest subarray ending with $X[i]$ that contains all equal values, and $len2$ is the length of the longest subarray ending with $X[i]$ that contains at most two distinct values.)

- (A) (P) $len1+1$ (Q) $len2+1$
- (B) (P) 1 (Q) $len1+1$
- (C) (P) 1 (Q) $len2+1$
- (D) (P) $len2+1$ (Q) $len1+1$

- Q.43 Consider the following expression: $x[i] = (p + r) * -s[i] + u/w$. The following sequence shows the list of triples representing the given expression, with entries missing for triples (1), (3), and (6).

(0)	+	p	r
(1)			
(2)	uminus	(1)	
(3)			
(4)	/	u	w
(5)	+	(3)	(4)
(6)			
(7)	=	(6)	(5)

Which one of the following options fills in the missing entries CORRECTLY?

- (A) (1) = [] $s\ i$ (3) * (0) (2) (6) [] = $x\ i$
- (B) (1) [] = $s\ i$ (3) - (0) (2) (6) = [] $x\ (5)$
- (C) (1) = [] $s\ i$ (3) * (0) (2) (6) [] = $x\ (5)$
- (D) (1) [] = $s\ i$ (3) - (0) (2) (6) = [] $x\ i$

Q.44 Let x and y be random variables, not necessarily independent, that take real values in the interval $[0,1]$. Let $z = xy$ and let the mean values of x, y, z be $\bar{x}, \bar{y}, \bar{z}$, respectively. Which one of the following statements is TRUE?

(A) $\bar{z} = \bar{x}\bar{y}$

(B) $\bar{z} \leq \bar{x}\bar{y}$

(C) $\bar{z} \geq \bar{x}\bar{y}$

(D) $\bar{z} \leq \bar{x}$

Q.45 The relation schema, $\text{Person}(\underline{\text{pid}}, \text{city})$, describes the city of residence for every person uniquely identified by pid . The following relational algebra operators are available: selection, projection, cross product, and rename.

To find the list of cities where at least 3 persons reside, using the above operators, the minimum number of cross product operations that must be used is

(A) 1

(B) 2

(C) 3

(D) 4

- Q.46 Consider a multi-threaded program with two threads T1 and T2. The threads share two semaphores: s1 (initialized to 1) and s2 (initialized to 0). The threads also share a global variable x (initialized to 0). The threads execute the code shown below.

<pre>// code of T1 wait(s1); x = x+1; print(x); wait(s2); signal(s1);</pre>	<pre>// code of T2 wait(s1); x = x+1; print(x); signal(s2); signal(s1);</pre>
---	---

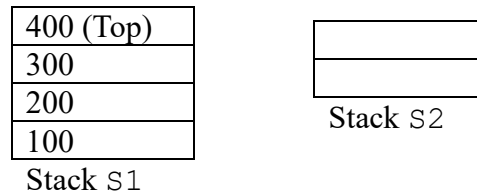
Which of the following outcomes is/are possible when threads T1 and T2 execute concurrently?

- (A) T1 runs first and prints 1, T2 runs next and prints 2
- (B) T2 runs first and prints 1, T1 runs next and prints 2
- (C) T1 runs first and prints 1, T2 does not print anything (deadlock)
- (D) T2 runs first and prints 1, T1 does not print anything (deadlock)

- Q.47 Let A be an $n \times n$ matrix over the set of all real numbers $\mathbb{R}$. Let B be a matrix obtained from A by swapping two rows. Which of the following statements is/are TRUE?

- (A) The determinant of B is the negative of the determinant of A
- (B) If A is invertible, then B is also invertible
- (C) If A is symmetric, then B is also symmetric
- (D) If the trace of A is zero, then the trace of B is also zero

- Q.48 Let S_1 and S_2 be two stacks. S_1 has capacity of 4 elements. S_2 has capacity of 2 elements. S_1 already has 4 elements: 100, 200, 300, and 400, whereas S_2 is empty, as shown below.



Only the following three operations are available:

`PushToS2`: Pop the top element from S_1 and push it on S_2 .

`PushToS1`: Pop the top element from S_2 and push it on S_1 .

`GenerateOutput`: Pop the top element from S_1 and output it to the user.

Note that the pop operation is not allowed on an empty stack and the push operation is not allowed on a full stack.

Which of the following output sequences can be generated by using the above operations?

- (A) 100, 200, 400, 300
- (B) 200, 300, 400, 100
- (C) 400, 200, 100, 300
- (D) 300, 200, 400, 100

Q.49 Which of the following is/are EQUAL to 224 in radix-5 (*i.e.*, base-5) notation?

- (A) 64 in radix-10
- (B) 100 in radix-8
- (C) 50 in radix-16
- (D) 121 in radix-7

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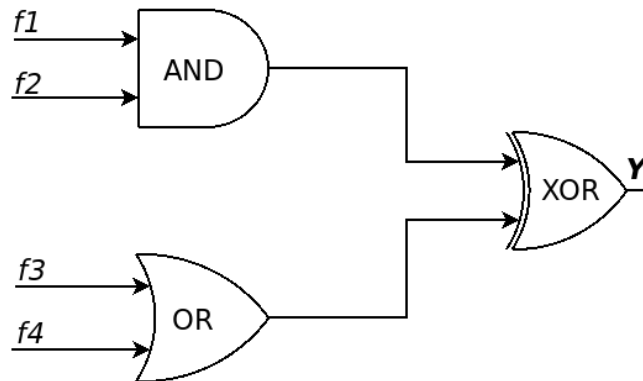
Q.50 Consider 4-variable functions f_1 , f_2 , f_3 , f_4 expressed in sum-of-minterms form as given below.

$$f_1 = \sum(0,2,3,5,7,8,11,13)$$

$$f_2 = \sum(1,3,5,7,11,13,15)$$

$$f_3 = \sum(0,1,4,11)$$

$$f_4 = \sum(0,2,6,13)$$



With respect to the circuit given above, which of the following options is/are CORRECT?

- (A) $Y = \sum(0,1,2,11,13)$
- (B) $Y = \Pi(3,4,5,6,7,8,9,10,12,14,15)$
- (C) $Y = \sum(0,1,2,3,4,5,6,7)$
- (D) $Y = \Pi(8,9,10,11,12,13,14,15)$

Q.51 Let G be an undirected connected graph in which every edge has a positive integer weight. Suppose that every spanning tree in G has *even* weight. Which of the following statements is/are TRUE for *every* such graph G ?

- (A) All edges in G have even weight
- (B) All edges in G have even weight **OR** all edges in G have odd weight
- (C) In each cycle C in G , all edges in C have even weight
- (D) In each cycle C in G , either all edges in C have even weight **OR** all edges in C have odd weight

Q.52 Consider a context-free grammar G with the following 3 rules.

$$S \rightarrow aS, S \rightarrow aSbS, S \rightarrow c$$

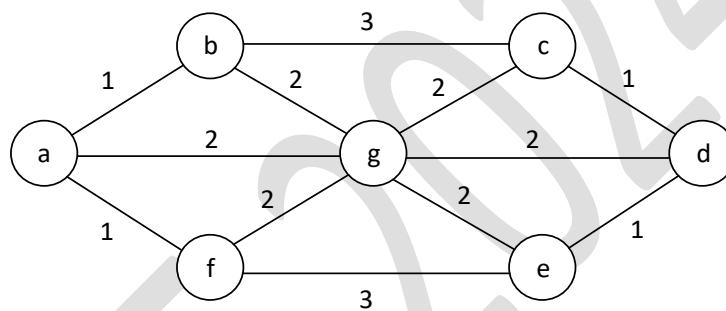
Let $w \in L(G)$. Let $n_a(w)$, $n_b(w)$, $n_c(w)$ denote the number of times a, b, c occur in w , respectively. Which of the following statements is/are TRUE?

- (A) $n_a(w) > n_b(w)$
- (B) $n_a(w) > n_c(w) - 2$
- (C) $n_c(w) = n_b(w) + 1$
- (D) $n_c(w) = n_b(w) * 2$

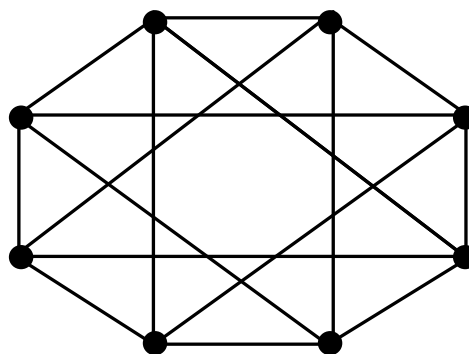
- Q.53 Consider a disk with the following specifications: rotation speed of 6000 RPM, average seek time of 5 milliseconds, 500 sectors/track, 512-byte sectors. A file has content stored in 3000 sectors located *randomly* on the disk. Assuming average rotational latency, the total time (*in seconds, rounded off to 2 decimal places*) to read the entire file from the disk is _____
- Q.54 Consider a TCP connection operating at a point of time with the congestion window of size 12 MSS (Maximum Segment Size), when a timeout occurs due to packet loss. Assuming that all the segments transmitted in the next two RTTs (Round Trip Time) are acknowledged correctly, the congestion window size (*in MSS*) during the third RTT will be _____
- Q.55 Consider an Ethernet segment with a transmission speed of 10^8 bits/sec and a maximum segment length of 500 meters. If the speed of propagation of the signal in the medium is 2×10^8 meters/sec, then the minimum frame size (*in bits*) required for collision detection is _____
- Q.56 A functional dependency $F: X \rightarrow Y$ is termed as a *useful functional dependency* if and only if it satisfies all the following three conditions:
- X is not the empty set.
 - Y is not the empty set.
 - Intersection of X and Y is the empty set.
- For a relation R with 4 attributes, the total number of possible useful functional dependencies is _____
- Q.57 A processor with 16 general purpose registers uses a 32-bit instruction format. The instruction format consists of an opcode field, an addressing mode field, two register operand fields, and a 16-bit scalar field. If 8 addressing modes are to be supported, the maximum number of unique opcodes possible for every addressing mode is _____

- Q.58 A non-pipelined instruction execution unit operating at 2 GHz takes an average of 6 cycles to execute an instruction of a program P. The unit is then redesigned to operate on a 5-stage pipeline at 2 GHz. Assume that the ideal throughput of the pipelined unit is 1 instruction per cycle. In the execution of program P, 20% instructions incur an average of 2 cycles stall due to data hazards and 20% instructions incur an average of 3 cycles stall due to control hazards. The speedup (*rounded off to one decimal place*) obtained by the pipelined design over the non-pipelined design is _____

- Q.59 The number of distinct minimum-weight spanning trees of the following graph is _____



- Q.60 The chromatic number of a graph is the minimum number of colours used in a *proper* colouring of the graph. The chromatic number of the following graph is _____



- Q.61 A processor uses a 32-bit instruction format and supports byte-addressable memory access. The ISA of the processor has 150 distinct instructions. The instructions are equally divided into two types, namely R-type and I-type, whose formats are shown below.

R-type Instruction Format:

OPCODE	UNUSED	DST Register	SRC Register1	SRC Register 2
--------	--------	--------------	---------------	----------------

I-type Instruction Format:

OPCODE	DST Register	SRC Register	# Immediate value/address
--------	--------------	--------------	---------------------------

In the OPCODE, 1 bit is used to distinguish between I-type and R-type instructions and the remaining bits indicate the operation. The processor has 50 architectural registers, and all register fields in the instructions are of equal size.

Let X be the number of bits used to encode the UNUSED field, Y be the number of bits used to encode the OPCODE field, and Z be the number of bits used to encode the immediate value/address field. The value of $X + 2Y + Z$ is _____

- Q.62 Let L_1 be the language represented by the regular expression $b^*ab^*(ab^*ab^*)^*$ and $L_2 = \{w \in (a+b)^* \mid |w| \leq 4\}$, where $|w|$ denotes the length of string w . The number of strings in L_2 which are also in L_1 is _____

- Q.63 Let Z_n be the group of integers $\{0, 1, 2, \dots, n-1\}$ with addition modulo n as the group operation. The number of elements in the group $Z_2 \times Z_3 \times Z_4$ that are their own inverses is _____

- Q.64 Consider a 32-bit system with 4 KB page size and page table entries of size 4 bytes each. Assume $1 \text{ KB} = 2^{10}$ bytes. The OS uses a 2-level page table for memory management, with the page table containing an outer page directory and an inner page table. The OS allocates a page for the outer page directory upon process creation. The OS uses demand paging when allocating memory for the inner page table, i.e., a page of the inner page table is allocated only if it contains at least one valid page table entry.

An active process in this system accesses 2000 unique pages during its execution, and none of the pages are swapped out to disk. After it completes the page accesses, let X denote the minimum and Y denote the maximum number of pages across the two levels of the page table of the process.

The value of $X+Y$ is _____

- Q.65 Consider the following augmented grammar, which is to be parsed with a SLR parser. The set of terminals is $\{a,b,c,d,\#, @\}$

$$\begin{aligned} S' &\rightarrow S \\ S &\rightarrow SS \mid Aa \mid bAc \mid Bc \mid bBa \\ A &\rightarrow d\# \\ B &\rightarrow @ \end{aligned}$$

Let $I_0 = \text{CLOSURE}(\{S' \rightarrow \bullet S\})$. The number of items in the set $\text{GOTO}(I_0, S)$ is _____

GATE 2023

Single session

65 questions | 69 pages

Paper 5 of 16

General Aptitude (GA)**Q.1 – Q.5 Carry ONE mark Each**

Q.1	We reached the station late, and _____ missed the train.
(A)	near
(B)	nearly
(C)	utterly
(D)	mostly
Q.2	Kind : _____ : : Often : Frequently (By word meaning)
(A)	Mean
(B)	Type
(C)	Cruel
(D)	Kindly

Q.3	A series of natural numbers $F_1, F_2, F_3, F_4, F_5, F_6, F_7, \dots$ obeys $F_{n+1} = F_n + F_{n-1}$ for all integers $n \geq 2$. If $F_6 = 37$, and $F_7 = 60$, then what is F_1 ?
(A)	4
(B)	5
(C)	8
(D)	9

Q.4	A survey for a certain year found that 90% of pregnant women received medical care at least once before giving birth. Of these women, 60% received medical care from doctors, while 40% received medical care from other healthcare providers. Given this information, which one of the following statements can be inferred with <i>certainty</i>?
(A)	More than half of the pregnant women received medical care at least once from a doctor.
(B)	Less than half of the pregnant women received medical care at least once from a doctor.
(C)	More than half of the pregnant women received medical care at most once from a doctor.
(D)	Less than half of the pregnant women received medical care at most once from a doctor.
Q.5	Looking at the surface of a smooth 3-dimensional object from the outside, which one of the following options is TRUE?
(A)	The surface of the object must be concave everywhere.
(B)	The surface of the object must be convex everywhere.
(C)	The surface of the object may be concave in some places and convex in other places.
(D)	The object can have edges, but no corners.

Q.6 – Q.10 Carry TWO marks Each

Q.6	The country of Zombieland is in distress since more than 75% of its working population is suffering from serious health issues. Studies conducted by competent health experts concluded that a complete lack of physical exercise among its working population was one of the leading causes of their health issues. As one of the measures to address the problem, the Government of Zombieland has decided to provide monetary incentives to those who ride bicycles to work. Based only on the information provided above, which one of the following statements can be logically inferred with <i>certainty</i>?
(A)	All the working population of Zombieland will henceforth ride bicycles to work.
(B)	Riding bicycles will ensure that all of the working population of Zombieland is free of health issues.
(C)	The health experts suggested to the Government of Zombieland to declare riding bicycles as mandatory.
(D)	The Government of Zombieland believes that riding bicycles is a form of physical exercise.

Q.7	Consider two functions of time (t), $f(t) = 0.01 t^2$ $g(t) = 4 t$ where $0 < t < \infty$. Now consider the following two statements: (i) For some $t > 0$, $g(t) > f(t)$. (ii) There exists a T, such that $f(t) > g(t)$ for all $t > T$. Which one of the following options is TRUE?
(A)	only (i) is correct
(B)	only (ii) is correct
(C)	both (i) and (ii) are correct
(D)	neither (i) nor (ii) is correct

Q.8	Which one of the following sentence sequences creates a coherent narrative? (i) Once on the terrace, on her way to her small room in the corner, she notices the man right away. (ii) She begins to pant by the time she has climbed all the stairs. (iii) Mina has bought vegetables and rice at the market, so her bags are heavy. (iv) He was leaning against the parapet, watching the traffic below.
(A)	(i), (ii), (iv), (iii)
(B)	(ii), (iii), (i), (iv)
(C)	(iv), (ii), (i), (iii)
(D)	(iii), (ii), (i), (iv)
Q.9	$f(x)$ and $g(y)$ are functions of x and y , respectively, and $f(x) = g(y)$ for all real values of x and y . Which one of the following options is <i>necessarily</i> TRUE for all x and y ?
(A)	$f(x) = 0$ and $g(y) = 0$
(B)	$f(x) = g(y) = \text{constant}$
(C)	$f(x) \neq \text{constant}$ and $g(y) \neq \text{constant}$
(D)	$f(x) + g(y) = f(x) - g(y)$

Q.10	Which one of the options best describes the transformation of the 2-dimensional figure P to Q , and then to R , as shown?
(A)	<i>Operation 1:</i> A clockwise rotation by 90° about an axis perpendicular to the plane of the figure <i>Operation 2:</i> A reflection along a horizontal line
(B)	<i>Operation 1:</i> A counter clockwise rotation by 90° about an axis perpendicular to the plane of the figure <i>Operation 2:</i> A reflection along a horizontal line
(C)	<i>Operation 1:</i> A clockwise rotation by 90° about an axis perpendicular to the plane of the figure <i>Operation 2:</i> A reflection along a vertical line
(D)	<i>Operation 1:</i> A counter clockwise rotation by 180° about an axis perpendicular to the plane of the figure <i>Operation 2:</i> A reflection along a vertical line

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Q.11 – Q.35 Carry ONE mark each.

Q.11	Consider the following statements regarding the front-end and back-end of a compiler. S1: The front-end includes phases that are independent of the target hardware. S2: The back-end includes phases that are specific to the target hardware. S3: The back-end includes phases that are specific to the programming language used in the source code. Identify the CORRECT option.
(A)	Only S1 is TRUE.
(B)	Only S1 and S2 are TRUE.
(C)	S1 , S2 , and S3 are all TRUE.
(D)	Only S1 and S3 are TRUE.

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Q.12	Which one of the following sequences when stored in an array at locations $A[1], \dots, A[10]$ forms a max-heap?
(A)	23, 17, 10, 6, 13, 14, 1, 5, 7, 12
(B)	23, 17, 14, 7, 13, 10, 1, 5, 6, 12
(C)	23, 17, 14, 6, 13, 10, 1, 5, 7, 15
(D)	23, 14, 17, 1, 10, 13, 16, 12, 7, 5

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Q.13	Let SLLdel be a function that deletes a node in a singly-linked list given a pointer to the node and a pointer to the head of the list. Similarly, let DLLdel be another function that deletes a node in a doubly-linked list given a pointer to the node and a pointer to the head of the list. Let n denote the number of nodes in each of the linked lists. Which one of the following choices is TRUE about the worst-case time complexity of SLLdel and DLLdel?
(A)	SLLdel is $O(1)$ and DLLdel is $O(n)$
(B)	Both SLLdel and DLLdel are $O(\log(n))$
(C)	Both SLLdel and DLLdel are $O(1)$
(D)	SLLdel is $O(n)$ and DLLdel is $O(1)$

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Q.14	Consider the Deterministic Finite-state Automaton (DFA) $\mathcal{A}$ shown below. The DFA runs on the alphabet $\{0, 1\}$, and has the set of states $\{s, p, q, r\}$, with s being the start state and p being the only final state. <pre> graph LR s((s)) -- 1 --> p(((p))) s -- 0 --> r((r)) p -- 0 --> p p -- 1 --> q((q)) q -- 1 --> p q -- 0 --> r r -- "0,1" --> r style s fill:none,stroke:none style p fill:none,stroke:none style q fill:none,stroke:none style r fill:none,stroke:none </pre> Which one of the following regular expressions correctly describes the language accepted by $\mathcal{A}$?
(A)	$1(0^*11)^*$
(B)	$0(0 + 1)^*$
(C)	$1(0 + 11)^*$
(D)	$1(110^*)^*$

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Q.15	The Lucas sequence L_n is defined by the recurrence relation: $L_n = L_{n-1} + L_{n-2}, \quad \text{for } n \geq 3,$ with $L_1 = 1$ and $L_2 = 3$. Which one of the options given is TRUE?
(A)	$L_n = \left(\frac{1 + \sqrt{5}}{2}\right)^n + \left(\frac{1 - \sqrt{5}}{2}\right)^n$
(B)	$L_n = \left(\frac{1 + \sqrt{5}}{2}\right)^n - \left(\frac{1 - \sqrt{5}}{3}\right)^n$
(C)	$L_n = \left(\frac{1 + \sqrt{5}}{2}\right)^n + \left(\frac{1 - \sqrt{5}}{3}\right)^n$
(D)	$L_n = \left(\frac{1 + \sqrt{5}}{2}\right)^n - \left(\frac{1 - \sqrt{5}}{2}\right)^n$

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Q.16	Which one of the options given below refers to the degree (or arity) of a relation in relational database systems?
(A)	Number of attributes of its relation schema.
(B)	Number of tuples stored in the relation.
(C)	Number of entries in the relation.
(D)	Number of distinct domains of its relation schema.

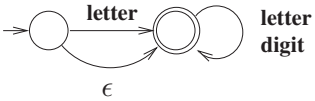
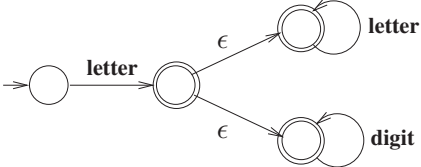
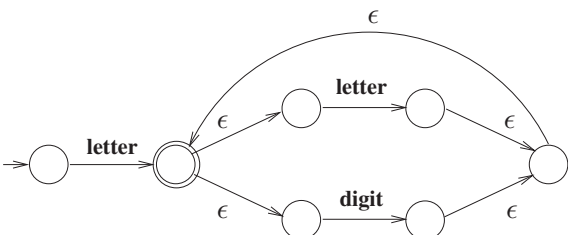
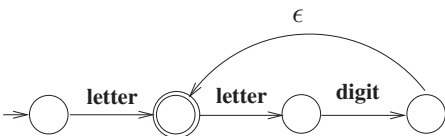
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Q.17	Suppose two hosts are connected by a point-to-point link and they are configured to use Stop-and-Wait protocol for reliable data transfer. Identify in which one of the following scenarios, the utilization of the link is the lowest.
(A)	Longer link length and lower transmission rate
(B)	Longer link length and higher transmission rate
(C)	Shorter link length and lower transmission rate
(D)	Shorter link length and higher transmission rate

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Q.18	Let $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{bmatrix}.$ Let $\det(A)$ and $\det(B)$ denote the determinants of the matrices A and B, respectively. Which one of the options given below is TRUE?
(A)	$\det(A) = \det(B)$
(B)	$\det(B) = -\det(A)$
(C)	$\det(A) = 0$
(D)	$\det(AB) = \det(A) + \det(B)$

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Q.19	Consider the following definition of a lexical token id for an identifier in a programming language, using extended regular expressions: letter $\rightarrow$ [A-Za-z] digit $\rightarrow$ [0-9] id $\rightarrow$ letter (letter digit)* Which one of the following Non-deterministic Finite-state Automata with ϵ-transitions accepts the set of valid identifiers? (A double-circle denotes a final state)
(A)	
(B)	
(C)	
(D)	

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Q.20	An algorithm has to store several keys generated by an adversary in a hash table. The adversary is malicious who tries to maximize the number of collisions. Let k be the number of keys, m be the number of slots in the hash table, and $k > m$. Which one of the following is the best hashing strategy to counteract the adversary?
(A)	Division method, i.e., use the hash function $h(k) = k \bmod m$.
(B)	Multiplication method, i.e., use the hash function $h(k) = \lfloor m(kA - \lfloor kA \rfloor) \rfloor$, where A is a carefully chosen constant.
(C)	Universal hashing method.
(D)	If k is a prime number, use Division method. Otherwise, use Multiplication method.

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Q.21	The output of a 2-input multiplexer is connected back to one of its inputs as shown in the figure. Match the functional equivalence of this circuit to one of the following options.
(A)	D Flip-flop
(B)	D Latch
(C)	Half-adder
(D)	Demultiplexer

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Q.22	Which one or more of the following need to be saved on a context switch from one thread (T1) of a process to another thread (T2) of the same process?
(A)	Page table base register
(B)	Stack pointer
(C)	Program counter
(D)	General purpose registers

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Q.23	Which one or more of the following options guarantee that a computer system will transition from user mode to kernel mode?
(A)	Function Call
(B)	malloc Call
(C)	Page Fault
(D)	System Call

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Q.24	Which of the following statements is/are CORRECT?
(A)	The intersection of two regular languages is regular.
(B)	The intersection of two context-free languages is context-free.
(C)	The intersection of two recursive languages is recursive.
(D)	The intersection of two recursively enumerable languages is recursively enumerable.

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Q.25	Which of the following statements is/are INCORRECT about the OSPF (Open Shortest Path First) routing protocol used in the Internet?
(A)	OSPF implements Bellman-Ford algorithm to find shortest paths.
(B)	OSPF uses Dijkstra's shortest path algorithm to implement least-cost path routing.
(C)	OSPF is used as an inter-domain routing protocol.
(D)	OSPF implements hierarchical routing.

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Q.26	Geetha has a conjecture about integers, which is of the form $\forall x \left(P(x) \implies \exists y Q(x, y) \right),$ where P is a statement about integers, and Q is a statement about pairs of integers. Which of the following (one or more) option(s) would <i>imply</i> Geetha's conjecture?
(A)	$\exists x \left(P(x) \wedge \forall y Q(x, y) \right)$
(B)	$\forall x \forall y Q(x, y)$
(C)	$\exists y \forall x \left(P(x) \implies Q(x, y) \right)$
(D)	$\exists x \left(P(x) \wedge \exists y Q(x, y) \right)$

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Q.27	Which one or more of the following CPU scheduling algorithms can potentially cause starvation?
(A)	First-in First-Out
(B)	Round Robin
(C)	Priority Scheduling
(D)	Shortest Job First

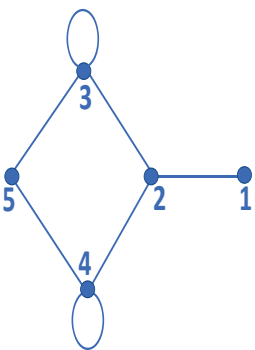
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Q.28	Let $f(x) = x^3 + 15x^2 - 33x - 36$ be a real-valued function. Which of the following statements is/are TRUE?
(A)	$f(x)$ does not have a local maximum.
(B)	$f(x)$ has a local maximum.
(C)	$f(x)$ does not have a local minimum.
(D)	$f(x)$ has a local minimum.

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Q.29	Let f and g be functions of natural numbers given by $f(n) = n$ and $g(n) = n^2$. Which of the following statements is/are TRUE?
(A)	$f \in O(g)$
(B)	$f \in \Omega(g)$
(C)	$f \in o(g)$
(D)	$f \in \Theta(g)$

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Q.30	Let A be the adjacency matrix of the graph with vertices $\{1, 2, 3, 4, 5\}$.  Let $\lambda_1, \lambda_2, \lambda_3, \lambda_4,$ and λ_5 be the five eigenvalues of A. Note that these eigenvalues need not be distinct. The value of $\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 + \lambda_5 =$ _____.

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Q.31	The value of the definite integral $\int_{-3}^3 \int_{-2}^2 \int_{-1}^1 (4x^2y - z^3) \, dz \, dy \, dx$ is _____. (Rounded off to the nearest integer)

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Q.32	A particular number is written as 132 in radix-4 representation. The same number in radix-5 representation is _____.

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Q.33	Consider a 3-stage pipelined processor having a delay of 10 ns (nanoseconds), 20 ns, and 14 ns, for the first, second, and the third stages, respectively. Assume that there is no other delay and the processor does not suffer from any pipeline hazards. Also assume that one instruction is fetched every cycle. The total execution time for executing 100 instructions on this processor is _____ ns.

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Q.34	A keyboard connected to a computer is used at a rate of 1 keystroke per second. The computer system polls the keyboard every 10 ms (milli seconds) to check for a keystroke and consumes $100\ \mu\text{s}$ (micro seconds) for each poll. If it is determined after polling that a key has been pressed, the system consumes an additional $200\ \mu\text{s}$ to process the keystroke. Let T_1 denote the fraction of a second spent in polling and processing a keystroke. In an alternative implementation, the system uses interrupts instead of polling. An interrupt is raised for every keystroke. It takes a total of 1 ms for servicing an interrupt and processing a keystroke. Let T_2 denote the fraction of a second spent in servicing the interrupt and processing a keystroke. The ratio $\frac{T_1}{T_2}$ is _____. (Rounded off to one decimal place)

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Q.35	The integer value printed by the ANSI-C program given below is _____. <pre>#include<stdio.h> int funcp(){ static int x = 1; x++; return x; } int main(){ int x,y; x = funcp(); y = funcp()+x; printf("%d\n", (x+y)); return 0; }</pre>

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Q.36 – Q.65 Carry TWO mark each.

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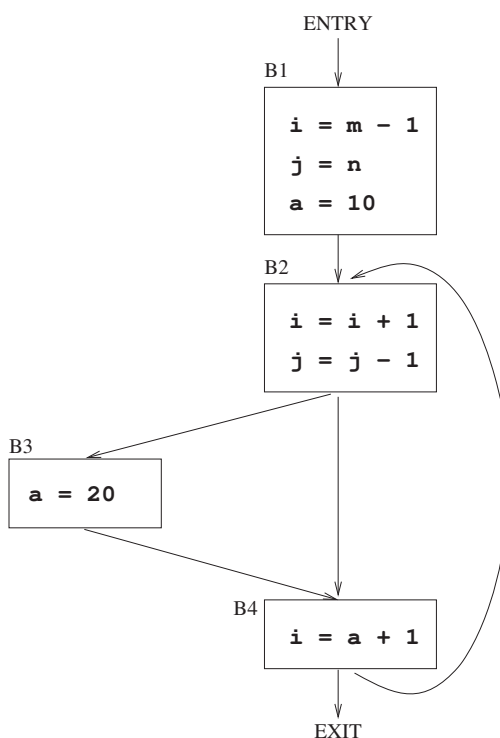
Q.36	Consider the following program: <pre> int main() int f1() int f2(int X) int f3() { f1(); { return(1); { f3(); { return(5); f2(2); } f3(); } return(0); } } } </pre> Which one of the following options represents the activation tree corresponding to the main function?
(A)	<pre> main / \ f1 f2 f3 / \ f3 f2 / \ f3 f1 </pre>
(B)	<pre> main / \ f1 f2 f3 / \ f3 f1 </pre>
(C)	<pre> main / f1 / f2 / \ f3 f1 </pre>
(D)	<pre> main / \ f1 f2 f3 / \ / f3 f2 f1 </pre>

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Q.37

Consider the control flow graph shown.



Which one of the following choices correctly lists the set of *live* variables at the exit point of each basic block?

- (A) B1: {}, B2: {a}, B3: {a}, B4: {a}
- (B) B1: {i, j}, B2: {a}, B3: {a}, B4: {i}
- (C) B1: {a, i, j}, B2: {a, i, j}, B3: {a, i}, B4: {a}
- (D) B1: {a, i, j}, B2: {a, j}, B3: {a, j}, B4: {a, i, j}

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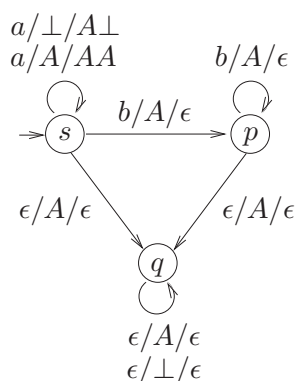
Q.38	Consider the two functions <code>incr</code> and <code>decr</code> shown below. <pre> incr(){ wait(s); X = X+1; signal(s); } incr(){ wait(s); X = X-1; signal(s); } </pre> There are 5 threads each invoking <code>incr</code> once, and 3 threads each invoking <code>decr</code> once, on the same shared variable <code>X</code>. The initial value of <code>X</code> is 10. Suppose there are two implementations of the semaphore <code>s</code>, as follows: I-1: <code>s</code> is a binary semaphore initialized to 1. I-2: <code>s</code> is a counting semaphore initialized to 2. Let <code>V1</code>, <code>V2</code> be the values of <code>X</code> at the end of execution of all the threads with implementations I-1, I-2, respectively. Which one of the following choices corresponds to the minimum possible values of <code>V1</code>, <code>V2</code>, respectively?
(A)	15, 7
(B)	7, 7
(C)	12, 7
(D)	12, 8

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Q.39	Consider the context-free grammar G below $\begin{aligned} S &\rightarrow aSb \mid X \\ X &\rightarrow aX \mid Xb \mid a \mid b, \end{aligned}$ where S and X are non-terminals, and a and b are terminal symbols. The starting non-terminal is S. Which one of the following statements is CORRECT?
(A)	The language generated by G is $(a + b)^*$
(B)	The language generated by G is $a^*(a + b)b^*$
(C)	The language generated by G is $a^*b^*(a + b)$
(D)	The language generated by G is not a regular language

Q.40

Consider the pushdown automaton (PDA) P below, which runs on the input alphabet $\{a, b\}$, has stack alphabet $\{\perp, A\}$, and has three states $\{s, p, q\}$, with s being the start state. A transition from state u to state v , labelled $c/X/\gamma$, where c is an input symbol or ϵ , X is a stack symbol, and γ is a string of stack symbols, represents the fact that in state u , the PDA can read c from the input, with X on the top of its stack, pop X from the stack, push in the string γ on the stack, and go to state v . In the initial configuration, the stack has only the symbol $\perp$ in it. The PDA accepts by empty stack.



Which one of the following options correctly describes the language accepted by P ?

(A) $\{a^m b^n \mid 1 \leq m \text{ and } n < m\}$

(B) $\{a^m b^n \mid 0 \leq n \leq m\}$

(C) $\{a^m b^n \mid 0 \leq m \text{ and } 0 \leq n\}$

(D) $\{a^m \mid 0 \leq m\} \cup \{b^n \mid 0 \leq n\}$

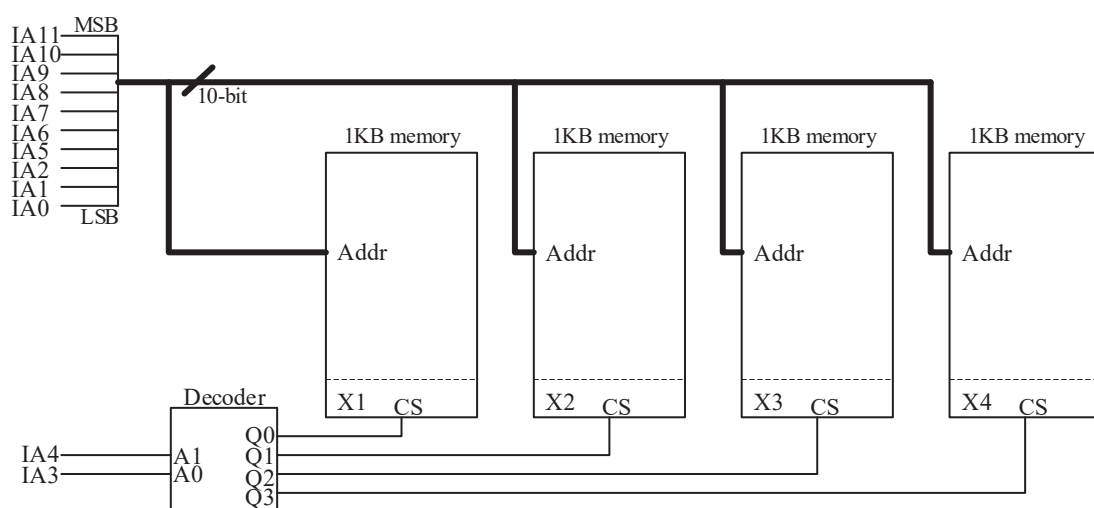
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Q.41	Consider the given C-code and its corresponding assembly code, with a few operands U1–U4 being unknown. Some useful information as well as the semantics of each unique assembly instruction is annotated as inline comments in the code. The memory is byte-addressable. <table border="1" data-bbox="331 457 1461 982"> <thead> <tr> <th>//C-code</th><th>;assembly-code (; indicates comments)</th></tr> </thead> <tbody> <tr> <td></td><td>;r1-r5 are 32-bit integer registers</td></tr> <tr> <td></td><td>;initialize r1=0, r2=10</td></tr> <tr> <td></td><td>;initialize r3, r4 with base address of a, b</td></tr> <tr> <td>int a[10], b[10], i;</td><td>L01: jeq r1, r2, end ;if(r1==r2) goto end</td></tr> <tr> <td>// int is 32-bit</td><td>L02: lw r5, 0(r4) ;r5 <- Memory[r4+0]</td></tr> <tr> <td>for (i=0; i<10;i++)</td><td>L03: shl r5, r5, U1 ;r5 <- r5 << U1</td></tr> <tr> <td>a[i] = b[i] * 8;</td><td>L04: sw r5, 0(r3) ;Memory[r3+0] <- r5</td></tr> <tr> <td></td><td>L05: add r3, r3, U2 ;r3 <- r3+U2</td></tr> <tr> <td></td><td>L06: add r4, r4, U3</td></tr> <tr> <td></td><td>L07: add r1, r1, 1</td></tr> <tr> <td></td><td>L08: jmp U4 ;goto U4</td></tr> <tr> <td></td><td>L09: end</td></tr> </tbody> </table> Which one of the following options is a CORRECT replacement for operands in the position (U1, U2, U3, U4) in the above assembly code?	//C-code	;assembly-code (; indicates comments)		;r1-r5 are 32-bit integer registers		;initialize r1=0, r2=10		;initialize r3, r4 with base address of a, b	int a[10], b[10], i;	L01: jeq r1, r2, end ;if(r1==r2) goto end	// int is 32-bit	L02: lw r5, 0(r4) ;r5 <- Memory[r4+0]	for (i=0; i<10;i++)	L03: shl r5, r5, U1 ;r5 <- r5 << U1	a[i] = b[i] * 8;	L04: sw r5, 0(r3) ;Memory[r3+0] <- r5		L05: add r3, r3, U2 ;r3 <- r3+U2		L06: add r4, r4, U3		L07: add r1, r1, 1		L08: jmp U4 ;goto U4		L09: end
//C-code	;assembly-code (; indicates comments)																										
	;r1-r5 are 32-bit integer registers																										
	;initialize r1=0, r2=10																										
	;initialize r3, r4 with base address of a, b																										
int a[10], b[10], i;	L01: jeq r1, r2, end ;if(r1==r2) goto end																										
// int is 32-bit	L02: lw r5, 0(r4) ;r5 <- Memory[r4+0]																										
for (i=0; i<10;i++)	L03: shl r5, r5, U1 ;r5 <- r5 << U1																										
a[i] = b[i] * 8;	L04: sw r5, 0(r3) ;Memory[r3+0] <- r5																										
	L05: add r3, r3, U2 ;r3 <- r3+U2																										
	L06: add r4, r4, U3																										
	L07: add r1, r1, 1																										
	L08: jmp U4 ;goto U4																										
	L09: end																										
(A)	(8, 4, 1, L02)																										
(B)	(3, 4, 4, L01)																										
(C)	(8, 1, 1, L02)																										
(D)	(3, 1, 1, L01)																										

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Q.42

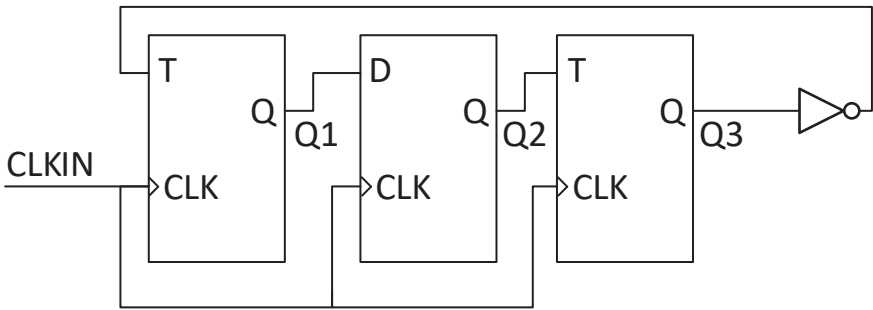
A 4 kilobyte (KB) byte-addressable memory is realized using four 1 KB memory blocks. Two input address lines (IA4 and IA3) are connected to the chip select (CS) port of these memory blocks through a decoder as shown in the figure. The remaining ten input address lines from IA11–IA0 are connected to the address port of these blocks. The chip select (CS) is active high.



The input memory addresses (IA11–IA0), in decimal, for the starting locations (Addr=0) of each block (indicated as X1, X2, X3, X4 in the figure) are among the options given below. Which one of the following options is CORRECT?

- (A) (0, 1, 2, 3)
- (B) (0, 1024, 2048, 3072)
- (C) (0, 8, 16, 24)
- (D) (0, 0, 0, 0)

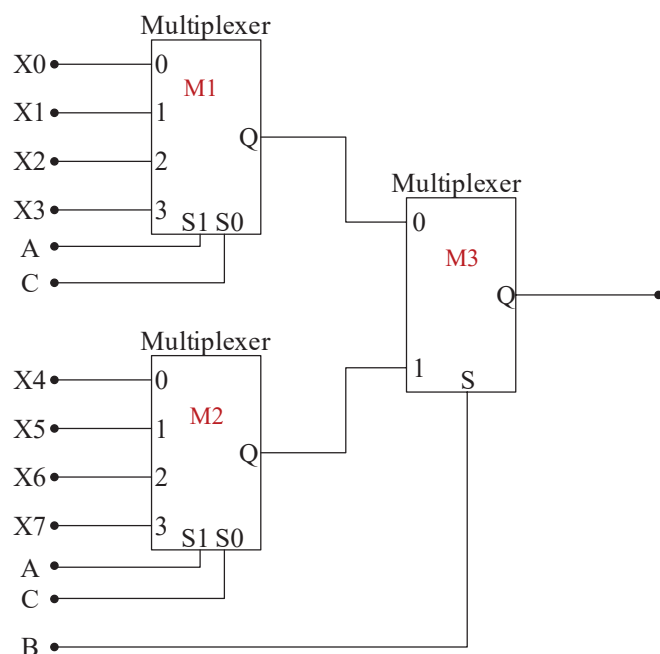
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Q.43	Consider a sequential digital circuit consisting of T flip-flops and D flip-flops as shown in the figure. CLKIN is the clock input to the circuit. At the beginning, Q1, Q2 and Q3 have values 0, 1 and 1, respectively.  Which one of the given values of (Q1, Q2, Q3) can NEVER be obtained with this digital circuit?
(A)	(0, 0, 1)
(B)	(1, 0, 0)
(C)	(1, 0, 1)
(D)	(1, 1, 1)

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Q.44

A Boolean digital circuit is composed using two 4-input multiplexers (M1 and M2) and one 2-input multiplexer (M3) as shown in the figure. X0–X7 are the inputs of the multiplexers M1 and M2 and could be connected to either 0 or 1. The select lines of the multiplexers are connected to Boolean variables A, B and C as shown.



Which one of the following set of values of (X0, X1, X2, X3, X4, X5, X6, X7) will realise the Boolean function $\overline{A} + \overline{A}.C + A.\overline{B}.C$?

(A) (1, 1, 0, 0, 1, 1, 1, 0)

(B) (1, 1, 0, 0, 1, 1, 0, 1)

(C) (1, 1, 0, 1, 1, 1, 0, 0)

(D) (0, 0, 1, 1, 0, 1, 1, 1)

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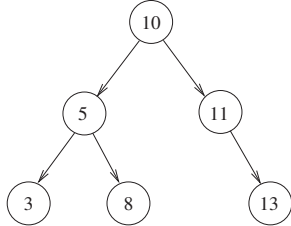
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Q.45	Consider the IEEE-754 single precision floating point numbers $P=0xC1800000$ and $Q=0x3F5C2EF4$. Which one of the following corresponds to the product of these numbers (i.e., $P \times Q$), represented in the IEEE-754 single precision format?
(A)	$0x404C2EF4$
(B)	$0x405C2EF4$
(C)	$0xC15C2EF4$
(D)	$0xC14C2EF4$

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Q.46	Let A be a priority queue for maintaining a set of elements. Suppose A is implemented using a max-heap data structure. The operation $\text{EXTRACT-MAX}(A)$ extracts and deletes the maximum element from A. The operation $\text{INSERT}(A, key)$ inserts a new element key in A. The properties of a max-heap are preserved at the end of each of these operations. When A contains n elements, which one of the following statements about the worst case running time of these two operations is TRUE?
(A)	Both $\text{EXTRACT-MAX}(A)$ and $\text{INSERT}(A, key)$ run in $O(1)$.
(B)	Both $\text{EXTRACT-MAX}(A)$ and $\text{INSERT}(A, key)$ run in $O(\log(n))$.
(C)	$\text{EXTRACT-MAX}(A)$ runs in $O(1)$ whereas $\text{INSERT}(A, key)$ runs in $O(n)$.
(D)	$\text{EXTRACT-MAX}(A)$ runs in $O(1)$ whereas $\text{INSERT}(A, key)$ runs in $O(\log(n))$.

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Q.47	Consider the C function <code>foo</code> and the binary tree shown. <pre> typedef struct node { int val; struct node *left, *right; } node; int foo(node *p) { int retval; if (p == NULL) return 0; else { retval = p->val + foo(p->left) + foo(p->right); printf("%d ", retval); return retval; } } </pre>  <pre> graph TD 10((10)) --> 5((5)) 10 --> 11((11)) 5 --> 3((3)) 5 --> 8((8)) 11 --> 13((13)) </pre> When <code>foo</code> is called with a pointer to the root node of the given binary tree, what will it print?
(A)	3 8 5 13 11 10
(B)	3 5 8 10 11 13
(C)	3 8 16 13 24 50
(D)	3 16 8 50 24 13

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Q.48	Let $U = \{1, 2, \dots, n\}$, where n is a large positive integer greater than 1000. Let k be a positive integer less than n. Let A, B be subsets of U with $A = B = k$ and $A \cap B = \emptyset$. We say that a permutation of U separates A from B if one of the following is true. - All members of A appear in the permutation before any of the members of B. - All members of B appear in the permutation before any of the members of A. How many permutations of U separate A from B?
(A)	$n!$
(B)	$\binom{n}{2k} (n - 2k)!$
(C)	$\binom{n}{2k} (n - 2k)! (k!)^2$
(D)	$2 \binom{n}{2k} (n - 2k)! (k!)^2$

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Q.49	Let $f : A \rightarrow B$ be an onto (or surjective) function, where A and B are nonempty sets. Define an equivalence relation $\sim$ on the set A as $a_1 \sim a_2 \text{ if } f(a_1) = f(a_2) ,$ where $a_1, a_2 \in A$. Let $\mathcal{E} = \{[x] : x \in A\}$ be the set of all the equivalence classes under $\sim$. Define a new mapping $F : \mathcal{E} \rightarrow B$ as $F([x]) = f(x), \text{ for all the equivalence classes } [x] \text{ in } \mathcal{E}.$ Which of the following statements is/are TRUE?
(A)	F is NOT well-defined.
(B)	F is an onto (or surjective) function.
(C)	F is a one-to-one (or injective) function.
(D)	F is a bijective function.

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Q.50	Suppose you are asked to design a new reliable byte-stream transport protocol like TCP. This protocol, named myTCP, runs over a 100 Mbps network with Round Trip Time of 150 milliseconds and the maximum segment lifetime of 2 minutes. Which of the following is/are valid lengths of the Sequence Number field in the myTCP header?
(A)	30 bits
(B)	32 bits
(C)	34 bits
(D)	36 bits

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Q.51	Let X be a set and 2^X denote the powerset of X. Define a binary operation Δ on 2^X as follows: $A\Delta B = (A - B) \cup (B - A) .$ Let $H = (2^X, \Delta)$. Which of the following statements about H is/are correct?
(A)	H is a group.
(B)	Every element in H has an inverse, but H is NOT a group.
(C)	For every $A \in 2^X$, the inverse of A is the complement of A .
(D)	For every $A \in 2^X$, the inverse of A is A .

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Q.52	Suppose in a web browser, you click on the <code>www.gate-2023.in</code> URL. The browser cache is empty. The IP address for this URL is not cached in your local host, so a DNS lookup is triggered (by the local DNS server deployed on your local host) over the 3-tier DNS hierarchy in an iterative mode. No resource records are cached anywhere across all DNS servers. Let RTT denote the round trip time between your local host and DNS servers in the DNS hierarchy. The round trip time between the local host and the web server hosting <code>www.gate-2023.in</code> is also equal to RTT. The HTML file associated with the URL is small enough to have negligible transmission time and negligible rendering time by your web browser, which references 10 equally small objects on the same web server. Which of the following statements is/are CORRECT about the minimum elapsed time between clicking on the URL and your browser fully rendering it?
(A)	7 RTTs, in case of non-persistent HTTP with 5 parallel TCP connections.
(B)	5 RTTs, in case of persistent HTTP with pipelining.
(C)	9 RTTs, in case of non-persistent HTTP with 5 parallel TCP connections.
(D)	6 RTTs, in case of persistent HTTP with pipelining.

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Q.53	Consider a random experiment where two fair coins are tossed. Let A be the event that denotes HEAD on both the throws, B be the event that denotes HEAD on the first throw, and C be the event that denotes HEAD on the second throw. Which of the following statements is/are TRUE?
(A)	A and B are independent.
(B)	A and C are independent.
(C)	B and C are independent.
(D)	$\text{Prob}(B C) = \text{Prob}(B)$

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Q.54	Consider functions Function_1 and Function_2 expressed in pseudocode as follows: Function_1 <pre> while $n > 1$ do for $i = 1$ to n do $x = x + 1$; end for $n = \lfloor n/2 \rfloor$; end while </pre> Function_2 <pre> for $i = 1$ to $100 * n$ do $x = x + 1$; end for </pre> Let $f_1(n)$ and $f_2(n)$ denote the number of times the statement “$x = x + 1$” is executed in Function_1 and Function_2, respectively. Which of the following statements is/are TRUE?
(A)	$f_1(n) \in \Theta(f_2(n))$
(B)	$f_1(n) \in o(f_2(n))$
(C)	$f_1(n) \in \omega(f_2(n))$
(D)	$f_1(n) \in O(n)$

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Q.55	Let G be a simple, finite, undirected graph with vertex set $\{v_1, \dots, v_n\}$. Let $\Delta(G)$ denote the maximum degree of G and let $\mathbb{N} = \{1, 2, \dots\}$ denote the set of all possible colors. Color the vertices of G using the following greedy strategy: for $i = 1, \dots, n$ $\text{color}(v_i) \leftarrow \min\{j \in \mathbb{N} : \text{no neighbour of } v_i \text{ is colored } j\}$ Which of the following statements is/are TRUE?
(A)	This procedure results in a proper vertex coloring of G .
(B)	The number of colors used is at most $\Delta(G) + 1$.
(C)	The number of colors used is at most $\Delta(G)$.
(D)	The number of colors used is equal to the chromatic number of G .

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Q.56	Let $U = \{1, 2, 3\}$. Let 2^U denote the powerset of U. Consider an undirected graph G whose vertex set is 2^U. For any $A, B \in 2^U$, (A, B) is an edge in G if and only if (i) $A \neq B$, and (ii) either $A \subsetneq B$ or $B \subsetneq A$. For any vertex A in G, the set of all possible orderings in which the vertices of G can be visited in a Breadth First Search (BFS) starting from A is denoted by $\mathcal{B}(A)$. If $\emptyset$ denotes the empty set, then the cardinality of $\mathcal{B}(\emptyset)$ is _____.

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Q.57	Consider the following two-dimensional array D in the C programming language, which is stored in row-major order: <pre>int D[128][128];</pre> Demand paging is used for allocating memory and each physical page frame holds 512 elements of the array D. The Least Recently Used (LRU) page-replacement policy is used by the operating system. A total of 30 physical page frames are allocated to a process which executes the following code snippet: <pre>for (int i = 0; i < 128; i++) for (int j = 0; j < 128; j++) D[j][i] *= 10;</pre> The number of page faults generated during the execution of this code snippet is _____.

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Q.58	Consider a computer system with 57-bit virtual addressing using multi-level tree-structured page tables with L levels for virtual to physical address translation. The page size is 4 KB (1 KB = 1024 B) and a page table entry at any of the levels occupies 8 bytes. The value of L is _____.

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Q.59	Consider a sequence a of elements $a_0 = 1$, $a_1 = 5$, $a_2 = 7$, $a_3 = 8$, $a_4 = 9$, and $a_5 = 2$. The following operations are performed on a stack S and a queue Q, both of which are initially empty. I: push the elements of a from a_0 to a_5 in that order into S. II: enqueue the elements of a from a_0 to a_5 in that order into Q. III: pop an element from S. IV: dequeue an element from Q. V: pop an element from S. VI: dequeue an element from Q. VII: dequeue an element from Q and push the same element into S. VIII: Repeat operation VII three times. IX: pop an element from S. X: pop an element from S. The top element of S after executing the above operations is _____.

Q.60	Consider the syntax directed translation given by the following grammar and semantic rules. Here N, I, F and B are non-terminals. N is the starting non-terminal, and $\#$, 0 and 1 are lexical tokens corresponding to input letters “$\#$”, “0” and “1”, respectively. $X.val$ denotes the synthesized attribute (a numeric value) associated with a non-terminal X. I_1 and F_1 denote occurrences of I and F on the right hand side of a production, respectively. For the tokens 0 and 1, $0.val = 0$ and $1.val = 1$. $ \begin{aligned} N &\rightarrow I \# F & N.val &= I.val + F.val \\ I &\rightarrow I_1 B & I.val &= (2 I_1.val) + B.val \\ I &\rightarrow B & I.val &= B.val \\ F &\rightarrow B F_1 & F.val &= \frac{1}{2}(B.val + F_1.val) \\ F &\rightarrow B & F.val &= \frac{1}{2}B.val \\ B &\rightarrow 0 & B.val &= 0.val \\ B &\rightarrow 1 & B.val &= 1.val \end{aligned} $ The value computed by the translation scheme for the input string 10#011 is _____. (Rounded off to three decimal places)

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Q.61

Consider the following table named **Student** in a relational database. The primary key of this table is rollNum.

Student

<u>rollNum</u>	name	gender	marks
1	Naman	M	62
2	Aliya	F	70
3	Aliya	F	80
4	James	M	82
5	Swati	F	65

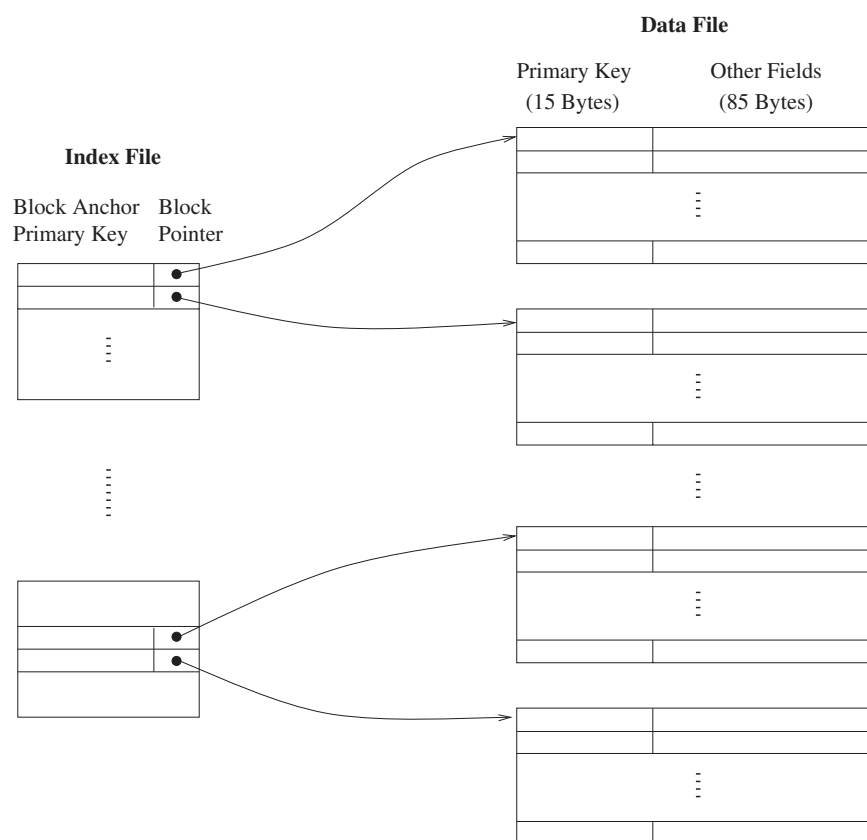
The SQL query below is executed on this database.

```
SELECT *  
FROM Student  
WHERE gender = 'F' AND  
      marks > 65;
```

The number of rows returned by the query is _____.

Q.62

Consider a database of fixed-length records, stored as an ordered file. The database has 25,000 records, with each record being 100 bytes, of which the primary key occupies 15 bytes. The data file is block-aligned in that each data record is fully contained within a block. The database is indexed by a primary index file, which is also stored as a block-aligned ordered file. The figure below depicts this indexing scheme.



Suppose the block size of the file system is 1024 bytes, and a pointer to a block occupies 5 bytes. The system uses binary search on the index file to search for a record with a given key. You may assume that a binary search on an index file of b blocks takes $\lceil \log_2 b \rceil$ block accesses in the worst case.

Given a key, the number of block accesses required to identify the block in the data file that may contain a record with the key, in the worst case, is _____.

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Q.63	Consider the language L over the alphabet $\{0, 1\}$, given below: $L = \{w \in \{0, 1\}^* \mid w \text{ does not contain three or more consecutive 1's}\}.$ The minimum number of states in a Deterministic Finite-State Automaton (DFA) for L is _____.

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Q.64	An 8-way set associative cache of size 64 KB (1 KB = 1024 bytes) is used in a system with 32-bit address. The address is sub-divided into TAG, INDEX, and BLOCK OFFSET. The number of bits in the TAG is _____.

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Q.65	The forwarding table of a router is shown below.																		
	<table><tr><th>Subnet Number</th><th>Subnet Mask</th><th>Interface ID</th></tr><tr><td>200.150.0.0</td><td>255.255.0.0</td><td>1</td></tr><tr><td>200.150.64.0</td><td>255.255.224.0</td><td>2</td></tr><tr><td>200.150.68.0</td><td>255.255.255.0</td><td>3</td></tr><tr><td>200.150.68.64</td><td>255.255.255.224</td><td>4</td></tr><tr><td>Default</td><td></td><td>0</td></tr></table>	Subnet Number	Subnet Mask	Interface ID	200.150.0.0	255.255.0.0	1	200.150.64.0	255.255.224.0	2	200.150.68.0	255.255.255.0	3	200.150.68.64	255.255.255.224	4	Default		0
	Subnet Number	Subnet Mask	Interface ID																
	200.150.0.0	255.255.0.0	1																
	200.150.64.0	255.255.224.0	2																
	200.150.68.0	255.255.255.0	3																
	200.150.68.64	255.255.255.224	4																
Default		0																	
A packet addressed to a destination address 200.150.68.118 arrives at the router. It will be forwarded to the interface with ID _____.																			

GATE 2022

Part 1 of 2

65 questions | 41 pages

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**GATE 2022 General Aptitude (GA)****Q.1 – Q.5 Carry ONE mark each.**

Q.1	The _____ is too high for it to be considered _____.
(A)	fair / fare
(B)	faer / fair
(C)	fare / fare
(D)	fare / fair

Q.2	A function $y(x)$ is defined in the interval $[0, 1]$ on the x-axis as $y(x) = \begin{cases} 2 & \text{if } 0 \leq x < \frac{1}{3} \\ 3 & \text{if } \frac{1}{3} \leq x < \frac{3}{4} \\ 1 & \text{if } \frac{3}{4} \leq x \leq 1 \end{cases}$ Which one of the following is the area under the curve for the interval $[0, 1]$ on the x-axis?
(A)	$\frac{5}{6}$
(B)	$\frac{6}{5}$
(C)	$\frac{13}{6}$
(D)	$\frac{6}{13}$



Q.3	Let r be a root of the equation $x^2 + 2x + 6 = 0$. Then the value of the expression $(r + 2)(r + 3)(r + 4)(r + 5)$ is
(A)	51
(B)	−51
(C)	126
(D)	−126

Q.4	Given below are four statements. Statement 1: All students are inquisitive. Statement 2: Some students are inquisitive. Statement 3: No student is inquisitive. Statement 4: Some students are not inquisitive. From the given four statements, find the two statements that CANNOT BE TRUE simultaneously, assuming that there is at least one student in the class.
(A)	Statement 1 and Statement 3
(B)	Statement 1 and Statement 2
(C)	Statement 2 and Statement 4
(D)	Statement 3 and Statement 4

Q.5	A palindrome is a word that reads the same forwards and backwards. In a game of words, a player has the following two plates painted with letters. A D From the additional plates given in the options, which one of the combinations of additional plates would allow the player to construct a five-letter palindrome. The player should use all the five plates exactly once. The plates can be rotated in their plane.
(A)	D d J
(B)	R A R
(C)	Z E D
(D)	I 7 Y

Q. 6 – Q. 10 Carry TWO marks each.

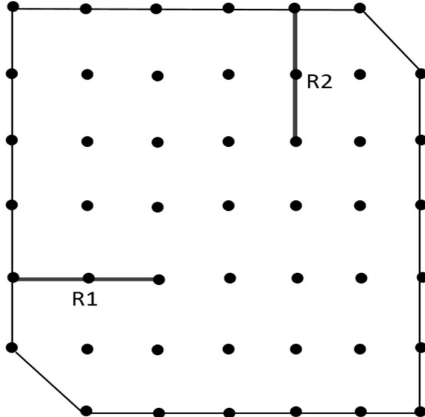
Q.6	Some people believe that “what gets measured, improves”. Some others believe that “what gets measured, gets gamed”. One possible reason for the difference in the beliefs is the work culture in organizations. In organizations with good work culture, metrics help improve outcomes. However, the same metrics are counterproductive in organizations with poor work culture. Which one of the following is the CORRECT logical inference based on the information in the above passage?
(A)	Metrics are useful in organizations with poor work culture
(B)	Metrics are useful in organizations with good work culture
(C)	Metrics are always counterproductive in organizations with good work culture
(D)	Metrics are never useful in organizations with good work culture

Q.7	In a recently conducted national entrance test, boys constituted 65% of those who appeared for the test. Girls constituted the remaining candidates and they accounted for 60% of the qualified candidates. Which one of the following is the correct logical inference based on the information provided in the above passage?
(A)	Equal number of boys and girls qualified
(B)	Equal number of boys and girls appeared for the test
(C)	The number of boys who appeared for the test is less than the number of girls who appeared
(D)	The number of boys who qualified the test is less than the number of girls who qualified



Q.8	A box contains five balls of same size and shape. Three of them are green coloured balls and two of them are orange coloured balls. Balls are drawn from the box one at a time. If a green ball is drawn, it is not replaced. If an orange ball is drawn, it is replaced with another orange ball. First ball is drawn. What is the probability of getting an orange ball in the next draw?
(A)	$\frac{1}{2}$
(B)	$\frac{8}{25}$
(C)	$\frac{19}{50}$
(D)	$\frac{23}{50}$

Q.9	The corners and mid-points of the sides of a triangle are named using the distinct letters P, Q, R, S, T and U, but not necessarily in the same order. Consider the following statements: • The line joining P and R is parallel to the line joining Q and S. • P is placed on the side opposite to the corner T. • S and U cannot be placed on the same side. Which one of the following statements is correct based on the above information?
(A)	P cannot be placed at a corner
(B)	S cannot be placed at a corner
(C)	U cannot be placed at a mid-point
(D)	R cannot be placed at a corner

Q.10	A plot of land must be divided between four families. They want their individual plots to be similar in shape, not necessarily equal in area. The land has equally spaced poles, marked as dots in the below figure. Two ropes, R1 and R2, are already present and cannot be moved. What is the least number of additional straight ropes needed to create the desired plots? A single rope can pass through three poles that are aligned in a straight line. 
(A) 2	
(B) 4	
(C) 5	
(D) 3	

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Q.11 – Q.22 Multiple Choice Questions (MCQ), carry ONE mark each.

Q.11	Which one of the following statements is TRUE for all positive functions $f(n)$?
(A)	$f(n^2) = \theta(f(n)^2)$, when $f(n)$ is a polynomial
(B)	$f(n^2) = o(f(n)^2)$
(C)	$f(n^2) = O(f(n)^2)$, when $f(n)$ is an exponential function
(D)	$f(n^2) = \Omega(f(n)^2)$
Q.12	Which one of the following regular expressions correctly represents the language of the finite automaton given below?
(A)	$ab^*bab^* + ba^*aba^*$
(B)	$(ab^*b)^*ab^* + (ba^*a)^*ba^*$
(C)	$(ab^*b + ba^*a)^*(a^* + b^*)$
(D)	$(ba^*a + ab^*b)^*(ab^* + ba^*)$

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Q.13	Which one of the following statements is TRUE?
(A)	The $LALR(1)$ parser for a grammar G cannot have reduce-reduce conflict if the $LR(1)$ parser for G does not have reduce-reduce conflict.
(B)	Symbol table is accessed only during the lexical analysis phase.
(C)	Data flow analysis is necessary for run-time memory management.
(D)	$LR(1)$ parsing is sufficient for deterministic context-free languages.
Q.14	In a relational data model, which one of the following statements is TRUE?
(A)	A relation with only two attributes is always in BCNF.
(B)	If all attributes of a relation are prime attributes, then the relation is in BCNF.
(C)	Every relation has at least one non-prime attribute.
(D)	BCNF decompositions preserve functional dependencies.



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Q.15	Consider the problem of reversing a singly linked list. To take an example, given the linked list below, the reversed linked list should look like Which one of the following statements is TRUE about the time complexity of algorithms that solve the above problem in $O(1)$ space?
(A)	The best algorithm for the problem takes $\theta(n)$ time in the worst case.
(B)	The best algorithm for the problem takes $\theta(n \log n)$ time in the worst case.
(C)	The best algorithm for the problem takes $\theta(n^2)$ time in the worst case.
(D)	It is not possible to reverse a singly linked list in $O(1)$ space.
Q.16	Suppose we are given n keys, m hash table slots, and two simple uniform hash functions h_1 and h_2. Further suppose our hashing scheme uses h_1 for the odd keys and h_2 for the even keys. What is the expected number of keys in a slot?
(A)	$\frac{m}{n}$
(B)	$\frac{n}{m}$
(C)	$\frac{2n}{m}$
(D)	$\frac{n}{2m}$

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Q.17	Which one of the following facilitates transfer of bulk data from hard disk to main memory with the highest throughput?
(A)	DMA based I/O transfer
(B)	Interrupt driven I/O transfer
(C)	Polling based I/O transfer
(D)	Programmed I/O transfer
Q.18	Let R1 and R2 be two 4-bit registers that store numbers in 2's complement form. For the operation $R1+R2$, which one of the following values of R1 and R2 gives an arithmetic overflow?
(A)	$R1 = 1011$ and $R2 = 1110$
(B)	$R1 = 1100$ and $R2 = 1010$
(C)	$R1 = 0011$ and $R2 = 0100$
(D)	$R1 = 1001$ and $R2 = 1111$

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Q.19	Consider the following threads, T_1 , T_2 , and T_3 executing on a single processor, synchronized using three binary semaphore variables, S_1 , S_2 , and S_3 , operated upon using standard <code>wait()</code> and <code>signal()</code> . The threads can be context switched in any order and at any time.		
	T_1 <pre>while(true){ wait(S_3); print("C"); signal(S_2); }</pre>	T_2 <pre>while(true){ wait(S_1); print("B"); signal(S_3); }</pre>	T_3 <pre>while(true){ wait(S_2); print("A"); signal(S_1); }</pre>
	Which initialization of the semaphores would print the sequence BCABCABCA....?		
(A)	$S_1 = 1; S_2 = 1; S_3 = 1$		
(B)	$S_1 = 1; S_2 = 1; S_3 = 0$		
(C)	$S_1 = 1; S_2 = 0; S_3 = 0$		
(D)	$S_1 = 0; S_2 = 1; S_3 = 1$		

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Q.20	Consider the following two statements with respect to the matrices $A_{m \times n}$, $B_{n \times m}$, $C_{n \times n}$ and $D_{n \times n}$. Statement 1: $tr(AB) = tr(BA)$ Statement 2: $tr(CD) = tr(DC)$ where $tr()$ represents the trace of a matrix. Which one of the following holds?
(A)	Statement 1 is correct and Statement 2 is wrong.
(B)	Statement 1 is wrong and Statement 2 is correct.
(C)	Both Statement 1 and Statement 2 are correct.
(D)	Both Statement 1 and Statement 2 are wrong.

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Q.21	What is printed by the following ANSI C program? <pre>#include<stdio.h> int main(int argc, char *argv[]) { int x = 1, z[2] = {10, 11}; int *p = NULL; p = &x; *p = 10; p = &z[1]; *(&z[0] + 1) += 3; printf("%d, %d, %d\n", x, z[0], z[1]); return 0; }</pre>
(A)	1, 10, 11
(B)	1, 10, 14
(C)	10, 14, 11
(D)	10, 10, 14

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Q.22	Consider an enterprise network with two Ethernet segments, a web server and a firewall, connected via three routers as shown below.
	The diagram illustrates an enterprise network topology. At the top, a 'To Internet' connection leads to a 'Firewall'. The Firewall is connected to a 'Router' on the left and another 'Router' on the right. The 'Web Server' is connected to the 'Router' on the right. The 'Router' on the left is connected to an 'Ethernet' segment, which is represented by a central switch with three computers connected to it. The 'Router' on the right is connected to another 'Ethernet' segment, also represented by a central switch with three computers connected to it. The two 'Router' units are connected to each other, forming a backbone between the two Ethernet segments. What is the number of subnets inside the enterprise network?
(A)	3
(B)	12
(C)	6
(D)	8

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Q.23 – Q.27 Multiple Select Questions (MSQ), carry ONE mark each.

Q.23	Which of the following statements is/are TRUE?
(A)	Every subset of a recursively enumerable language is recursive.
(B)	If a language L and its complement $\bar{L}$ are both recursively enumerable, then L must be recursive.
(C)	Complement of a context-free language must be recursive.
(D)	If L_1 and L_2 are regular, then $L_1 \cap L_2$ must be deterministic context-free.
Q.24	Let WB and WT be two set associative cache organizations that use LRU algorithm for cache block replacement. WB is a write back cache and WT is a write through cache. Which of the following statements is/are FALSE?
(A)	Each cache block in WB and WT has a dirty bit.
(B)	Every write hit in WB leads to a data transfer from cache to main memory.
(C)	Eviction of a block from WT will not lead to data transfer from cache to main memory.
(D)	A read miss in WB will never lead to eviction of a dirty block from WB.

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Q.25	Consider the following three relations in a relational database. $Employee(\underline{eId}, Name), Brand(\underline{bId}, bName), Own(\underline{eId}, \underline{bId})$ Which of the following relational algebra expressions return the set of $eIds$ who own all the brands?
(A)	$\Pi_{eId}(\Pi_{eId, bId}(Own) / \Pi_{bId}(Brand))$
(B)	$\Pi_{eId}(Own) - \Pi_{eId}((\Pi_{eId}(Own) \times \Pi_{bId}(Brand)) - \Pi_{eId, bId}(Own))$
(C)	$\Pi_{eId}(\Pi_{eId, bId}(Own) / \Pi_{bId}(Own))$
(D)	$\Pi_{eId}((\Pi_{eId}(Own) \times \Pi_{bId}(Own)) / \Pi_{bId}(Brand))$
Q.26	Which of the following statements is/are TRUE with respect to deadlocks?
(A)	Circular wait is a necessary condition for the formation of deadlock.
(B)	In a system where each resource has more than one instance, a cycle in its wait-for graph indicates the presence of a deadlock.
(C)	If the current allocation of resources to processes leads the system to unsafe state, then deadlock will necessarily occur.
(D)	In the resource-allocation graph of a system, if every edge is an assignment edge, then the system is not in deadlock state.

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Q.27	Which of the following statements is/are TRUE for a group G ?
(A)	If for all $x, y \in G$, $(xy)^2 = x^2y^2$, then G is commutative.
(B)	If for all $x \in G$, $x^2 = 1$, then G is commutative. Here, 1 is the identity element of G .
(C)	If the order of G is 2, then G is commutative.
(D)	If G is commutative, then a subgroup of G need not be commutative.

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Q.28 – Q.35 Numerical Answer Type (NAT), carry ONE mark each.

Q.28	Suppose a binary search tree with 1000 distinct elements is also a complete binary tree. The tree is stored using the array representation of binary heap trees. Assuming that the array indices start with 0, the 3 rd largest element of the tree is stored at index _____.
Q.29	Consider the augmented grammar with $\{+, *, (,), id\}$ as the set of terminals. $S' \rightarrow S$ $S \rightarrow S + R \mid R$ $R \rightarrow R * P \mid P$ $P \rightarrow (S) \mid id$ If I_0 is the set of two $LR(0)$ items $\{[S' \rightarrow S.], [S \rightarrow S. + R]\}$, then $goto(closure(I_0), +)$ contains exactly _____ items.
Q.30	Consider a simple undirected graph of 10 vertices. If the graph is disconnected, then the maximum number of edges it can have is _____.
Q.31	Consider a relation $R(A, B, C, D, E)$ with the following three functional dependencies. $AB \rightarrow C$; $BC \rightarrow D$; $C \rightarrow E$; The number of superkeys in the relation R is _____.
Q.32	The number of arrangements of six identical balls in three identical bins is _____.

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Q.33	A cache memory that has a hit rate of 0.8 has an access latency 10 ns and miss penalty 100 ns. An optimization is done on the cache to reduce the miss rate. However, the optimization results in an increase of cache access latency to 15 ns, whereas the miss penalty is not affected. The minimum hit rate (<i>rounded off to two decimal places</i>) needed after the optimization such that it should not increase the average memory access time is _____.
Q.34	The value of the following limit is _____. $\lim_{x \rightarrow 0^+} \frac{\sqrt{x}}{1 - e^{2\sqrt{x}}}$
Q.35	Consider the resolution of the domain name <code>www.gate.org.in</code> by a DNS resolver. Assume that no resource records are cached anywhere across the DNS servers and that iterative query mechanism is used in the resolution. The number of DNS query-response pairs involved in completely resolving the domain name is _____.

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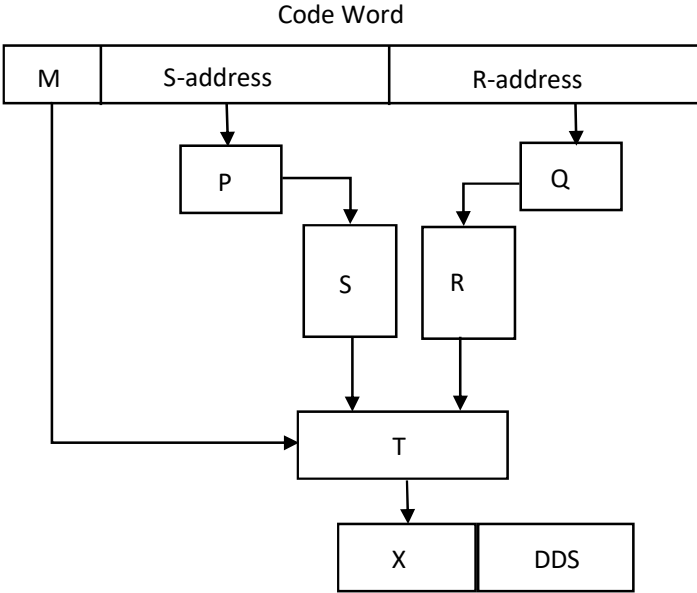
Q.36 – Q.45 Multiple Choice Questions (MCQ), carry TWO marks each.

Q.36	Which one of the following is the closed form for the generating function of the sequence $\{a_n\}_{n \geq 0}$ defined below? $a_n = \begin{cases} n+1, & n \text{ is odd} \\ 1, & \text{otherwise} \end{cases}$
(A)	$\frac{x(1+x^2)}{(1-x^2)^2} + \frac{1}{1-x}$
(B)	$\frac{x(3-x^2)}{(1-x^2)^2} + \frac{1}{1-x}$
(C)	$\frac{2x}{(1-x^2)^2} + \frac{1}{1-x}$
(D)	$\frac{x}{(1-x^2)^2} + \frac{1}{1-x}$
Q.37	Consider a simple undirected unweighted graph with at least three vertices. If A is the adjacency matrix of the graph, then the number of 3-cycles in the graph is given by the trace of
(A)	A^3
(B)	A^3 divided by 2
(C)	A^3 divided by 3
(D)	A^3 divided by 6

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Q.38	Which one of the following statements is FALSE?
(A)	The TLB performs an associative search in parallel on all its valid entries using page number of incoming virtual address.
(B)	If the virtual address of a word given by CPU has a TLB hit, but the subsequent search for the word results in a cache miss, then the word will always be present in the main memory.
(C)	The memory access time using a given inverted page table is always same for all incoming virtual addresses.
(D)	In a system that uses hashed page tables, if two distinct virtual addresses V1 and V2 map to the same value while hashing, then the memory access time of these addresses will not be the same.
Q.39	Let $R_i(z)$ and $W_i(z)$ denote read and write operations on a data element z by a transaction T_i, respectively. Consider the schedule S with four transactions. $S: R_4(x)R_2(x)R_3(x)R_1(y)W_1(y)W_2(x)W_3(y)R_4(y)$ Which one of the following serial schedules is conflict equivalent to S?
(A)	$T_1 \rightarrow T_3 \rightarrow T_4 \rightarrow T_2$
(B)	$T_1 \rightarrow T_4 \rightarrow T_3 \rightarrow T_2$
(C)	$T_4 \rightarrow T_1 \rightarrow T_3 \rightarrow T_2$
(D)	$T_3 \rightarrow T_1 \rightarrow T_4 \rightarrow T_2$

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Q.40	Consider a digital display system (DDS) shown in the figure that displays the contents of register X. A 16-bit code word is used to load a word in X, either from S or from R. S is a 1024-word memory segment and R is a 32-word register file. Based on the value of mode bit M, T selects an input word to load in X. P and Q interface with the corresponding bits in the code word to choose the addressed word. Which one of the following represents the functionality of P, Q, and T?
	
(A)	P is 10:1 multiplexer; Q is 5:1 multiplexer; T is 2:1 multiplexer
(B)	P is 10:2¹⁰ decoder; Q is 5:2⁵ decoder; T is 2:1 encoder
(C)	P is 10:2¹⁰ decoder; Q is 5:2⁵ decoder; T is 2:1 multiplexer
(D)	P is 1:10 de-multiplexer; Q is 1:5 de-multiplexer; T is 2:1 multiplexer

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Q.41	Consider three floating point numbers A, B and C stored in registers R_A, R_B and R_C, respectively as per IEEE-754 single precision floating point format. The 32-bit content stored in these registers (in hexadecimal form) are as follows. <table><tr><td>$R_A = 0xC1400000$</td><td>$R_B = 0x42100000$</td><td>$R_C = 0x41400000$</td></tr></table> Which one of the following is FALSE?	$R_A = 0xC1400000$	$R_B = 0x42100000$	$R_C = 0x41400000$
$R_A = 0xC1400000$	$R_B = 0x42100000$	$R_C = 0x41400000$		
(A)	$A + C = 0$			
(B)	$C = A + B$			
(C)	$B = 3C$			
(D)	$(B - C) > 0$			
Q.42	Consider four processes P, Q, R, and S scheduled on a CPU as per round robin algorithm with a time quantum of 4 units. The processes arrive in the order P, Q, R, S, all at time $t = 0$. There is exactly one context switch from S to Q, exactly one context switch from R to Q, and exactly two context switches from Q to R. There is no context switch from S to P. Switching to a ready process after the termination of another process is also considered a context switch. Which one of the following is NOT possible as CPU burst time (in time units) of these processes?			
(A)	$P = 4, Q = 10, R = 6, S = 2$			
(B)	$P = 2, Q = 9, R = 5, S = 1$			
(C)	$P = 4, Q = 12, R = 5, S = 4$			
(D)	$P = 3, Q = 7, R = 7, S = 3$			

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Q.43	What is printed by the following ANSI C program? <pre>#include<stdio.h> int main(int argc, char *argv[]) { int a[3][3][3] = {{1, 2, 3, 4, 5, 6, 7, 8, 9}, {10, 11, 12, 13, 14, 15, 16, 17, 18}, {19, 20, 21, 22, 23, 24, 25, 26, 27}}; int i = 0, j = 0, k = 0; for(i = 0; i < 3; i++){ for(k = 0; k < 3; k++) printf("%d ", a[i][j][k]); printf("\n"); } return 0; }</pre>
(A)	<pre>1 2 3 10 11 12 19 20 21</pre>
(B)	<pre>1 4 7 10 13 16 19 22 25</pre>
(C)	<pre>1 2 3 4 5 6 7 8 9</pre>
(D)	<pre>1 2 3 13 14 15 25 26 27</pre>

GATE 2022 Computer Science and Information Technology (CS)

Q.44 What is printed by the following ANSI C program?

```
#include<stdio.h>

int main(int argc, char *argv[]){

    char a = 'P';

    char b = 'x';

    char c = (a & b) + '*';

    char d = (a | b) - '-';

    char e = (a ^ b) + '+';

    printf("%c %c %c\n", c, d, e);

    return 0;

}
```

ASCII encoding for relevant characters is given below

A	B	C	...	Z
65	66	67	...	90

a	b	c	...	z
97	98	99	...	122

*	+	-
42	43	45

(A) z K S

(B) 122 75 83

(C) * - +

(D) P x +

GATE 2022 Computer Science and Information Technology (CS)

Q.45	Consider solving the following system of simultaneous equations using LU decomposition. $\begin{aligned}x_1 + x_2 - 2x_3 &= 4 \\x_1 + 3x_2 - x_3 &= 7 \\2x_1 + x_2 - 5x_3 &= 7\end{aligned}$ where L and U are denoted as $L = \begin{pmatrix} L_{11} & 0 & 0 \\ L_{21} & L_{22} & 0 \\ L_{31} & L_{32} & L_{33} \end{pmatrix}, \quad U = \begin{pmatrix} U_{11} & U_{12} & U_{13} \\ 0 & U_{22} & U_{23} \\ 0 & 0 & U_{33} \end{pmatrix}$ Which one of the following is the correct combination of values for L_{32}, U_{33}, and x_1?
(A)	$L_{32} = 2, U_{33} = -\frac{1}{2}, x_1 = -1$
(B)	$L_{32} = 2, U_{33} = 2, x_1 = -1$
(C)	$L_{32} = -\frac{1}{2}, U_{33} = 2, x_1 = 0$
(D)	$L_{32} = -\frac{1}{2}, U_{33} = -\frac{1}{2}, x_1 = 0$

GATE 2022 Computer Science and Information Technology (CS)

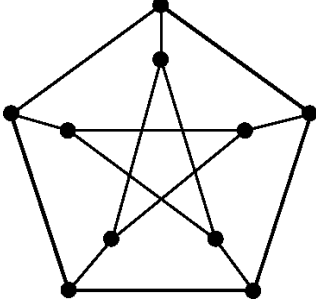
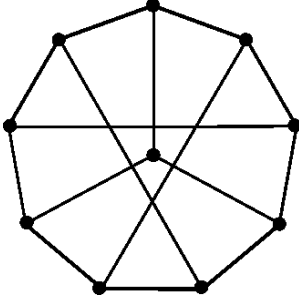
Q.46 – Q.555 Multiple Select Questions (MSQ), carry TWO marks each.

Q.46	Which of the following is/are undecidable?
(A)	Given two Turing machines M_1 and M_2 , decide if $L(M_1) = L(M_2)$.
(B)	Given a Turing machine M , decide if $L(M)$ is regular.
(C)	Given a Turing machine M , decide if M accepts all strings.
(D)	Given a Turing machine M , decide if M takes more than 1073 steps on every string.
Q.47	Consider the following languages: $L_1 = \{a^n w a^n \mid w \in \{a, b\}^*\}$ $L_2 = \{w x w^R \mid w, x \in \{a, b\}^*, w , x > 0\}$ Note that w^R is the reversal of the string w. Which of the following is/are TRUE?
(A)	L_1 and L_2 are regular.
(B)	L_1 and L_2 are context-free.
(C)	L_1 is regular and L_2 is context-free.
(D)	L_1 and L_2 are context-free but not regular.

GATE 2022 Computer Science and Information Technology (CS)

Q.48	Consider the following languages: $L_1 = \{ww \mid w \in \{a,b\}^*\}$ $L_2 = \{a^n b^n c^m \mid m, n \geq 0\}$ $L_3 = \{a^m b^n c^n \mid m, n \geq 0\}$ Which of the following statements is/are FALSE?
(A)	L_1 is not context-free but L_2 and L_3 are deterministic context-free.
(B)	Neither L_1 nor L_2 is context-free.
(C)	L_2 , L_3 and $L_2 \cap L_3$ all are context-free.
(D)	Neither L_1 nor its complement is context-free.
Q.49	Consider a simple undirected weighted graph G, all of whose edge weights are distinct. Which of the following statements about the minimum spanning trees of G is/are TRUE?
(A)	The edge with the second smallest weight is always part of any minimum spanning tree of G .
(B)	One or both of the edges with the third smallest and the fourth smallest weights are part of any minimum spanning tree of G .
(C)	Suppose $S \subseteq V$ be such that $S \neq \emptyset$ and $S \neq V$. Consider the edge with the minimum weight such that one of its vertices is in S and the other in $V \setminus S$. Such an edge will always be part of any minimum spanning tree of G .
(D)	G can have multiple minimum spanning trees.

GATE 2022 Computer Science and Information Technology (CS)

Q.50	The following simple undirected graph is referred to as the Peterson graph.  Which of the following statements is/are TRUE?
(A)	The chromatic number of the graph is 3.
(B)	The graph has a Hamiltonian path.
(C)	The following graph is isomorphic to the Peterson graph. 
(D)	The size of the largest independent set of the given graph is 3. (A subset of vertices of a graph form an independent set if no two vertices of the subset are adjacent.)

GATE 2022 Computer Science and Information Technology (CS)

Q.51	Consider the following recurrence: $\begin{aligned} f(1) &= 1; \\ f(2n) &= 2f(n) - 1, \text{ for } n \geq 1; \\ f(2n+1) &= 2f(n) + 1, \text{ for } n \geq 1. \end{aligned}$ Then, which of the following statements is/are TRUE?
(A)	$f(2^n - 1) = 2^n - 1$
(B)	$f(2^n) = 1$
(C)	$f(5 \cdot 2^n) = 2^{n+1} + 1$
(D)	$f(2^n + 1) = 2^n + 1$
Q.52	Which of the properties hold for the adjacency matrix A of a simple undirected unweighted graph having n vertices?
(A)	The diagonal entries of A^2 are the degrees of the vertices of the graph.
(B)	If the graph is connected, then none of the entries of $A^{n-1} + I_n$ can be zero.
(C)	If the sum of all the elements of A is at most $2(n-1)$, then the graph must be acyclic.
(D)	If there is at least a 1 in each of A 's rows and columns, then the graph must be connected.



GATE 2022 Computer Science and Information Technology (CS)

Q.53	Which of the following is/are the eigenvector(s) for the matrix given below? $\begin{pmatrix} -9 & -6 & -2 & -4 \\ -8 & -6 & -3 & -1 \\ 20 & 15 & 8 & 5 \\ 32 & 21 & 7 & 12 \end{pmatrix}$
(A)	$\begin{pmatrix} -1 \\ 1 \\ 0 \\ 1 \end{pmatrix}$
(B)	$\begin{pmatrix} 1 \\ 0 \\ -1 \\ 0 \end{pmatrix}$
(C)	$\begin{pmatrix} -1 \\ 0 \\ 2 \\ 2 \end{pmatrix}$
(D)	$\begin{pmatrix} 0 \\ 1 \\ -3 \\ 0 \end{pmatrix}$

GATE 2022 Computer Science and Information Technology (CS)

Q.54	Consider a system with 2 KB direct mapped data cache with a block size of 64 bytes. The system has a physical address space of 64 KB and a word length of 16 bits. During the execution of a program, four data words P, Q, R, and S are accessed in that order 10 times (<i>i.e.</i> , PQRSPQRS...). Hence, there are 40 accesses to data cache altogether. Assume that the data cache is initially empty and no other data words are accessed by the program. The addresses of the first bytes of P, Q, R, and S are 0xA248, 0xC28A, 0xCA8A, and 0xA262, respectively. For the execution of the above program, which of the following statements is/are TRUE with respect to the data cache?
(A)	Every access to S is a hit.
(B)	Once P is brought to the cache it is never evicted.
(C)	At the end of the execution only R and S reside in the cache.
(D)	Every access to R evicts Q from the cache.

GATE 2022 Computer Science and Information Technology (CS)

Q.55	Consider routing table of an organization’s router shown below:				
		Subnet Number	Subnet Mask	Next Hop	
		12.20.164.0	255.255.252.0	R1	
		12.20.170.0	255.255.254.0	R2	
		12.20.168.0	255.255.254.0	Interface 0	
		12.20.166.0	255.255.254.0	Interface 1	
		default		R3	
	Which of the following prefixes in CIDR notation can be collectively used to correctly aggregate all of the subnets in the routing table?				
(A)	12.20.164.0/20				
(B)	12.20.164.0/22				
(C)	12.20.164.0/21				
(D)	12.20.168.0/22				


GATE 2022 Computer Science and Information Technology (CS)
Q.56 – Q.65 Numerical Answer Type (NAT), carry TWO marks each.

Q.56 Consider the relational database with the following four schemas and their respective instances.

Student (sNo, sName, dNo) Dept (dNo, dName)

Course (cNo, cName, dNo) Register (sNo, cNo)

Student		
sNo	sName	dNo
S01	James	D01
S02	Rocky	D01
S03	Jackson	D02
S04	Jane	D01
S05	Milli	D02

Dept	
dNo	dName
D01	CSE
D02	EEE

	Course	
	cNo	dNo
	C11	DS
	C12	OS
	C21	DE
	C22	PT
	C23	CV

Register	
sNo	cNo
S01	C11
S01	C12
S02	C11
S03	C21
S03	C22
S03	C23
S04	C11
S04	C12
S05	C11
S05	C21

SQL Query:

```
SELECT * FROM Student AS S WHERE NOT EXIST
  (SELECT cNo FROM Course WHERE dNo = "D01"
    EXCEPT
    SELECT cNo FROM Register WHERE sNo = S.sNo)
```

The number of rows returned by the above SQL query is_____.

Q.57 Consider a network with three routers P, Q, R shown in the figure below. All the links have cost of unity.

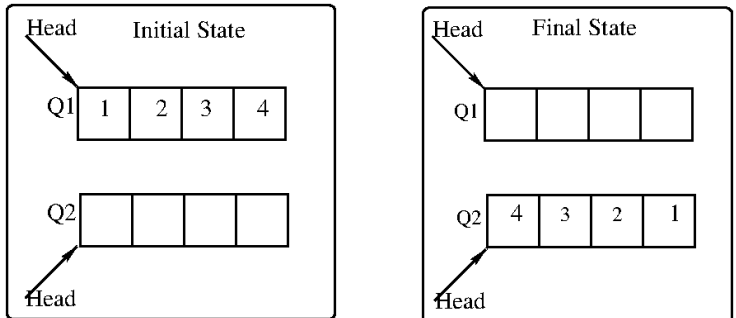
```
graph LR; P[Router P] --- Q[Router Q]; Q --- R[Router R];
```

The routers exchange distance vector routing information and have converged on the routing tables, after which the link Q–R fails. Assume that P and Q send out routing updates at random times, each at the same average rate. The probability of a routing loop formation (*rounded off to one decimal place*) between P and Q, leading to count-to-infinity problem, is_____.

GATE 2022 Computer Science and Information Technology (CS)

Q.58	Let $G(V, E)$ be a directed graph, where $V = \{1, 2, 3, 4, 5\}$ is the set of vertices and E is the set of directed edges, as defined by the following adjacency matrix A. $A[i][j] = \begin{cases} 1, & 1 \leq j \leq i \leq 5 \\ 0, & \text{otherwise} \end{cases}$ $A[i][j] = 1$ indicates a directed edge from node i to node j. A <i>directed spanning tree</i> of G, rooted at $r \in V$, is defined as a subgraph T of G such that the undirected version of T is a tree, and T contains a directed path from r to every other vertex in V. The number of such directed spanning trees rooted at vertex 5 is _____.
Q.59	Consider a 100 Mbps link between an earth station (sender) and a satellite (receiver) at an altitude of 2100 km. The signal propagates at a speed of 3×10^8 m/s. The time taken (in milliseconds, <i>rounded off to two decimal places</i>) for the receiver to completely receive a packet of 1000 bytes transmitted by the sender is _____.
Q.60	Consider the data transfer using TCP over a 1 Gbps link. Assuming that the maximum segment lifetime (MSL) is set to 60 seconds, the minimum number of bits required for the sequence number field of the TCP header, to prevent the sequence number space from wrapping around during the MSL is _____.
Q.61	A processor X_1 operating at 2 GHz has a standard 5-stage RISC instruction pipeline having a base CPI (cycles per instruction) of one without any pipeline hazards. For a given program P that has 30% branch instructions, control hazards incur 2 cycles stall for every branch. A new version of the processor X_2 operating at same clock frequency has an additional branch predictor unit (BPU) that completely eliminates stalls for correctly predicted branches. There is neither any savings nor any additional stalls for wrong predictions. There are no structural hazards and data hazards for X_1 and X_2. If the BPU has a prediction accuracy of 80%, the speed up (<i>rounded off to two decimal places</i>) obtained by X_2 over X_1 in executing P is _____.

GATE 2022 Computer Science and Information Technology (CS)

Q.62	Consider the queues Q_1 containing four elements and Q_2 containing none (shown as the Initial State in the figure). The only operations allowed on these two queues are $\text{Enqueue}(Q, \text{element})$ and $\text{Dequeue}(Q)$. The minimum number of Enqueue operations on Q_1 required to place the elements of Q_1 in Q_2 in reverse order (shown as the Final State in the figure) without using any additional storage is _____.
	
Q.63	Consider two files systems A and B, that use contiguous allocation and linked allocation, respectively. A file of size 100 blocks is already stored in A and also in B. Now, consider inserting a new block in the middle of the file (between 50th and 51st block), whose data is already available in the memory. Assume that there are enough free blocks at the end of the file and that the file control blocks are already in memory. Let the number of disk accesses required to insert a block in the middle of the file in A and B are n_A and n_B, respectively, then the value of $n_A + n_B$ is _____.
Q.64	Consider a demand paging system with four page frames (initially empty) and LRU page replacement policy. For the following page reference string 7, 2, 7, 3, 2, 5, 3, 4, 6, 7, 7, 1, 5, 6, 1 the page fault rate, defined as the ratio of number of page faults to the number of memory accesses (<i>rounded off to one decimal place</i>) is _____.



GATE 2022 Computer Science and Information Technology (CS)

Q.65	Consider the following grammar along with translation rules. $S \rightarrow S_1 \# T \quad \{S_{val} = S_{1val} * T_{val}\}$ $S \rightarrow T \quad \{S_{val} = T_{val}\}$ $T \rightarrow T_1 \% R \quad \{T_{val} = T_{1val} \div R_{val}\}$ $T \rightarrow R \quad \{T_{val} = R_{val}\}$ $R \rightarrow id \quad \{R_{val} = id_{val}\}$ Here # and % are operators and <i>id</i> is a token that represents an integer and id_{val} represents the corresponding integer value. The set of non-terminals is $\{S, T, R, P\}$ and a subscripted non-terminal indicates an instance of the non-terminal. Using this translation scheme, the computed value of S_{val} for root of the parse tree for the expression $20\#10\%5\#8\%2\%2$ is _____.

GATE 2022

Part 2 of 2

official answer key (not a question paper) | 2 pages

Paper 7 of 16

Q. No.	Session	Question Type	Subject Name	Key/Range	Mark
1	1	MCQ	GA	D	1
2	1	MCQ	GA	C	1
3	1	MCQ	GA	D	1
4	1	MCQ	GA	A	1
5	1	MCQ	GA	B	1
6	1	MCQ	GA	B	2
7	1	MCQ	GA	D	2
8	1	MCQ	GA	D	2
9	1	MCQ	GA	B	2
10	1	MCQ	GA	D	2
11	1	MCQ	CS	A	1
12	1	MCQ	CS	D	1
13	1	MCQ	CS	D	1
14	1	MCQ	CS	A	1
15	1	MCQ	CS	A	1
16	1	MCQ	CS	B	1
17	1	MCQ	CS	A	1
18	1	MCQ	CS	B	1
19	1	MCQ	CS	C	1
20	1	MCQ	CS	C	1
21	1	MCQ	CS	D	1
22	1	MCQ	CS	C	1
23	1	MSQ	CS	B,C,D	1
24	1	MSQ	CS	A,B,D	1
25	1	MSQ	CS	A,B	1
26	1	MSQ	CS	A,D	1
27	1	MSQ	CS	A,B,C	1
28	1	NAT	CS	509 to 509	1
29	1	NAT	CS	5 to 5	1
30	1	NAT	CS	36 to 36	1
31	1	NAT	CS	8 to 8	1
32	1	NAT	CS	7 to 7	1
33	1	NAT	CS	0.85 to 0.85	1
34	1	NAT	CS	-0.5 to -0.5	1
35	1	NAT	CS	4 to 4	1
36	1	MCQ	CS	A	2
37	1	MCQ	CS	D	2
38	1	MCQ	CS	C	2
39	1	MCQ	CS	A	2
40	1	MCQ	CS	C	2
41	1	MCQ	CS	B	2
42	1	MCQ	CS	D	2
43	1	MCQ	CS	A	2
44	1	MCQ	CS	A	2



45	1	MCQ	CS	D	2
46	1	MSQ	CS	A,B,C	2
47	1	MSQ	CS	A,B,C	2
48	1	MSQ	CS	B,C,D	2
49	1	MSQ	CS	A,B,C	2
50	1	MSQ	CS	A,B,C	2
51	1	MSQ	CS	A,B,C	2
52	1	MSQ	CS	A	2
53	1	MSQ	CS	A,C,D	2
54	1	MSQ	CS	A,B,D	2
55	1	MSQ	CS	B,D	2
56	1	NAT	CS	2 to 2	2
57	1	NAT	CS	0.5 to 0.5	2
58	1	NAT	CS	24 to 24	2
59	1	NAT	CS	7.07 to 7.09	2
60	1	NAT	CS	33 to 33	2
61	1	NAT	CS	1.42 to 1.45	2
62	1	NAT	CS	0 to 0	2
63	1	NAT	CS	153 to 153	2
64	1	NAT	CS	0.6 to 0.6	2
65	1	NAT	CS	80 to 80	2

GATE 2021

Set 1

65 questions | 40 pages

Paper 8 of 16

Computer Science and Information Technology (CS, Set-1)

General Aptitude (GA)

Q.1 – Q.5 Multiple Choice Question (MCQ), carry ONE mark each (for each wrong answer: – 1/3).

Q.1	The ratio of boys to girls in a class is 7 to 3. Among the options below, an acceptable value for the total number of students in the class is:
(A)	21
(B)	37
(C)	50
(D)	73



Computer Science and Information Technology (CS, Set-1)

Q.2	A polygon is convex if, for every pair of points, P and Q belonging to the polygon, the line segment PQ lies completely inside or on the polygon. Which one of the following is <u>NOT</u> a convex polygon?
(A)	
(B)	
(C)	
(D)	

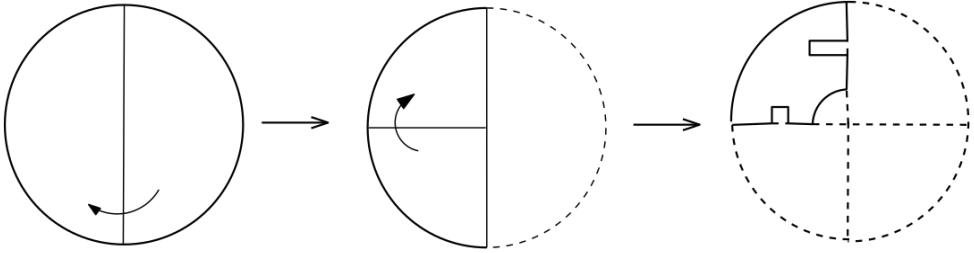
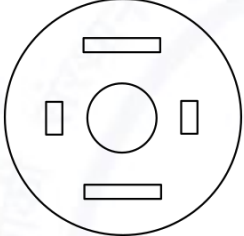
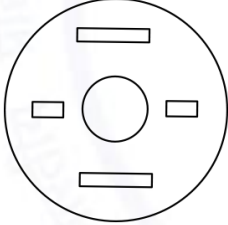
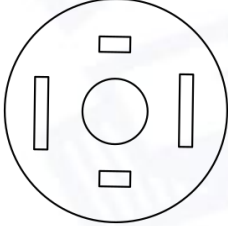
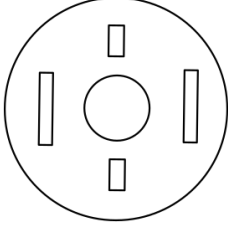


Computer Science and Information Technology (CS, Set-1)

Q.3	Consider the following sentences: (i) Everybody in the class is prepared for the exam. (ii) Babu invited Danish to his home because he enjoys playing chess. Which of the following is the CORRECT observation about the above two sentences?
(A)	(i) is grammatically correct and (ii) is unambiguous
(B)	(i) is grammatically incorrect and (ii) is unambiguous
(C)	(i) is grammatically correct and (ii) is ambiguous
(D)	(i) is grammatically incorrect and (ii) is ambiguous



Computer Science and Information Technology (CS, Set-1)

Q.4	 A circular sheet of paper is folded along the lines in the directions shown. The paper, after being punched in the final folded state as shown and unfolded in the reverse order of folding, will look like _____.
(A)	
(B)	
(C)	
(D)	

Computer Science and Information Technology (CS, Set-1)

Q.5	_____ is to <i>surgery</i> as <i>writer</i> is to _____ Which one of the following options maintains a similar logical relation in the above sentence?
(A)	Plan, outline
(B)	Hospital, library
(C)	Doctor, book
(D)	Medicine, grammar

Computer Science and Information Technology (CS, Set-1)

Q. 6 – Q. 10 Multiple Choice Question (MCQ), carry TWO marks each (for each wrong answer: – 2/3).

Q.6	We have 2 rectangular sheets of paper, M and N, of dimensions 6 cm x 1 cm each. Sheet M is rolled to form an open cylinder by bringing the short edges of the sheet together. Sheet N is cut into equal square patches and assembled to form the largest possible closed cube. Assuming the ends of the cylinder are closed, the ratio of the volume of the cylinder to that of the cube is _____
(A)	$\frac{\pi}{2}$
(B)	$\frac{3}{\pi}$
(C)	$\frac{9}{\pi}$
(D)	3π



Computer Science and Information Technology (CS, Set-1)

Q.7

Items	Cost (₹)	Profit %	Marked Price (₹)
P	5,400	---	5,860
Q	---	25	10,000

Details of prices of two items P and Q are presented in the above table. The ratio of cost of item P to cost of item Q is 3:4. Discount is calculated as the difference between the marked price and the selling price. The profit percentage is calculated as the ratio of the difference between selling price and cost, to the cost (Profit % = $\frac{\text{Selling price} - \text{Cost}}{\text{Cost}} \times 100$).

The discount on item Q, as a percentage of its marked price, is _____

(A) 25

(B) 12.5

(C) 10

(D) 5



Computer Science and Information Technology (CS, Set-1)

Q.8	There are five bags each containing identical sets of ten distinct chocolates. One chocolate is picked from each bag. The probability that at least two chocolates are identical is _____
(A)	0.3024
(B)	0.4235
(C)	0.6976
(D)	0.8125



Computer Science and Information Technology (CS, Set-1)

Q.9	Given below are two statements 1 and 2, and two conclusions I and II. Statement 1: All bacteria are microorganisms. Statement 2: All pathogens are microorganisms. Conclusion I: Some pathogens are bacteria. Conclusion II: All pathogens are not bacteria. Based on the above statements and conclusions, which one of the following options is logically CORRECT?
(A)	Only conclusion I is correct
(B)	Only conclusion II is correct
(C)	Either conclusion I or II is correct.
(D)	Neither conclusion I nor II is correct.



Computer Science and Information Technology (CS, Set-1)

Q.10	Some people suggest anti-obesity measures (AOM) such as displaying calorie information in restaurant menus. Such measures sidestep addressing the core problems that cause obesity: poverty and income inequality. Which one of the following statements summarizes the passage?
(A)	The proposed AOM addresses the core problems that cause obesity.
(B)	If obesity reduces, poverty will naturally reduce, since obesity causes poverty.
(C)	AOM are addressing the core problems and are likely to succeed.
(D)	AOM are addressing the problem superficially.

Computer Science and Information Technology (CS, Set-1)

Computer Science and Information Technology (CS, Set-1)

Q.1 – Q.10 Multiple Choice Question (MCQ), carry ONE mark each (for each wrong answer: – 1/3).

Q.1	Suppose that L_1 is a regular language and L_2 is a context-free language. Which one of the following languages is NOT necessarily context-free?
(A)	$L_1 \cap L_2$
(B)	$L_1 \cdot L_2$
(C)	$L_1 - L_2$
(D)	$L_1 \cup L_2$

Q.2	Let P be an array containing n integers. Let t be the lowest upper bound on the number of comparisons of the array elements, required to find the minimum and maximum values in an arbitrary array of n elements. Which one of the following choices is correct?
(A)	$t > 2n - 2$
(B)	$t > 3 \left\lceil \frac{n}{2} \right\rceil$ and $t \leq 2n - 2$
(C)	$t > n$ and $t \leq 3 \left\lceil \frac{n}{2} \right\rceil$
(D)	$t > \lceil \log_2(n) \rceil$ and $t \leq n$



Computer Science and Information Technology (CS, Set-1)

Q.3	Consider the following three functions. $f_1 = 10^n \quad f_2 = n^{\log n} \quad f_3 = n^{\sqrt{n}}$ Which one of the following options arranges the functions in the increasing order of asymptotic growth rate?
(A)	f_3, f_2, f_1
(B)	f_2, f_1, f_3
(C)	f_1, f_2, f_3
(D)	f_2, f_3, f_1

Q.4	Consider the following statements. S_1 : The sequence of procedure calls corresponds to a preorder traversal of the activation tree. S_2 : The sequence of procedure returns corresponds to a postorder traversal of the activation tree. Which one of the following options is correct?
(A)	S_1 is true and S_2 is false
(B)	S_1 is false and S_2 is true
(C)	S_1 is true and S_2 is true
(D)	S_1 is false and S_2 is false



Computer Science and Information Technology (CS, Set-1)

Q.5	Consider the following statements. S_1 : Every SLR(1) grammar is unambiguous but there are certain unambiguous grammars that are not SLR(1). S_2 : For any context-free grammar, there is a parser that takes at most $O(n^3)$ time to parse a string of length n. Which one of the following options is correct?
(A)	S_1 is true and S_2 is false
(B)	S_1 is false and S_2 is true
(C)	S_1 is true and S_2 is true
(D)	S_1 is false and S_2 is false

Q.6	Let the representation of a number in base 3 be 210. What is the hexadecimal representation of the number?
(A)	15
(B)	21
(C)	D2
(D)	528



Computer Science and Information Technology (CS, Set-1)

Q.7	Let p and q be two propositions. Consider the following two formulae in propositional logic. $S_1 : (\neg p \wedge (p \vee q)) \rightarrow q$ $S_2 : q \rightarrow (\neg p \wedge (p \vee q))$ Which one of the following choices is correct?
(A)	Both S_1 and S_2 are tautologies.
(B)	S_1 is a tautology but S_2 is not a tautology.
(C)	S_1 is not a tautology but S_2 is a tautology.
(D)	Neither S_1 nor S_2 is a tautology.

Q.8	Consider the following two statements. $S_1 : \text{Destination MAC address of an ARP reply is a broadcast address.}$ $S_2 : \text{Destination MAC address of an ARP request is a broadcast address.}$ Which one of the following choices is correct?
(A)	Both S_1 and S_2 are true.
(B)	S_1 is true and S_2 is false.
(C)	S_1 is false and S_2 is true.
(D)	Both S_1 and S_2 are false.



Computer Science and Information Technology (CS, Set-1)

Q.9	Consider the following array. <table><tr><td>23</td><td>32</td><td>45</td><td>69</td><td>72</td><td>73</td><td>89</td><td>97</td></tr></table> Which algorithm out of the following options uses the least number of comparisons (among the array elements) to sort the above array in ascending order?	23	32	45	69	72	73	89	97
23	32	45	69	72	73	89	97		
(A)	Selection sort								
(B)	Mergesort								
(C)	Insertion sort								
(D)	Quicksort using the last element as pivot								

Q.10	A binary search tree T contains n distinct elements. What is the time complexity of picking an element in T that is smaller than the maximum element in T?
(A)	$\Theta(n \log n)$
(B)	$\Theta(n)$
(C)	$\Theta(\log n)$
(D)	$\Theta(1)$



Computer Science and Information Technology (CS, Set-1)

Q.11 – Q.15 Multiple Select Question (MSQ), carry ONE mark each (no negative marks).

Q.11	In the context of operating systems, which of the following statements is/are correct with respect to paging?
(A)	Paging helps solve the issue of external fragmentation.
(B)	Page size has no impact on internal fragmentation.
(C)	Paging incurs memory overheads.
(D)	Multi-level paging is necessary to support pages of different sizes.

Q.12	Let $\langle M \rangle$ denote an encoding of an automaton M . Suppose that $\Sigma = \{0, 1\}$. Which of the following languages is/are NOT recursive?
(A)	$L = \{\langle M \rangle \mid M \text{ is a DFA such that } L(M) = \emptyset\}$
(B)	$L = \{\langle M \rangle \mid M \text{ is a DFA such that } L(M) = \Sigma^*\}$
(C)	$L = \{\langle M \rangle \mid M \text{ is a PDA such that } L(M) = \emptyset\}$
(D)	$L = \{\langle M \rangle \mid M \text{ is a PDA such that } L(M) = \Sigma^*\}$

Computer Science and Information Technology (CS, Set-1)

Q.13	Suppose a database system crashes again while recovering from a previous crash. Assume checkpointing is not done by the database either during the transactions or during recovery. Which of the following statements is/are correct?
(A)	The same undo and redo list will be used while recovering again.
(B)	The system cannot recover any further.
(C)	All the transactions that are already undone and redone will not be recovered again.
(D)	The database will become inconsistent.

Q.14	Which of the following standard C library functions will <i>always</i> invoke a system call when executed from a single-threaded process in a UNIX/Linux operating system?
(A)	<code>exit</code>
(B)	<code>malloc</code>
(C)	<code>sleep</code>
(D)	<code>strlen</code>

Computer Science and Information Technology (CS, Set-1)

Q.15	Consider a linear list based directory implementation in a file system. Each directory is a list of nodes, where each node contains the file name along with the file metadata, such as the list of pointers to the data blocks. Consider a given directory <code>foo</code>. Which of the following operations will necessarily require a full scan of <code>foo</code> for successful completion?
(A)	Creation of a new file in <code>foo</code>
(B)	Deletion of an existing file from <code>foo</code>
(C)	Renaming of an existing file in <code>foo</code>
(D)	Opening of an existing file in <code>foo</code>



Computer Science and Information Technology (CS, Set-1)

Q.16 – Q.25 Numerical Answer Type (NAT), carry ONE mark each (no negative marks).

Q.16	In an undirected connected planar graph G , there are eight vertices and five faces. The number of edges in G is _____.
------	---

Q.17	Consider the following undirected graph with edge weights as shown: The number of minimum-weight spanning trees of the graph is _____.
------	--

Q.18	The lifetime of a component of a certain type is a random variable whose probability density function is exponentially distributed with parameter 2. For a randomly picked component of this type, the probability that its lifetime exceeds the expected lifetime (rounded to 2 decimal places) is _____.
------	--

Q.19	There are 6 jobs with distinct difficulty levels, and 3 computers with distinct processing speeds. Each job is assigned to a computer such that: - The fastest computer gets the toughest job and the slowest computer gets the easiest job. - Every computer gets at least one job. The number of ways in which this can be done is _____.
------	--



Computer Science and Information Technology (CS, Set-1)

Q.20	Consider the following expression. $\lim_{x \rightarrow -3} \frac{\sqrt{2x+22} - 4}{x+3}$ The value of the above expression (rounded to 2 decimal places) is _____.
Q.21	Consider the following sequence of operations on an empty stack. <code>push(54); push(52); pop(); push(55); push(62); s = pop();</code> Consider the following sequence of operations on an empty queue. <code>enqueue(21); enqueue(24); dequeue(); enqueue(28); enqueue(32); q = dequeue();</code> The value of <code>s + q</code> is _____.
Q.22	Consider a computer system with a byte-addressable primary memory of size 2^{32} bytes. Assume the computer system has a direct-mapped cache of size 32 KB ($1 \text{ KB} = 2^{10}$ bytes), and each cache block is of size 64 bytes. The size of the tag field is _____ bits.
Q.23	A relation $r(A, B)$ in a relational database has 1200 tuples. The attribute A has integer values ranging from 6 to 20, and the attribute B has integer values ranging from 1 to 20. Assume that the attributes A and B are independently distributed. The estimated number of tuples in the output of $\sigma_{(A>10) \vee (B=18)}(r)$ is _____.



Computer Science and Information Technology (CS, Set-1)

Q.24	Consider the following representation of a number in IEEE 754 single-precision floating point format with a bias of 127. $S : 1 \quad E : 10000001 \quad F : 11110000000000000000000$ Here S, E and F denote the sign, exponent and fraction components of the floating point representation. The decimal value corresponding to the above representation (rounded to 2 decimal places) is _____.
Q.25	Three processes arrive at time zero with CPU bursts of 16, 20 and 10 milliseconds. If the scheduler has prior knowledge about the length of the CPU bursts, the minimum achievable average waiting time for these three processes in a non-preemptive scheduler (rounded to nearest integer) is _____ milliseconds.

Computer Science and Information Technology (CS, Set-1)

Q.26 – Q.39 Multiple Choice Question (MCQ), carry TWO mark each (for each wrong answer: – 2/3).

Q.26	Consider the following grammar (that admits a series of declarations, followed by expressions) and the associated syntax directed translation (SDT) actions, given as pseudo-code: $P \rightarrow D^* E^*$ $D \rightarrow \text{int ID} \{\text{record that ID.lexeme is of type int}\}$ $D \rightarrow \text{bool ID} \{\text{record that ID.lexeme is of type bool}\}$ $E \rightarrow E_1 + E_2 \{\text{check that } E_1.\text{type} = E_2.\text{type} = \text{int}; \text{ set } E.\text{type} := \text{int}\}$ $E \rightarrow !E_1 \{\text{check that } E_1.\text{type} = \text{bool}; \text{ set } E.\text{type} := \text{bool}\}$ $E \rightarrow \text{ID} \{\text{set } E.\text{type} := \text{int}\}$ With respect to the above grammar, which one of the following choices is correct?
(A)	The actions can be used to correctly type-check any syntactically correct program.
(B)	The actions can be used to type-check syntactically correct integer variable declarations and integer expressions.
(C)	The actions can be used to type-check syntactically correct boolean variable declarations and boolean expressions.
(D)	The actions will lead to an infinite loop.

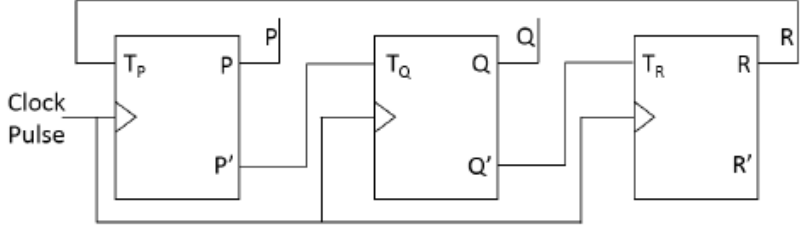


Computer Science and Information Technology (CS, Set-1)

Q.27	The following relation records the age of 500 employees of a company, where <i>empNo</i> (indicating the employee number) is the key: $empAge(\underline{empNo}, age)$ Consider the following relational algebra expression: $\Pi_{empNo}(empAge \bowtie_{(age > age1)} \rho_{empNo1, age1}(empAge))$ What does the above expression generate?
(A)	Employee numbers of only those employees whose age is the maximum.
(B)	Employee numbers of only those employees whose age is more than the age of exactly one other employee.
(C)	Employee numbers of all employees whose age is not the minimum.
(D)	Employee numbers of all employees whose age is the minimum.



Computer Science and Information Technology (CS, Set-1)

Q.28	Consider a 3-bit counter, designed using T flip-flops, as shown below:  Assuming the initial state of the counter given by PQR as 000, what are the next three states?
(A)	011, 101, 000
(B)	001, 010, 111
(C)	011, 101, 111
(D)	001, 010, 000

Q.29	Assume that a 12-bit Hamming codeword consisting of 8-bit data and 4 check bits is $d_8d_7d_6d_5c_8d_4d_3d_2c_4d_1c_2c_1$, where the data bits and the check bits are given in the following tables: <table><tr><th colspan="8">Data bits</th></tr><tr><th>d_8</th><th>d_7</th><th>d_6</th><th>d_5</th><th>d_4</th><th>d_3</th><th>d_2</th><th>d_1</th></tr><tr><td>1</td><td>1</td><td>0</td><td>x</td><td>0</td><td>1</td><td>0</td><td>1</td></tr></table> <table><tr><th colspan="4">Check bits</th></tr><tr><th>c_8</th><th>c_4</th><th>c_2</th><th>c_1</th></tr><tr><td>y</td><td>0</td><td>1</td><td>0</td></tr></table> Which one of the following choices gives the correct values of x and y?	Data bits								d_8	d_7	d_6	d_5	d_4	d_3	d_2	d_1	1	1	0	x	0	1	0	1	Check bits				c_8	c_4	c_2	c_1	y	0	1	0
Data bits																																					
d_8	d_7	d_6	d_5	d_4	d_3	d_2	d_1																														
1	1	0	x	0	1	0	1																														
Check bits																																					
c_8	c_4	c_2	c_1																																		
y	0	1	0																																		
(A)	x is 0 and y is 0.																																				
(B)	x is 0 and y is 1.																																				
(C)	x is 1 and y is 0.																																				
(D)	x is 1 and y is 1.																																				



Computer Science and Information Technology (CS, Set-1)

Q.30	Consider the following recurrence relation. $T(n) = \begin{cases} T(n/2) + T(2n/5) + 7n & \text{if } n > 0 \\ 1 & \text{if } n = 0 \end{cases}$ Which one of the following options is correct?
(A)	$T(n) = \Theta(n^{5/2})$
(B)	$T(n) = \Theta(n \log n)$
(C)	$T(n) = \Theta(n)$
(D)	$T(n) = \Theta((\log n)^{5/2})$





Computer Science and Information Technology (CS, Set-1)

Q.31

Consider the following context-free grammar where the set of terminals is $\{a, b, c, d, f\}$.

$$S \rightarrow daT \mid Rf$$
$$T \rightarrow aS \mid baT \mid \epsilon$$
$$R \rightarrow caTR \mid \epsilon$$

The following is a partially-filled LL(1) parsing table.

	a	b	c	d	f	$\$$
S			①	$S \rightarrow daT$	②	
T	$T \rightarrow aS$	$T \rightarrow baT$	③		$T \rightarrow \epsilon$	④
R			$R \rightarrow caTR$		$R \rightarrow \epsilon$	

Which one of the following choices represents the correct combination for the numbered cells in the parsing table (“blank” denotes that the corresponding cell is empty)?

(A) ① $S \rightarrow Rf$ ② $S \rightarrow Rf$ ③ $T \rightarrow \epsilon$ ④ $T \rightarrow \epsilon$

(B) ① blank ② $S \rightarrow Rf$ ③ $T \rightarrow \epsilon$ ④ $T \rightarrow \epsilon$

(C) ① $S \rightarrow Rf$ ② blank ③ blank ④ $T \rightarrow \epsilon$

(D) ① blank ② $S \rightarrow Rf$ ③ blank ④ blank



Computer Science and Information Technology (CS, Set-1)

Q.32	Let $r_i(z)$ and $w_i(z)$ denote read and write operations respectively on a data item z by a transaction T_i. Consider the following two schedules. $S_1 : r_1(x) \ r_1(y) \ r_2(x) \ r_2(y) \ w_2(y) \ w_1(x)$ $S_2 : r_1(x) \ r_2(x) \ r_2(y) \ w_2(y) \ r_1(y) \ w_1(x)$ Which one of the following options is correct?
(A)	S_1 is conflict serializable, and S_2 is not conflict serializable.
(B)	S_1 is not conflict serializable, and S_2 is conflict serializable.
(C)	Both S_1 and S_2 are conflict serializable.
(D)	Neither S_1 nor S_2 is conflict serializable.

Q.33	Consider the relation $R(P, Q, S, T, X, Y, Z, W)$ with the following functional dependencies. $PQ \rightarrow X; \ P \rightarrow YX; \ Q \rightarrow Y; \ Y \rightarrow ZW$ Consider the decomposition of the relation R into the constituent relations according to the following two decomposition schemes. $D_1 : \quad R = [(P, Q, S, T); (P, T, X); (Q, Y); (Y, Z, W)]$ $D_2 : \quad R = [(P, Q, S); (T, X); (Q, Y); (Y, Z, W)]$ Which one of the following options is correct?
(A)	D_1 is a lossless decomposition, but D_2 is a lossy decomposition.
(B)	D_1 is a lossy decomposition, but D_2 is a lossless decomposition.
(C)	Both D_1 and D_2 are lossless decompositions.
(D)	Both D_1 and D_2 are lossy decompositions.



Computer Science and Information Technology (CS, Set-1)

Q.34	Let G be a group of order 6, and H be a subgroup of G such that $1 < H < 6$. Which one of the following options is correct?
(A)	Both G and H are always cyclic.
(B)	G may not be cyclic, but H is always cyclic.
(C)	G is always cyclic, but H may not be cyclic.
(D)	Both G and H may not be cyclic.

Q.35	Consider the two statements. S_1 : There exist random variables X and Y such that $\left(\mathbb{E}[(X - \mathbb{E}(X))(Y - \mathbb{E}(Y))] \right)^2 > \text{Var}[X] \text{Var}[Y]$ S_2 : For all random variables X and Y, $\text{Cov}[X, Y] = \mathbb{E}[X - \mathbb{E}[X] Y - \mathbb{E}[Y]]$ Which one of the following choices is correct?
(A)	Both S_1 and S_2 are true.
(B)	S_1 is true, but S_2 is false.
(C)	S_1 is false, but S_2 is true.
(D)	Both S_1 and S_2 are false.



Computer Science and Information Technology (CS, Set-1)

Q.36	Let $G = (V, E)$ be an undirected unweighted connected graph. The <i>diameter</i> of G is defined as: $\text{diam}(G) = \max_{u,v \in V} \{\text{the length of shortest path between } u \text{ and } v\}$ Let M be the adjacency matrix of G. Define graph G_2 on the same set of vertices with adjacency matrix N, where $N_{ij} = \begin{cases} 1 & \text{if } M_{ij} > 0 \text{ or } P_{ij} > 0, \text{ where } P = M^2 \\ 0 & \text{otherwise} \end{cases}$ Which one of the following statements is true?
(A)	$\text{diam}(G_2) \leq \lceil \text{diam}(G)/2 \rceil$
(B)	$\lceil \text{diam}(G)/2 \rceil < \text{diam}(G_2) < \text{diam}(G)$
(C)	$\text{diam}(G_2) = \text{diam}(G)$
(D)	$\text{diam}(G) < \text{diam}(G_2) \leq 2 \text{ diam}(G)$



Computer Science and Information Technology (CS, Set-1)

Q.37	Consider the following ANSI C program. <pre>#include <stdio.h> int main() { int i, j, count; count = 0; i = 0; for (j = -3; j <= 3; j++) { if ((j >= 0) && (i++)) count = count + j; } count = count + i; printf("%d", count); return 0; }</pre> Which one of the following options is correct?
(A)	The program will not compile successfully.
(B)	The program will compile successfully and output 10 when executed.
(C)	The program will compile successfully and output 8 when executed.
(D)	The program will compile successfully and output 13 when executed.



Computer Science and Information Technology (CS, Set-1)

Q.38	Consider the following language. $L = \{w \in \{0,1\}^* \mid w \text{ ends with the substring } 011\}$ Which one of the following deterministic finite automata accepts L?
(A)	<pre> graph LR start((start)) -- 1 --> 0((0)) start -- 0 --> 1((1)) 0 -- 0 --> 1 0 -- 1 --> 2((2)) 1 -- 0 --> 2 1 -- 1 --> 3(((3))) 2 -- 0 --> 2 2 -- 1 --> 3 3 -- 0 --> 3 3 -- 1 --> 0 </pre>
(B)	<pre> graph LR start((start)) -- 1 --> 0((0)) start -- 0 --> 1((1)) 0 -- 0 --> 1 0 -- 1 --> 2((2)) 1 -- 0 --> 2 1 -- 1 --> 3(((3))) 2 -- 0 --> 2 2 -- 1 --> 3 3 -- "0,1" --> 3 </pre>
(C)	<pre> graph LR start((start)) -- 1 --> 0((0)) start -- 0 --> 1((1)) 0 -- 0 --> 1 0 -- 1 --> 2((2)) 1 -- 0 --> 2 1 -- 1 --> 3(((3))) 2 -- 0 --> 2 2 -- 1 --> 3 3 -- 0 --> 0 3 -- 1 --> 2 </pre>
(D)	<pre> graph LR start((start)) -- 1 --> 0((0)) start -- 0 --> 1((1)) 0 -- 0 --> 1 0 -- 1 --> 2((2)) 1 -- 0 --> 2 1 -- 1 --> 3(((3))) 2 -- 0 --> 2 2 -- 1 --> 3 3 -- 0 --> 1 3 -- 1 --> 0 </pre>



Computer Science and Information Technology (CS, Set-1)

Q.39	For a Turing machine M, $\langle M \rangle$ denotes an encoding of M. Consider the following two languages. $L_1 = \{ \langle M \rangle \mid M \text{ takes more than 2021 steps on all inputs} \}$ $L_2 = \{ \langle M \rangle \mid M \text{ takes more than 2021 steps on some input} \}$ Which one of the following options is correct?
(A)	Both L_1 and L_2 are decidable.
(B)	L_1 is decidable and L_2 is undecidable.
(C)	L_1 is undecidable and L_2 is decidable.
(D)	Both L_1 and L_2 are undecidable.





Computer Science and Information Technology (CS, Set-1)

Q.40 – Q.47 Multiple Select Question (MSQ), carry TWO mark each (no negative marks).

Q.40	Define R_n to be the maximum amount earned by cutting a rod of length n meters into one or more pieces of integer length and selling them. For $i > 0$, let $p[i]$ denote the selling price of a rod whose length is i meters. Consider the array of prices: $p[1] = 1, p[2] = 5, p[3] = 8, p[4] = 9, p[5] = 10, p[6] = 17, p[7] = 18$ Which of the following statements is/are correct about R_7?
(A)	$R_7 = 18$
(B)	$R_7 = 19$
(C)	R_7 is achieved by three different solutions.
(D)	R_7 cannot be achieved by a solution consisting of three pieces.

Q.41	An <i>articulation point</i> in a connected graph is a vertex such that removing the vertex and its incident edges disconnects the graph into two or more connected components. Let T be a DFS tree obtained by doing DFS in a connected undirected graph G. Which of the following options is/are correct?
(A)	Root of T can never be an articulation point in G .
(B)	Root of T is an articulation point in G if and only if it has 2 or more children.
(C)	A leaf of T can be an articulation point in G .
(D)	If u is an articulation point in G such that x is an ancestor of u in T and y is a descendent of u in T , then all paths from x to y in G must pass through u .



Computer Science and Information Technology (CS, Set-1)

Q.42	Consider the following Boolean expression. $F = (X + Y + Z)(\overline{X} + Y)(\overline{Y} + Z)$ Which of the following Boolean expressions is/are equivalent to $\overline{F}$ (complement of F)?
(A)	$(\overline{X} + \overline{Y} + \overline{Z})(X + \overline{Y})(Y + \overline{Z})$
(B)	$X\overline{Y} + \overline{Z}$
(C)	$(X + \overline{Z})(\overline{Y} + \overline{Z})$
(D)	$X\overline{Y} + Y\overline{Z} + \overline{X}\overline{Y}\overline{Z}$

Q.43	A relation R is said to be <i>circular</i> if aRb and bRc together imply cRa. Which of the following options is/are correct?
(A)	If a relation S is reflexive and symmetric, then S is an equivalence relation.
(B)	If a relation S is circular and symmetric, then S is an equivalence relation.
(C)	If a relation S is reflexive and circular, then S is an equivalence relation.
(D)	If a relation S is transitive and circular, then S is an equivalence relation.



Computer Science and Information Technology (CS, Set-1)

Q.44	A TCP server application is programmed to listen on port number P on host S. A TCP client is connected to the TCP server over the network. Consider that while the TCP connection was active, the server machine S crashed and rebooted. Assume that the client does not use the TCP keepalive timer. Which of the following behaviors is/are possible?
(A)	If the client was waiting to receive a packet, it may wait indefinitely.
(B)	The TCP server application on S can listen on P after reboot.
(C)	If the client sends a packet after the server reboot, it will receive a RST segment.
(D)	If the client sends a packet after the server reboot, it will receive a FIN segment.

Q.45	Consider two hosts P and Q connected through a router R. The maximum transfer unit (MTU) value of the link between P and R is 1500 bytes, and between R and Q is 820 bytes. A TCP segment of size 1400 bytes was transferred from P to Q through R, with IP identification value as 0x1234. Assume that the IP header size is 20 bytes. Further, the packet is allowed to be fragmented, i.e., Don't Fragment (DF) flag in the IP header is <i>not</i> set by P. Which of the following statements is/are correct?
(A)	Two fragments are created at R and the IP datagram size carrying the second fragment is 620 bytes.
(B)	If the second fragment is lost, R will resend the fragment with the IP identification value 0x1234.
(C)	If the second fragment is lost, P is required to resend the whole TCP segment.
(D)	TCP destination port can be determined by analysing <i>only</i> the second fragment.



Computer Science and Information Technology (CS, Set-1)

Q.46	Consider the following pseudocode, where S is a semaphore initialized to 5 in line#2 and $counter$ is a shared variable initialized to 0 in line#1. Assume that the increment operation in line#7 is <i>not</i> atomic. <pre> 1. int counter = 0; 2. Semaphore S = init(5); 3. void parop(void) 4. { 5. wait(S); 6. wait(S); 7. counter++; 8. signal(S); 9. signal(S); 10.} </pre> If five threads execute the function <code>parop</code> concurrently, which of the following program behavior(s) is/are possible?
(A)	The value of <code>counter</code> is 5 after all the threads successfully complete the execution of <code>parop</code> .
(B)	The value of <code>counter</code> is 1 after all the threads successfully complete the execution of <code>parop</code> .
(C)	The value of <code>counter</code> is 0 after all the threads successfully complete the execution of <code>parop</code> .
(D)	There is a deadlock involving all the threads.



Computer Science and Information Technology (CS, Set-1)

Q.47	Consider a <i>dynamic</i> hashing approach for 4-bit integer keys: 1. There is a main hash table of size 4. 2. The 2 least significant bits of a key is used to index into the main hash table. 3. Initially, the main hash table entries are empty. 4. Thereafter, when more keys are hashed into it, to resolve collisions, the set of all keys corresponding to a main hash table entry is organized as a binary tree that grows on demand. 5. First, the 3rd least significant bit is used to divide the keys into left and right subtrees. 6. To resolve more collisions, each node of the binary tree is further sub-divided into left and right subtrees based on the 4th least significant bit. 7. A split is done only if it is needed, i.e., only when there is a collision. Consider the following state of the hash table. <pre> graph TD subgraph HashTable [Hash Table] direction LR S00[00] S01[01] S10[10] S11[11] end S00 --> E[empty] S01 --> N01(()) S10 --> N10(()) S11 --> N11(()) N01 -- 0 --> C01_0(()) N01 -- 1 --> C01_1(()) N10 -- 0 --> C10_0(()) N10 -- 1 --> C10_1(()) N11 --> C11_0(()) </pre> Which of the following sequences of key insertions can cause the above state of the hash table (assume the keys are in decimal notation)?
(A)	5, 9, 4, 13, 10, 7
(B)	9, 5, 10, 6, 7, 1
(C)	10, 9, 6, 7, 5, 13
(D)	9, 5, 13, 6, 10, 14



Computer Science and Information Technology (CS, Set-1)

Q.48 – Q.55 Numerical Answer Type (NAT), carry TWO mark each (no negative marks).

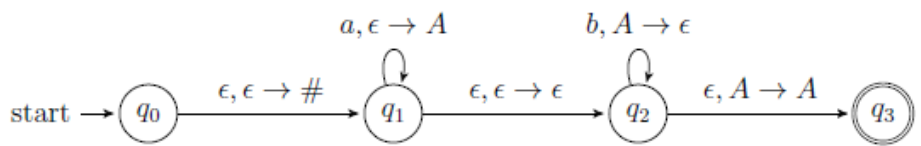
Q.48	Consider the following ANSI C function: <pre>int SimpleFunction(int Y[], int n, int x) { int total = Y[0], loopIndex; for (loopIndex = 1; loopIndex <= n - 1; loopIndex++) total = x * total + Y[loopIndex]; return total; }</pre> Let Z be an array of 10 elements with $Z[i]=1$, for all i such that $0 \leq i \leq 9$. The value returned by <code>SimpleFunction(Z, 10, 2)</code> is _____.
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Q.49	Consider the sliding window flow-control protocol operating between a sender and a receiver over a full-duplex error-free link. Assume the following: • The time taken for processing the data frame by the receiver is negligible. • The time taken for processing the acknowledgement frame by the sender is negligible. • The sender has infinite number of frames available for transmission. • The size of the data frame is 2,000 bits and the size of the acknowledgement frame is 10 bits. • The link data rate in each direction is 1 Mbps ($= 10^6$ bits per second). • One way propagation delay of the link is 100 milliseconds. The minimum value of the sender's window size in terms of the number of frames, (rounded to the nearest integer) needed to achieve a link utilization of 50% is _____.
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Q.50	Consider the following C code segment: <pre>a = b + c; e = a + 1; d = b + c; f = d + 1; g = e + f;</pre> In a compiler, this code segment is represented internally as a directed acyclic graph (DAG). The number of nodes in the DAG is _____.
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Computer Science and Information Technology (CS, Set-1)

Q.51	In a pushdown automaton $P = (Q, \Sigma, \Gamma, \delta, q_0, F)$, a transition of the form, $\begin{array}{c} (p) \xrightarrow{a, X \rightarrow Y} (q) \end{array}$ where $p, q \in Q$, $a \in \Sigma \cup \{\epsilon\}$, and $X, Y \in \Gamma \cup \{\epsilon\}$, represents $(q, Y) \in \delta(p, a, X).$ Consider the following pushdown automaton over the input alphabet $\Sigma = \{a, b\}$ and stack alphabet $\Gamma = \{\#, A\}$.  The number of strings of length 100 accepted by the above pushdown automaton is _____.
Q.52	Consider the following matrix. $\begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}$ The largest eigenvalue of the above matrix is _____.
Q.53	A five-stage pipeline has stage delays of 150, 120, 150, 160 and 140 nanoseconds. The registers that are used between the pipeline stages have a delay of 5 nanoseconds each. The total time to execute 100 independent instructions on this pipeline, assuming there are no pipeline stalls, is _____ nanoseconds.

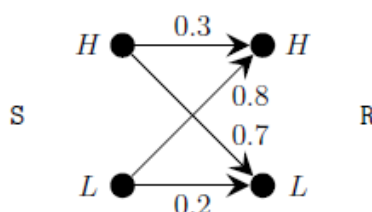


Computer Science and Information Technology (CS, Set-1)

Q.54

A sender (S) transmits a signal, which can be one of the two kinds: H and L with probabilities 0.1 and 0.9 respectively, to a receiver (R).

In the graph below, the weight of edge (u, v) is the probability of receiving v when u is transmitted, where $u, v \in \{H, L\}$. For example, the probability that the received signal is L given the transmitted signal was H , is 0.7.



If the received signal is H , the probability that the transmitted signal was H (rounded to 2 decimal places) is _____.

Q.55

Consider the following instruction sequence where registers R1, R2 and R3 are general purpose and $\text{MEMORY}[X]$ denotes the content at the memory location X .

Instruction	Semantics	Instruction Size (bytes)
MOV R1, (5000)	$R1 \leftarrow \text{MEMORY}[5000]$	4
MOV R2, (R3)	$R2 \leftarrow \text{MEMORY}[R3]$	4
ADD R2, R1	$R2 \leftarrow R1 + R2$	2
MOV (R3), R2	$\text{MEMORY}[R3] \leftarrow R2$	4
INC R3	$R3 \leftarrow R3 + 1$	2
DEC R1	$R1 \leftarrow R1 - 1$	2
BNZ 1004	Branch if not zero to the given absolute address	2
HALT	Stop	1

Assume that the content of the memory location 5000 is 10, and the content of the register R3 is 3000. The content of each of the memory locations from 3000 to 3010 is 50. The instruction sequence starts from the memory location 1000. All the numbers are in decimal format. Assume that the memory is byte addressable.

After the execution of the program, the content of memory location 3010 is _____.

END OF QUESTION PAPER

GATE 2021

Set 2

65 questions | 43 pages

Paper 9 of 16

Computer Science and Information Technology (CS, Set-2)

General Aptitude (GA)

Q.1 – Q.5 Multiple Choice Question (MCQ), carry ONE mark each (for each wrong answer: – 1/3).

Q.1	Gauri said that she can play the keyboard _____ her sister.
(A)	as well as
(B)	as better as
(C)	as nicest as
(D)	as worse as



Computer Science and Information Technology (CS, Set-2)

Q.2	A transparent square sheet shown above is folded along the dotted line. The folded sheet will look like _____.
(A)	
(B)	
(C)	
(D)	



Computer Science and Information Technology (CS, Set-2)

Q.3	If θ is the angle, in degrees, between the longest diagonal of the cube and any one of the edges of the cube, then, $\cos \theta =$
(A)	$\frac{1}{2}$
(B)	$\frac{1}{\sqrt{3}}$
(C)	$\frac{1}{\sqrt{2}}$
(D)	$\frac{\sqrt{3}}{2}$

Q.4	If $\left(x - \frac{1}{2}\right)^2 - \left(x - \frac{3}{2}\right)^2 = x + 2$, then the value of x is:
(A)	2
(B)	4
(C)	6
(D)	8



Computer Science and Information Technology (CS, Set-2)

Q.5	Pen : Write :: Knife : _____ Which one of the following options maintains a similar logical relation in the above?
(A)	Vegetables
(B)	Sharp
(C)	Cut
(D)	Blunt



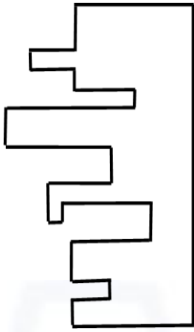
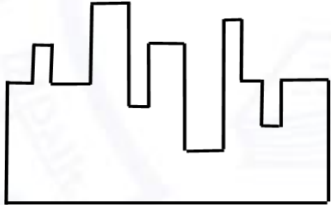
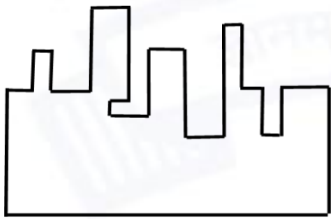
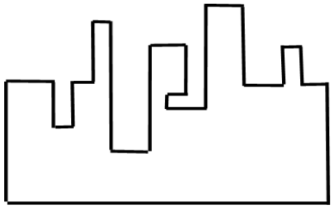
Computer Science and Information Technology (CS, Set-2)

Q. 6 – Q. 10 Multiple Choice Question (MCQ), carry TWO marks each (for each wrong answer: – 2/3).

Q.6	Listening to music during exercise improves exercise performance and reduces discomfort. Scientists researched whether listening to music while studying can help students learn better and the results were inconclusive. Students who needed external stimulation for studying fared worse while students who did not need any external stimulation benefited from music. Which one of the following statements is the CORRECT inference of the above passage?
(A)	Listening to music has no effect on learning and a positive effect on physical exercise.
(B)	Listening to music has a clear positive effect both on physical exercise and on learning.
(C)	Listening to music has a clear positive effect on physical exercise. Music has a positive effect on learning only in some students.
(D)	Listening to music has a clear positive effect on learning in all students. Music has a positive effect only in some students who exercise.



Computer Science and Information Technology (CS, Set-2)

Q.7	 A jigsaw puzzle has 2 pieces. One of the pieces is shown above. Which one of the given options for the missing piece when assembled will form a rectangle? The piece can be moved, rotated or flipped to assemble with the above piece.
(A)	
(B)	
(C)	
(D)	

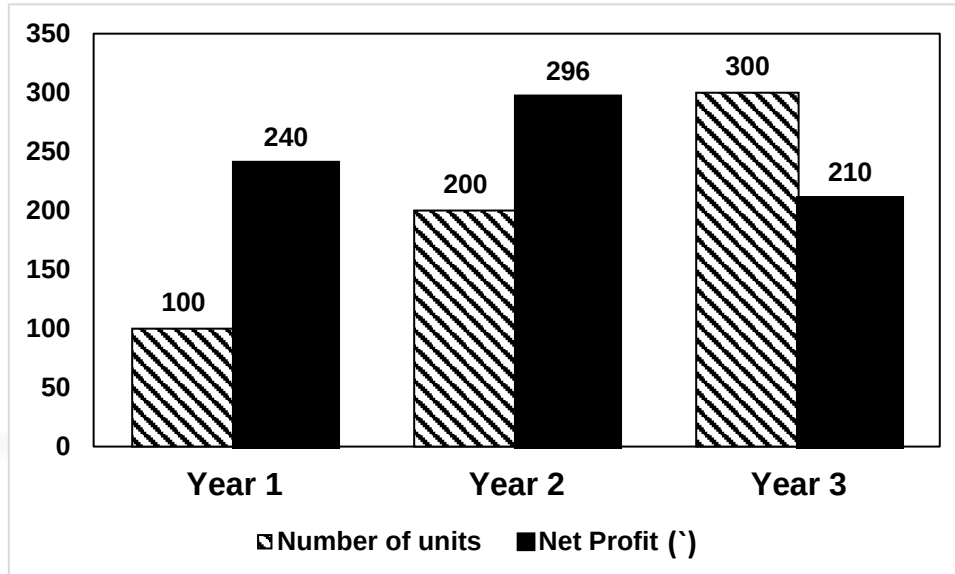


Computer Science and Information Technology (CS, Set-2)

Q.8	The number of students in three classes is in the ratio 3:13:6. If 18 students are added to each class, the ratio changes to 15:35:21. The total number of students in all the three classes in the beginning was:
(A)	22
(B)	66
(C)	88
(D)	110



Q.9



The number of units of a product sold in three different years and the respective net profits are presented in the figure above. The cost/unit in Year 3 was ₹1, which was half the cost/unit in Year 2. The cost/unit in Year 3 was one-third of the cost/unit in Year 1. Taxes were paid on the selling price at 10%, 13% and 15% respectively for the three years. Net profit is calculated as the difference between the selling price and the sum of cost and taxes paid in that year.

The ratio of the selling price in Year 2 to the selling price in Year 3 is _____.

A) 4:3

(B) 1:1

(C) 3:4

(D) 1:2

Computer Science and Information Technology (CS, Set-2)

Q.10	Six students P, Q, R, S, T and U, with distinct heights, compare their heights and make the following observations. Observation I: S is taller than R. Observation II: Q is the shortest of all. Observation III: U is taller than only one student. Observation IV: T is taller than S but is not the tallest. The number of students that are taller than R is the same as the number of students shorter than ____.
(A)	T
(B)	R
(C)	S
(D)	P

Computer Science and Information Technology (CS, Set-2)

Computer Science and Information Technology (CS, Set-2)

Q.1 – Q.10 Multiple Choice Question (MCQ), carry ONE mark each (for each wrong answer: – 1/3).

Q.1	Let G be a connected undirected weighted graph. Consider the following two statements. S_1 : There exists a minimum weight edge in G which is present in every minimum spanning tree of G. S_2 : If every edge in G has distinct weight, then G has a unique minimum spanning tree. Which one of the following options is correct?
(A)	Both S_1 and S_2 are true.
(B)	S_1 is true and S_2 is false.
(C)	S_1 is false and S_2 is true.
(D)	Both S_1 and S_2 are false.

Q.2	Let H be a binary min-heap consisting of n elements implemented as an array. What is the worst case time complexity of an optimal algorithm to find the maximum element in H?
(A)	$\Theta(1)$
(B)	$\Theta(\log n)$
(C)	$\Theta(n)$
(D)	$\Theta(n \log n)$



Computer Science and Information Technology (CS, Set-2)

Q.3	Consider the following ANSI C program: <pre>int main() { Integer x; return 0; }</pre> Which one of the following phases in a seven-phase C compiler will throw an error?
(A)	Lexical analyzer
(B)	Syntax analyzer
(C)	Semantic analyzer
(D)	Machine dependent optimizer

Q.4	The format of the single-precision floating-point representation of a real number as per the IEEE 754 standard is as follows: <table><tr><td>sign</td><td>exponent</td><td>mantissa</td></tr></table> Which one of the following choices is correct with respect to the <i>smallest</i> normalized positive number represented using the standard?	sign	exponent	mantissa
sign	exponent	mantissa		
(A)	exponent = 00000000 and mantissa = 000000000000000000000000			
(B)	exponent = 00000000 and mantissa = 000000000000000000000001			
(C)	exponent = 00000001 and mantissa = 000000000000000000000000			
(D)	exponent = 00000001 and mantissa = 000000000000000000000001			



Computer Science and Information Technology (CS, Set-2)

Q.5	Which one of the following circuits implements the Boolean function given below? $f(x, y, z) = m_0 + m_1 + m_3 + m_4 + m_5 + m_6$, where m_i is the i^{th} minterm.
(A)	
(B)	
(C)	
(D)	



Computer Science and Information Technology (CS, Set-2)

Q.6	Consider the following statements S1 and S2 about the relational data model: S1: A relation scheme can have at most one foreign key. S2: A foreign key in a relation scheme R cannot be used to refer to tuples of R. Which one of the following choices is correct?
(A)	Both S1 and S2 are true.
(B)	S1 is true and S2 is false.
(C)	S1 is false and S2 is true.
(D)	Both S1 and S2 are false.

Q.7	Consider the three-way handshake mechanism followed during TCP connection establishment between hosts P and Q. Let X and Y be two random 32-bit starting sequence numbers chosen by P and Q respectively. Suppose P sends a TCP connection request message to Q with a TCP segment having SYN bit = 1, SEQ number = X, and ACK bit = 0. Suppose Q accepts the connection request. Which one of the following choices represents the information present in the TCP segment header that is sent by Q to P?
(A)	SYN bit = 1, SEQ number = X+1, ACK bit = 0, ACK number = Y, FIN bit = 0
(B)	SYN bit = 0, SEQ number = X+1, ACK bit = 0, ACK number = Y, FIN bit = 1
(C)	SYN bit = 1, SEQ number = Y, ACK bit = 1, ACK number = X+1, FIN bit = 0
(D)	SYN bit = 1, SEQ number = Y, ACK bit = 1, ACK number = X, FIN bit = 0



Computer Science and Information Technology (CS, Set-2)

Q.8	What is the worst-case number of arithmetic operations performed by recursive binary search on a sorted array of size n ?
(A)	$\Theta(\sqrt{n})$
(B)	$\Theta(\log_2(n))$
(C)	$\Theta(n^2)$
(D)	$\Theta(n)$

Q.9	Let $L \subseteq \{0, 1\}^*$ be an arbitrary regular language accepted by a minimal DFA with k states. Which one of the following languages must necessarily be accepted by a minimal DFA with k states?
(A)	$L - \{01\}$
(B)	$L \cup \{01\}$
(C)	$\{0, 1\}^* - L$
(D)	$L \cdot L$



Computer Science and Information Technology (CS, Set-2)

Q.10	Consider the following ANSI C program. <pre>#include <stdio.h> int main(){ int arr[4][5]; int i, j; for (i=0; i<4; i++){ for (j=0; j<5; j++){ arr[i][j] = 10*i + j; } } printf("%d", *(arr[1] + 9)); return 0; }</pre> What is the output of the above program?
(A)	14
(B)	20
(C)	24
(D)	30

Computer Science and Information Technology (CS, Set-2)

Q.11 – Q.15 Multiple Select Question (MSQ), carry ONE mark each (no negative marks).

Q.11	Consider the following sets, where $n \geq 2$: S_1: Set of all $n \times n$ matrices with entries from the set $\{a, b, c\}$ S_2: Set of all functions from the set $\{0, 1, 2, \dots, n^2 - 1\}$ to the set $\{0, 1, 2\}$ Which of the following choice(s) is/are correct?
(A)	There does not exist a bijection from S_1 to S_2 .
(B)	There exists a surjection from S_1 to S_2 .
(C)	There exists a bijection from S_1 to S_2 .
(D)	There does not exist an injection from S_1 to S_2 .

Q.12	Let L_1 be a regular language and L_2 be a context-free language. Which of the following languages is/are context-free?
(A)	$L_1 \cap \overline{L_2}$
(B)	$\overline{\overline{L_1} \cup \overline{L_2}}$
(C)	$L_1 \cup (L_2 \cup \overline{L_2})$
(D)	$(L_1 \cap L_2) \cup (\overline{L_1} \cap L_2)$



Computer Science and Information Technology (CS, Set-2)

Q.13	In the context of compilers, which of the following is/are NOT an intermediate representation of the source program?
(A)	Three address code
(B)	Abstract Syntax Tree (AST)
(C)	Control Flow Graph (CFG)
(D)	Symbol table

Q.14	Which of the following statement(s) is/are correct in the context of CPU scheduling?
(A)	Turnaround time includes waiting time.
(B)	The goal is to only maximize CPU utilization and minimize throughput.
(C)	Round-robin policy can be used even when the CPU time required by each of the processes is not known apriori.
(D)	Implementing preemptive scheduling needs hardware support.



Computer Science and Information Technology (CS, Set-2)

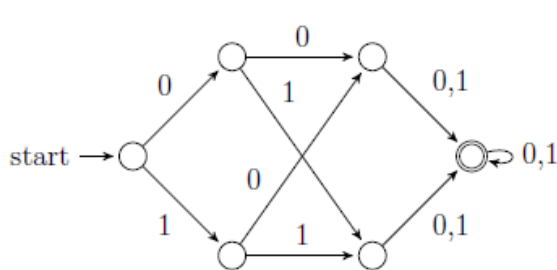
Q.15	Choose the correct choice(s) regarding the following propositional logic assertion S: $S : ((P \wedge Q) \rightarrow R) \rightarrow ((P \wedge Q) \rightarrow (Q \rightarrow R))$
(A)	S is neither a tautology nor a contradiction.
(B)	S is a tautology.
(C)	S is a contradiction.
(D)	The antecedent of S is logically equivalent to the consequent of S .



Computer Science and Information Technology (CS, Set-2)

Q.16 – Q.25 Numerical Answer Type (NAT), carry ONE mark each (no negative marks).

Q.16	Consider a complete binary tree with 7 nodes. Let A denote the set of first 3 elements obtained by performing Breadth-First Search (BFS) starting from the root. Let B denote the set of first 3 elements obtained by performing Depth-First Search (DFS) starting from the root. The value of $ A - B $ is _____.
------	---

Q.17	Consider the following deterministic finite automaton (DFA).  The number of strings of length 8 accepted by the above automaton is _____.
------	--

Q.18	If x and y are two decimal digits and $(0.1101)_2 = (0.8xy5)_{10}$, the decimal value of $x + y$ is _____.
------	---

Q.19	Consider a set-associative cache of size 2KB ($1\text{KB} = 2^{10}$ bytes) with cache block size of 64 bytes. Assume that the cache is byte-addressable and a 32-bit address is used for accessing the cache. If the width of the tag field is 22 bits, the associativity of the cache is _____.
------	---



Computer Science and Information Technology (CS, Set-2)

Q.20	Consider a computer system with DMA support. The DMA module is transferring one 8-bit character in one CPU cycle from a device to memory through <i>cycle stealing</i> at regular intervals. Consider a 2 MHz processor. If 0.5% processor cycles are used for DMA, the data transfer rate of the device is _____ bits per second.
------	--

Q.21	A data file consisting of 1,50,000 student-records is stored on a hard disk with block size of 4096 bytes. The data file is sorted on the primary key RollNo. The size of a record pointer for this disk is 7 bytes. Each student-record has a candidate key attribute called ANum of size 12 bytes. Suppose an index file with records consisting of two fields, ANum value and the record pointer to the corresponding student record, is built and stored on the same disk. Assume that the records of data file and index file are not split across disk blocks. The number of blocks in the index file is _____.
------	---

Q.22	For a given biased coin, the probability that the outcome of a toss is a head is 0.4. This coin is tossed 1,000 times. Let X denote the random variable whose value is the number of times that head appeared in these 1,000 tosses. The standard deviation of X (rounded to 2 decimal places) is _____.
------	--

Q.23	Consider the following ANSI C function: <pre>int SomeFunction(int x, int y) { if ((x == 1) (y == 1)) return 1; if (x == y) return x; if (x > y) return SomeFunction(x - y, y); if (y > x) return SomeFunction(x, y - x); }</pre> The value returned by SomeFunction(15, 255) is _____.
------	---



Computer Science and Information Technology (CS, Set-2)

Q.24	Suppose that P is a 4×5 matrix such that every solution of the equation $Px = 0$ is a scalar multiple of $[2 \ 5 \ 4 \ 3 \ 1]^T$. The rank of P is _____.
------	---

Q.25	Suppose that $f: \mathbb{R} \rightarrow \mathbb{R}$ is a continuous function on the interval $[-3, 3]$ and a differentiable function in the interval $(-3, 3)$ such that for every x in the interval, $f'(x) \leq 2$. If $f(-3) = 7$, then $f(3)$ is at most _____.
------	---



Computer Science and Information Technology (CS, Set-2)

Q.26 – Q.39 Multiple Choice Question (MCQ), carry TWO mark each (for each wrong answer: – 2/3).

Q.26	Consider the string <code>abbccddeee</code>. Each letter in the string must be assigned a binary code satisfying the following properties: 1. For any two letters, the code assigned to one letter must not be a prefix of the code assigned to the other letter. 2. For any two letters of the same frequency, the letter which occurs earlier in the dictionary order is assigned a code whose length is at most the length of the code assigned to the other letter. Among the set of all binary code assignments which satisfy the above two properties, what is the minimum length of the encoded string?
(A)	21
(B)	23
(C)	25
(D)	30



Computer Science and Information Technology (CS, Set-2)

Q.27	Assume a two-level inclusive cache hierarchy, L1 and L2, where L2 is the larger of the two. Consider the following statements. S_1 : Read misses in a write through L1 cache do not result in writebacks of dirty lines to the L2. S_2 : Write allocate policy <i>must</i> be used in conjunction with write through caches and no-write allocate policy is used with writeback caches. Which of the following statements is correct?
(A)	S_1 is true and S_2 is false
(B)	S_1 is false and S_2 is true
(C)	S_1 is true and S_2 is true
(D)	S_1 is false and S_2 is false



Computer Science and Information Technology (CS, Set-2)

Q.28	Suppose we want to design a synchronous circuit that processes a string of 0's and 1's. Given a string, it produces another string by replacing the first 1 in any subsequence of consecutive 1's by a 0. Consider the following example. Input sequence: 00100011000011100 Output sequence: 00000001000001100 A <i>Mealy Machine</i> is a state machine where both the next state and the output are functions of the present state and the current input. The above mentioned circuit can be designed as a two-state Mealy machine. The states in the Mealy machine can be represented using Boolean values 0 and 1. We denote the current state, the next state, the next incoming bit, and the output bit of the Mealy machine by the variables s, t, b and y respectively. Assume the initial state of the Mealy machine is 0. What are the Boolean expressions corresponding to t and y in terms of s and b?
(A)	$t = s + b$ $y = sb$
(B)	$t = b$ $y = sb$
(C)	$t = b$ $y = s\bar{b}$
(D)	$t = s + b$ $y = s\bar{b}$

Computer Science and Information Technology (CS, Set-2)

Q.29	In an examination, a student can choose the order in which two questions (QuesA and QuesB) must be attempted. - If the first question is answered wrong, the student gets zero marks. - If the first question is answered correctly and the second question is not answered correctly, the student gets the marks only for the first question. - If both the questions are answered correctly, the student gets the sum of the marks of the two questions. The following table shows the probability of correctly answering a question and the marks of the question respectively. <table><tr><th>question</th><th>probability of answering correctly</th><th>marks</th></tr><tr><td>QuesA</td><td>0.8</td><td>10</td></tr><tr><td>QuesB</td><td>0.5</td><td>20</td></tr></table> Assuming that the student always wants to maximize her expected marks in the examination, in which order should she attempt the questions and what is the expected marks for that order (assume that the questions are independent)?	question	probability of answering correctly	marks	QuesA	0.8	10	QuesB	0.5	20
question	probability of answering correctly	marks								
QuesA	0.8	10								
QuesB	0.5	20								
(A)	First QuesA and then QuesB. Expected marks 14.									
(B)	First QuesB and then QuesA. Expected marks 14.									
(C)	First QuesB and then QuesA. Expected marks 22.									
(D)	First QuesA and then QuesB. Expected marks 16.									



Computer Science and Information Technology (CS, Set-2)

Q.30	Consider the following ANSI C code segment: <pre> z = x + 3 + y->f1 + y->f2; for (i = 0; i < 200; i = i + 2){ if (z > i) { p = p + x + 3; q = q + y->f1; } else { p = p + y->f2; q = q + x + 3; } } </pre> Assume that the variable y points to a struct (allocated on the heap) containing two fields f1 and f2, and the local variables x, y, z, p, q, and i are allotted registers. Common sub-expression elimination (CSE) optimization is applied on the code. The number of addition and dereference operations (of the form y->f1 or y->f2) in the optimized code, respectively, are:
(A)	403 and 102
(B)	203 and 2
(C)	303 and 102
(D)	303 and 2



Computer Science and Information Technology (CS, Set-2)

Q.31	The relation scheme given below is used to store information about the employees of a company, where <code>empId</code> is the key and <code>deptId</code> indicates the department to which the employee is assigned. Each employee is assigned to exactly one department. <code>emp(empId, name, gender, salary, deptId)</code> Consider the following SQL query: <pre>select deptId, count(*) from emp where gender = "female" and salary > (select avg(salary) from emp) group by deptId;</pre> The above query gives, for each department in the company, the number of female employees whose salary is greater than the average salary of
(A)	employees in the department.
(B)	employees in the company.
(C)	female employees in the department.
(D)	female employees in the company.



Computer Science and Information Technology (CS, Set-2)

Q.32	Let S be the following schedule of operations of three transactions T_1, T_2 and T_3 in a relational database system: $R_2(Y), R_1(X), R_3(Z), R_1(Y), W_1(X), R_2(Z), W_2(Y), R_3(X), W_3(Z)$ Consider the statements P and Q below: P: S is conflict-serializable. Q: If T_3 commits before T_1 finishes, then S is recoverable. Which one of the following choices is correct?
(A)	Both P and Q are true.
(B)	P is true and Q is false.
(C)	P is false and Q is true.
(D)	Both P and Q are false.

Q.33	A bag has r red balls and b black balls. All balls are identical except for their colours. In a trial, a ball is randomly drawn from the bag, its colour is noted and the ball is placed back into the bag along with another ball of the same colour. Note that the number of balls in the bag will increase by one, after the trial. A sequence of four such trials is conducted. Which one of the following choices gives the probability of drawing a red ball in the fourth trial?
(A)	$\frac{r}{r+b}$
(B)	$\frac{r}{r+b+3}$
(C)	$\frac{r+3}{r+b+3}$
(D)	$\left(\frac{r}{r+b}\right) \left(\frac{r+1}{r+b+1}\right) \left(\frac{r+2}{r+b+2}\right) \left(\frac{r+3}{r+b+3}\right)$



Computer Science and Information Technology (CS, Set-2)

Q.34	Consider the cyclic redundancy check (CRC) based error detecting scheme having the generator polynomial $X^3 + X + 1$. Suppose the message $m_4m_3m_2m_1m_0 = 11000$ is to be transmitted. Check bits $c_2c_1c_0$ are appended at the end of the message by the transmitter using the above CRC scheme. The transmitted bit string is denoted by $m_4m_3m_2m_1m_0c_2c_1c_0$. The value of the checkbit sequence $c_2c_1c_0$ is
(A)	101
(B)	110
(C)	100
(D)	111



Computer Science and Information Technology (CS, Set-2)

Q.35	Consider the following ANSI C program: <pre>#include <stdio.h> #include <stdlib.h> struct Node{ int value; struct Node *next;}; int main(){ struct Node *boxE, *head, *boxN; int index = 0; boxE = head = (struct Node *) malloc(sizeof(struct Node)); head->value = index; for (index = 1; index <= 3; index++){ boxN = (struct Node *) malloc(sizeof(struct Node)); boxE->next = boxN; boxN->value = index; boxE = boxN; } for (index = 0; index <= 3; index++) { printf("Value at index %d is %d\n", index, head->value); head = head->next; printf("Value at index %d is %d\n", index+1, head->value); } }</pre> Which one of the statements below is correct about the program?
(A)	Upon execution, the program creates a linked-list of five nodes.
(B)	Upon execution, the program goes into an infinite loop.
(C)	It has a missing return which will be reported as an error by the compiler.
(D)	It dereferences an uninitialized pointer that may result in a run-time error.



Computer Science and Information Technology (CS, Set-2)

Q.36	Consider the following two statements about regular languages: S_1: Every infinite regular language contains an undecidable language as a subset. S_2: Every finite language is regular. Which one of the following choices is correct?
(A)	Only S_1 is true.
(B)	Only S_2 is true.
(C)	Both S_1 and S_2 are true.
(D)	Neither S_1 nor S_2 is true.

Q.37	For two n-dimensional real vectors P and Q, the operation $s(P, Q)$ is defined as follows: $s(P, Q) = \sum_{i=1}^n (P[i] \cdot Q[i])$ Let $\mathcal{L}$ be a set of 10-dimensional non-zero real vectors such that for every pair of distinct vectors $P, Q \in \mathcal{L}$, $s(P, Q) = 0$. What is the maximum cardinality possible for the set $\mathcal{L}$?
(A)	9
(B)	10
(C)	11
(D)	100



Computer Science and Information Technology (CS, Set-2)

Q.38	For a statement S in a program, in the context of liveness analysis, the following sets are defined: $USE(S)$: the set of variables used in S $IN(S)$: the set of variables that are live at the entry of S $OUT(S)$: the set of variables that are live at the exit of S Consider a basic block that consists of two statements, S_1 followed by S_2. Which one of the following statements is correct?
(A)	$OUT(S_1) = IN(S_2)$
(B)	$OUT(S_1) = IN(S_1) \cup USE(S_1)$
(C)	$OUT(S_1) = IN(S_2) \cup OUT(S_2)$
(D)	$OUT(S_1) = USE(S_1) \cup IN(S_2)$

Q.39	For constants $a \geq 1$ and $b > 1$, consider the following recurrence defined on the non-negative integers: $T(n) = a \cdot T\left(\frac{n}{b}\right) + f(n)$ Which one of the following options is correct about the recurrence $T(n)$?
(A)	If $f(n)$ is $n \log_2(n)$, then $T(n)$ is $\Theta(n \log_2(n))$.
(B)	If $f(n)$ is $\frac{n}{\log_2(n)}$, then $T(n)$ is $\Theta(\log_2(n))$.
(C)	If $f(n)$ is $O(n^{\log_b(a)-\epsilon})$ for some $\epsilon > 0$, then $T(n)$ is $\Theta(n^{\log_b(a)})$.
(D)	If $f(n)$ is $\Theta(n^{\log_b(a)})$, then $T(n)$ is $\Theta(n^{\log_b(a)})$.



Computer Science and Information Technology (CS, Set-2)

Q.40 – Q.47 Multiple Select Question (MSQ), carry TWO mark each (no negative marks).

Q.40	Suppose the following functional dependencies hold on a relation U with attributes P, Q, R, S, and T: $P \rightarrow QR$ $RS \rightarrow T$ Which of the following functional dependencies can be inferred from the above functional dependencies?
(A)	$PS \rightarrow T$
(B)	$R \rightarrow T$
(C)	$P \rightarrow R$
(D)	$PS \rightarrow Q$

Q.41	For a string w, we define w^R to be the reverse of w. For example, if $w = 01101$ then $w^R = 10110$. Which of the following languages is/are context-free?
(A)	$\{wxw^Rx^R \mid w, x \in \{0, 1\}^*\}$
(B)	$\{ww^Rxx^R \mid w, x \in \{0, 1\}^*\}$
(C)	$\{wxw^R \mid w, x \in \{0, 1\}^*\}$
(D)	$\{wx x^R w^R \mid w, x \in \{0, 1\}^*\}$



Computer Science and Information Technology (CS, Set-2)

Q.42	Consider the following multi-threaded code segment (in a mix of C and pseudo-code), invoked by two processes P1 and P2, and each of the processes spawns two threads T1 and T2: <pre> int x = 0; // global Lock L1; // global main() { create a thread to execute foo(); // Thread T1 create a thread to execute foo(); // Thread T2 wait for the two threads to finish execution; print (x);} foo() { int y = 0; Acquire L1; x = x + 1; y = y + 1; Release L1; print (y);} </pre> Which of the following statement(s) is/are correct?
(A)	Both P1 and P2 will print the value of x as 2.
(B)	At least one of P1 and P2 will print the value of x as 4.
(C)	At least one of the threads will print the value of y as 2.
(D)	Both T1 and T2, in both the processes, will print the value of y as 1.



Computer Science and Information Technology (CS, Set-2)

Q.43	Consider a computer system with multiple shared resource types, with one instance per resource type. Each instance can be owned by only one process at a time. Owning and freeing of resources are done by holding a global lock (L). The following scheme is used to own a resource instance : <pre> function OWNRESOURCE(Resource R) Acquire lock L // a global lock if R is available then Acquire R Release lock L else if R is owned by another process P then Terminate P, after releasing all resources owned by P Acquire R Restart P Release lock L end if end if end function </pre> Which of the following choice(s) about the above scheme is/are correct?
(A)	The scheme ensures that deadlocks will not occur.
(B)	The scheme may lead to live-lock.
(C)	The scheme may lead to starvation.
(D)	The scheme violates the mutual exclusion property.

Computer Science and Information Technology (CS, Set-2)

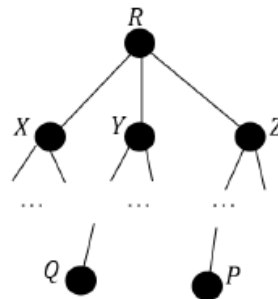
Q.44	If the numerical value of a 2-byte unsigned integer on a little endian computer is 255 more than that on a big endian computer, which of the following choices represent(s) the unsigned integer on a little endian computer?
(A)	0x6665
(B)	0x0001
(C)	0x4243
(D)	0x0100



Computer Science and Information Technology (CS, Set-2)

Q.45

Consider a computer network using the distance vector routing algorithm in its network layer. The partial topology of the network is as shown below.



The objective is to find the shortest-cost path from the router R to routers P and Q . Assume that R does not initially know the shortest routes to P and Q . Assume that R has three neighbouring routers denoted as X , Y , and Z . During one iteration, R measures its distance to its neighbours X , Y , and Z as 3, 2, and 5, respectively. Router R gets routing vectors from its neighbours that indicate that the distance to router P from routers X , Y , and Z are 7, 6, and 5, respectively. The routing vector also indicates that the distance to router Q from routers X , Y , and Z are 4, 6, and 8, respectively. Which of the following statement(s) is/are correct with respect to the new routing table of R , after updation during this iteration?

- (A) The distance from R to P will be stored as 10.
- (B) The distance from R to Q will be stored as 7.
- (C) The next hop router for a packet from R to P is Y .
- (D) The next hop router for a packet from R to Q is Z .



Computer Science and Information Technology (CS, Set-2)

Q.46	Consider the following directed graph: Which of the following is/are correct about the graph?
(A)	The graph does not have a topological order.
(B)	A depth-first traversal starting at vertex S classifies three directed edges as back edges.
(C)	The graph does not have a strongly connected component.
(D)	For each pair of vertices u and v , there is a directed path from u to v .



Computer Science and Information Technology (CS, Set-2)

Q.47	Which of the following regular expressions represent(s) the set of all binary numbers that are divisible by three? Assume that the string ϵ is divisible by three.
(A)	$(0 + 1(01^*0)^*1)^*$
(B)	$(0 + 11 + 10(1 + 00)^*01)^*$
(C)	$(0^*(1(01^*0)^*1)^*)^*$
(D)	$(0 + 11 + 11(1 + 00)^*00)^*$





Computer Science and Information Technology (CS, Set-2)

Q.48 – Q.55 Numerical Answer Type (NAT), carry TWO mark each (no negative marks).

Q.48 Consider a three-level page table to translate a 39-bit virtual address to a physical address as shown below.

<----- 39-bit virtual address ----->

Level 1	Level 2	Level 3	Page
offset	offset	offset	offset
9 bits	9 bits	9 bits	12 bits

The page size is 4KB ($1\text{KB} = 2^{10}$ bytes) and page table entry size at every level is 8 bytes. A process P is currently using 2GB ($1\text{GB} = 2^{30}$ bytes) virtual memory which is mapped to 2GB of physical memory. The minimum amount of memory required for the page table of P across all levels is _____ KB.

Q.49 Consider the following ANSI C program.

```
#include <stdio.h>

int foo(int x, int y, int q)
{
    if ((x <= 0) && (y <= 0))
        return q;
    if (x <= 0)
        return foo(x, y-q, q);
    if (y <= 0)
        return foo(x-q, y, q);
    return foo(x, y-q, q) + foo(x-q, y, q);
}

int main()
{
    int r = foo(15,15,10);
    printf("%d", r);
    return 0;
}
```

The output of the program upon execution is _____.



Computer Science and Information Technology (CS, Set-2)

Q.50	Let S be a set consisting of 10 elements. The number of tuples of the form (A, B) such that A and B are subsets of S , and $A \subseteq B$ is _____.
------	--

Q.51	Consider the following augmented grammar with $\{\#, @, <, >, a, b, c\}$ as the set of terminals. $\begin{aligned} S' &\rightarrow S \\ S &\rightarrow S \# c S \\ S &\rightarrow S S \\ S &\rightarrow S @ \\ S &\rightarrow < S > \\ S &\rightarrow a \\ S &\rightarrow b \\ S &\rightarrow c \end{aligned}$ Let $I_0 = \text{CLOSURE}(\{S' \rightarrow \bullet S\})$. The number of items in the set $\text{GOTO}(\text{GOTO}(I_0, <), <)$ is _____.
------	--

Q.52	Consider a Boolean function $f(w, x, y, z)$ such that $\begin{aligned} f(w, 0, 0, z) &= 1 \\ f(1, x, 1, z) &= x + z \\ f(w, 1, y, z) &= wz + y \end{aligned}$ The number of literals in the minimal sum-of-products expression of f is _____.
------	---



Computer Science and Information Technology (CS, Set-2)

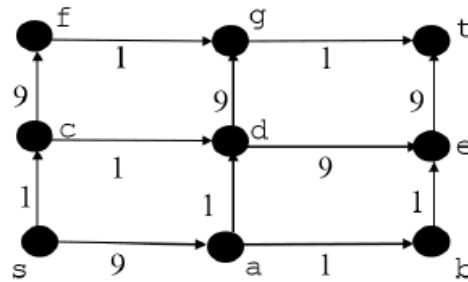
Q.53	Consider a pipelined processor with 5 stages, Instruction Fetch (IF), Instruction Decode (ID), Execute (EX), Memory Access (MEM), and Write Back (WB). Each stage of the pipeline, except the EX stage, takes one cycle. Assume that the ID stage merely decodes the instruction and the register read is performed in the EX stage. The EX stage takes one cycle for ADD instruction and two cycles for MUL instruction. Ignore pipeline register latencies. Consider the following sequence of 8 instructions: ADD, MUL, ADD, MUL, ADD, MUL, ADD, MUL Assume that every MUL instruction is data-dependent on the ADD instruction just before it and every ADD instruction (except the first ADD) is data-dependent on the MUL instruction just before it. The <i>Speedup</i> is defined as follows: $\text{Speedup} = \frac{\text{Execution time without operand forwarding}}{\text{Execution time with operand forwarding}}$ The <i>Speedup</i> achieved in executing the given instruction sequence on the pipelined processor (rounded to 2 decimal places) is _____.
Q.54	Consider a network using the pure ALOHA medium access control protocol, where each frame is of length 1,000 bits. The channel transmission rate is 1 Mbps (= 10^6 bits per second). The aggregate number of transmissions across all the nodes (including new frame transmissions and retransmitted frames due to collisions) is modelled as a Poisson process with a rate of 1,000 frames per second. Throughput is defined as the average number of frames successfully transmitted per second. The throughput of the network (rounded to the nearest integer) is _____.



Computer Science and Information Technology (CS, Set-2)

Q.55

In a directed acyclic graph with a source vertex s , the *quality-score* of a directed path is defined to be the product of the weights of the edges on the path. Further, for a vertex v other than s , the quality-score of v is defined to be the maximum among the quality-scores of all the paths from s to v . The quality-score of s is assumed to be 1.



The sum of the quality-scores of all the vertices in the graph shown above is _____.

END OF QUESTION PAPER

GATE 2020

Single session

65 questions | 17 pages

Paper 10 of 16

CS: Computer Sc. and Information Technology

GA - General Aptitude

Q1 - Q5 carry one mark each.

Q.No. 1 Raman is confident of speaking English_____six months as he has been practising regularly_____the last three weeks.

- (A) during, for
- (B) for, since
- (C) for, in
- (D) within, for

Q.No. 2 His knowledge of the subject was excellent but his classroom performance was_____.

- (A) extremely poor
- (B) good
- (C) desirable
- (D) praiseworthy

Q.No. 3 Select the word that fits the analogy:

Cook : Cook :: Fly : _____

- (A) Flyer
- (B) Flying
- (C) Flew
- (D) Fighter

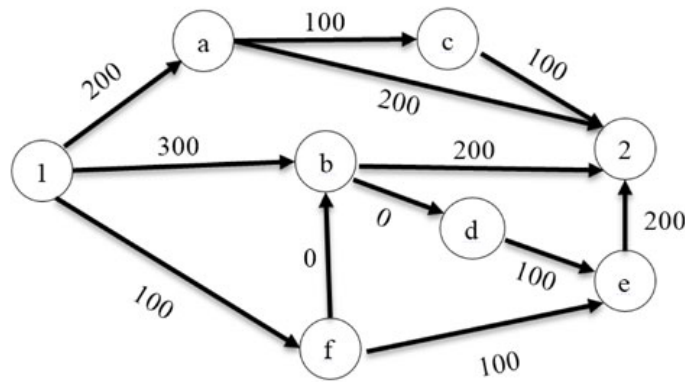
Q.No. 4 The dawn of the 21st century witnessed the melting glaciers oscillating between giving too much and too little to billions of people who depend on them for fresh water. The UN climate report estimates that without deep cuts to man-made emissions, at least 30% of the northern hemisphere's surface permafrost could melt by the end of the century. Given this situation of imminent global exodus of billions of people displaced by rising seas, nation-states need to rethink their carbon footprint for political concerns, if not for environmental ones.

Which one of the following statements can be inferred from the given passage?

- (A) Nation-states do not have environmental concerns.
- (B) Nation-states are responsible for providing fresh water to billions of people.
- (C) Billions of people are responsible for man-made emissions.
- (D) Billions of people are affected by melting glaciers.

Q.No. 5

There are multiple routes to reach from node 1 to node 2, as shown in the network.



The cost of travel on an edge between two nodes is given in rupees. Nodes 'a', 'b', 'c', 'd', 'e', and 'f' are toll booths. The toll price at toll booths marked 'a' and 'e' is Rs. 200, and is Rs. 100 for the other toll booths. Which is the cheapest route from node 1 to node 2?

- (A) 1-a-c-2
- (B) 1-f-b-2
- (C) 1-b-2
- (D) 1-f-e-2

Q6 - Q10 carry two marks each.

Q.No. 6 Goods and Services Tax (GST) is an indirect tax introduced in India in 2017 that is imposed on the supply of goods and services, and it subsumes all indirect taxes except few. It is a destination-based tax imposed on goods and services used, and it is not imposed at the point of origin from where goods come. GST also has a few components specific to state governments, central government and Union Territories (UTs).

Which one of the following statements can be inferred from the given passage?

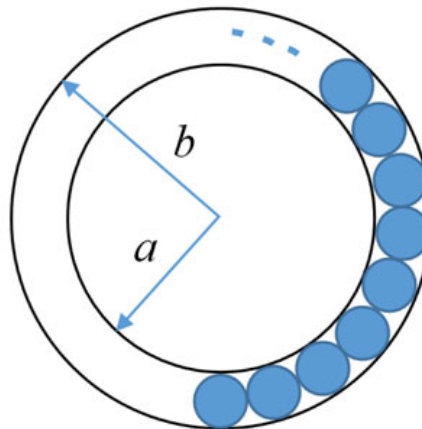
- (A) GST is imposed on the production of goods and services.
- (B) GST includes all indirect taxes.
- (C) GST does not have a component specific to UT.
- (D) GST is imposed at the point of usage of goods and services.

Q.No. 7 If $P = 3$, $R = 27$, $T = 243$, then $Q + S = \underline{\hspace{2cm}}$.

- (A) 40
- (B) 80
- (C) 90
- (D) 110

Q.No. 8

The figure below shows an annular ring with outer and inner radii as b and a , respectively. The annular space has been painted in the form of blue colour circles touching the outer and inner periphery of annular space. If maximum n number of circles can be painted, then the unpainted area available in annular space is_____.

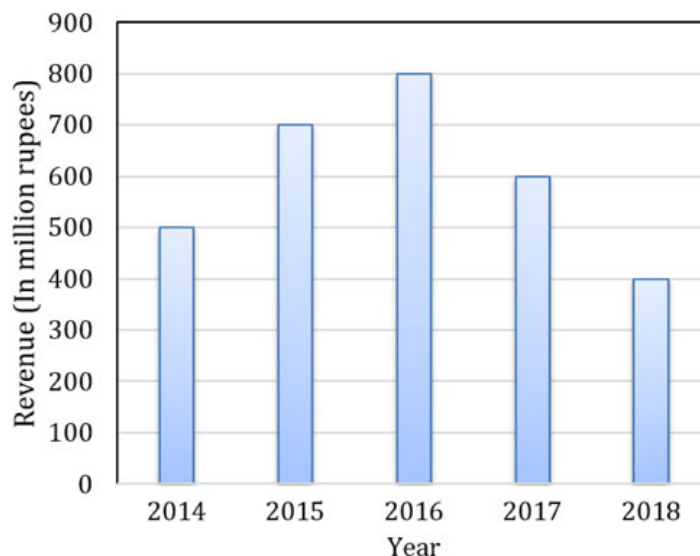


- (A) $\pi \left[(b^2 - a^2) - \frac{n}{4}(b - a)^2 \right]$
 (B) $\pi [(b^2 - a^2) - n(b - a)^2]$
 (C) $\pi \left[(b^2 - a^2) + \frac{n}{4}(b - a)^2 \right]$
 (D) $\pi [(b^2 - a^2) + n(b - a)^2]$

Q.No. 9 Two straight lines are drawn perpendicular to each other in X-Y plane. If α and β are the acute angles the straight lines make with the X-axis, then $\alpha + \beta$ is_____.

- (A) 60°
 (B) 90°
 (C) 120°
 (D) 180°

Q.No. 10 The total revenue of a company during 2014-2018 is shown in the bar graph. If the total expenditure of the company in each year is 500 million rupees, then the aggregate profit or loss (in percentage) on the total expenditure of the company during 2014-2018 is _____.



- (A) 16.67 % profit
 (B) 16.67 % loss
 (C) 20 % profit

- (D) 20 % loss

CS: Computer Sc. and Information Technology

Q1 - Q25 carry one mark each.

Q.No. 1 Consider the functions

- I. e^{-x}
- II. $x^2 - \sin x$
- III. $\sqrt{x^3 + 1}$

Which of the above functions is/are increasing everywhere in $[0,1]$?

- (A) III only
- (B) II only
- (C) II and III only
- (D) I and III only

Q.No. 2 For parameters a and b , both of which are $\omega(1)$, $T(n) = T(n^{1/a}) + 1$, and $T(b) = 1$.

Then $T(n)$ is

- (A) $\Theta(\log_a \log_b n)$
- (B) $\Theta(\log_{ab} n)$
- (C) $\Theta(\log_b \log_a n)$
- (D) $\Theta(\log_2 \log_2 n)$

Q.No. 3 Consider the following statements.

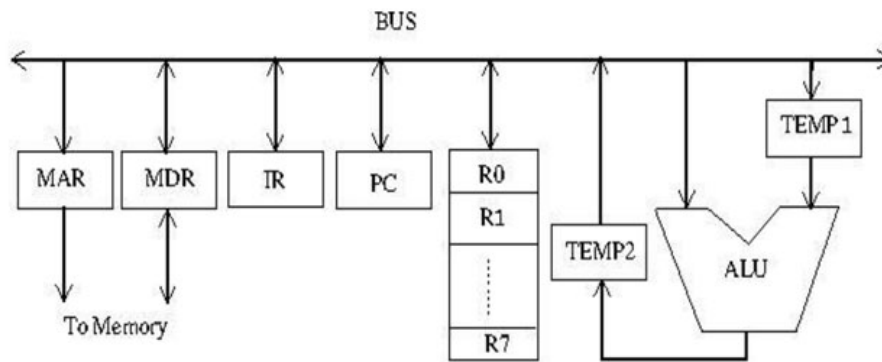
- I. Daisy chaining is used to assign priorities in attending interrupts.
- II. When a device raises a vectored interrupt, the CPU does polling to identify the source of interrupt.
- III. In polling, the CPU periodically checks the status bits to know if any device needs its attention.
- IV. During DMA, both the CPU and DMA controller can be bus masters at the same time.

Which of the above statements is/are TRUE?

- (A) I and II only
- (B) I and IV only
- (C) I and III only
- (D) III only

Q.No. 4

Consider the following data path diagram.



Consider an instruction: $R0 \leftarrow R1 + R2$. The following steps are used to execute it over the given data path. Assume that PC is incremented appropriately. The subscripts r and w indicate read and write operations, respectively.

1. $R2_r, TEMP1_r, ALU_{add}, TEMP2_w$
2. $R1_r, TEMP1_w$
3. PC_r, MAR_w, MEM_r
4. $TEMP2_r, R0_w$
5. MDR_r, IR_w

Which one of the following is the correct order of execution of the above steps?

- (A) 2, 1, 4, 5, 3
- (B) 1, 2, 4, 3, 5
- (C) 3, 5, 2, 1, 4
- (D) 3, 5, 1, 2, 4

Q.No. 5 The preorder traversal of a binary search tree is 15, 10, 12, 11, 20, 18, 16, 19.

Which one of the following is the postorder traversal of the tree?

- (A) 10, 11, 12, 15, 16, 18, 19, 20
- (B) 11, 12, 10, 16, 19, 18, 20, 15
- (C) 20, 19, 18, 16, 15, 12, 11, 10
- (D) 19, 16, 18, 20, 11, 12, 10, 15

Q.No. 6 What is the worst case time complexity of inserting n^2 elements into an AVL-tree with n elements initially?

- (A) $\theta(n^4)$
- (B) $\theta(n^2)$
- (C) $\theta(n^2 \log n)$
- (D) $\theta(n^3)$

Q.No. 7 Which one of the following regular expressions represents the set of all binary strings with an odd number of 1's?

- (A) $((0 + 1)^* 1 (0 + 1)^* 1)^* 10^*$
- (B) $(0^* 10^* 10^*)^* 0^* 1$

- (C) $10^*(0^*10^*10^*)^*$
(D) $(0^*10^*10^*)^*10^*$

Q.No. 8 Consider the following statements.

- I. If $L_1 \cup L_2$ is regular, then both L_1 and L_2 must be regular.
II. The class of regular languages is closed under infinite union.

Which of the above statements is/are TRUE?

- (A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

Q.No. 9 Consider the following statements.

- I. Symbol table is accessed only during lexical analysis and syntax analysis.
II. Compilers for programming languages that support recursion necessarily need heap storage for memory allocation in the run-time environment.
III. Errors violating the condition '*any variable must be declared before its use*' are detected during syntax analysis.

Which of the above statements is/are TRUE?

- (A) I only
(B) I and III only
(C) II only
(D) None of I, II, and III

Q.No. 10 Consider the language $L = \{a^n \mid n \geq 0\} \cup \{a^n b^n \mid n \geq 0\}$ and the following statements.

- I. L is deterministic context-free.
II. L is context-free but not deterministic context-free.
III. L is not $LL(k)$ for any k .

Which of the above statements is/are TRUE?

- (A) I only
(B) II only
(C) I and III only
(D) III only

Q.No. 11 Consider allocation of memory to a new process. Assume that none of the existing holes in the memory will exactly fit the process's memory requirement. Hence, a new hole of smaller size will be created if allocation is made in any of the existing holes. Which one of the following statements is TRUE?

- (A) The hole created by first fit is always larger than the hole created by next fit.

- (B) The hole created by worst fit is always larger than the hole created by first fit.
 (C) The hole created by best fit is never larger than the hole created by first fit.
 (D) The hole created by next fit is never larger than the hole created by best fit.

Q.No. 12 Consider the following statements about process state transitions for a system using preemptive scheduling.

- I. A running process can move to ready state.
 II. A ready process can move to running state.
 III. A blocked process can move to running state.
 IV. A blocked process can move to ready state.

Which of the above statements are TRUE?

- (A) I, II, and III only
 (B) II and III only
 (C) I, II, and IV only
 (D) I, II, III, and IV

Q.No. 13 Consider a relational database containing the following schemas.

Catalogue		
<u>sno</u>	<u>pno</u>	cost
S1	P1	150
S1	P2	50
S1	P3	100
S2	P4	200
S2	P5	250
S3	P1	250
S3	P2	150
S3	P5	300
S3	P4	250

Suppliers		
<u>sno</u>	sname	location
S1	M/s Royal furniture	Delhi
S2	M/s Balaji furniture	Bangalore
S3	M/s Premium furniture	Chennai

Parts		
<u>pno</u>	pname	part_spec
P1	Table	Wood
P2	Chair	Wood
P3	Table	Steel
P4	Almirah	Steel
P5	Almirah	Wood

The primary key of each table is indicated by underlining the constituent fields.

```
SELECT s.sno, s.sname
FROM Suppliers s, Catalogue c
WHERE s.sno = c.sno AND
      cost > (SELECT AVG (cost)
              FROM Catalogue
              WHERE pno = 'P4'
              GROUP BY pno);
```

The number of rows returned by the above SQL query is

- (A) 4
 (B) 5
 (C) 0

(D) 2

Q.No. 14 Which one of the following is used to represent the supporting many-one relationships of a weak entity set in an entity-relationship diagram?

- (A) Diamonds with double/bold border
- (B) Rectangles with double/bold border
- (C) Ovals with double/bold border
- (D) Ovals that contain underlined identifiers

Q.No. 15 Consider the following statements about the functionality of an IP based router.

- I. A router does not modify the IP packets during forwarding.
- II. It is not necessary for a router to implement any routing protocol.
- III. A router should reassemble IP fragments if the MTU of the outgoing link is larger than the size of the incoming IP packet.

Which of the above statements is/are TRUE?

- (A) I and II only
- (B) I only
- (C) II and III only
- (D) II only

Q.No. 16 What is the worst case time complexity of inserting n elements into an empty linked list, if the linked list needs to be maintained in sorted order?

- (A) $\Theta(n)$
- (B) $\Theta(n \log n)$
- (C) $\Theta(n^2)$
- (D) $\Theta(1)$

Q.No. 17 Let $\mathcal{R}$ be the set of all binary relations on the set $\{1,2,3\}$. Suppose a relation is chosen from $\mathcal{R}$ at random. The probability that the chosen relation is reflexive (round off to 3 decimal places) is _____.

Q.No. 18 Let G be a group of 35 elements. Then the largest possible size of a subgroup of G other than G itself is _____.

Q.No. 19 A multiplexer is placed between a group of 32 registers and an accumulator to regulate data movement such that at any given point in time the content of only one register will move to the accumulator. The minimum number of select lines needed for the multiplexer is _____.

Q.No. 20 If there are m input lines and n output lines for a decoder that is used to uniquely address a byte addressable 1 KB RAM, then the minimum value of $m + n$ is _____.

Q.No. 21

A direct mapped cache memory of 1 MB has a block size of 256 bytes. The cache has an access time of 3 ns and a hit rate of 94%. During a cache miss, it takes 20 ns to bring the first word of a block from the main memory, while each subsequent word takes 5 ns. The word size is 64 bits. The average memory access time in ns (round off to 1 decimal place) is _____.

Q.No. 22 Consider the following C program.

```
#include <stdio.h>

int main() {
    int a[4][5]={{1, 2, 3, 4, 5},
                 {6, 7, 8, 9, 10},
                 {11, 12, 13, 14, 15},
                 {16, 17, 18, 19, 20}};

    printf("%d\n", *((a**a+2)+3));

    return(0);
}
```

The output of the program is _____.

Q.No. 23 Consider a double hashing scheme in which the primary hash function is $h_1(k) = k \bmod 23$, and the secondary hash function is $h_2(k) = 1 + (k \bmod 19)$. Assume that the table size is 23. Then the address returned by probe 1 in the probe sequence (assume that the probe sequence begins at probe 0) for key value $k = 90$ is _____.

Q.No. 24 Consider the following grammar.

$$S \rightarrow aSB \mid d$$

$$B \rightarrow b$$

The number of reduction steps taken by a bottom-up parser while accepting the string *aaadb* is _____.

Q.No. 25 Assume that you have made a request for a web page through your web browser to a web server. Initially the browser cache is empty. Further, the browser is configured to send HTTP requests in non-persistent mode. The web page contains text and five very small images. The minimum number of TCP connections required to display the web page completely in your browser is _____.

Q26 - Q55 carry two marks each.

Q.No. 26

Which of the following languages are undecidable? Note that $\langle M \rangle$ indicates encoding of the Turing machine M .

$$L_1 = \{\langle M \rangle \mid L(M) = \emptyset\}$$

$$L_2 = \{\langle M, w, q \rangle \mid M \text{ on input } w \text{ reaches state } q \text{ in exactly 100 steps}\}$$

$$L_3 = \{\langle M \rangle \mid L(M) \text{ is not recursive}\}$$

$$L_4 = \{\langle M \rangle \mid L(M) \text{ contains at least 21 members}\}$$

- (A) L_1, L_3 , and L_4 only
- (B) L_1 and L_3 only
- (C) L_2 and L_3 only
- (D) L_2, L_3 , and L_4 only

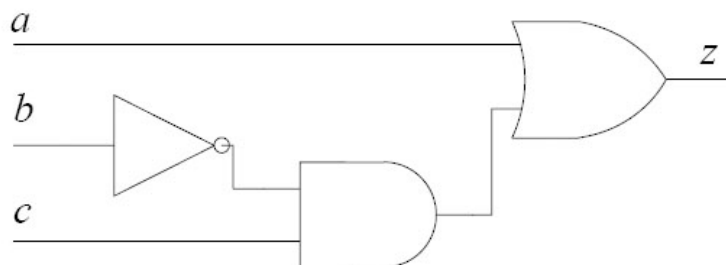
Q.No. 27 Let A and B be two $n \times n$ matrices over real numbers. Let $\text{rank}(M)$ and $\det(M)$ denote the rank and determinant of a matrix M , respectively. Consider the following statements.

- I. $\text{rank}(AB) = \text{rank}(A) \text{rank}(B)$
- II. $\det(AB) = \det(A) \det(B)$
- III. $\text{rank}(A + B) \leq \text{rank}(A) + \text{rank}(B)$
- IV. $\det(A + B) \leq \det(A) + \det(B)$

Which of the above statements are TRUE?

- (A) I and II only
- (B) I and IV only
- (C) II and III only
- (D) III and IV only

Q.No. 28 Consider the Boolean function $z(a, b, c)$.



Which one of the following minterm lists represents the circuit given above?

- (A) $z = \sum (0, 1, 3, 7)$
- (B) $z = \sum (1, 4, 5, 6, 7)$
- (C) $z = \sum (2, 4, 5, 6, 7)$

(D) $z = \sum (2, 3, 5)$

Q.No. 29 Consider three registers R1, R2, and R3 that store numbers in IEEE-754 single precision floating point format. Assume that R1 and R2 contain the values (in hexadecimal notation) 0x42200000 and 0xC1200000, respectively.

If $R3 = \frac{R1}{R2}$, what is the value stored in R3?

- (A) 0x40800000
- (B) 0xC0800000
- (C) 0x83400000
- (D) 0xC8500000

Q.No. 30 A computer system with a word length of 32 bits has a 16 MB byte-addressable main memory and a 64 KB, 4-way set associative cache memory with a block size of 256 bytes. Consider the following four physical addresses represented in hexadecimal notation.

$$A1 = 0x42C8A4, \quad A2 = 0x546888, \quad A3 = 0x6A289C, \quad A4 = 0x5E4880$$

Which one of the following is TRUE?

- (A) A1 and A4 are mapped to different cache sets.
- (B) A2 and A3 are mapped to the same cache set.
- (C) A3 and A4 are mapped to the same cache set.
- (D) A1 and A3 are mapped to the same cache set.

Q.No. 31 Let $G = (V, E)$ be a weighted undirected graph and let T be a Minimum Spanning Tree (MST) of G maintained using adjacency lists. Suppose a new weighted edge $(u, v) \in V \times V$ is added to G . The worst case time complexity of determining if T is still an MST of the resultant graph is

- (A) $\Theta(|E| + |V|)$
- (B) $\Theta(|E||V|)$
- (C) $\Theta(|E| \log |V|)$
- (D) $\Theta(|V|)$

Q.No. 32 Consider the following languages.

$$L_1 = \{wxyx \mid w, x, y \in (0 + 1)^+\}$$

$$L_2 = \{xy \mid x, y \in (a + b)^*, |x| = |y|, x \neq y\}$$

Which one of the following is TRUE?

- (A) L_1 is regular and L_2 is context-free.
- (B) L_1 is context-free but not regular and L_2 is context-free.
- (C) Neither L_1 nor L_2 is context-free.

- (D) L_1 is context-free but L_2 is not context-free.

Q.No. 33 Consider the productions $A \rightarrow PQ$ and $A \rightarrow XY$. Each of the five non-terminals A, P, Q, X, and Y has two attributes: s is a synthesized attribute, and i is an inherited attribute. Consider the following rules.

Rule 1: $P.i = A.i + 2$, $Q.i = P.i + A.i$, and $A.s = P.s + Q.s$

Rule 2: $X.i = A.i + Y.s$ and $Y.i = X.s + A.i$

Which one of the following is TRUE?

- (A) Both Rule 1 and Rule 2 are L-attributed.
(B) Only Rule 1 is L-attributed.
(C) Only Rule 2 is L-attributed.
(D) Neither Rule 1 nor Rule 2 is L-attributed.

Q.No. 34 Each of a set of n processes executes the following code using two semaphores a and b initialized to 1 and 0, respectively. Assume that `count` is a shared variable initialized to 0 and not used in CODE SECTION P.

CODE SECTION P

```
wait(a); count=count+1;
if (count==n) signal(b);
signal(a); wait(b); signal(b);
```

CODE SECTION Q

What does the code achieve?

- (A) It ensures that no process executes CODE SECTION Q before every process has finished CODE SECTION P.
(B) It ensures that at most two processes are in CODE SECTION Q at any time.
(C) It ensures that all processes execute CODE SECTION P mutually exclusively.
(D) It ensures that at most $n-1$ processes are in CODE SECTION P at any time.

Q.No. 35 Consider the following five disk access requests of the form (request id, cylinder number) that are present in the disk scheduler queue at a given time.

(P, 155), (Q, 85), (R, 110), (S, 30), (T, 115)

Assume the head is positioned at cylinder 100. The scheduler follows Shortest Seek Time First scheduling to service the requests.

Which one of the following statements is FALSE?

- (A) T is serviced before P.
(B)

Q is serviced after S, but before T.

- (C) The head reverses its direction of movement between servicing of Q and P.
 (D) R is serviced before P.

Q.No. 36 Consider a relational table R that is in 3NF, but not in BCNF. Which one of the following statements is TRUE?

- (A) R has a nontrivial functional dependency $X \rightarrow A$, where X is not a superkey and A is a prime attribute.
 (B) R has a nontrivial functional dependency $X \rightarrow A$, where X is not a superkey and A is a non-prime attribute and X is not a proper subset of any key.
 (C) R has a nontrivial functional dependency $X \rightarrow A$, where X is not a superkey and A is a non-prime attribute and X is a proper subset of some key.
 (D) A cell in R holds a set instead of an atomic value.

Q.No. 37 Consider a schedule of transactions T_1 and T_2 :

T_1	RA			RC		WD		WB	Commit	
T_2		RB	WB		RD		WC			Commit

Here, RX stands for “Read(X)” and WX stands for “Write(X)”. Which one of the following schedules is conflict equivalent to the above schedule?

- (A)

T_1				RA	RC	WD	WB		Commit	
T_2	RB	WB	RD					WC		Commit
- (B)

T_1	RA	RC	WD	WB					Commit	
T_2					RB	WB	RD	WC		Commit
- (C)

T_1	RA	RC	WD				WB		Commit	
T_2				RB	WB	RD		WC		Commit
- (D)

T_1					RA	RC	WD	WB	Commit	
T_2	RB	WB	RD	WC						Commit

Q.No. 38

An organization requires a range of IP addresses to assign one to each of its 1500 computers. The organization has approached an Internet Service Provider (ISP) for this task. The ISP uses CIDR and serves the requests from the available IP address space 202.61.0.0/17. The ISP wants to assign an address space to the organization which will minimize the number of routing entries in the ISP's router using route aggregation. Which of the following address spaces are potential candidates from which the ISP can allot any one to the organization?

- I. 202.61.84.0/21
- II. 202.61.104.0/21
- III. 202.61.64.0/21
- IV. 202.61.144.0/21

- (A) I and II only
- (B) II and III only
- (C) III and IV only
- (D) I and IV only

Q.No. 39 Which one of the following predicate formulae is NOT logically valid?

Note that W is a predicate formula without any free occurrence of x .

- (A) $\forall x(p(x) \vee W) \equiv \forall x p(x) \vee W$
- (B) $\exists x(p(x) \wedge W) \equiv \exists x p(x) \wedge W$
- (C) $\forall x(p(x) \rightarrow W) \equiv \forall x p(x) \rightarrow W$
- (D) $\exists x(p(x) \rightarrow W) \equiv \forall x p(x) \rightarrow W$

Q.No. 40 Let $G = (V, E)$ be a directed, weighted graph with weight function $w: E \rightarrow \mathbb{R}$.

For some function $f: V \rightarrow \mathbb{R}$, for each edge $(u, v) \in E$, define $w'(u, v)$ as $w(u, v) + f(u) - f(v)$.

Which one of the options completes the following sentence so that it is TRUE?

“The shortest paths in G under w are shortest paths under w' too, _____”.

- (A) for every $f: V \rightarrow \mathbb{R}$
- (B) if and only if $\forall u \in V$, $f(u)$ is positive
- (C) if and only if $\forall u \in V$, $f(u)$ is negative
- (D) if and only if $f(u)$ is the distance from s to u in the graph obtained by adding a new vertex s to G and edges of zero weight from s to every vertex of G

Q.No. 41 In a balanced binary search tree with n elements, what is the worst case time complexity of reporting all elements in range $[a, b]$? Assume that the number of reported elements is k .

- (A) $\Theta(\log n)$
- (B) $\Theta(\log n + k)$
- (C) $\Theta(k \log n)$

(D) $\Theta(n \log k)$

Q.No. 42 The number of permutations of the characters in LILAC so that no character appears in its original position, if the two L's are indistinguishable, is _____.

Q.No. 43 Consider a non-pipelined processor operating at 2.5 GHz. It takes 5 clock cycles to complete an instruction. You are going to make a 5-stage pipeline out of this processor. Overheads associated with pipelining force you to operate the pipelined processor at 2 GHz. In a given program, assume that 30% are memory instructions, 60% are ALU instructions and the rest are branch instructions. 5% of the memory instructions cause stalls of 50 clock cycles each due to cache misses and 50% of the branch instructions cause stalls of 2 cycles each. Assume that there are no stalls associated with the execution of ALU instructions. For this program, the speedup achieved by the pipelined processor over the non-pipelined processor (round off to 2 decimal places) is _____.

Q.No. 44 A processor has 64 registers and uses 16-bit instruction format. It has two types of instructions: I-type and R-type. Each I-type instruction contains an opcode, a register name, and a 4-bit immediate value. Each R-type instruction contains an opcode and two register names. If there are 8 distinct I-type opcodes, then the maximum number of distinct R-type opcodes is _____.

Q.No. 45 For $n > 2$, let $a \in \{0,1\}^n$ be a non-zero vector. Suppose that x is chosen uniformly at random from $\{0,1\}^n$. Then, the probability that $\sum_{i=1}^n a_i x_i$ is an odd number is _____.

Q.No. 46 Consider the following C functions.

```
int fun1(int n) {
    static int i = 0;
    if (n > 0) {
        ++i;
        fun1(n-1);
    }
    return(i);
}
```

```
int fun2(int n) {
    static int i = 0;
    if (n > 0) {
        i = i + fun1(n);
        fun2(n-1);
    }
    return(i);
}
```

The return value of `fun2(5)` is _____.

Q.No. 47 Consider the array representation of a binary min-heap containing 1023 elements. The minimum number of comparisons required to find the maximum in the heap is _____.

Q.No. 48 Consider the following C functions.

```
int tob(int b, int* arr){
    int i;
    for(i=0; b>0; i++){
        if(b%2) arr[i]=1;
        else    arr[i]=0;
        b = b/2;
    }
    return(i);
}

int pp(int a,int b) {
    int arr[20];
    int i, tot = 1, ex, len;
    ex = a;
    len = tob(b,arr);
    for(i=0; i<len; i++){
        if(arr[i]==1)
            tot = tot * ex;
        ex = ex * ex;
    }
    return(tot);
}
```

The value returned by $pp(3, 4)$ is _____.

Q.No. 49 Consider a graph $G = (V, E)$, where $V = \{v_1, v_2, \dots, v_{100}\}$, $E = \{(v_i, v_j) | 1 \leq i < j \leq 100\}$, and weight of the edge (v_i, v_j) is $|i - j|$. The weight of minimum spanning tree of G is _____.

Q.No. 50 Consider the following set of processes, assumed to have arrived at time 0. Consider the CPU scheduling algorithms Shortest Job First (SJF) and Round Robin (RR). For RR, assume that the processes are scheduled in the order P_1, P_2, P_3, P_4 .

Processes	P_1	P_2	P_3	P_4
Burst time (in ms)	8	7	2	4

If the time quantum for RR is 4 ms, then the absolute value of the difference between the average turnaround times (in ms) of SJF and RR (round off to 2 decimal places) is _____.

Q.No. 51 Consider the following language.

$$L = \{x \in \{a, b\}^* | \text{number of } a\text{'s in } x \text{ is divisible by 2 but not divisible by 3}\}$$

The minimum number of states in a DFA that accepts L is _____.

Q.No. 52 Graph G is obtained by adding vertex s to $K_{3,4}$ and making s adjacent to every vertex of $K_{3,4}$. The minimum number of colours required to edge-colour G is _____.

- Q.No. 53 Consider a paging system that uses 1-level page table residing in main memory and a TLB for address translation. Each main memory access takes 100 ns and TLB lookup takes 20 ns. Each page transfer to/from the disk takes 5000 ns. Assume that the TLB hit ratio is 95%, page fault rate is 10%. Assume that for 20% of the total page faults, a dirty page has to be written back to disk before the required page is read in from disk. TLB update time is negligible. The average memory access time in ns (round off to 1 decimal places) is _____.
- Q.No. 54 Consider a database implemented using B+ tree for file indexing and installed on a disk drive with block size of 4 KB. The size of search key is 12 bytes and the size of tree/disk pointer is 8 bytes. Assume that the database has one million records. Also assume that no node of the B+ tree and no records are present initially in main memory. Consider that each record fits into one disk block. The minimum number of disk accesses required to retrieve any record in the database is _____.
- Q.No. 55 Consider a TCP connection between a client and a server with the following specifications: the round trip time is 6 ms, the size of the receiver advertised window is 50 KB, slow-start threshold at the client is 32 KB, and the maximum segment size is 2 KB. The connection is established at time $t = 0$. Assume that there are no timeouts and errors during transmission. Then the size of the congestion window (in KB) at time $t + 60$ ms after all acknowledgements are processed is _____.

GATE 2019

Single session

65 questions | 16 pages

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Q. 1 – Q. 5 carry one mark each.

Q.1 The expenditure on the project _____ as follows: equipment Rs.20 lakhs, salaries Rs.12 lakhs, and contingency Rs.3 lakhs.

- (A) break down (B) break (C) breaks down (D) breaks

Q.2 The search engine's business model _____ around the fulcrum of trust.

- (A) revolves (B) plays (C) sinks (D) bursts

Q.3 Two cars start at the same time from the same location and go in the same direction. The speed of the first car is 50 km/h and the speed of the second car is 60 km/h. The number of hours it takes for the distance between the two cars to be 20 km is ____.

- (A) 1 (B) 2 (C) 3 (D) 6

Q.4 Ten friends planned to share equally the cost of buying a gift for their teacher. When two of them decided not to contribute, each of the other friends had to pay Rs 150 more. The cost of the gift was Rs. ____.

- (A) 666 (B) 3000 (C) 6000 (D) 12000

Q.5 A court is to a judge as _____ is to a teacher.

- (A) a student (B) a punishment (C) a syllabus (D) a school

Q. 6 – Q. 10 carry two marks each.

Q.6 The police arrested four criminals – P, Q, R and S. The criminals knew each other. They made the following statements:

P says "Q committed the crime."

Q says "S committed the crime."

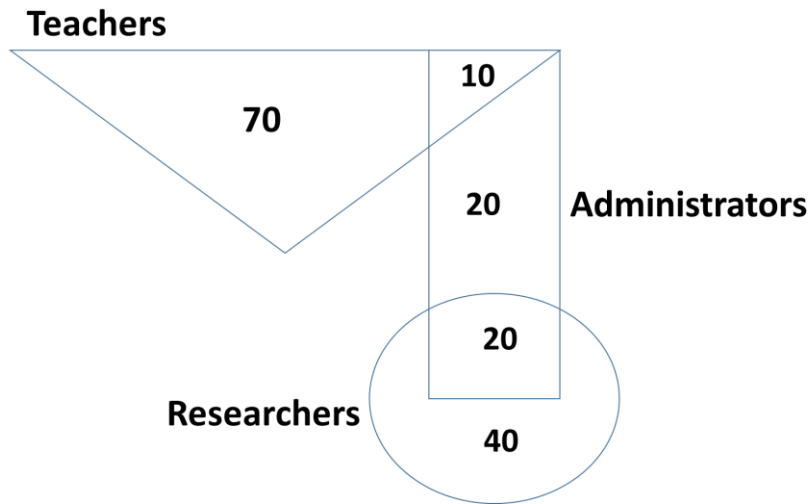
R says "I did not do it."

S says "What Q said about me is false."

Assume only one of the arrested four committed the crime and only one of the statements made above is true. Who committed the crime?

- (A) P (B) R (C) S (D) Q

- Q.7 In the given diagram, teachers are represented in the triangle, researchers in the circle and administrators in the rectangle. Out of the total number of the people, the percentage of administrators shall be in the range of _____.



- (A) 0 to 15 (B) 16 to 30 (C) 31 to 45 (D) 46 to 60
- Q.8 “A recent High Court judgement has sought to dispel the idea of begging as a disease — which leads to its stigmatization and criminalization — and to regard it as a symptom. The underlying disease is the failure of the state to protect citizens who fall through the social security net.”

Which one of the following statements can be inferred from the given passage?

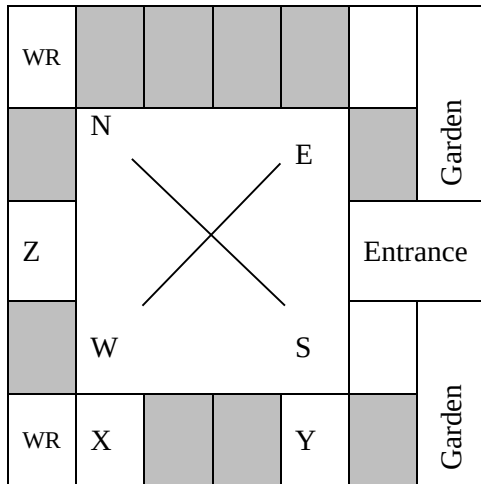
- (A) Beggars are lazy people who beg because they are unwilling to work
(B) Beggars are created because of the lack of social welfare schemes
(C) Begging is an offence that has to be dealt with firmly
(D) Begging has to be banned because it adversely affects the welfare of the state
- Q.9 In a college, there are three student clubs. Sixty students are only in the Drama club, 80 students are only in the Dance club, 30 students are only in the Maths club, 40 students are in both Drama and Dance clubs, 12 students are in both Dance and Maths clubs, 7 students are in both Drama and Maths clubs, and 2 students are in all the clubs. If 75% of the students in the college are not in any of these clubs, then the total number of students in the college is _____.

- (A) 1000 (B) 975 (C) 900 (D) 225

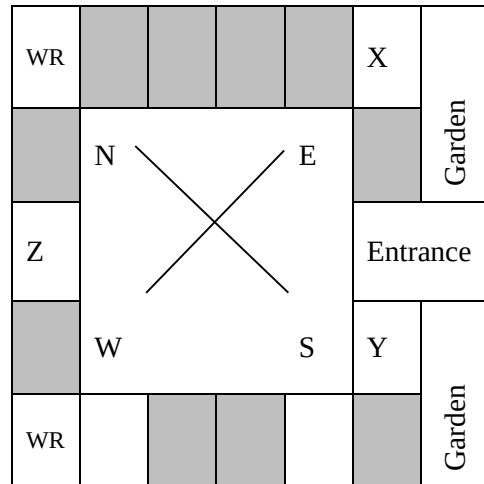
GATE 2019 General Aptitude (GA) Set-6

- Q.10 Three of the five students allocated to a hostel put in special requests to the warden. Given the floor plan of the vacant rooms, select the allocation plan that will accommodate all their requests.
Request by X: Due to pollen allergy, I want to avoid a wing next to the garden.
Request by Y: I want to live as far from the washrooms as possible, since I am very sensitive to smell.
Request by Z: I believe in Vaastu and so want to stay in the South-west wing.
The shaded rooms are already occupied. WR is washroom.

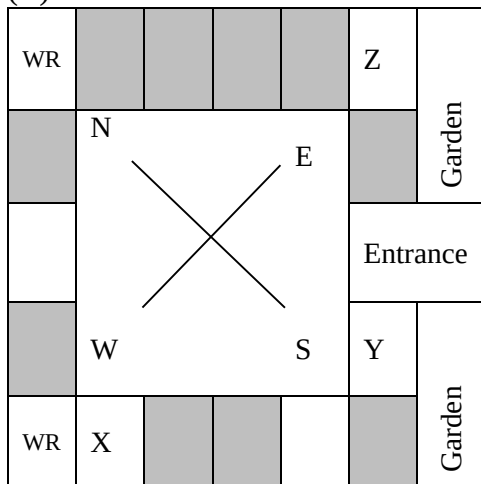
(A)



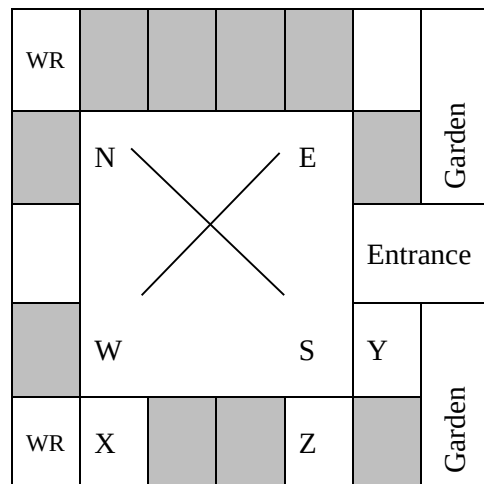
(B)



(C)



(D)

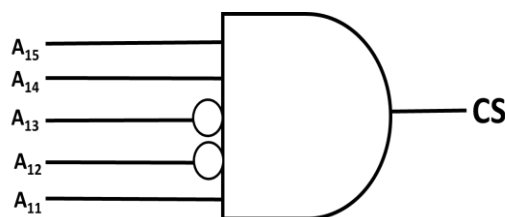


END OF THE QUESTION PAPER

- Q.1 A certain processor uses a fully associative cache of size 16 kB. The cache block size is 16 bytes. Assume that the main memory is byte addressable and uses a 32-bit address. How many bits are required for the *Tag* and the *Index* fields respectively in the addresses generated by the processor?

(A) 24 bits and 0 bits
(B) 28 bits and 4 bits
(C) 24 bits and 4 bits
(D) 28 bits and 0 bits

- Q.2 The chip select logic for a certain DRAM chip in a memory system design is shown below. Assume that the memory system has 16 address lines denoted by A_{15} to A_0 . What is the range of addresses (in hexadecimal) of the memory system that can get enabled by the chip select (CS) signal?



(A) C800 to CFFF
(B) CA00 to CAFF
(C) C800 to C8FF
(D) DA00 to DFFF

- Q.3 Which one of the following kinds of derivation is used by LR parsers?

(A) Leftmost
(B) Leftmost in reverse
(C) Rightmost
(D) Rightmost in reverse

- Q.4 In 16-bit 2's complement representation, the decimal number -28 is:

(A) 1111 1111 0001 1100
(B) 0000 0000 1110 0100
(C) 1111 1111 1110 0100
(D) 1000 0000 1110 0100

- Q.5 Let $U = \{1, 2, \dots, n\}$. Let $A = \{(x, X) | x \in X, X \subseteq U\}$. Consider the following two statements on $|A|$.

I. $|A| = n2^{n-1}$

II. $|A| = \sum_{k=1}^n k \binom{n}{k}$

Which of the above statements is/are TRUE?

(A) Only I
(B) Only II
(C) Both I and II
(D) Neither I nor II

Q.6 Which one of the following is NOT a valid identity?

(A) $(x \oplus y) \oplus z = x \oplus (y \oplus z)$

(B) $(x + y) \oplus z = x \oplus (y + z)$

(C) $x \oplus y = x + y$, if $xy = 0$

(D) $x \oplus y = (xy + x'y')'$

Q.7 If L is a regular language over $\Sigma = \{a, b\}$, which one of the following languages is NOT regular?

(A) $L \cdot L^R = \{xy \mid x \in L, y^R \in L\}$

(B) $\{ww^R \mid w \in L\}$

(C) $\text{Prefix}(L) = \{x \in \Sigma^* \mid \exists y \in \Sigma^* \text{ such that } xy \in L\}$

(D) $\text{Suffix}(L) = \{y \in \Sigma^* \mid \exists x \in \Sigma^* \text{ such that } xy \in L\}$

Q.8 Consider $Z = X - Y$, where X , Y and Z are all in sign-magnitude form. X and Y are each represented in n bits. To avoid overflow, the representation of Z would require a minimum of:

(A) n bits

(B) $n - 1$ bits

(C) $n + 1$ bits

(D) $n + 2$ bits

Q.9 Let X be a square matrix. Consider the following two statements on X .

I. X is invertible.

II. Determinant of X is non-zero.

Which one of the following is TRUE?

(A) I implies II; II does not imply I.

(B) II implies I; I does not imply II.

(C) I does not imply II; II does not imply I.

(D) I and II are equivalent statements.

Q.10 Let G be an arbitrary group. Consider the following relations on G :

$$R_1: \forall a, b \in G, a R_1 b \text{ if and only if } \exists g \in G \text{ such that } a = g^{-1}bg$$

$$R_2: \forall a, b \in G, a R_2 b \text{ if and only if } a = b^{-1}$$

Which of the above is/are equivalence relation/relation(s)?

(A) R_1 and R_2

(B) R_1 only

(C) R_2 only

(D) Neither R_1 nor R_2

Q.11 Consider the following two statements about database transaction schedules:

- I. Strict two-phase locking protocol generates conflict serializable schedules that are also recoverable.
- II. Timestamp-ordering concurrency control protocol with Thomas' Write Rule can generate view serializable schedules that are not conflict serializable.

Which of the above statements is/are TRUE?

- (A) I only (B) II only (C) Both I and II (D) Neither I nor II

Q.12 Let G be an undirected complete graph on n vertices, where $n > 2$. Then, the number of different Hamiltonian cycles in G is equal to

- (A) $n!$ (B) $(n - 1)!$ (C) 1 (D) $\frac{(n-1)!}{2}$

Q.13 Compute $\lim_{x \rightarrow 3} \frac{x^4 - 81}{2x^2 - 5x - 3}$

- (A) 1 (B) 53/12
(C) 108/7 (D) Limit does not exist

Q.14 Which one of the following statements is NOT correct about the B+ tree data structure used for creating an index of a relational database table?

- (A) B+ Tree is a height-balanced tree
- (B) Non-leaf nodes have pointers to data records
- (C) Key values in each node are kept in sorted order
- (D) Each leaf node has a pointer to the next leaf node

Q.15 For $\Sigma = \{a, b\}$, let us consider the regular language $L = \{x \mid x = a^{2+3k} \text{ or } x = b^{10+12k}, k \geq 0\}$. Which one of the following can be a pumping length (the constant guaranteed by the pumping lemma) for L ?

- (A) 3 (B) 5 (C) 9 (D) 24

Q.16 Which of the following protocol pairs can be used to send and retrieve e-mails (in that order)?

- (A) IMAP, POP3 (B) SMTP, POP3
- (C) SMTP, MIME (D) IMAP, SMTP

Q.17 The following C program is executed on a Unix/Linux system:

```
#include <unistd.h>
int main()
{
    int i;
    for (i=0; i<10; i++)
        if (i%2 == 0) fork();
    return 0;
}
```

The total number of child processes created is _____.

Q.18 Consider the following C program:

```
#include <stdio.h>
int jumble(int x, int y){
    x=2*x+y;
    return x;
}
int main(){
    int x=2, y=5;
    y=jumble(y,x);
    x=jumble(y,x);
    printf("%d \n", x);
    return 0;
}
```

The value printed by the program is _____.

Q.19 Consider the grammar given below:

$S \rightarrow Aa$
 $A \rightarrow BD$
 $B \rightarrow b \mid \epsilon$
 $D \rightarrow d \mid \epsilon$

Let a, b, d, and \$ be indexed as follows:

a	b	d	\$
3	2	1	0

Compute the FOLLOW set of the non-terminal B and write the index values for the symbols in the FOLLOW set in the descending order. (For example, if the FOLLOW set is {a, b, d, \$}, then the answer should be 3210)

Answer: _____.

- Q.20 An array of 25 distinct elements is to be sorted using quicksort. Assume that the pivot element is chosen uniformly at random. The probability that the pivot element gets placed in the worst possible location in the first round of partitioning (rounded off to 2 decimal places) is _____.
- Q.21 The value of $3^{51} \bmod 5$ is _____.
- Q.22 Two numbers are chosen independently and uniformly at random from the set $\{1, 2, \dots, 13\}$. The probability (rounded off to 3 decimal places) that their 4-bit (unsigned) binary representations have the same most significant bit is _____.
- Q.23 Consider three concurrent processes P1, P2 and P3 as shown below, which access a shared variable D that has been initialized to 100.

P1	P2	P3
:	:	:
:	:	:
$D = D + 20$	$D = D - 50$	$D = D + 10$
:	:	:
:	:	:

The processes are executed on a uniprocessor system running a time-shared operating system. If the minimum and maximum possible values of D after the three processes have completed execution are X and Y respectively, then the value of $Y - X$ is _____.

- Q.24 Consider the following C program:

```
#include <stdio.h>
int main(){
    int arr[]={1,2,3,4,5,6,7,8,9,0,1,2,5}, *ip=arr+4;
    printf("%d\n", ip[1]);
    return 0;
}
```

The number that will be displayed on execution of the program is _____.

- Q.25 Consider a sequence of 14 elements: $A = [-5, -10, 6, 3, -1, -2, 13, 4, -9, -1, 4, 12, -3, 0]$. The subsequence sum $S(i, j) = \sum_{k=i}^j A[k]$. Determine the maximum of $S(i, j)$, where $0 \leq i \leq j < 14$. (Divide and conquer approach may be used.)

Answer: _____.

- Q.26 Consider the following C function.

```
void convert(int n){
    if(n<0)
        printf("%d",n);
    else {
        convert(n/2);
        printf("%d",n%2);
    }
}
```

Which one of the following will happen when the function `convert` is called with any positive integer n as argument?

- (A) It will print the binary representation of n and terminate
 - (B) It will print the binary representation of n in the reverse order and terminate
 - (C) It will print the binary representation of n but will not terminate
 - (D) It will not print anything and will not terminate
- Q.27 Consider the following C program:

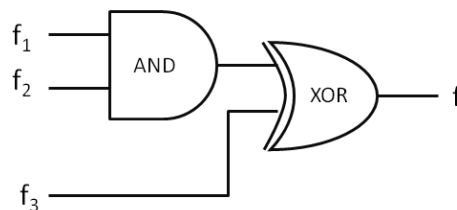
```
#include <stdio.h>
int r(){
    static int num=7;
    return num--;
}

int main(){
    for (r();r();r())
        printf("%d",r());
    return 0;
}
```

Which one of the following values will be displayed on execution of the programs?

- (A) 41 (B) 52 (C) 63 (D) 630

- Q.28 Consider three machines M, N, and P with IP addresses 100.10.5.2, 100.10.5.5, and 100.10.5.6 respectively. The subnet mask is set to 255.255.255.252 for all the three machines. Which one of the following is true?
- (A) M, N, and P all belong to the same subnet
 (B) Only M and N belong to the same subnet
 (C) Only N and P belong to the same subnet
 (D) M, N, and P belong to three different subnets
- Q.29 Suppose that in an IP-over-Ethernet network, a machine X wishes to find the MAC address of another machine Y in its subnet. Which one of the following techniques can be used for this?
- (A) X sends an ARP request packet to the local gateway's IP address which then finds the MAC address of Y and sends to X
 (B) X sends an ARP request packet to the local gateway's MAC address which then finds the MAC address of Y and sends to X
 (C) X sends an ARP request packet with broadcast MAC address in its local subnet
 (D) X sends an ARP request packet with broadcast IP address in its local subnet
- Q.30 Consider three 4-variable functions f_1 , f_2 , and f_3 , which are expressed in sum-of-minterms as
 $f_1 = \sum (0, 2, 5, 8, 14)$, $f_2 = \sum (2, 3, 6, 8, 14, 15)$, $f_3 = \sum (2, 7, 11, 14)$
 For the following circuit with one AND gate and one XOR gate, the output function f can be expressed as:



- (A) $\sum (7, 8, 11)$
 (B) $\sum (2, 7, 8, 11, 14)$
 (C) $\sum (2, 14)$
 (D) $\sum (0, 2, 3, 5, 6, 7, 8, 11, 14, 15)$
- Q.31 Which one of the following languages over $\Sigma = \{a, b\}$ is NOT context-free?
- (A) $\{ww^R \mid w \in \{a, b\}^*\}$
 (B) $\{wa^n b^n w^R \mid w \in \{a, b\}^*, n \geq 0\}$
 (C) $\{wa^n w^R b^n \mid w \in \{a, b\}^*, n \geq 0\}$
 (D) $\{a^n b^i \mid i \in \{n, 3n, 5n\}, n \geq 0\}$

- Q.32 Let the set of functional dependencies $F = \{QR \rightarrow S, R \rightarrow P, S \rightarrow Q\}$ hold on a relation schema $X = (PQRS)$. X is not in BCNF. Suppose X is decomposed into two schemas Y and Z , where $Y = (PR)$ and $Z = (QRS)$.

Consider the two statements given below.

- I. Both Y and Z are in BCNF
- II. Decomposition of X into Y and Z is dependency preserving and lossless

Which of the above statements is/are correct?

- (A) Both I and II (B) I only (C) II only (D) Neither I nor II

- Q.33 Assume that in a certain computer, the virtual addresses are 64 bits long and the physical addresses are 48 bits long. The memory is word addressable. The page size is 8 kB and the word size is 4 bytes. The Translation Look-aside Buffer (TLB) in the address translation path has 128 valid entries. At most how many distinct virtual addresses can be translated without any TLB miss?

- (A) 16×2^{10} (B) 256×2^{10} (C) 4×2^{20} (D) 8×2^{20}

- Q.34 Consider the following sets:

- S1. Set of all recursively enumerable languages over the alphabet $\{0,1\}$
- S2. Set of all syntactically valid C programs
- S3. Set of all languages over the alphabet $\{0,1\}$
- S4. Set of all non-regular languages over the alphabet $\{0,1\}$

Which of the above sets are uncountable?

- (A) S1 and S2 (B) S3 and S4 (C) S2 and S3 (D) S1 and S4

- Q.35 Consider the first order predicate formula φ :

$\forall x [(\forall z z|x \Rightarrow ((z = x) \vee (z = 1))) \Rightarrow \exists w (w > x) \wedge (\forall z z|w \Rightarrow ((w = z) \vee (z = 1)))]$

Here ' $a|b$ ' denotes that ' a divides b ', where a and b are integers. Consider the following sets:

- S1. $\{1,2,3, \dots, 100\}$
- S2. Set of all positive integers
- S3. Set of all integers

Which of the above sets satisfy φ ?

- (A) S1 and S2 (B) S1 and S3 (C) S2 and S3 (D) S1, S2 and S3

- Q.36 Consider the following grammar and the semantic actions to support the inherited type declaration attributes. Let X_1, X_2, X_3, X_4, X_5 , and X_6 be the placeholders for the non-terminals D, T, L or L_1 in the following table:

Production rule	Semantic action
$D \rightarrow T L$	$X_1.type = X_2.type$
$T \rightarrow \text{int}$	$T.type = \text{int}$
$T \rightarrow \text{float}$	$T.type = \text{float}$
$L \rightarrow L_1, \text{id}$	$X_3.type = X_4.type$ $\text{addType}(\text{id.entry}, X_5.type)$
$L \rightarrow \text{id}$	$\text{addType}(\text{id.entry}, X_6.type)$

Which one of the following are the appropriate choices for X_1, X_2, X_3 and X_4 ?

- (A) $X_1 = L, X_2 = T, X_3 = L_1, X_4 = L$ (B) $X_1 = T, X_2 = L, X_3 = L_1, X_4 = T$
 (C) $X_1 = L, X_2 = L, X_3 = L_1, X_4 = T$ (D) $X_1 = T, X_2 = L, X_3 = T, X_4 = L_1$
- Q.37 There are n unsorted arrays: $A_1, A_2, \dots, A_n$. Assume that n is odd. Each of $A_1, A_2, \dots, A_n$ contains n distinct elements. There are no common elements between any two arrays. The worst-case time complexity of computing the median of the medians of $A_1, A_2, \dots, A_n$ is

- (A) $O(n)$ (B) $O(n \log n)$ (C) $O(n^2)$ (D) $\Omega(n^2 \log n)$

- Q.38 Let G be any connected, weighted, undirected graph.

- I. G has a unique minimum spanning tree, if no two edges of G have the same weight.
 II. G has a unique minimum spanning tree, if, for every cut of G , there is a unique minimum-weight edge crossing the cut.

Which of the above two statements is/are TRUE?

- (A) I only (B) II only (C) Both I and II (D) Neither I nor II

Q.39 Consider the following snapshot of a system running n concurrent processes. Process i is holding X_i instances of a resource R , $1 \leq i \leq n$. Assume that all instances of R are currently in use. Further, for all i , process i can place a request for at most Y_i additional instances of R while holding the X_i instances it already has. Of the n processes, there are exactly two processes p and q such that $Y_p = Y_q = 0$. Which one of the following conditions guarantees that no other process apart from p and q can complete execution?

- (A) $X_p + X_q < \text{Min} \{Y_k \mid 1 \leq k \leq n, k \neq p, k \neq q\}$
- (B) $X_p + X_q < \text{Max} \{Y_k \mid 1 \leq k \leq n, k \neq p, k \neq q\}$
- (C) $\text{Min}(X_p, X_q) \geq \text{Min} \{Y_k \mid 1 \leq k \leq n, k \neq p, k \neq q\}$
- (D) $\text{Min}(X_p, X_q) \leq \text{Max} \{Y_k \mid 1 \leq k \leq n, k \neq p, k \neq q\}$

Q.40 Consider the following statements:

- I. The smallest element in a max-heap is always at a leaf node
- II. The second largest element in a max-heap is always a child of the root node
- III. A max-heap can be constructed from a binary search tree in $\Theta(n)$ time
- IV. A binary search tree can be constructed from a max-heap in $\Theta(n)$ time

Which of the above statements are TRUE?

- (A) I, II and III
- (B) I, II and IV
- (C) I, III and IV
- (D) II, III and IV

Q.41 Consider the following four processes with arrival times (in milliseconds) and their length of CPU bursts (in milliseconds) as shown below:

Process	P1	P2	P3	P4
Arrival time	0	1	3	4
CPU burst time	3	1	3	Z

These processes are run on a single processor using preemptive Shortest Remaining Time First scheduling algorithm. If the average waiting time of the processes is 1 millisecond, then the value of Z is _____.

Q.42 The index node (inode) of a Unix-like file system has 12 direct, one single-indirect and one double-indirect pointers. The disk block size is 4 kB, and the disk block address is 32-bits long. The maximum possible file size is (rounded off to 1 decimal place) _____ GB.

Q.43 Consider the augmented grammar given below:

$$S' \rightarrow S$$

$$S \rightarrow \langle L \rangle \mid id$$

$$L \rightarrow L, S \mid S$$

Let $I_0 = \text{CLOSURE}(\{[S' \rightarrow \bullet S]\})$. The number of items in the set $\text{GOTO}(I_0, \langle \rangle)$ is: _____.

Q.44 Consider the following matrix:

$$R = \begin{bmatrix} 1 & 2 & 4 & 8 \\ 1 & 3 & 9 & 27 \\ 1 & 4 & 16 & 64 \\ 1 & 5 & 25 & 125 \end{bmatrix}$$

The absolute value of the product of Eigen values of R is _____.

Q.45 A certain processor deploys a single-level cache. The cache block size is 8 words and the word size is 4 bytes. The memory system uses a 60-MHz clock. To service a cache miss, the memory controller first takes 1 cycle to accept the starting address of the block, it then takes 3 cycles to fetch all the eight words of the block, and finally transmits the words of the requested block at the rate of 1 word per cycle. The maximum bandwidth for the memory system when the program running on the processor issues a series of read operations is _____ $\times 10^6$ bytes/sec.

Q.46 Let T be a full binary tree with 8 leaves. (A full binary tree has every level full.) Suppose two leaves a and b of T are chosen uniformly and independently at random. The expected value of the distance between a and b in T (i.e., the number of edges in the unique path between a and b) is (rounded off to 2 decimal places) _____.

Q.47 Suppose Y is distributed uniformly in the open interval $(1,6)$. The probability that the polynomial $3x^2 + 6xY + 3Y + 6$ has only real roots is (rounded off to 1 decimal place) _____.

Q.48 Let Σ be the set of all bijections from $\{1, \dots, 5\}$ to $\{1, \dots, 5\}$, where id denotes the identity function, i.e. $id(j) = j, \forall j$. Let $\circ$ denote composition on functions. For a string $x = x_1 x_2 \dots x_n \in \Sigma^n, n \geq 0$, let $\pi(x) = x_1 \circ x_2 \circ \dots \circ x_n$. Consider the language $L = \{x \in \Sigma^* \mid \pi(x) = id\}$. The minimum number of states in any DFA accepting L is _____.

- Q.49 Consider that 15 machines need to be connected in a LAN using 8-port Ethernet switches. Assume that these switches do not have any separate uplink ports. The minimum number of switches needed is_____.
- Q.50 What is the minimum number of 2-input NOR gates required to implement a 4-variable function expressed in sum-of-minterms form as $f = \sum (0, 2, 5, 7, 8, 10, 13, 15)$? Assume that all the inputs and their complements are available. Answer: _____.
- Q.51 A relational database contains two tables Student and Performance as shown below:

Student	
Roll_no.	Student_name
1	Amit
2	Priya
3	Vinit
4	Rohan
5	Smita

Performance		
Roll_no.	Subject_code	Marks
1	A	86
1	B	95
1	C	90
2	A	89
2	C	92
3	C	80

The primary key of the Student table is Roll_no. For the Performance table, the columns Roll_no. and Subject_code together form the primary key. Consider the SQL query given below:

```
SELECT S.Student_name, sum(P.Marks)
FROM Student S, Performance P
WHERE P.Marks > 84
GROUP BY S.Student_name;
```

The number of rows returned by the above SQL query is _____.

- Q.52 Consider the following C program:

```
#include <stdio.h>
int main(){
    float sum = 0.0, j = 1.0, i = 2.0;
    while (i/j > 0.0625){
        j = j + j;
        sum = sum + i/j;
        printf("%f\n", sum);
    }
    return 0;
}
```

The number of times the variable sum will be printed, when the above program is executed, is _____.

Q.53 Consider the following C program:

```
#include <stdio.h>
int main()
{
    int a[] = {2, 4, 6, 8, 10};
    int i, sum = 0, *b = a + 4;
    for (i = 0; i < 5; i++)
        sum = sum + (*b - i) - *(b - i);
    printf ("%d\n", sum);
    return 0;
}
```

The output of the above C program is _____.

Q.54 In an RSA cryptosystem, the value of the public modulus parameter n is 3007. If it is also known that $\phi(n) = 2880$, where $\phi()$ denotes Euler's Totient Function, then the prime factor of n which is greater than 50 is _____.

Q.55 Consider the following relations $P(X,Y,Z)$, $Q(X,Y,T)$ and $R(Y,V)$.

P		
X	Y	Z
X1	Y1	Z1
X1	Y1	Z2
X2	Y2	Z2
X2	Y4	Z4

Q		
X	Y	T
X2	Y1	2
X1	Y2	5
X1	Y1	6
X3	Y3	1

R	
Y	V
Y1	V1
Y3	V2
Y2	V3
Y2	V2

How many tuples will be returned by the following relational algebra query?

$$\prod_X (\sigma_{(P.Y=R.Y \wedge R.V=V2)}(P \times R)) - \prod_X (\sigma_{(Q.Y=R.Y \wedge Q.T>2)}(Q \times R))$$

Answer: _____.

GATE 2018

Single session

65 questions | 23 pages

Paper 12 of 16

Q. 1 – Q. 5 carry one mark each.

Q.1 “From where are they bringing their books? _____ bringing _____ books from _____.”

The words that best fill the blanks in the above sentence are

- (A) Their, they’re, there
- (B) They’re, their, there
- (C) There, their, they’re
- (D) They’re, there, there

Q.2 “A _____ investigation can sometimes yield new facts, but typically organized ones are more successful.”

The word that best fills the blank in the above sentence is

- (A) meandering (B) timely (C) consistent (D) systematic

Q.3 The area of a square is d . What is the area of the circle which has the diagonal of the square as its diameter?

- (A) πd (B) πd^2 (C) $\frac{1}{4}\pi d^2$ (D) $\frac{1}{2}\pi d$

Q.4 What would be the smallest natural number which when divided either by 20 or by 42 or by 76 leaves a remainder of 7 in each case?

- (A) 3047 (B) 6047 (C) 7987 (D) 63847

Q.5 What is the missing number in the following sequence?

2, 12, 60, 240, 720, 1440, _____, 0

- (A) 2880 (B) 1440 (C) 720 (D) 0

Q. 6 – Q. 10 carry two marks each.

- Q.6 In appreciation of the social improvements completed in a town, a wealthy philanthropist decided to gift Rs 750 to each male senior citizen in the town and Rs 1000 to each female senior citizen. Altogether, there were 300 senior citizens eligible for this gift. However, only $\frac{8}{9}$ th of the eligible men and $\frac{2}{3}$ rd of the eligible women claimed the gift. How much money (in Rupees) did the philanthropist give away in total?

(A) 1,50,000 (B) 2,00,000
(C) 1,75,000 (D) 1,51,000

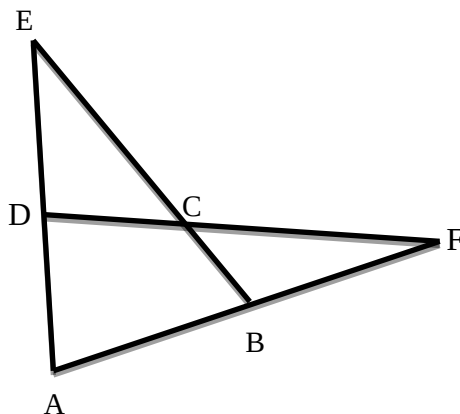
- Q.7 If $pqr \neq 0$ and $p^{-x} = \frac{1}{q}$, $q^{-y} = \frac{1}{r}$, $r^{-z} = \frac{1}{p}$, what is the value of the product xyz ?

(A) -1 (B) $\frac{1}{pqr}$ (C) 1 (D) pqr

- Q.8 In a party, 60% of the invited guests are male and 40% are female. If 80% of the invited guests attended the party and if all the invited female guests attended, what would be the ratio of males to females among the attendees in the party?

(A) 2:3 (B) 1:1 (C) 3:2 (D) 2:1

- Q.9 In the figure below, $\angle DEC + \angle BFC$ is equal to _____.



(A) $\angle BCD - \angle BAD$ (B) $\angle BAD + \angle BCF$
(C) $\angle BAD + \angle BCD$ (D) $\angle CBA + \angle ADC$

- Q.10 A six sided unbiased die with four green faces and two red faces is rolled seven times. Which of the following combinations is the most likely outcome of the experiment?
- (A) Three green faces and four red faces.
 - (B) Four green faces and three red faces.
 - (C) Five green faces and two red faces.
 - (D) Six green faces and one red face.

END OF THE QUESTION PAPER

Q. 1 – Q. 25 carry one mark each.

Q.1 Which one of the following is a closed form expression for the generating function of the sequence $\{a_n\}$, where $a_n = 2n + 3$ for all $n = 0, 1, 2, \dots$?

- (A) $\frac{3}{(1-x)^2}$ (B) $\frac{3x}{(1-x)^2}$ (C) $\frac{2-x}{(1-x)^2}$ (D) $\frac{3-x}{(1-x)^2}$

Q.2 Consider the following C program.

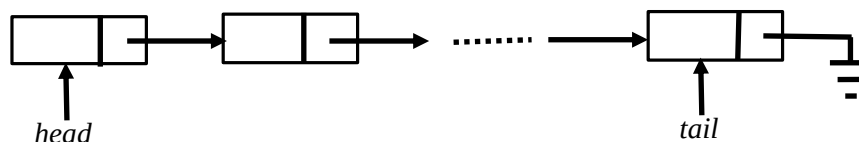
```
#include<stdio.h>
struct Ournode{
    char x,y,z;
};

int main(){
    struct Ournode p = {'1', '0', 'a'+2};
    struct Ournode *q = &p;
    printf ("%c, %c", *((char*)q+1), *((char*)q+2));
    return 0;
}
```

The output of this program is:

- (A) 0, c (B) 0, a+2 (C) '0', 'a+2' (D) '0', 'c'

Q.3 A queue is implemented using a non-circular singly linked list. The queue has a head pointer and a tail pointer, as shown in the figure. Let n denote the number of nodes in the queue. Let enqueue be implemented by inserting a new node at the head, and dequeue be implemented by deletion of a node from the tail.



Which one of the following is the time complexity of the most time-efficient implementation of enqueue and dequeue, respectively, for this data structure?

- (A) $\theta(1), \theta(1)$ (B) $\theta(1), \theta(n)$ (C) $\theta(n), \theta(1)$ (D) $\theta(n), \theta(n)$

Q.4 Let $\oplus$ and $\odot$ denote the Exclusive OR and Exclusive NOR operations, respectively. Which one of the following is NOT CORRECT?

- (A) $\overline{P \oplus Q} = P \odot Q$
- (B) $\bar{P} \oplus Q = P \odot Q$
- (C) $\bar{P} \oplus \bar{Q} = P \oplus Q$
- (D) $(P \oplus \bar{P}) \oplus Q = (P \odot \bar{P}) \odot \bar{Q}$

Q.5 Consider the following processor design characteristics.

- I. Register-to-register arithmetic operations only
- II. Fixed-length instruction format
- III. Hardwired control unit

Which of the characteristics above are used in the design of a RISC processor?

- (A) I and II only (B) II and III only (C) I and III only (D) I, II and III

Q.6 Let N be an NFA with n states. Let k be the number of states of a minimal DFA which is equivalent to N . Which one of the following is necessarily true?

- (A) $k \geq 2^n$ (B) $k \geq n$ (C) $k \leq n^2$ (D) $k \leq 2^n$

Q.7 The set of all recursively enumerable languages is

- (A) closed under complementation.
- (B) closed under intersection.
- (C) a subset of the set of all recursive languages.
- (D) an uncountable set.

Q.8 Which one of the following statements is FALSE?

- (A) Context-free grammar can be used to specify both lexical and syntax rules.
- (B) Type checking is done before parsing.
- (C) High-level language programs can be translated to different Intermediate Representations.
- (D) Arguments to a function can be passed using the program stack.

Q.9 The following are some events that occur after a device controller issues an interrupt while process L is under execution.

- (P) The processor pushes the process status of L onto the control stack.
- (Q) The processor finishes the execution of the current instruction.
- (R) The processor executes the interrupt service routine.
- (S) The processor pops the process status of L from the control stack.
- (T) The processor loads the new PC value based on the interrupt.

Which one of the following is the correct order in which the events above occur?

- (A) QPTRS (B) PTRSQ (C) TRPQS (D) QTPRS

Q.10 Consider a process executing on an operating system that uses demand paging. The average time for a memory access in the system is M units if the corresponding memory page is available in memory, and D units if the memory access causes a page fault. It has been experimentally measured that the average time taken for a memory access in the process is X units.

Which one of the following is the correct expression for the page fault rate experienced by the process?

- (A) $(D - M) / (X - M)$ (B) $(X - M) / (D - M)$
(C) $(D - X) / (D - M)$ (D) $(X - M) / (D - X)$

Q.11 In an Entity-Relationship (ER) model, suppose R is a many-to-one relationship from entity set $E1$ to entity set $E2$. Assume that $E1$ and $E2$ participate totally in R and that the cardinality of $E1$ is greater than the cardinality of $E2$.

Which one of the following is true about R ?

- (A) Every entity in $E1$ is associated with exactly one entity in $E2$.
- (B) Some entity in $E1$ is associated with more than one entity in $E2$.
- (C) Every entity in $E2$ is associated with exactly one entity in $E1$.
- (D) Every entity in $E2$ is associated with at most one entity in $E1$.

Q.12 Consider the following two tables and four queries in SQL.

Book (isbn, bname), Stock (isbn, copies)

- Query 1: SELECT B.isbn, S.copies
 FROM Book B INNER JOIN Stock S
 ON B.isbn = S.isbn;
- Query 2: SELECT B.isbn, S.copies
 FROM Book B LEFT OUTER JOIN Stock S
 ON B.isbn = S.isbn;
- Query 3: SELECT B.isbn, S.copies
 FROM Book B RIGHT OUTER JOIN Stock S
 ON B.isbn = S.isbn;
- Query 4: SELECT B.isbn, S.copies
 FROM Book B FULL OUTER JOIN Stock S
 ON B.isbn = S.isbn;

Which one of the queries above is certain to have an output that is a superset of the outputs of the other three queries?

- (A) Query 1 (B) Query 2 (C) Query 3 (D) Query 4

Q.13 Match the following:

<u>Field</u>	<u>Length in bits</u>
P. UDP Header's Port Number	I. 48
Q. Ethernet MAC Address	II. 8
R. IPv6 Next Header	III. 32
S. TCP Header's Sequence Number	IV. 16

- (A) P-III, Q-IV, R-II, S-I (B) P-II, Q-I, R-IV, S-III
(C) P-IV, Q-I, R-II, S-III (D) P-IV, Q-I, R-III, S-II

Q.14 Consider the following statements regarding the slow start phase of the TCP congestion control algorithm. Note that *cwnd* stands for the TCP congestion window and MSS denotes the Maximum Segment Size.

- (i) The *cwnd* increases by 2 MSS on every successful acknowledgment.
- (ii) The *cwnd* approximately doubles on every successful acknowledgement.
- (iii) The *cwnd* increases by 1 MSS every round trip time.
- (iv) The *cwnd* approximately doubles every round trip time.

Which one of the following is correct?

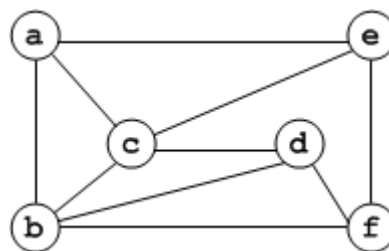
- (A) Only (ii) and (iii) are true
- (B) Only (i) and (iii) are true
- (C) Only (iv) is true
- (D) Only (i) and (iv) are true

Q.15 Two people, P and Q, decide to independently roll two identical dice, each with 6 faces, numbered 1 to 6. The person with the lower number wins. In case of a tie, they roll the dice repeatedly until there is no tie. Define a trial as a throw of the dice by P and Q. Assume that all 6 numbers on each dice are equi-probable and that all trials are independent. The probability (rounded to 3 decimal places) that one of them wins on the third trial is ____.

Q.16 The value of $\int_0^{\pi/4} x \cos(x^2) dx$ correct to three decimal places (assuming that $\pi = 3.14$) is ____.

Q.17 Consider a matrix $A = uv^T$ where $u = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$, $v = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$. Note that v^T denotes the transpose of v . The largest eigenvalue of A is ____.

Q.18 The chromatic number of the following graph is ____.



Q.19 Let G be a finite group on 84 elements. The size of a largest possible proper subgroup of G is _____.

Q.20 The postorder traversal of a binary tree is 8,9,6,7,4,5,2,3,1. The inorder traversal of the same tree is 8,6,9,4,7,2,5,1,3. The height of a tree is the length of the longest path from the root to any leaf. The height of the binary tree above is _____.

Q.21 Consider the following C program:

```
#include <stdio.h>

int counter = 0;

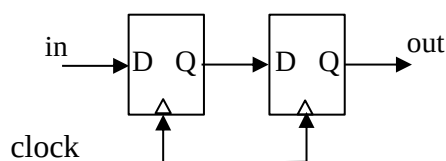
int calc (int a, int b) {
    int c;

    counter++;
    if (b==3) return (a*a*a);
    else {
        c = calc(a, b/3);
        return (c*c*c);
    }
}

int main () {
    calc(4, 81);
    printf ("%d", counter);
}
```

The output of this program is _____.

Q.22 Consider the sequential circuit shown in the figure, where both flip-flops used are positive edge-triggered D flip-flops.



The number of states in the state transition diagram of this circuit that have a transition back to the same state on some value of “in” is _____.

- Q.23 A 32-bit wide main memory unit with a capacity of 1 GB is built using $256\text{M} \times 4\text{-bit}$ DRAM chips. The number of rows of memory cells in the DRAM chip is 2^{14} . The time taken to perform one refresh operation is 50 nanoseconds. The refresh period is 2 milliseconds. The percentage (rounded to the closest integer) of the time available for performing the memory read/write operations in the main memory unit is _____.
- Q.24 Consider a system with 3 processes that share 4 instances of the same resource type. Each process can request a maximum of K instances. Resource instances can be requested and released only one at a time. The largest value of K that will always avoid deadlock is ____.
- Q.25 Consider a long-lived TCP session with an end-to-end bandwidth of 1 Gbps ($= 10^9$ bits-per-second). The session starts with a sequence number of 1234. The minimum time (in seconds, rounded to the closest integer) before this sequence number can be used again is _____.

Q. 26 – Q. 55 carry two marks each.

- Q.26 Consider a matrix $\mathbf{P}$ whose only eigenvectors are the multiples of $\begin{bmatrix} 1 \\ 4 \end{bmatrix}$.

Consider the following statements.

- (I) $\mathbf{P}$ does not have an inverse
- (II) $\mathbf{P}$ has a repeated eigenvalue
- (III) $\mathbf{P}$ cannot be diagonalized

Which one of the following options is correct?

- (A) Only I and III are necessarily true
- (B) Only II is necessarily true
- (C) Only I and II are necessarily true
- (D) Only II and III are necessarily true

Q.27 Let N be the set of natural numbers. Consider the following sets.

P : Set of Rational numbers (positive and negative)

Q : Set of functions from $\{0, 1\}$ to N

R : Set of functions from N to $\{0, 1\}$

S : Set of finite subsets of N .

Which of the sets above are countable?

- (A) Q and S only (B) P and S only (C) P and R only (D) P , Q and S only

Q.28 Consider the first-order logic sentence

$$\varphi \equiv \exists s \exists t \exists u \forall v \forall w \forall x \forall y \psi(s, t, u, v, w, x, y)$$

where $\psi(s, t, u, v, w, x, y)$ is a quantifier-free first-order logic formula using only predicate symbols, and possibly equality, but no function symbols. Suppose φ has a model with a universe containing 7 elements.

Which one of the following statements is necessarily true?

- (A) There exists at least one model of φ with universe of size less than or equal to 3.
(B) There exists no model of φ with universe of size less than or equal to 3.
(C) There exists no model of φ with universe of size greater than 7.
(D) Every model of φ has a universe of size equal to 7.

Q.29 Consider the following C program:

```
#include<stdio.h>

void fun1(char *s1, char *s2){
    char *tmp;
    tmp = s1;
    s1 = s2;
    s2 = tmp;
}

void fun2(char **s1, char **s2){
    char *tmp;
    tmp = *s1;
    *s1 = *s2;
    *s2 = tmp;
}

int main(){
    char *str1 = "Hi", *str2 = "Bye";
    fun1(str1, str2);    printf("%s %s ", str1, str2);
    fun2(&str1, &str2); printf("%s %s", str1, str2);
    return 0;
}
```

The output of the program above is

- | | |
|-------------------|-------------------|
| (A) Hi Bye Bye Hi | (B) Hi Bye Hi Bye |
| (C) Bye Hi Hi Bye | (D) Bye Hi Bye Hi |

Q.30 Let G be a simple undirected graph. Let T_D be a depth first search tree of G . Let T_B be a breadth first search tree of G . Consider the following statements.

- (I) No edge of G is a cross edge with respect to T_D . (A cross edge in G is between two nodes neither of which is an ancestor of the other in T_D .)
- (II) For every edge (u,v) of G , if u is at depth i and v is at depth j in T_B , then $|i - j| = 1$.

Which of the statements above must necessarily be true?

- | | |
|-------------------|----------------------|
| (A) I only | (B) II only |
| (C) Both I and II | (D) Neither I nor II |

- Q.31 Assume that multiplying a matrix G_1 of dimension $p \times q$ with another matrix G_2 of dimension $q \times r$ requires pqr scalar multiplications. Computing the product of n matrices $G_1 G_2 G_3 \dots G_n$ can be done by parenthesizing in different ways. Define $G_i G_{i+1}$ as an **explicitly computed pair** for a given parenthesization if they are directly multiplied. For example, in the matrix multiplication chain $G_1 G_2 G_3 G_4 G_5 G_6$ using parenthesization $(G_1(G_2 G_3))(G_4(G_5 G_6))$, $G_2 G_3$ and $G_5 G_6$ are the only explicitly computed pairs.

Consider a matrix multiplication chain $F_1 F_2 F_3 F_4 F_5$, where matrices F_1, F_2, F_3, F_4 and F_5 are of dimensions $2 \times 25, 25 \times 3, 3 \times 16, 16 \times 1$ and 1×1000 , respectively. In the parenthesization of $F_1 F_2 F_3 F_4 F_5$ that minimizes the total number of scalar multiplications, the explicitly computed pairs is/are

- (A) $F_1 F_2$ and $F_3 F_4$ only
(B) $F_2 F_3$ only
(C) $F_3 F_4$ only
(D) $F_1 F_2$ and $F_4 F_5$ only
- Q.32 Consider the following C code. Assume that unsigned long int type length is 64 bits.

```
unsigned long int fun(unsigned long int n){
    unsigned long int i, j = 0, sum = 0;
    for (i = n; i > 1; i = i/2) j++;
    for ( ; j > 1; j = j/2) sum++;
    return(sum);
}
```

The value returned when we call fun with the input 2^{40} is

- (A) 4
(B) 5
(C) 6
(D) 40
- Q.33 Consider the unsigned 8-bit fixed point binary number representation below,

$b_7 \ b_6 \ b_5 \ b_4 \ b_3 . b_2 \ b_1 \ b_0$

where the position of the binary point is between b_3 and b_2 . Assume b_7 is the most significant bit. Some of the decimal numbers listed below **cannot** be represented **exactly** in the above representation:

(i) 31.500 (ii) 0.875 (iii) 12.100 (iv) 3.001

Which one of the following statements is true?

- (A) None of (i), (ii), (iii), (iv) can be exactly represented
(B) Only (ii) cannot be exactly represented
(C) Only (iii) and (iv) cannot be exactly represented
(D) Only (i) and (ii) cannot be exactly represented

Q.34 The size of the physical address space of a processor is 2^P bytes. The word length is 2^W bytes. The capacity of cache memory is 2^N bytes. The size of each cache block is 2^M words. For a K -way set-associative cache memory, the length (in number of bits) of the tag field is

- (A) $P - N - \log_2 K$ (B) $P - N + \log_2 K$
 (C) $P - N - M - W - \log_2 K$ (D) $P - N - M - W + \log_2 K$

Q.35 Consider the following languages:

- I. $\{a^m b^n c^p d^q \mid m + p = n + q, \text{ where } m, n, p, q \geq 0\}$
 II. $\{a^m b^n c^p d^q \mid m = n \text{ and } p = q, \text{ where } m, n, p, q \geq 0\}$
 III. $\{a^m b^n c^p d^q \mid m = n = p \text{ and } p \neq q, \text{ where } m, n, p, q \geq 0\}$
 IV. $\{a^m b^n c^p d^q \mid mn = p + q, \text{ where } m, n, p, q \geq 0\}$

Which of the languages above are context-free?

- (A) I and IV only (B) I and II only
 (C) II and III only (D) II and IV only

Q.36 Consider the following problems. $L(G)$ denotes the language generated by a grammar G . $L(M)$ denotes the language accepted by a machine M .

- (I) For an unrestricted grammar G and a string w , whether $w \in L(G)$
 (II) Given a Turing machine M , whether $L(M)$ is regular
 (III) Given two grammars G_1 and G_2 , whether $L(G_1) = L(G_2)$
 (IV) Given an NFA N , whether there is a deterministic PDA P such that N and P accept the same language.

Which one of the following statements is correct?

- (A) Only I and II are undecidable (B) Only III is undecidable
 (C) Only II and IV are undecidable (D) Only I, II and III are undecidable

- Q.37 A lexical analyzer uses the following patterns to recognize three tokens T_1 , T_2 , and T_3 over the alphabet $\{a,b,c\}$.

$$T_1: a? (b|c)^* a$$

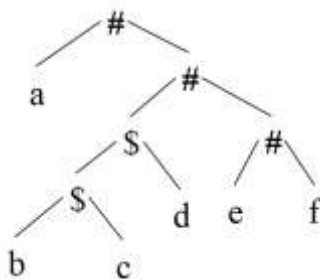
$$T_2: b? (a|c)^* b$$

$$T_3: c? (b|a)^* c$$

Note that 'x?' means 0 or 1 occurrence of the symbol x. Note also that the analyzer outputs the token that matches the longest possible prefix.

If the string *bbaacabc* is processed by the analyzer, which one of the following is the sequence of tokens it outputs?

- (A) $T_1 T_2 T_3$ (B) $T_1 T_1 T_3$ (C) $T_2 T_1 T_3$ (D) $T_3 T_3$
- Q.38 Consider the following parse tree for the expression $a\#b\$c\$d\#e\#f$, involving two binary operators $\$$ and $\#$.



Which one of the following is correct for the given parse tree?

- (A) $\$$ has higher precedence and is left associative; $\#$ is right associative
 (B) $\#$ has higher precedence and is left associative; $\$$ is right associative
 (C) $\$$ has higher precedence and is left associative; $\#$ is left associative
 (D) $\#$ has higher precedence and is right associative; $\$$ is left associative

- Q.39 In a system, there are three types of resources: E , F and G . Four processes P_0 , P_1 , P_2 and P_3 execute concurrently. At the outset, the processes have declared their maximum resource requirements using a matrix named Max as given below. For example, $\text{Max}[P_2, F]$ is the maximum number of instances of F that P_2 would require. The number of instances of the resources allocated to the various processes at any given state is given by a matrix named Allocation.

Consider a state of the system with the Allocation matrix as shown below, and in which 3 instances of E and 3 instances of F are the only resources available.

Allocation				Max			
	E	F	G		E	F	G
P_0	1	0	1	P_0	4	3	1
P_1	1	1	2	P_1	2	1	4
P_2	1	0	3	P_2	1	3	3
P_3	2	0	0	P_3	5	4	1

From the perspective of deadlock avoidance, which one of the following is true?

- (A) The system is in *safe* state.
- (B) The system is not in *safe* state, but would be *safe* if one more instance of E were available
- (C) The system is not in *safe* state, but would be *safe* if one more instance of F were available
- (D) The system is not in *safe* state, but would be *safe* if one more instance of G were available

- Q.40 Consider the following solution to the producer-consumer synchronization problem. The shared buffer size is N . Three semaphores *empty*, *full* and *mutex* are defined with respective initial values of 0, N and 1. Semaphore *empty* denotes the number of available slots in the buffer, for the consumer to read from. Semaphore *full* denotes the number of available slots in the buffer, for the producer to write to. The placeholder variables, denoted by P , Q , R , and S , in the code below can be assigned either *empty* or *full*. The valid semaphore operations are: `wait()` and `signal()`.

Producer:	Consumer:
<pre>do{ wait(P); wait(mutex); //Add item to buffer signal(mutex); signal(Q); }while(1);</pre>	<pre>do{ wait(R); wait(mutex); //Consume item from buffer signal(mutex); signal(S); }while(1);</pre>

Which one of the following assignments to P , Q , R and S will yield the correct solution?

- (A) P : *full*, Q : *full*, R : *empty*, S : *empty*
 (B) P : *empty*, Q : *empty*, R : *full*, S : *full*
 (C) P : *full*, Q : *empty*, R : *empty*, S : *full*
 (D) P : *empty*, Q : *full*, R : *full*, S : *empty*
- Q.41 Consider the relations $r(A, B)$ and $s(B, C)$, where $s.B$ is a primary key and $r.B$ is a foreign key referencing $s.B$. Consider the query

$$Q: r \bowtie (\sigma_{B < 5}(s))$$

Let *LOJ* denote the natural left outer-join operation. Assume that r and s contain no null values.

Which one of the following queries is NOT equivalent to Q ?

- (A) $\sigma_{B < 5}(r \bowtie s)$ (B) $\sigma_{B < 5}(r \text{ LOJ } s)$
 (C) $r \text{ LOJ } (\sigma_{B < 5}(s))$ (D) $\sigma_{B < 5}(r) \text{ LOJ } s$

- Q.42 Consider the following four relational schemas. For each schema, all non-trivial functional dependencies are listed. The underlined attributes are the respective primary keys.

Schema I: *Registration* (rollno, *courses*)

Field '*courses*' is a set-valued attribute containing the set of courses a student has registered for.

Non-trivial functional dependency:

rollno $\rightarrow$ *courses*

Schema II: *Registration* (rollno, courseid, *email*)

Non-trivial functional dependencies:

rollno, *courseid* $\rightarrow$ *email*

email $\rightarrow$ *rollno*

Schema III: *Registration* (rollno, courseid, *marks*, *grade*)

Non-trivial functional dependencies:

rollno, *courseid* $\rightarrow$ *marks*, *grade*

marks $\rightarrow$ *grade*

Schema IV: *Registration* (rollno, courseid, *credit*)

Non-trivial functional dependencies:

rollno, *courseid* $\rightarrow$ *credit*

courseid $\rightarrow$ *credit*

Which one of the relational schemas above is in 3NF but not in BCNF?

- (A) Schema I (B) Schema II (C) Schema III (D) Schema IV

- Q.43 Let G be a graph with $100!$ vertices, with each vertex labelled by a distinct permutation of the numbers $1, 2, \dots, 100$. There is an edge between vertices u and v if and only if the label of u can be obtained by swapping two adjacent numbers in the label of v . Let y denote the degree of a vertex in G , and z denote the number of connected components in G . Then, $y + 10z = \underline{\hspace{2cm}}$.

- Q.44 Consider Guwahati (G) and Delhi (D) whose temperatures can be classified as high (H), medium (M) and low (L). Let $P(H_G)$ denote the probability that Guwahati has high temperature. Similarly, $P(M_G)$ and $P(L_G)$ denotes the probability of Guwahati having medium and low temperatures respectively. Similarly, we use $P(H_D)$, $P(M_D)$ and $P(L_D)$ for Delhi.

The following table gives the conditional probabilities for Delhi's temperature given Guwahati's temperature.

	H_D	M_D	L_D
H_G	0.40	0.48	0.12
M_G	0.10	0.65	0.25
L_G	0.01	0.50	0.49

Consider the first row in the table above. The first entry denotes that if Guwahati has high temperature (H_G) then the probability of Delhi also having a high temperature (H_D) is 0.40; i.e., $P(H_D|H_G) = 0.40$. Similarly, the next two entries are $P(M_D|H_G) = 0.48$ and $P(L_D|H_G) = 0.12$. Similarly for the other rows.

If it is known that $P(H_G) = 0.2$, $P(M_G) = 0.5$, and $P(L_G) = 0.3$, then the probability (correct to two decimal places) that Guwahati has high temperature given that Delhi has high temperature is _____.

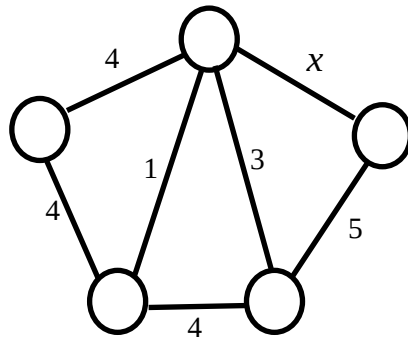
- Q.45 Consider the following program written in pseudo-code. Assume that x and y are integers.

```
Count(x, y) {
    if (y != 1) {
        if (x != 1) {
            print("*");
            Count(x/2, y);
        }
        else {
            y = y-1;
            Count(1024, y);
        }
    }
}
```

The number of times that the print statement is executed by the call $\text{Count}(1024, 1024)$ is _____.

- Q.46 The number of possible min-heaps containing each value from $\{1, 2, 3, 4, 5, 6, 7\}$ exactly once is _____.

Q.47 Consider the following undirected graph G :



Choose a value for x that will maximize the number of minimum weight spanning trees (MWSTs) of G . The number of MWSTs of G for this value of x is _____.

Q.48 Consider the weights and values of items listed below. Note that there is only one unit of each item.

Item number	Weight (in Kgs)	Value (in Rupees)
1	10	60
2	7	28
3	4	20
4	2	24

The task is to pick a subset of these items such that their total weight is no more than 11 Kgs and their total value is maximized. Moreover, no item may be split. The total value of items picked by an optimal algorithm is denoted by V_{opt} . A greedy algorithm sorts the items by their value-to-weight ratios in descending order and packs them greedily, starting from the first item in the ordered list. The total value of items picked by the greedy algorithm is denoted by V_{greedy} .

The value of $V_{\text{opt}} - V_{\text{greedy}}$ is _____.

Q.49 Consider the minterm list form of a Boolean function F given below.

$$F(P, Q, R, S) = \sum m(0, 2, 5, 7, 9, 11) + d(3, 8, 10, 12, 14)$$

Here, m denotes a minterm and d denotes a don't care term. The number of essential prime implicants of the function F is _____.

- Q.50 The instruction pipeline of a RISC processor has the following stages: Instruction Fetch (IF), Instruction Decode (ID), Operand Fetch (OF), Perform Operation (PO) and Writeback (WB). The IF, ID, OF and WB stages take 1 clock cycle each for every instruction. Consider a sequence of 100 instructions. In the PO stage, 40 instructions take 3 clock cycles each, 35 instructions take 2 clock cycles each, and the remaining 25 instructions take 1 clock cycle each. Assume that there are no data hazards and no control hazards.

The number of clock cycles required for completion of execution of the sequence of instructions is _____.

- Q.51 A processor has 16 integer registers (R0, R1, .. , R15) and 64 floating point registers (F0, F1, ... , F63). It uses a 2-byte instruction format. There are four categories of instructions: Type-1, Type-2, Type-3, and Type-4. Type-1 category consists of four instructions, each with 3 integer register operands (3Rs). Type-2 category consists of eight instructions, each with 2 floating point register operands (2Fs). Type-3 category consists of fourteen instructions, each with one integer register operand and one floating point register operand (1R+1F). Type-4 category consists of N instructions, each with a floating point register operand (1F).

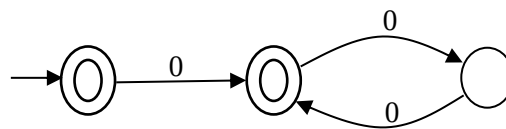
The maximum value of N is _____.

- Q.52 Given a language L , define L^i as follows:

$$L^0 = \{\epsilon\}$$

$$L^i = L^{i-1} \cdot L \text{ for all } i > 0$$

The order of a language L is defined as the smallest k such that $L^k = L^{k+1}$. Consider the language L_1 (over alphabet 0) accepted by the following automaton.



The order of L_1 is _____.

- Q.53 Consider a storage disk with 4 platters (numbered as 0, 1, 2 and 3), 200 cylinders (numbered as 0, 1, ... , 199), and 256 sectors per track (numbered as 0, 1, ... , 255). The following 6 disk requests of the form [sector number, cylinder number, platter number] are received by the disk controller at the same time:

[120, 72, 2] , [180, 134, 1] , [60, 20, 0] , [212, 86, 3] , [56, 116, 2] , [118, 16, 1]

Currently the head is positioned at sector number 100 of cylinder 80, and is moving towards higher cylinder numbers. The average power dissipation in moving the head over 100 cylinders is 20 milliwatts and for reversing the direction of the head movement once is 15 milliwatts. Power dissipation associated with rotational latency and switching of head between different platters is negligible.

The total power consumption in milliwatts to satisfy all of the above disk requests using the Shortest Seek Time First disk scheduling algorithm is _____.

- Q.54 Consider an IP packet with a length of 4,500 bytes that includes a 20-byte IPv4 header and a 40-byte TCP header. The packet is forwarded to an IPv4 router that supports a Maximum Transmission Unit (MTU) of 600 bytes. Assume that the length of the IP header in all the outgoing fragments of this packet is 20 bytes. Assume that the fragmentation offset value stored in the first fragment is 0.

The fragmentation offset value stored in the third fragment is _____.

- Q.55 Consider a simple communication system where multiple nodes are connected by a shared broadcast medium (like Ethernet or wireless). The nodes in the system use the following carrier-sense based medium access protocol. A node that receives a packet to transmit will carrier-sense the medium for 5 units of time. If the node does not detect any other transmission in this duration, it starts transmitting its packet in the next time unit. If the node detects another transmission, it waits until this other transmission finishes, and then begins to carrier-sense for 5 time units again. Once they start to transmit, nodes do not perform any collision detection and continue transmission even if a collision occurs. All transmissions last for 20 units of time. Assume that the transmission signal travels at the speed of 10 meters per unit time in the medium.

Assume that the system has two nodes P and Q, located at a distance d meters from each other. P starts transmitting a packet at time $t=0$ after successfully completing its carrier-sense phase. Node Q has a packet to transmit at time $t=0$ and begins to carrier-sense the medium.

The maximum distance d (in meters, rounded to the closest integer) that allows Q to successfully avoid a collision between its proposed transmission and P's ongoing transmission is _____.

END OF THE QUESTION PAPER

GATE 2017

Set 1

65 questions | 27 pages

Paper 13 of 16

Graduate Aptitude Test in Engineering 2017

Question Paper Name:

Computer Science and Information Technology 11th Feb 2017 Session 1

Subject Name:

Computer Science and Information Technology

Duration:

180

Total Marks:

100



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Indian Institute of Technology Roorkee



Question Number : 1**Correct : 1 Wrong : -0.33**

The statement $(\neg p) \Rightarrow (\neg q)$ is logically equivalent to which of the statements below?

- I. $p \Rightarrow q$
- II. $q \Rightarrow p$
- III. $(\neg q) \vee p$
- IV. $(\neg p) \vee q$

- (A) I only (B) I and IV only (C) II only (D) II and III only

Question Number : 2**Correct : 1 Wrong : -0.33**

Consider the first-order logic sentence $F: \forall x(\exists y R(x, y))$. Assuming non-empty logical domains, which of the sentences below are *implied* by F ?

- I. $\exists y(\exists x R(x, y))$
- II. $\exists y(\forall x R(x, y))$
- III. $\forall y(\exists x R(x, y))$
- IV. $\neg \exists x(\forall y \neg R(x, y))$

- (A) IV only (B) I and IV only (C) II only (D) II and III only

Question Number : 3**Correct : 1 Wrong : -0.33**

Let $c_1, \dots, c_n$ be scalars, not all zero, such that $\sum_{i=1}^n c_i a_i = 0$ where a_i are column vectors in $\mathbf{R}^n$.

Consider the set of linear equations

$$Ax = b$$

where $A = [a_1, \dots, a_n]$ and $b = \sum_{i=1}^n a_i$. The set of equations has

- (A) a unique solution at $x = J_n$ where J_n denotes a n -dimensional vector of all 1
- (B) no solution
- (C) infinitely many solutions
- (D) finitely many solutions

Question Number : 4**Correct : 1 Wrong : -0.33**

Consider the following functions from positive integers to real numbers:

$$10, \sqrt{n}, n, \log_2 n, \frac{100}{n}.$$

The CORRECT arrangement of the above functions in increasing order of asymptotic complexity is:

- (A) $\log_2 n, \frac{100}{n}, 10, \sqrt{n}, n$
- (B) $\frac{100}{n}, 10, \log_2 n, \sqrt{n}, n$
- (C) $10, \frac{100}{n}, \sqrt{n}, \log_2 n, n$
- (D) $\frac{100}{n}, \log_2 n, 10, \sqrt{n}, n$

Question Number : 5**Correct : 1 Wrong : -0.33**

Consider the following table:

Algorithms	Design Paradigms
(P) Kruskal	(i) Divide and Conquer
(Q) Quicksort	(ii) Greedy
(R) Floyd-Warshall	(iii) Dynamic Programming

Match the algorithms to the design paradigms they are based on.

- (A) $(P) \leftrightarrow (ii), (Q) \leftrightarrow (iii), (R) \leftrightarrow (i)$
(B) $(P) \leftrightarrow (iii), (Q) \leftrightarrow (i), (R) \leftrightarrow (ii)$
(C) $(P) \leftrightarrow (ii), (Q) \leftrightarrow (i), (R) \leftrightarrow (iii)$
(D) $(P) \leftrightarrow (i), (Q) \leftrightarrow (ii), (R) \leftrightarrow (iii)$

Question Number : 6**Correct : 1 Wrong : -0.33**

Let T be a binary search tree with 15 nodes. The minimum and maximum possible heights of T are:
Note: The height of a tree with a single node is 0.

- (A) 4 and 15 respectively
(B) 3 and 14 respectively
(C) 4 and 14 respectively
(D) 3 and 15 respectively

Question Number : 7**Correct : 1 Wrong : -0.33**

The n -bit fixed-point representation of an unsigned real number X uses f bits for the fraction part. Let $i = n - f$. The range of decimal values for X in this representation is

- (A) 2^{-f} to 2^i (B) 2^{-f} to $(2^i - 2^{-f})$ (C) 0 to 2^i (D) 0 to $(2^i - 2^{-f})$

Question Number : 8**Correct : 1 Wrong : -0.33**

Consider the C code fragment given below.

```
typedef struct node {
    int data;
    node* next;
} node;

void join(node* m, node* n) {
    node* p = n;
    while (p->next != NULL) {
        p = p->next;
    }
    p->next = m;
}
```

Assuming that m and n point to valid NULL-terminated linked lists, invocation of join will

- (A) append list m to the end of list n for all inputs.
- (B) either cause a null pointer dereference or append list m to the end of list n.
- (C) cause a null pointer dereference for all inputs.
- (D) append list n to the end of list m for all inputs.

Question Number : 9**Correct : 1 Wrong : -0.33**

When two 8-bit numbers $A_7 \cdots A_0$ and $B_7 \cdots B_0$ in 2's complement representation (with A_0 and B_0 as the least significant bits) are added using a **ripple-carry adder**, the sum bits obtained are $S_7 \cdots S_0$ and the carry bits are $C_7 \cdots C_0$. An overflow is said to have occurred if

- (A) the carry bit C_7 is 1
- (B) all the carry bits $(C_7, \cdots, C_0)$ are 1
- (C) $(A_7 \cdot B_7 \cdot \overline{S_7} + \overline{A_7} \cdot \overline{B_7} \cdot S_7)$ is 1
- (D) $(A_0 \cdot B_0 \cdot \overline{S_0} + \overline{A_0} \cdot \overline{B_0} \cdot S_0)$ is 1

Question Number : 10**Correct : 1 Wrong : -0.33**

Consider the following context-free grammar over the alphabet $\Sigma = \{a, b, c\}$ with S as the start symbol:

$$\begin{aligned} S &\rightarrow abScT \mid abcT \\ T &\rightarrow bT \mid b \end{aligned}$$

Which one of the following represents the language generated by the above grammar?

- (A) $\{(ab)^n(cb)^n \mid n \geq 1\}$
- (B) $\{(ab)^n cb^{m_1} cb^{m_2} \dots cb^{m_n} \mid n, m_1, m_2, \dots, m_n \geq 1\}$
- (C) $\{(ab)^n (cb^m)^n \mid m, n \geq 1\}$
- (D) $\{(ab)^n (cb^n)^m \mid m, n \geq 1\}$

Question Number : 11**Correct : 1 Wrong : -0.33**

Consider the C struct defined below:

```
struct data {  
    int marks [100];  
    char grade;  
    int cnumber;  
};  
struct data student;
```

The base address of `student` is available in register `R1`. The field `student.grade` can be accessed efficiently using

- (A) Post-increment addressing mode, $(R1)+$
- (B) Pre-decrement addressing mode, $-(R1)$
- (C) Register direct addressing mode, `R1`
- (D) Index addressing mode, $X(R1)$, where X is an offset represented in 2's complement 16-bit representation.

Consider the following intermediate program in three address code

$$\begin{aligned}p &= a - b \\q &= p * c \\p &= u * v \\q &= p + q\end{aligned}$$

Which one of the following corresponds to a *static single assignment* form of the above code?

(A)
$$\begin{aligned}p_1 &= a - b \\q_1 &= p_1 * c \\p_1 &= u * v \\q_1 &= p_1 + q_1\end{aligned}$$

(B)
$$\begin{aligned}p_3 &= a - b \\q_4 &= p_3 * c \\p_4 &= u * v \\q_5 &= p_4 + q_4\end{aligned}$$

(C)
$$\begin{aligned}p_1 &= a - b \\q_1 &= p_2 * c \\p_3 &= u * v \\q_2 &= p_4 + q_3\end{aligned}$$

(D)
$$\begin{aligned}p_1 &= a - b \\q_1 &= p * c \\p_2 &= u * v \\q_2 &= p + q\end{aligned}$$

Consider the following C code:

```
#include <stdio.h>
int *assignval(int *x, int val) {
    *x = val;
    return x;
}

void main () {
    int *x = malloc(sizeof(int));
    if(NULL == x) return;
    x = assignval(x,0);
    if(x) {
        x = (int *)malloc(sizeof(int));
        if(NULL == x) return;
        x = assignval(x,10);
    }
    printf("%d\n", *x);
    free(x);
}
```

The code suffers from which one of the following problems:

- (A) compiler error as the return of `malloc` is not typecast appropriately
- (B) compiler error because the comparison should be made as `x == NULL` and not as shown
- (C) compiles successfully but execution may result in dangling pointer
- (D) compiles successfully but execution may result in memory leak

Consider a TCP client and a TCP server running on two different machines. After completing data transfer, the TCP client calls `close` to terminate the connection and a FIN segment is sent to the TCP server. Server-side TCP responds by sending an ACK, which is received by the client-side TCP. As per the TCP connection state diagram (RFC 793), in which state does the client-side TCP connection wait for the FIN from the server-side TCP?

- (A) LAST-ACK
- (B) TIME-WAIT
- (C) FIN-WAIT-1
- (D) FIN-WAIT-2

Question Number : 15**Correct : 1 Wrong : -0.33**

A sender S sends a message m to receiver R, which is digitally signed by S with its private key. In this scenario, one or more of the following security violations can take place.

- (I) S can launch a birthday attack to replace m with a fraudulent message.
- (II) A third party attacker can launch a birthday attack to replace m with a fraudulent message.
- (III) R can launch a birthday attack to replace m with a fraudulent message.

Which of the following are possible security violations?

- (A) (I) and (II) only
- (B) (I) only
- (C) (II) only
- (D) (II) and (III) only

Question Number : 16**Correct : 1 Wrong : -0.33**

The following functional dependencies hold true for the relational schema $R\{V, W, X, Y, Z\}$:

$$\begin{aligned}V &\rightarrow W \\ VW &\rightarrow X \\ Y &\rightarrow VX \\ Y &\rightarrow Z\end{aligned}$$

Which of the following is irreducible equivalent for this set of functional dependencies?

- | | | | | | | | |
|-----|-------------------|-----|-------------------|-----|-------------------|-----|-------------------|
| (A) | $V \rightarrow W$ | (B) | $V \rightarrow W$ | (C) | $V \rightarrow W$ | (D) | $V \rightarrow W$ |
| | $V \rightarrow X$ | | $W \rightarrow X$ | | $V \rightarrow X$ | | $W \rightarrow X$ |
| | $Y \rightarrow V$ | | $Y \rightarrow V$ | | $Y \rightarrow V$ | | $Y \rightarrow V$ |
| | $Y \rightarrow Z$ | | $Y \rightarrow Z$ | | $Y \rightarrow X$ | | $Y \rightarrow X$ |
| | | | | | $Y \rightarrow Z$ | | $Y \rightarrow Z$ |

Question Number : 17**Correct : 1 Wrong : -0.33**

Consider the following grammar:

$\begin{aligned} P &\rightarrow xQRS \\ Q &\rightarrow yz \mid z \\ R &\rightarrow w \mid \varepsilon \\ S &\rightarrow y \end{aligned}$
--

What is FOLLOW(Q)?

- (A) $\{R\}$ (B) $\{w\}$ (C) $\{w, y\}$ (D) $\{w, \$\}$

Question Number : 18**Correct : 1 Wrong : -0.33**

Threads of a process share

- (A) global variables but not heap.
(B) heap but not global variables.
(C) neither global variables nor heap.
(D) both heap and global variables.

Question Number : 19**Correct : 1 Wrong : 0**

Let X be a Gaussian random variable with mean 0 and variance σ^2 . Let $Y = \max(X, 0)$ where $\max(a, b)$ is the maximum of a and b . The median of Y is _____.

Question Number : 20**Correct : 1 Wrong : 0**

Let T be a tree with 10 vertices. The sum of the degrees of all the vertices in T is _____.

Question Number : 21**Correct : 1 Wrong : 0**

Consider the Karnaugh map given below, where X represents “don't care” and blank represents 0.

$\begin{array}{c} \backslash \\ ba \end{array}$ dc	00	01	11	10
00		X	X	
01	1			X
11	1			1
10		X	X	

Assume for all inputs (a, b, c, d) , the respective complements $(\bar{a}, \bar{b}, \bar{c}, \bar{d})$ are also available. The above logic is implemented using 2-input NOR gates only. The minimum number of gates required is _____.

Question Number : 22**Correct : 1 Wrong : 0**

Consider the language L given by the regular expression $(a+b)^*b(a+b)$ over the alphabet $\{a, b\}$. The smallest number of states needed in a deterministic finite-state automaton (DFA) accepting L is _____.

Question Number : 23**Correct : 1 Wrong : 0**

Consider a database that has the relation schema EMP (EmpId, EmpName, and DeptName). An instance of the schema EMP and a SQL query on it are given below.

EMP		
EmpId	EmpName	DeptName
1	XYA	AA
2	XYB	AA
3	XYC	AA
4	XYD	AA
5	XYE	AB
6	XYF	AB
7	XYG	AB
8	XYH	AC
9	XYI	AC
10	XYJ	AC
11	XYK	AD
12	XYL	AD
13	XYM	AE

```
SELECT AVG(EC.Num)
FROM EC
WHERE (DeptName, Num) IN
      (SELECT DeptName, COUNT(EmpId) AS
        EC(DeptName, Num)
       FROM EMP
        GROUP BY DeptName)
```

The output of executing the SQL query is _____.

Question Number : 24**Correct : 1 Wrong : 0**

Consider the following CPU processes with arrival times (in milliseconds) and length of CPU bursts (in milliseconds) as given below :

Process	Arrival time	Burst time
P1	0	7
P2	3	3
P3	5	5
P4	6	2

If the pre-emptive shortest remaining time first scheduling algorithm is used to schedule the processes, then the average waiting time across all processes is _____ milliseconds.

Question Number : 25**Correct : 1 Wrong : 0**

Consider a two-level cache hierarchy with L1 and L2 caches. An application incurs 1.4 memory accesses per instruction on average. For this application, the miss rate of L1 cache is 0.1; the L2 cache experiences, on average, 7 misses per 1000 instructions. The miss rate of L2 expressed correct to two decimal places is _____.

Question Number : 26**Correct : 2 Wrong : -0.66**

Let $G = (V, E)$ be any connected undirected edge-weighted graph. The weights of the edges in E are positive and distinct. Consider the following statements:

- (I) Minimum Spanning Tree of G is always unique.
- (II) Shortest path between any two vertices of G is always unique.

Which of the above statements is/are necessarily true?

- (A) (I) only
- (B) (II) only
- (C) both (I) and (II)
- (D) neither (I) nor (II)

Question Number : 27**Correct : 2 Wrong : -0.66**

A multithreaded program P executes with x number of threads and uses y number of locks for ensuring mutual exclusion while operating on shared memory locations. All locks in the program are *non-reentrant*, i.e., if a thread holds a lock l , then it cannot re-acquire lock l without releasing it. If a thread is unable to acquire a lock, it blocks until the lock becomes available. The *minimum* value of x and the *minimum* value of y together for which execution of P can result in a deadlock are:

- | | |
|--------------------|--------------------|
| (A) $x = 1, y = 2$ | (B) $x = 2, y = 1$ |
| (C) $x = 2, y = 2$ | (D) $x = 1, y = 1$ |

Question Number : 28**Correct : 2 Wrong : -0.66**

The value of $\lim_{x \rightarrow 1} \frac{x^7 - 2x^5 + 1}{x^3 - 3x^2 + 2}$

- (A) is 0
- (B) is -1
- (C) is 1
- (D) does not exist

Question Number : 29**Correct : 2 Wrong : -0.66**

Let p , q , and r be propositions and the expression $(p \rightarrow q) \rightarrow r$ be a contradiction. Then, the expression $(r \rightarrow p) \rightarrow q$ is

- (A) a tautology.
- (B) a contradiction.
- (C) always TRUE when p is FALSE.
- (D) always TRUE when q is TRUE.

Question Number : 30**Correct : 2 Wrong : -0.66**

Let u and v be two vectors in $\mathbf{R}^2$ whose Euclidean norms satisfy $\|u\| = 2\|v\|$. What is the value of α such that $w = u + \alpha v$ bisects the angle between u and v ?

- (A) 2
- (B) 1/2
- (C) 1
- (D) -1/2

Question Number : 31**Correct : 2 Wrong : -0.66**

Let A be $n \times n$ real valued square symmetric matrix of rank 2 with $\sum_{i=1}^n \sum_{j=1}^n A_{ij}^2 = 50$. Consider the following statements.

- (I) One eigenvalue must be in $[-5, 5]$
- (II) The eigenvalue with the largest magnitude must be strictly greater than 5

Which of the above statements about eigenvalues of A is/are necessarily CORRECT?

- (A) Both (I) and (II)
- (B) (I) only
- (C) (II) only
- (D) Neither (I) nor (II)

Question Number : 32**Correct : 2 Wrong : -0.66**

A computer network uses polynomials over $GF(2)$ for error checking with 8 bits as information bits and uses $x^3 + x + 1$ as the generator polynomial to generate the check bits. In this network, the message 01011011 is transmitted as

(A) 01011011010

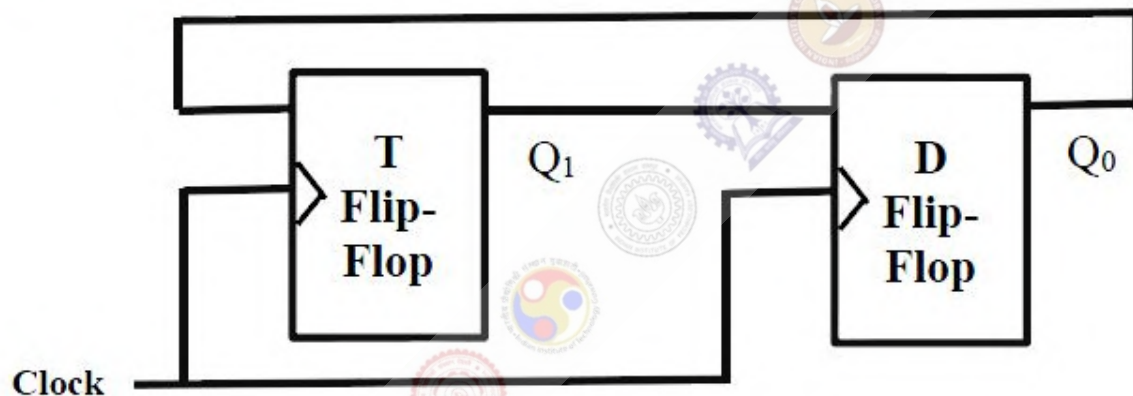
(B) 01011011011

(C) 01011011101

(D) 01011011100

Question Number : 33**Correct : 2 Wrong : -0.66**

Consider a combination of T and D flip-flops connected as shown below. The output of the D flip-flop is connected to the input of the T flip-flop and the output of the T flip-flop is connected to the input of the D flip-flop.



Initially, both Q_0 and Q_1 are set to 1 (before the 1st clock cycle). The outputs

- (A) $Q_1 Q_0$ after the 3rd cycle are 11 and after the 4th cycle are 00 respectively
- (B) $Q_1 Q_0$ after the 3rd cycle are 11 and after the 4th cycle are 01 respectively
- (C) $Q_1 Q_0$ after the 3rd cycle are 00 and after the 4th cycle are 11 respectively
- (D) $Q_1 Q_0$ after the 3rd cycle are 01 and after the 4th cycle are 01 respectively

Question Number : 34**Correct : 2 Wrong : -0.66**

If G is a grammar with productions

$$S \rightarrow SaS \mid aSb \mid bSa \mid SS \mid \epsilon$$

where S is the start variable, then which one of the following strings is not generated by G ?

(A) $abab$ (B) $aaab$ (C) $abbaa$ (D) $babba$

Question Number : 35**Correct : 2 Wrong : -0.66**

Consider the following two functions.

```
void fun1(int n) {  
    if(n == 0) return;  
    printf("%d", n);  
    fun2(n - 2);  
    printf("%d", n);  
}
```

```
void fun2(int n) {  
    if(n == 0) return;  
    printf("%d", n);  
    fun1(++n);  
    printf("%d", n);  
}
```

The output printed when fun1 (5) is called is

- (A) 53423122233445
(C) 53423122132435

- (B) 53423120112233
(D) 53423120213243

Question Number : 36**Correct : 2 Wrong : -0.66**

Consider the C functions foo and bar given below:

```
int foo(int val) {  
    int x = 0;  
    while(val > 0) {  
        x = x + foo(val--);  
    }  
    return val;  
}
```

```
int bar(int val) {  
    int x = 0;  
    while(val > 0) {  
        x = x + bar(val-1);  
    }  
    return val;  
}
```

Invocations of foo (3) and bar (3) will result in :

- (A) Return of 6 and 6 respectively.
(B) Infinite loop and abnormal termination respectively.
(C) Abnormal termination and infinite loop respectively.
(D) Both terminating abnormally.

Question Number : 37**Correct : 2 Wrong : -0.66**

Consider the context-free grammars over the alphabet $\{a, b, c\}$ given below. S and T are non-terminals.

$$G_1: S \rightarrow aSb | T, T \rightarrow cT | \epsilon$$

$$G_2: S \rightarrow bSa | T, T \rightarrow cT | \epsilon$$

The language $L(G_1) \cap L(G_2)$ is

- (A) Finite. (B) Not finite but regular.
(C) Context-Free but not regular. (D) Recursive but not context-free.

Question Number : 38**Correct : 2 Wrong : -0.66**

Consider the following languages over the alphabet $\Sigma = \{a, b, c\}$.
Let $L_1 = \{a^n b^n c^m \mid m, n \geq 0\}$ and $L_2 = \{a^m b^n c^n \mid m, n \geq 0\}$.

Which of the following are context-free languages?

- I. $L_1 \cup L_2$
II. $L_1 \cap L_2$

- (A) I only (B) II only
(C) I and II (D) Neither I nor II

Question Number : 39**Correct : 2 Wrong : -0.66**

Let A and B be finite alphabets and let $\#$ be a symbol outside both A and B . Let f be a total function from A^* to B^* . We say f is *computable* if there exists a Turing machine M which given an input x in A^* , always halts with $f(x)$ on its tape. Let L_f denote the language $\{x\#f(x) \mid x \in A^*\}$. Which of the following statements is true:

- (A) f is computable if and only if L_f is recursive.
(B) f is computable if and only if L_f is recursively enumerable.
(C) If f is computable then L_f is recursive, but not conversely.
(D) If f is computable then L_f is recursively enumerable, but not conversely.

Question Number : 40**Correct : 2 Wrong : -0.66**

Recall that Belady's anomaly is that the page-fault rate may *increase* as the number of allocated frames increases. Now, consider the following statements:

S1: *Random page replacement* algorithm (where a page chosen at random is replaced) suffers from Belady's anomaly

S2: *LRU page replacement* algorithm suffers from Belady's anomaly

Which of the following is CORRECT?

(A) S1 is true, S2 is true

(B) S1 is true, S2 is false

(C) S1 is false, S2 is true

(D) S1 is false, S2 is false

Question Number : 41**Correct : 2 Wrong : -0.66**

Consider a database that has the relation schemas EMP(EmpId, EmpName, DeptId), and DEPT(DeptName, DeptId). Note that the DeptId can be permitted to be NULL in the relation EMP. Consider the following queries on the database expressed in tuple relational calculus.

(I) $\{t \mid \exists u \in \text{EMP}(t[\text{EmpName}] = u[\text{EmpName}] \wedge \forall v \in \text{DEPT}(t[\text{DeptId}] \neq v[\text{DeptId}]))\}$

(II) $\{t \mid \exists u \in \text{EMP}(t[\text{EmpName}] = u[\text{EmpName}] \wedge \exists v \in \text{DEPT}(t[\text{DeptId}] \neq v[\text{DeptId}]))\}$

(III) $\{t \mid \exists u \in \text{EMP}(t[\text{EmpName}] = u[\text{EmpName}] \wedge \exists v \in \text{DEPT}(t[\text{DeptId}] = v[\text{DeptId}]))\}$

Which of the above queries are safe?

(A) (I) and (II) only

(B) (I) and (III) only

(C) (II) and (III) only

(D) (I), (II) and (III)

Question Number : 42**Correct : 2 Wrong : -0.66**

In a database system, unique timestamps are assigned to each transaction using Lamport's logical clock. Let $TS(T_1)$ and $TS(T_2)$ be the timestamps of transactions T_1 and T_2 respectively. Besides, T_1 holds a lock on the resource R , and T_2 has requested a conflicting lock on the same resource R . The following algorithm is used to prevent deadlocks in the database system assuming that a killed transaction is restarted with the same timestamp.

```
if  $TS(T_2) < TS(T_1)$  then  
     $T_1$  is killed  
else  $T_2$  waits.
```

Assume any transaction that is not killed terminates eventually. Which of the following is TRUE about the database system that uses the above algorithm to prevent deadlocks?

- (A) The database system is both deadlock-free and starvation-free.
- (B) The database system is deadlock-free, but not starvation-free.
- (C) The database system is starvation-free, but not deadlock-free.
- (D) The database system is neither deadlock-free nor starvation-free.

Question Number : 43**Correct : 2 Wrong : 0**

Consider the following grammar:

```
stmt    -> if expr then expr else expr; stmt |    
expr    -> term relop term | term  
term    -> id | number  
id       -> a | b | c  
number  -> [0-9]
```

where **relop** is a relational operator (e.g., $<$, $>$, ...), ϵ refers to the empty statement, and **if**, **then**, **else** are terminals.

Consider a program P following the above grammar containing ten **if** terminals. The number of control flow paths in P is _____. For example, the program

if e_1 **then** e_2 **else** e_3

has 2 control flow paths, $e_1 \rightarrow e_2$ and $e_1 \rightarrow e_3$.

Question Number : 44

Correct : 2 Wrong : 0

In a RSA cryptosystem, a participant A uses two prime numbers $p = 13$ and $q = 17$ to generate her public and private keys. If the public key of A is 35, then the private key of A is _____.

Question Number : 45

Correct : 2 Wrong : 0

The values of parameters for the Stop-and-Wait ARQ protocol are as given below:

- Bit rate of the transmission channel = 1 Mbps.
- Propagation delay from sender to receiver = 0.75 ms.
- Time to process a frame = 0.25 ms.
- Number of bytes in the information frame = 1980.
- Number of bytes in the acknowledge frame = 20.
- Number of overhead bytes in the information frame = 20.

Assume that there are no transmission errors. Then, the transmission efficiency (expressed in percentage) of the Stop-and-Wait ARQ protocol for the above parameters is _____ (correct to 2 decimal places).

Question Number : 46**Correct : 2 Wrong : 0**

Consider a database that has the relation schema CR(StudentName, CourseName). An instance of the schema CR is as given below.

CR	
StudentName	CourseName
SA	CA
SA	CB
SA	CC
SB	CB
SB	CC
SC	CA
SC	CB
SC	CC
SD	CA
SD	CB
SD	CC
SD	CD
SE	CD
SE	CA
SE	CB
SF	CA
SF	CB
SF	CC

The following query is made on the database.

$$T1 \leftarrow \pi_{CourseName}(\sigma_{StudentName='SA'}(CR))$$

$$T2 \leftarrow CR \div T1$$

The number of rows in $T2$ is _____.

Question Number : 47**Correct : 2 Wrong : 0**

The number of integers between 1 and 500 (both inclusive) that are divisible by 3 or 5 or 7 is _____.

Question Number : 48**Correct : 2 Wrong : 0**

Let A be an array of 31 numbers consisting of a sequence of 0's followed by a sequence of 1's. The problem is to find the smallest index i such that $A[i]$ is 1 by probing the minimum number of locations in A . The *worst case* number of probes performed by an *optimal* algorithm is _____.

Question Number : 49**Correct : 2 Wrong : 0**

Consider a RISC machine where each instruction is exactly 4 bytes long. Conditional and unconditional branch instructions use PC-relative addressing mode with Offset specified in bytes to the target location of the branch instruction. Further the Offset is always with respect to the address of the next instruction in the program sequence. Consider the following instruction sequence

<u>Instr. No.</u>	<u>Instruction</u>
i :	add R2, R3, R4
i+1 :	sub R5, R6, R7
i+2 :	cmp R1, R9, R10
i+3 :	beq R1, Offset

If the target of the branch instruction is i, then the decimal value of the Offset is _____.

Question Number : 50**Correct : 2 Wrong : 0**

Instruction execution in a processor is divided into 5 stages, *Instruction Fetch* (IF), *Instruction Decode* (ID), *Operand Fetch* (OF), *Execute* (EX), and *Write Back* (WB). These stages take **5, 4, 20, 10, and 3 nanoseconds (ns)** respectively. A pipelined implementation of the processor requires buffering between each pair of consecutive stages with a delay of **2 ns**. Two pipelined implementations of the processor are contemplated:

- (i) a naive pipeline implementation (NP) with 5 stages and
- (ii) an efficient pipeline (EP) where the OF stage is divided into stages OF1 and OF2 with execution times of **12 ns** and **8 ns** respectively.

The speedup (correct to two decimal places) achieved by EP over NP in executing 20 independent instructions with no hazards is _____.

Question Number : 51**Correct : 2 Wrong : 0**

Consider a 2-way set associative cache with 256 blocks and uses LRU replacement. Initially the cache is empty. Conflict misses are those misses which occur due to contention of multiple blocks for the same cache set. Compulsory misses occur due to first time access to the block. The following sequence of accesses to memory blocks

(0, 128, 256, 128, 0, 128, 256, 128, 1, 129, 257, 129, 1, 129, 257, 129)

is repeated 10 times. The number of *conflict misses* experienced by the cache is _____.

Question Number : 52**Correct : 2 Wrong : 0**

Consider the expression $(a-1)*(((b+c)/3)+d)$. Let X be the minimum number of registers required by an *optimal* code generation (without any register spill) algorithm for a load/store architecture, in which (i) *only load and store instructions can have memory operands* and (ii) *arithmetic instructions can have only register or immediate operands*. The value of X is _____.

Question Number : 53**Correct : 2 Wrong : 0**

Consider the following C program.

```
#include <stdio.h>
#include <string.h>

void printlength(char *s, char *t) {
    unsigned int c = 0;
    int len = ((strlen(s) - strlen(t)) > c) ? strlen(s) : strlen(t);
    printf("%d\n", len);
}

void main() {
    char *x = "abc";
    char *y = "defgh";
    printlength(x, y);
}
```

Recall that `strlen` is defined in `string.h` as returning a value of type `size_t`, which is an unsigned int. The output of the program is _____.

Question Number : 54**Correct : 2 Wrong : 0**

A cache memory unit with capacity of N words and block size of B words is to be designed. If it is designed as a direct mapped cache, the length of the TAG field is 10 bits. If the cache unit is now designed as a 16-way set-associative cache, the length of the TAG field is _____ bits.

Question Number : 55**Correct : 2 Wrong : 0**

The output of executing the following C program is _____ .

```
#include <stdio.h>

int total(int v) {
    static int count = 0;
    while(v) {
        count += v&1;
        v >>= 1;
    }
    return count;
}

void main() {
    static int x = 0;
    int i = 5;
    for(; i > 0; i--) {
        x = x + total(i);
    }
    printf("%d\n", x);
}
```

Question Number : 56**Correct : 1 Wrong : -0.33**

After Rajendra Chola returned from his voyage to Indonesia, he _____ to visit the temple in Thanjavur.

- (A) was wishing (B) is wishing (C) wished (D) had wished

Question Number : 57**Correct : 1 Wrong : -0.33**

Research in the workplace reveals that people work for many reasons _____.

- (A) money beside (B) beside money (C) money besides (D) besides money

Question Number : 58**Correct : 1 Wrong : -0.33**

Rahul, Murali, Srinivas and Arul are seated around a square table. Rahul is sitting to the left of Murali. Srinivas is sitting to the right of Arul. Which of the following pairs are seated opposite each other?

- (A) Rahul and Murali (B) Srinivas and Arul
(C) Srinivas and Murali (D) Srinivas and Rahul

Question Number : 59**Correct : 1 Wrong : -0.33**

Find the smallest number y such that $y \times 162$ is a perfect cube.

- (A) 24 (B) 27 (C) 32 (D) 36

Question Number : 60**Correct : 1 Wrong : -0.33**

The probability that a k -digit number does NOT contain the digits 0, 5, or 9 is

- (A) 0.3^k (B) 0.6^k (C) 0.7^k (D) 0.9^k

Question Number : 61**Correct : 2 Wrong : -0.66**

“The hold of the nationalist imagination on our colonial past is such that anything inadequately or improperly nationalist is just not history.”

Which of the following statements best reflects the author’s opinion?

- (A) Nationalists are highly imaginative.
- (B) History is viewed through the filter of nationalism.
- (C) Our colonial past never happened.
- (D) Nationalism has to be both adequately and properly imagined.

Question Number : 62**Correct : 2 Wrong : -0.66**

Six people are seated around a circular table. There are at least two men and two women. There are at least three right-handed persons. Every woman has a left-handed person to her immediate right. None of the women are right-handed. The number of women at the table is

- (A) 2
- (B) 3
- (C) 4
- (D) Cannot be determined

Question Number : 63**Correct : 2 Wrong : -0.66**

The expression $\frac{(x+y)-|x-y|}{2}$ is equal to

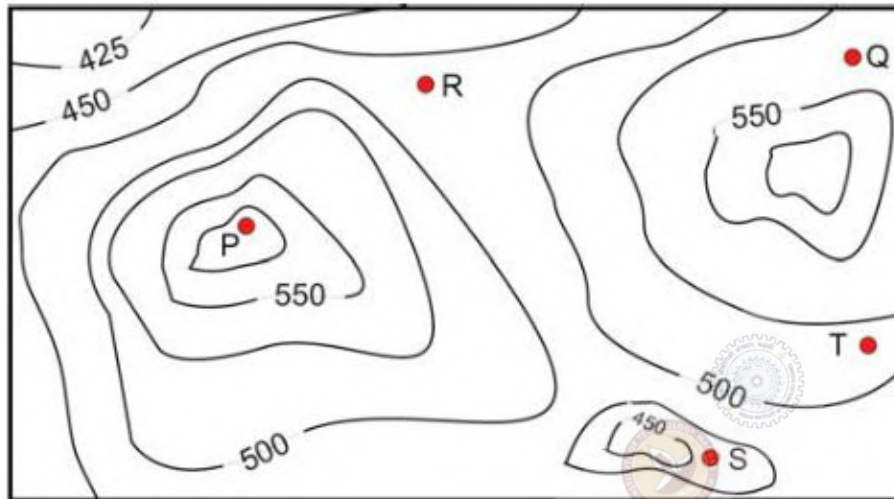
- (A) the maximum of x and y
- (B) the minimum of x and y
- (C) 1
- (D) none of the above

Question Number : 64**Correct : 2 Wrong : -0.66**

Arun, Gulab, Neel and Shweta must choose one shirt each from a pile of four shirts coloured red, pink, blue and white respectively. Arun dislikes the colour red and Shweta dislikes the colour white. Gulab and Neel like all the colours. In how many different ways can they choose the shirts so that no one has a shirt with a colour he or she dislikes?

- (A) 21
- (B) 18
- (C) 16
- (D) 14

A contour line joins locations having the same height above the mean sea level. The following is a contour plot of a geographical region. Contour lines are shown at 25 m intervals in this plot. If in a flood, the water level rises to 525 m, which of the villages P, Q, R, S, T get submerged?



(A) P, Q

(B) P, Q, T

(C) R, S, T

(D) Q, R, S

GATE 2017

Set 2

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Paper 14 of 16

Graduate Aptitude Test in Engineering 2017

Question Paper Name:

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Question Number : 1**Correct : 1 Wrong : -0.33**

The representation of the value of a 16-bit unsigned integer X in hexadecimal number system is BCA9. The representation of the value of X in octal number system is

- (A) 571244 (B) 736251 (C) 571247 (D) 136251

Question Number : 2**Correct : 1 Wrong : -0.33**

Match the following:

(P) <code>static char var;</code>	(i) Sequence of memory locations to store addresses
(Q) <code>m = malloc(10);</code> <code>m = NULL;</code>	(ii) A variable located in data section of memory
(R) <code>char *ptr[10];</code>	(iii) Request to allocate a CPU register to store data
(S) <code>register int var1;</code>	(iv) A lost memory which cannot be freed

- (A) $P \rightarrow (ii), Q \rightarrow (iv), R \rightarrow (i), S \rightarrow (iii)$ (B) $P \rightarrow (ii), Q \rightarrow (i), R \rightarrow (iv), S \rightarrow (iii)$
 (C) $P \rightarrow (ii), Q \rightarrow (iv), R \rightarrow (iii), S \rightarrow (i)$ (D) $P \rightarrow (iii), Q \rightarrow (iv), R \rightarrow (i), S \rightarrow (ii)$

Question Number : 3**Correct : 1 Wrong : -0.33**

Match the algorithms with their time complexities:

<u>Algorithm</u>	<u>Time complexity</u>
(P) Towers of Hanoi with n disks	(i) $\Theta(n^2)$
(Q) Binary search given n sorted numbers	(ii) $\Theta(n \log n)$
(R) Heap sort given n numbers at the worst case	(iii) $\Theta(2^n)$
(S) Addition of two $n \times n$ matrices	(iv) $\Theta(\log n)$

- (A) $P \rightarrow (iii), Q \rightarrow (iv), R \rightarrow (i), S \rightarrow (ii)$
 (B) $P \rightarrow (iv), Q \rightarrow (iii), R \rightarrow (i), S \rightarrow (ii)$
 (C) $P \rightarrow (iii), Q \rightarrow (iv), R \rightarrow (ii), S \rightarrow (i)$
 (D) $P \rightarrow (iv), Q \rightarrow (iii), R \rightarrow (ii), S \rightarrow (i)$

Question Number : 4**Correct : 1 Wrong : -0.33**

Let L_1, L_2 be any two context-free languages and R be any regular language. Then which of the following is/are CORRECT?

- I. $L_1 \cup L_2$ is context-free.
- II. $\overline{L_1}$ is context-free.
- III. $L_1 - R$ is context-free.
- IV. $L_1 \cap L_2$ is context-free.

(A) I, II and IV only (B) I and III only (C) II and IV only (D) I only

Question Number : 5**Correct : 1 Wrong : -0.33**

Match the following according to input (from the left column) to the compiler phase (in the right column) that processes it:

(P) Syntax tree	(i) Code generator
(Q) Character stream	(ii) Syntax analyzer
(R) Intermediate representation	(iii) Semantic analyzer
(S) Token stream	(iv) Lexical analyzer

- (A) $P \rightarrow (ii), Q \rightarrow (iii), R \rightarrow (iv), S \rightarrow (i)$
- (B) $P \rightarrow (ii), Q \rightarrow (i), R \rightarrow (iii), S \rightarrow (iv)$
- (C) $P \rightarrow (iii), Q \rightarrow (iv), R \rightarrow (i), S \rightarrow (ii)$
- (D) $P \rightarrow (i), Q \rightarrow (iv), R \rightarrow (ii), S \rightarrow (iii)$

Question Number : 6**Correct : 1 Wrong : -0.33**

Which of the following statements about parser is/are CORRECT?

- I. Canonical LR is more powerful than SLR.
- II. SLR is more powerful than LALR.
- III. SLR is more powerful than Canonical LR.

(A) I only (B) II only (C) III only (D) II and III only

Question Number : 7

Correct : 1 Wrong : -0.33

Which of the following is/are shared by all the threads in a process?

- I. Program counter
- II. Stack
- III. Address space
- IV. Registers

(A) I and II only (B) III only (C) IV only (D) III and IV only

Question Number : 8

Correct : 1 Wrong : -0.33

In a file allocation system, which of the following allocation scheme(s) can be used if no external fragmentation is allowed?

- I. Contiguous
- II. Linked
- III. Indexed

(A) I and III only (B) II only (C) III only (D) II and III only

Question Number : 9

Correct : 1 Wrong : -0.33

Consider the following statements about the routing protocols, Routing Information Protocol (RIP) and Open Shortest Path First (OSPF) in an IPv4 network.

- I: RIP uses distance vector routing
- II: RIP packets are sent using UDP
- III: OSPF packets are sent using TCP
- IV: OSPF operation is based on link-state routing

Which of the statements above are CORRECT?

- (A) I and IV only
- (B) I, II and III only
- (C) I, II and IV only
- (D) II, III and IV only

Question Number : 10**Correct : 1 Wrong : -0.33**

If $f(x) = R \sin\left(\frac{\pi x}{2}\right) + S$, $f'\left(\frac{1}{2}\right) = \sqrt{2}$ and $\int_0^1 f(x) dx = \frac{2R}{\pi}$, then the constants R and S are, respectively

- (A) $\frac{2}{\pi}$ and $\frac{16}{\pi}$ (B) $\frac{2}{\pi}$ and 0
(C) $\frac{4}{\pi}$ and 0 (D) $\frac{4}{\pi}$ and $\frac{16}{\pi}$

Question Number : 11**Correct : 1 Wrong : -0.33**

Let p, q, r denote the statements “It is raining”, “It is cold”, and “It is pleasant”, respectively. Then the statement “It is not raining and it is pleasant, and it is not pleasant only if it is raining and it is cold” is represented by

- (A) $(\neg p \wedge r) \wedge (\neg r \rightarrow (p \wedge q))$ (B) $(\neg p \wedge r) \wedge ((p \wedge q) \rightarrow \neg r)$
(C) $(\neg p \wedge r) \vee ((p \wedge q) \rightarrow \neg r)$ (D) $(\neg p \wedge r) \vee (r \rightarrow (p \wedge q))$

Question Number : 12**Correct : 1 Wrong : -0.33**

Given the following binary number in 32-bit (single precision) IEEE-754 format:

00111110011011010000000000000000

The decimal value closest to this floating-point number is

- (A) 1.45×10^1 (B) 1.45×10^{-1} (C) 2.27×10^{-1} (D) 2.27×10^1

Question Number : 13**Correct : 1 Wrong : -0.33**

A circular queue has been implemented using a singly linked list where each node consists of a value and a single pointer pointing to the next node. We maintain exactly two external pointers **FRONT** and **REAR** pointing to the front node and the rear node of the queue, respectively. Which of the following statements is/are CORRECT for such a circular queue, so that insertion and deletion operations can be performed in $O(1)$ time?

- I. Next pointer of front node points to the rear node.
II. Next pointer of rear node points to the front node.

- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

Question Number : 14**Correct : 1 Wrong : -0.33**

Consider the following function implemented in C:

```
void printxy(int x, int y) {  
    int *ptr;  
    x = 0;  
    ptr = &x;  
    y = *ptr;  
    *ptr = 1;  
    printf("%d, %d", x, y);  
}
```

The output of invoking `printxy(1, 1)` is

(A) 0,0

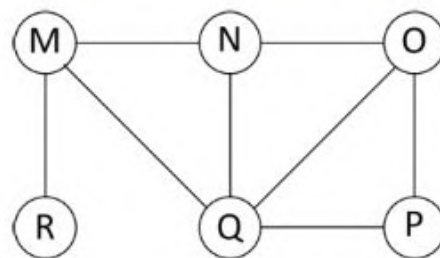
(B) 0,1

(C) 1,0

(D) 1,1

Question Number : 15**Correct : 1 Wrong : -0.33**

The Breadth First Search (BFS) algorithm has been implemented using the queue data structure. Which one of the following is a possible order of visiting the nodes in the graph below?



(A) MNOPQR

(B) NQMPOR

(C) QMNROP

(D) POQNMR

Question Number : 16**Correct : 1 Wrong : -0.33**

Identify the language generated by the following grammar, where S is the start variable.

$$\begin{aligned} S &\rightarrow XY \\ X &\rightarrow aX \mid a \\ Y &\rightarrow aYb \mid \epsilon \end{aligned}$$

- (A) $\{a^m b^n \mid m \geq n, n > 0\}$
(C) $\{a^m b^n \mid m > n, n \geq 0\}$

- (B) $\{a^m b^n \mid m \geq n, n \geq 0\}$
(D) $\{a^m b^n \mid m > n, n > 0\}$

Question Number : 17**Correct : 1 Wrong : -0.33**

An ER model of a database consists of entity types A and B. These are connected by a relationship R which does not have its own attribute. Under which one of the following conditions, can the relational table for R be merged with that of A?

- (A) Relationship R is one-to-many and the participation of A in R is total.
(B) Relationship R is one-to-many and the participation of A in R is partial.
(C) Relationship R is many-to-one and the participation of A in R is total.
(D) Relationship R is many-to-one and the participation of A in R is partial.

Question Number : 18**Correct : 1 Wrong : -0.33**

Consider socket API on a Linux machine that supports connected UDP sockets. A connected UDP socket is a UDP socket on which **connect** function has already been called. Which of the following statements is/are CORRECT?

- I. A connected UDP socket can be used to communicate with multiple peers simultaneously.
II. A process can successfully call **connect** function again for an already connected UDP socket.
- (A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

Question Number : 19**Correct : 1 Wrong : 0**

Consider the following tables T1 and T2.

T1		T2	
P	Q	R	S
2	2	2	2
3	8	8	3
7	3	3	2
5	8	9	7
6	9	5	7
8	5	7	2
9	8		

In table T1, **P** is the primary key and **Q** is the foreign key referencing **R** in table T2 with on-delete cascade and on-update cascade. In table T2, **R** is the primary key and **S** is the foreign key referencing **P** in table T1 with on-delete set NULL and on-update cascade. In order to delete record (3,8) from table T1, the number of additional records that need to be deleted from table T1 is _____.

Question Number : 20**Correct : 1 Wrong : 0**

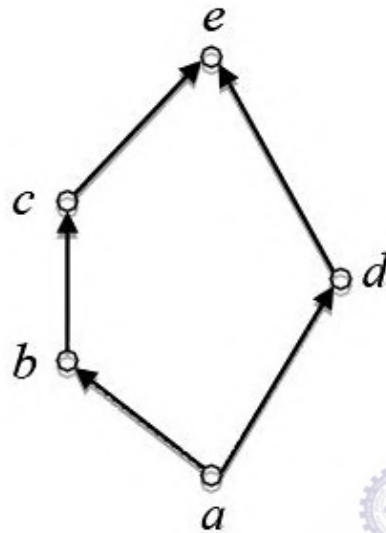
The maximum number of IPv4 router addresses that can be listed in the record route (RR) option field of an IPv4 header is _____.

Question Number : 21**Correct : 1 Wrong : 0**

Consider the set $X = \{a, b, c, d, e\}$ under the partial ordering

$$R = \{(a, a), (a, b), (a, c), (a, d), (a, e), (b, b), (b, c), (b, e), (c, c), (c, e), (d, d), (d, e), (e, e)\}.$$

The Hasse diagram of the partial order (X, R) is shown below.



The minimum number of ordered pairs that need to be added to R to make (X, R) a lattice is _____.

Question Number : 22**Correct : 1 Wrong : 0**

Let $P = \begin{bmatrix} 1 & 1 & -1 \\ 2 & -3 & 4 \\ 3 & -2 & 3 \end{bmatrix}$ and $Q = \begin{bmatrix} -1 & -2 & -1 \\ 6 & 12 & 6 \\ 5 & 10 & 5 \end{bmatrix}$ be two matrices.

Then the rank of $P + Q$ is _____.

Question Number : 23**Correct : 1 Wrong : 0**

G is an undirected graph with n vertices and 25 edges such that each vertex of G has degree at least 3. Then the maximum possible value of n is _____.

Question Number : 24**Correct : 1 Wrong : 0**

Consider a quadratic equation $x^2 - 13x + 36 = 0$ with coefficients in a base b . The solutions of this equation in the same base b are $x = 5$ and $x = 6$. Then $b =$ _____.

Question Number : 25**Correct : 1 Wrong : 0**

The minimum possible number of states of a deterministic finite automaton that accepts the regular language $L = \{w_1aw_2 \mid w_1, w_2 \in \{a, b\}^*, |w_1| = 2, |w_2| \geq 3\}$ is _____.

Question Number : 26**Correct : 2 Wrong : -0.66**

P and Q are considering to apply for a job. The probability that P applies for the job is $\frac{1}{4}$, the probability that P applies for the job given that Q applies for the job is $\frac{1}{2}$, and the probability that Q applies for the job given that P applies for the job is $\frac{1}{3}$. Then the probability that P does not apply for the job given that Q does not apply for the job is

- (A) $\frac{4}{5}$ (B) $\frac{5}{6}$ (C) $\frac{7}{8}$ (D) $\frac{11}{12}$

Question Number : 27**Correct : 2 Wrong : -0.66**

If w, x, y, z are Boolean variables, then which one of the following is INCORRECT?

- (A) $wx + w(x + y) + x(x + y) = x + wy$
(B) $w\bar{x}(y + \bar{z}) + \bar{w}x = \bar{w} + x + \bar{y}z$
(C) $(w\bar{x}(y + x\bar{z}) + \bar{w}\bar{x})y = x\bar{y}$
(D) $(w + y)(wxy + wyz) = wxy + wyz$

Question Number : 28**Correct : 2 Wrong : -0.66**

Given $f(w, x, y, z) = \sum_m(0,1,2,3,7,8,10) + \sum_d(5,6,11,15)$, where d represents the *don't-care* condition in Karnaugh maps. Which of the following is a minimum product-of-sums (POS) form of $f(w, x, y, z)$?

(A) $f = (\bar{w} + \bar{z})(\bar{x} + z)$

(B) $f = (\bar{w} + z)(x + z)$

(C) $f = (w + z)(\bar{x} + z)$

(D) $f = (w + \bar{z})(\bar{x} + z)$

Question Number : 29**Correct : 2 Wrong : -0.66**

In a two-level cache system, the access times of L_1 and L_2 caches are 1 and 8 clock cycles, respectively. The miss penalty from the L_2 cache to main memory is 18 clock cycles. The miss rate of L_1 cache is twice that of L_2 . The average memory access time (AMAT) of this cache system is 2 cycles. The miss rates of L_1 and L_2 respectively are:

(A) 0.111 and 0.056

(B) 0.056 and 0.111

(C) 0.0892 and 0.1784

(D) 0.1784 and 0.0892

Question Number : 30**Correct : 2 Wrong : -0.66**

Consider the recurrence function

$$T(n) = \begin{cases} 2T(\sqrt{n}) + 1, & n > 2 \\ 2, & 0 < n \leq 2 \end{cases}$$

Then $T(n)$ in terms of Θ notation is

(A) $\Theta(\log \log n)$

(B) $\Theta(\log n)$

(C) $\Theta(\sqrt{n})$

(D) $\Theta(n)$

Question Number : 31**Correct : 2 Wrong : -0.66**

For any discrete random variable X , with probability mass function

$P(X = j) = p_j, p_j \geq 0, j \in \{0, \dots, N\}$, and $\sum_{j=0}^N p_j = 1$, define the polynomial function

$g_X(z) = \sum_{j=0}^N p_j z^j$. For a certain discrete random variable Y , there exists a scalar $\beta \in [0, 1]$ such

that $g_Y(z) = (1 - \beta + \beta z)^N$. The expectation of Y is

- (A) $N\beta(1 - \beta)$
- (B) $N\beta$
- (C) $N(1 - \beta)$
- (D) Not expressible in terms of N and β alone

Question Number : 32**Correct : 2 Wrong : -0.66**

Consider the following expression grammar G :

$$\begin{aligned} E &\rightarrow E - T \mid T \\ T &\rightarrow T + F \mid F \\ F &\rightarrow (E) \mid id \end{aligned}$$

Which of the following grammars is not left recursive, but is equivalent to G ?

(A)
$$\begin{aligned} E &\rightarrow E - T \mid T \\ T &\rightarrow T + F \mid F \\ F &\rightarrow (E) \mid id \end{aligned}$$

(B)
$$\begin{aligned} E &\rightarrow TE' \\ E' &\rightarrow -TE' \mid \epsilon \\ T &\rightarrow T + F \mid F \\ F &\rightarrow (E) \mid id \end{aligned}$$

(C)
$$\begin{aligned} E &\rightarrow TX \\ X &\rightarrow -TX \mid \epsilon \\ T &\rightarrow FY \\ Y &\rightarrow +FY \mid \epsilon \\ F &\rightarrow (E) \mid id \end{aligned}$$

(D)
$$\begin{aligned} E &\rightarrow TX \mid (TX) \\ X &\rightarrow -TX \mid +TX \mid \epsilon \\ T &\rightarrow id \end{aligned}$$

Question Number : 33**Correct : 2 Wrong : -0.66**

A system shares 9 tape drives. The current allocation and maximum requirement of tape drives for three processes are shown below:

Process	Current Allocation	Maximum Requirement
P1	3	7
P2	1	6
P3	3	5

Which of the following best describes current state of the system?

- (A) Safe, Deadlocked (B) Safe, Not Deadlocked
(C) Not Safe, Deadlocked (D) Not Safe, Not Deadlocked

Question Number : 34**Correct : 2 Wrong : -0.66**

Consider a binary code that consists of only four valid codewords as given below:

00000, 01011, 10101, 11110

Let the minimum Hamming distance of the code be p and the maximum number of erroneous bits that can be corrected by the code be q . Then the values of p and q are

- (A) $p=3$ and $q=1$
(B) $p=3$ and $q=2$
(C) $p=4$ and $q=1$
(D) $p=4$ and $q=2$

Question Number : 35**Correct : 2 Wrong : -0.66**

Consider two hosts X and Y , connected by a single direct link of rate 10^6 bits/sec. The distance between the two hosts is 10,000 km and the propagation speed along the link is 2×10^8 m/sec. Host X sends a file of 50,000 bytes as one large message to host Y continuously. Let the transmission and propagation delays be p milliseconds and q milliseconds, respectively. Then the values of p and q are

- (A) $p=50$ and $q=100$
(B) $p=50$ and $q=400$
(C) $p=100$ and $q=50$
(D) $p=400$ and $q=50$

Question Number : 36**Correct : 2 Wrong : -0.66**

The pre-order traversal of a binary search tree is given by 12, 8, 6, 2, 7, 9, 10, 16, 15, 19, 17, 20. Then the post-order traversal of this tree is:

- (A) 2, 6, 7, 8, 9, 10, 12, 15, 16, 17, 19, 20 (B) 2, 7, 6, 10, 9, 8, 15, 17, 20, 19, 16, 12
(C) 7, 2, 6, 8, 9, 10, 20, 17, 19, 15, 16, 12 (D) 7, 6, 2, 10, 9, 8, 15, 16, 17, 20, 19, 12

Question Number : 37**Correct : 2 Wrong : -0.66**

Consider the C program fragment below which is meant to divide x by y using repeated subtractions. The variables x , y , q and r are all unsigned int.

```
while (r >= y) {  
    r = r - y;  
    q = q + 1;  
}
```

Which of the following conditions on the variables x , y , q and r before the execution of the fragment will ensure that the loop terminates in a state satisfying the condition $x == (y * q + r)$?

- (A) $(q == r) \ \&\& \ (r == 0)$
(B) $(x > 0) \ \&\& \ (r == x) \ \&\& \ (y > 0)$
(C) $(q == 0) \ \&\& \ (r == x) \ \&\& \ (y > 0)$
(D) $(q == 0) \ \&\& \ (y > 0)$

Question Number : 38**Correct : 2 Wrong : -0.66**

Consider the following C function.

```
int fun(int n) {  
    int i, j;  
    for(i = 1; i <= n; i++) {  
        for(j = 1; j < n; j += i) {  
            printf(" %d %d", i, j);  
        }  
    }  
}
```

Time complexity of fun in terms of Θ notation is

- (A) $\Theta(n\sqrt{n})$ (B) $\Theta(n^2)$ (C) $\Theta(n \log n)$ (D) $\Theta(n^2 \log n)$

Question Number : 39**Correct : 2 Wrong : -0.66**

Let δ denote the transition function and $\hat{\delta}$ denote the extended transition function of the ϵ -NFA whose transition table is given below:

δ	ϵ	a	b
$\rightarrow q_0$	$\{q_2\}$	$\{q_1\}$	$\{q_0\}$
q_1	$\{q_2\}$	$\{q_2\}$	$\{q_3\}$
q_2	$\{q_0\}$	$\emptyset$	$\emptyset$
q_3	$\emptyset$	$\emptyset$	$\{q_2\}$

Then $\hat{\delta}(q_2, aba)$ is

- (A) $\emptyset$ (B) $\{q_0, q_1, q_3\}$ (C) $\{q_0, q_1, q_2\}$ (D) $\{q_0, q_2, q_3\}$

Question Number : 40**Correct : 2 Wrong : -0.66**

Consider the following languages.

$$L_1 = \{a^p \mid p \text{ is a prime number}\}$$

$$L_2 = \{a^n b^m c^{2m} \mid n \geq 0, m \geq 0\}$$

$$L_3 = \{a^n b^n c^{2n} \mid n \geq 0\}$$

$$L_4 = \{a^n b^n \mid n \geq 1\}$$

Which of the following are CORRECT?

- I. L_1 is context-free but not regular.
 II. L_2 is not context-free.
 III. L_3 is not context-free but recursive.
 IV. L_4 is deterministic context-free.

- (A) I, II and IV only (B) II and III only (C) I and IV only (D) III and IV only

Question Number : 41

Correct : 2 Wrong : -0.66

Let $L(R)$ be the language represented by regular expression R . Let $L(G)$ be the language generated by a context free grammar G . Let $L(M)$ be the language accepted by a Turing machine M . Which of the following decision problems are undecidable?

- I. Given a regular expression R and a string w , is $w \in L(R)$?
- II. Given a context-free grammar G , is $L(G) = \emptyset$?
- III. Given a context-free grammar G , is $L(G) = \Sigma^*$ for some alphabet Σ ?
- IV. Given a Turing machine M and a string w , is $w \in L(M)$?

(A) I and IV only (B) II and III only (C) II, III and IV only (D) III and IV only

Question Number : 42

Correct : 2 Wrong : -0.66

The next state table of a 2-bit saturating up-counter is given below.

Q_1	Q_0	Q_1^+	Q_0^+
0	0	0	1
0	1	1	0
1	0	1	1
1	1	1	1

The counter is built as a synchronous sequential circuit using T flip-flops. The expressions for T_1 and T_0 are

- (A) $T_1 = Q_1 Q_0, \quad T_0 = \bar{Q}_1 \bar{Q}_0$
- (B) $T_1 = \bar{Q}_1 Q_0, \quad T_0 = \bar{Q}_1 + \bar{Q}_0$
- (C) $T_1 = Q_1 + Q_0, \quad T_0 = \bar{Q}_1 + \bar{Q}_0$
- (D) $T_1 = \bar{Q}_1 Q_0, \quad T_0 = Q_1 + Q_0$

Consider the following snippet of a C program. Assume that `swap (&x, &y)` exchanges the contents of `x` and `y`.

```
int main() {  
    int array[] = {3, 5, 1, 4, 6, 2};  
    int done = 0;  
    int i;  
  
    while (done == 0) {  
        done = 1;  
        for (i=0; i<=4; i++) {  
            if (array[i] < array[i+1]) {  
                swap(&array[i], &array[i+1]);  
                done = 0;  
            }  
        }  
        for (i=5; i>=1; i--) {  
            if (array[i] > array[i-1]) {  
                swap(&array[i], &array[i-1]);  
                done = 0;  
            }  
        }  
    }  
    printf("%d", array[3]);  
}
```

The output of the program is _____.

Question Number : 44**Correct : 2 Wrong : 0**

Two transactions T_1 and T_2 are given as

$$T_1: r_1(X)w_1(X)r_1(Y)w_1(Y)$$

$$T_2: r_2(Y)w_2(Y)r_2(Z)w_2(Z)$$

where $r_i(V)$ denotes a *read* operation by transaction T_i on a variable V and $w_i(V)$ denotes a *write* operation by transaction T_i on a variable V . The total number of conflict serializable schedules that can be formed by T_1 and T_2 is _____.

Question Number : 45**Correct : 2 Wrong : 0**

The read access times and the hit ratios for different caches in a memory hierarchy are as given below.

Cache	Read access time (in nanoseconds)	Hit ratio
I-cache	2	0.8
D-cache	2	0.9
L2-cache	8	0.9

The read access time of main memory is 90 nanoseconds. Assume that the caches use the referred-word-first read policy and the write back policy. Assume that all the caches are direct mapped caches. Assume that the dirty bit is always 0 for all the blocks in the caches. In execution of a program, 60% of memory reads are for instruction fetch and 40% are for memory operand fetch. The average read access time in nanoseconds (up to 2 decimal places) is _____.

Consider the following database table named *top_scorer*.

top_scorer

player	country	goals
Klose	Germany	16
Ronaldo	Brazil	15
G Müller	Germany	14
Fontaine	France	13
Pelé	Brazil	12
Klinsmann	Germany	11
Kocsis	Hungary	11
Batistuta	Argentina	10
Cubillas	Peru	10
Lato	Poland	10
Lineker	England	10
T Müller	Germany	10
Rahn	Germany	10

Consider the following SQL query:

```
SELECT ta.player FROM top_scorer AS ta
WHERE ta.goals >ALL (SELECT tb.goals
                     FROM top_scorer AS tb
                     WHERE tb.country = 'Spain')
AND ta.goals >ANY (SELECT tc.goals
                   FROM top_scorer AS tc
                   WHERE tc.country = 'Germany')
```

The number of tuples returned by the above SQL query is _____.

If the ordinary generating function of a sequence $\{a_n\}_{n=0}^{\infty}$ is $\frac{1+z}{(1-z)^3}$, then $a_3 - a_0$ is equal to _____.

Question Number : 48**Correct : 2 Wrong : 0**

If a random variable X has a Poisson distribution with mean 5, then the expectation $E[(X + 2)^2]$ equals _____.

Question Number : 49**Correct : 2 Wrong : 0**

In a B+ tree, if the search-key value is 8 bytes long, the block size is 512 bytes and the block pointer size is 2 bytes, then the maximum order of the B+ tree is _____.

Question Number : 50**Correct : 2 Wrong : 0**

A message is made up entirely of characters from the set $X = \{P, Q, R, S, T\}$. The table of probabilities for each of the characters is shown below:

Character	Probability
P	0.22
Q	0.34
R	0.17
S	0.19
T	0.08
Total	1.00

If a message of 100 characters over X is encoded using Huffman coding, then the expected length of the encoded message in bits is _____.

Question Number : 51**Correct : 2 Wrong : 0**

Consider the set of processes with arrival time (in milliseconds), CPU burst time (in milliseconds), and priority (0 is the highest priority) shown below. None of the processes have I/O burst time.

Process	Arrival Time	Burst Time	Priority
P_1	0	11	2
P_2	5	28	0
P_3	12	2	3
P_4	2	10	1
P_5	9	16	4

The average waiting time (in milliseconds) of all the processes using preemptive priority scheduling algorithm is _____.

Question Number : 52**Correct : 2 Wrong : 0**

If the characteristic polynomial of a 3×3 matrix M over $\mathbb{R}$ (the set of real numbers) is $\lambda^3 - 4\lambda^2 + a\lambda + 30$, $a \in \mathbb{R}$, and one eigenvalue of M is 2, then the largest among the absolute values of the eigenvalues of M is _____.

Question Number : 53**Correct : 2 Wrong : 0**

Consider a machine with a byte addressable main memory of 2^{32} bytes divided into blocks of size 32 bytes. Assume that a direct mapped cache having 512 cache lines is used with this machine. The size of the tag field in bits is _____.

Question Number : 54

Correct : 2 Wrong : 0

Consider the following C Program.

```
#include<stdio.h>
int main() {
    int m = 10;
    int n, n1;
    n = ++m;
    n1 = m++;
    n--;
    --n1;
    n -= n1;
    printf("%d", n);
    return 0;
}
```

The output of the program is _____.

Question Number : 55

Correct : 2 Wrong : 0

Consider the following C Program.

```
#include<stdio.h>
#include<string.h>
int main() {
    char* c = "GATECSIT2017";
    char* p = c;
    printf("%d", (int)strlen(c+2[p]-6[p]-1));
    return 0;
}
```

The output of the program is _____.

Question Number : 56

Correct : 1 Wrong : -0.33

Choose the option with words that are not synonyms.

(A) aversion, dislike

(B) luminous, radiant

(C) plunder, loot

(D) yielding, resistant

Question Number : 57

Correct : 1 Wrong : -0.33

Saturn is _____ to be seen on a clear night with the naked eye.

(A) enough bright

(B) bright enough

(C) as enough bright

(D) bright as enough

Question Number : 58

Correct : 1 Wrong : -0.33

There are five buildings called V, W, X, Y and Z in a row (not necessarily in that order). V is to the West of W. Z is to the East of X and the West of V. W is to the West of Y. Which is the building in the middle?

(A) V

(B) W

(C) X

(D) Y

Question Number : 59

Correct : 1 Wrong : -0.33

A test has twenty questions worth 100 marks in total. There are two types of questions. Multiple choice questions are worth 3 marks each and essay questions are worth 11 marks each. How many multiple choice questions does the exam have?

(A) 12

(B) 15

(C) 18

(D) 19

Question Number : 60**Correct : 1 Wrong : -0.33**

There are 3 red socks, 4 green socks and 3 blue socks. You choose 2 socks. The probability that they are of the same colour is

- (A) $1/5$ (B) $7/30$ (C) $1/4$ (D) $4/15$

Question Number : 61**Correct : 2 Wrong : -0.66**

“We lived in a culture that denied any merit to literary works, considering them important only when they were handmaidens to something seemingly more urgent – namely ideology. This was a country where all gestures, even the most private, were interpreted in political terms.”

The author’s belief that ideology is not as important as literature is revealed by the word:

- (A) ‘culture’ (B) ‘seemingly’ (C) ‘urgent’ (D) ‘political’

Question Number : 62**Correct : 2 Wrong : -0.66**

There are three boxes. One contains apples, another contains oranges and the last one contains both apples and oranges. All three are known to be incorrectly labelled. If you are permitted to open just one box and then pull out and inspect only one fruit, which box would you open to determine the contents of all three boxes?

- (A) The box labelled ‘Apples’ (B) The box labelled ‘Apples and Oranges’
(C) The box labelled ‘Oranges’ (D) Cannot be determined

Question Number : 63**Correct : 2 Wrong : -0.66**

X is a 30 digit number starting with the digit 4 followed by the digit 7. Then the number X^3 will have

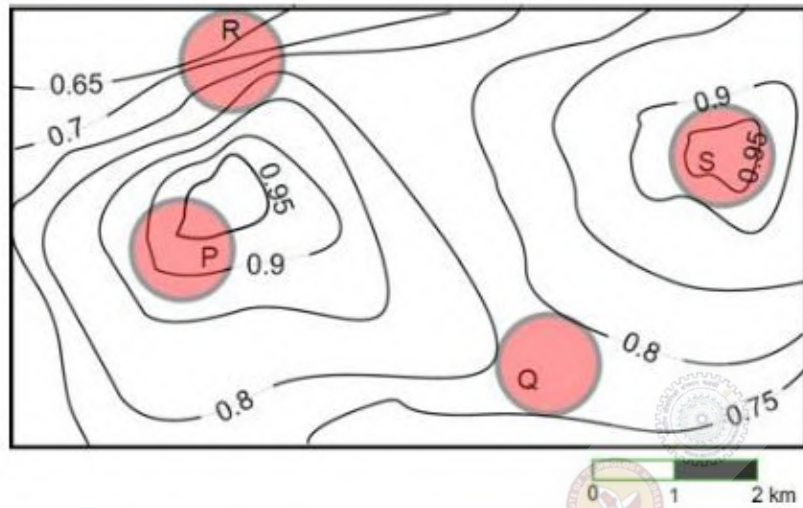
- (A) 90 digits (B) 91 digits (C) 92 digits (D) 93 digits

Question Number : 64**Correct : 2 Wrong : -0.66**

The number of roots of $e^x + 0.5x^2 - 2 = 0$ in the range $[-5, 5]$ is

- (A) 0 (B) 1 (C) 2 (D) 3

An air pressure contour line joins locations in a region having the same atmospheric pressure. The following is an air pressure contour plot of a geographical region. Contour lines are shown at 0.05 bar intervals in this plot.



If the possibility of a thunderstorm is given by how fast air pressure rises or drops over a region, which of the following regions is most likely to have a thunderstorm?

- (A) P (B) Q (C) R (D) S

GATE 2016

Set 1

65 questions | 20 pages

Paper 15 of 16

Q. 1 – Q. 5 carry one mark each.

Q.1 Out of the following four sentences, select the most suitable sentence with respect to grammar and usage.

- (A) I will not leave the place until the minister does not meet me.
- (B) I will not leave the place until the minister doesn't meet me.
- (C) I will not leave the place until the minister meet me.
- (D) I will not leave the place until the minister meets me.

Q.2 A rewording of something written or spoken is a _____.

- (A) paraphrase
- (B) paradox
- (C) paradigm
- (D) paraffin

Q.3 Archimedes said, "Give me a lever long enough and a fulcrum on which to place it, and I will move the world."

The sentence above is an example of a _____ statement.

- (A) figurative
- (B) collateral
- (C) literal
- (D) figurine

Q.4 If 'relftaga' means carefree, 'otaga' means careful and 'fertaga' means careless, which of the following could mean 'aftercare'?

- (A) zentaga
- (B) tagafer
- (C) tagazen
- (D) relffer

Q.5 A cube is built using 64 cubic blocks of side one unit. After it is built, one cubic block is removed from every corner of the cube. The resulting surface area of the body (in square units) after the removal is _____.

- (A) 56
- (B) 64
- (C) 72
- (D) 96

Q. 6 – Q. 10 carry two marks each.

- Q.6 A shaving set company sells 4 different types of razors, Elegance, Smooth, Soft and Executive. Elegance sells at Rs. 48, Smooth at Rs. 63, Soft at Rs. 78 and Executive at Rs. 173 per piece. The table below shows the numbers of each razor sold in each quarter of a year.

Quarter \ Product	Elegance	Smooth	Soft	Executive
Q1	27300	20009	17602	9999
Q2	25222	19392	18445	8942
Q3	28976	22429	19544	10234
Q4	21012	18229	16595	10109

Which product contributes the greatest fraction to the revenue of the company in that year?

- (A) Elegance (B) Executive (C) Smooth (D) Soft

- Q.7 Indian currency notes show the denomination indicated in at least seventeen languages. If this is not an indication of the nation's diversity, nothing else is.

Which of the following can be logically inferred from the above sentences?

- (A) India is a country of exactly seventeen languages.
 (B) Linguistic pluralism is the only indicator of a nation's diversity.
 (C) Indian currency notes have sufficient space for all the Indian languages.
 (D) Linguistic pluralism is strong evidence of India's diversity.

- Q.8 Consider the following statements relating to the level of poker play of four players **P**, **Q**, **R** and **S**.

- I. **P** always beats **Q**
 II. **R** always beats **S**
 III. **S** loses to **P** only sometimes
 IV. **R** always loses to **Q**

Which of the following can be logically inferred from the above statements?

- (i) **P** is likely to beat all the three other players
 (ii) **S** is the absolute worst player in the set

- (A) (i) only (B) (ii) only (C) (i) and (ii) (D) neither (i) nor (ii)

- Q.9 If $f(x) = 2x^7 + 3x - 5$, which of the following is a factor of $f(x)$?

- (A) (x^3+8) (B) $(x-1)$ (C) $(2x-5)$ (D) $(x+1)$

Q.10 In a process, the number of cycles to failure decreases exponentially with an increase in load. At a load of 80 units, it takes 100 cycles for failure. When the load is halved, it takes 10000 cycles for failure. The load for which the failure will happen in 5000 cycles is _____.

(A) 40.00

(B) 46.02

(C) 60.01

(D) 92.02

END OF THE QUESTION PAPER

Q. 1 - Q. 25 carry one mark each.

Q.1 Let p, q, r, s represent the following propositions.

p : $x \in \{8, 9, 10, 11, 12\}$

q : x is a composite number

r : x is a perfect square

s : x is a prime number

The integer $x \geq 2$ which satisfies $\neg((p \Rightarrow q) \wedge (\neg r \vee \neg s))$ is _____ .

Q.2 Let a_n be the number of n -bit strings that do NOT contain two consecutive 1s. Which one of the following is the recurrence relation for a_n ?

(A) $a_n = a_{n-1} + 2a_{n-2}$

(B) $a_n = a_{n-1} + a_{n-2}$

(C) $a_n = 2a_{n-1} + a_{n-2}$

(D) $a_n = 2a_{n-1} + 2a_{n-2}$

Q.3

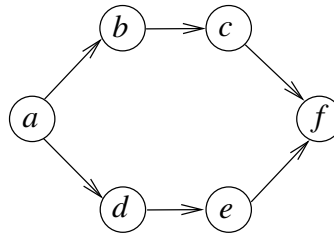
$$\lim_{x \rightarrow 4} \frac{\sin(x-4)}{x-4} = \text{_____}.$$

Q.4 A probability density function on the interval $[a, 1]$ is given by $1/x^2$ and outside this interval the value of the function is zero. The value of a is _____ .

Q.5 Two eigenvalues of a 3×3 real matrix P are $(2 + \sqrt{-1})$ and 3. The determinant of P is _____ .

- Q.6** Consider the Boolean operator # with the following properties:
 $x \# 0 = x$, $x \# 1 = \bar{x}$, $x \# x = 0$ and $x \# \bar{x} = 1$. Then $x \# y$ is equivalent to
- (A) $x\bar{y} + \bar{x}y$
 - (B) $x\bar{y} + \bar{x}\bar{y}$
 - (C) $\bar{x}y + xy$
 - (D) $xy + \bar{x}\bar{y}$
- Q.7** The 16-bit 2's complement representation of an integer is 1111 1111 1111 0101; its decimal representation is _____.
- Q.8** We want to design a synchronous counter that counts the sequence 0-1-0-2-0-3 and then repeats. The minimum number of J-K flip-flops required to implement this counter is _____.
- Q.9** A processor can support a maximum memory of 4 GB, where the memory is word-addressable (a word consists of two bytes). The size of the address bus of the processor is at least _____ bits.
- Q.10** A queue is implemented using an array such that ENQUEUE and DEQUEUE operations are performed efficiently. Which one of the following statements is **CORRECT** (n refers to the number of items in the queue)?
- (A) Both operations can be performed in $O(1)$ time
 - (B) At most one operation can be performed in $O(1)$ time but the worst case time for the other operation will be $\Omega(n)$
 - (C) The worst case time complexity for both operations will be $\Omega(n)$
 - (D) Worst case time complexity for both operations will be $\Omega(\log n)$

Q.11 Consider the following directed graph:



The number of different topological orderings of the vertices of the graph is _____ .

Q.12 Consider the following C program.

```
void f(int, short);  
void main()  
{  
    int i = 100;  
    short s = 12;  
    short *p = &s;  
    _____ ;    // call to f()  
}
```

Which one of the following expressions, when placed in the blank above, will **NOT** result in a type checking error?

- (A) `f(s, *s)`
- (B) `i = f(i, s)`
- (C) `f(i, *s)`
- (D) `f(i, *p)`

Q.13 The worst case running times of *Insertion sort*, *Merge sort* and *Quick sort*, respectively, are:

- (A) $\Theta(n \log n)$, $\Theta(n \log n)$, and $\Theta(n^2)$
- (B) $\Theta(n^2)$, $\Theta(n^2)$, and $\Theta(n \log n)$
- (C) $\Theta(n^2)$, $\Theta(n \log n)$, and $\Theta(n \log n)$
- (D) $\Theta(n^2)$, $\Theta(n \log n)$, and $\Theta(n^2)$

Q.14 Let G be a weighted connected undirected graph with distinct positive edge weights. If every edge weight is increased by the same value, then which of the following statements is/are **TRUE**?

P: Minimum spanning tree of G does not change

Q: Shortest path between any pair of vertices does not change

- (A) P only
- (B) Q only
- (C) Neither P nor Q
- (D) Both P and Q

Q.15 Consider the following C program.

```
#include<stdio.h>
void mystery(int *ptrb, int *ptrb) {
    int *temp;
    temp = ptrb;
    ptrb = ptrb;
    ptrb = temp;
}
int main() {
    int a=2016, b=0, c=4, d=42;
    mystery(&a, &b);
    if (a < c)
        mystery(&c, &a);
    mystery(&a, &d);
    printf("%d\n", a);
}
```

The output of the program is _____ .

Q.16 Which of the following languages is generated by the given grammar?

$$S \longrightarrow aS \mid bS \mid \varepsilon$$

- (A) $\{a^n b^m \mid n, m \geq 0\}$
- (B) $\{w \in \{a, b\}^* \mid w \text{ has equal number of a's and b's}\}$
- (C) $\{a^n \mid n \geq 0\} \cup \{b^n \mid n \geq 0\} \cup \{a^n b^n \mid n \geq 0\}$
- (D) $\{a, b\}^*$

Q.17 Which of the following decision problems are undecidable?

- I. Given NFAs N_1 and N_2 , is $L(N_1) \cap L(N_2) = \Phi$?
- II. Given a CFG $G = (N, \Sigma, P, S)$ and a string $x \in \Sigma^*$, does $x \in L(G)$?
- III. Given CFGs G_1 and G_2 , is $L(G_1) = L(G_2)$?
- IV. Given a TM M , is $L(M) = \Phi$?

- (A) I and IV only
- (B) II and III only
- (C) III and IV only
- (D) II and IV only

Q.18 Which one of the following regular expressions represents the language: *the set of all binary strings having two consecutive 0s and two consecutive 1s*?

- (A) $(0+1)^*0011(0+1)^* + (0+1)^*1100(0+1)^*$
- (B) $(0+1)^*(00(0+1)^*11 + 11(0+1)^*00)(0+1)^*$
- (C) $(0+1)^*00(0+1)^* + (0+1)^*11(0+1)^*$
- (D) $00(0+1)^*11 + 11(0+1)^*00$

Q.19 Consider the following code segment.

```
x = u - t;  
y = x * v;  
x = y + w;  
y = t - z;  
y = x * y;
```

The minimum number of *total* variables required to convert the above code segment to *static single assignment* form is _____.

- Q.20** Consider an arbitrary set of CPU-bound processes with unequal CPU burst lengths submitted at the same time to a computer system. Which one of the following process scheduling algorithms would minimize the average waiting time in the ready queue?
- (A) Shortest remaining time first
 - (B) Round-robin with time quantum less than the shortest CPU burst
 - (C) Uniform random
 - (D) Highest priority first with priority proportional to CPU burst length
- Q.21** Which of the following is **NOT** a superkey in a relational schema with attributes V, W, X, Y, Z and primary key VY ?
- (A) $VXYZ$
 - (B) $VWXZ$
 - (C) $VWXY$
 - (D) $VWXYZ$
- Q.22** Which one of the following is **NOT** a part of the ACID properties of database transactions?
- (A) Atomicity
 - (B) Consistency
 - (C) Isolation
 - (D) Deadlock-freedom

Q.23 A database of research articles in a journal uses the following schema.

(VOLUME, NUMBER, STARTPAGE, ENDPAGE, TITLE, YEAR, PRICE)

The primary key is (VOLUME, NUMBER, STARTPAGE, ENDPAGE) and the following functional dependencies exist in the schema.

(VOLUME, NUMBER, STARTPAGE, ENDPAGE) $\rightarrow$ TITLE

(VOLUME, NUMBER) $\rightarrow$ YEAR

(VOLUME, NUMBER, STARTPAGE, ENDPAGE) $\rightarrow$ PRICE

The database is redesigned to use the following schemas.

(VOLUME, NUMBER, STARTPAGE, ENDPAGE, TITLE, PRICE)

(VOLUME, NUMBER, YEAR)

Which is the weakest normal form that the new database satisfies, but the old one does not?

- (A) 1NF
- (B) 2NF
- (C) 3NF
- (D) BCNF

Q.24 Which one of the following protocols is **NOT** used to resolve one form of address to another one?

- (A) DNS
- (B) ARP
- (C) DHCP
- (D) RARP

Q.25 Which of the following is/are example(s) of stateful application layer protocols?

- (i) HTTP
- (ii) FTP
- (iii) TCP
- (iv) POP3

- (A) (i) and (ii) only
- (B) (ii) and (iii) only
- (C) (ii) and (iv) only
- (D) (iv) only

Q. 26 - Q. 55 carry two marks each.

Q.26 The coefficient of x^{12} in $(x^3 + x^4 + x^5 + x^6 + \dots)^3$ is _____ .

Q.27 Consider the recurrence relation $a_1 = 8$, $a_n = 6n^2 + 2n + a_{n-1}$. Let $a_{99} = K \times 10^4$. The value of K is _____ .

Q.28 A function $f : \mathbb{N}^+ \rightarrow \mathbb{N}^+$, defined on the set of positive integers $\mathbb{N}^+$, satisfies the following properties:

$$\begin{aligned} f(n) &= f(n/2) && \text{if } n \text{ is even} \\ f(n) &= f(n+5) && \text{if } n \text{ is odd} \end{aligned}$$

Let $R = \{i \mid \exists j : f(j) = i\}$ be the set of distinct values that f takes. The maximum possible size of R is _____ .

Q.29 Consider the following experiment.

Step 1. Flip a fair coin twice.

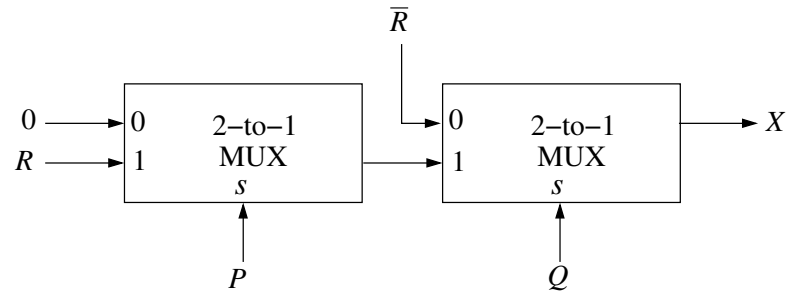
Step 2. If the outcomes are (TAILS, HEADS) then output Y and stop.

Step 3. If the outcomes are either (HEADS, HEADS) or (HEADS, TAILS), then output N and stop.

Step 4. If the outcomes are (TAILS, TAILS), then go to Step 1.

The probability that the output of the experiment is Y is (up to two decimal places) _____ .

Q.30 Consider the two cascaded 2-to-1 multiplexers as shown in the figure.



The minimal sum of products form of the output X is

- (A) $\bar{P}\bar{Q} + PQR$
- (B) $\bar{P}Q + QR$
- (C) $PQ + \bar{P}\bar{Q}R$
- (D) $\bar{Q}\bar{R} + PQR$

Q.31 The size of the data count register of a DMA controller is 16 bits. The processor needs to transfer a file of 29,154 kilobytes from disk to main memory. The memory is byte addressable. The minimum number of times the DMA controller needs to get the control of the system bus from the processor to transfer the file from the disk to main memory is _____.

Q.32 The stage delays in a 4-stage pipeline are 800, 500, 400 and 300 picoseconds. The first stage (with delay 800 picoseconds) is replaced with a functionally equivalent design involving two stages with respective delays 600 and 350 picoseconds. The throughput increase of the pipeline is _____ percent.

Q.33 Consider a carry lookahead adder for adding two n -bit integers, built using gates of fan-in at most two. The time to perform addition using this adder is

- (A) $\Theta(1)$
- (B) $\Theta(\log(n))$
- (C) $\Theta(\sqrt{n})$
- (D) $\Theta(n)$

Q.34 The following function computes the maximum value contained in an integer array `p[]` of size `n` (`n >= 1`).

```
int max(int *p, int n) {  
    int a=0, b=n-1;  
  
    while (_____) {  
        if (p[a] <= p[b]) { a = a+1; }  
        else                { b = b-1; }  
    }  
  
    return p[a];  
}
```

The missing loop condition is

- (A) `a != n`
- (B) `b != 0`
- (C) `b > (a + 1)`
- (D) `b != a`

Q.35 What will be the output of the following C program?

```
void count(int n){  
    static int d=1;  
  
    printf("%d ", n);  
    printf("%d ", d);  
    d++;  
    if(n>1) count(n-1);  
    printf("%d ", d);  
}
```

```
void main(){  
    count(3);  
}
```

- (A) 3 1 2 2 1 3 4 4 4
- (B) 3 1 2 1 1 1 2 2 2
- (C) 3 1 2 2 1 3 4
- (D) 3 1 2 1 1 1 2

- Q.36** What will be the output of the following pseudo-code when parameters are passed by reference and dynamic scoping is assumed?

```
a=3;
void n(x) {x = x * a; print(x);}
void m(y) {a = 1; a = y - a; n(a); print(a);}
void main() {m(a);}
```

- (A) 6, 2
(B) 6, 6
(C) 4, 2
(D) 4, 4
- Q.37** An operator `delete(i)` for a binary heap data structure is to be designed to delete the item in the *i*-th node. Assume that the heap is implemented in an array and *i* refers to the *i*-th index of the array. If the heap tree has depth *d* (number of edges on the path from the root to the farthest leaf), then what is the time complexity to re-fix the heap efficiently after the removal of the element?
- (A) $O(1)$
(B) $O(d)$ but not $O(1)$
(C) $O(2^d)$ but not $O(d)$
(D) $O(d2^d)$ but not $O(2^d)$
- Q.38** Consider the weighted undirected graph with 4 vertices, where the weight of edge $\{i, j\}$ is given by the entry W_{ij} in the matrix *W*.

$$W = \begin{bmatrix} 0 & 2 & 8 & 5 \\ 2 & 0 & 5 & 8 \\ 8 & 5 & 0 & x \\ 5 & 8 & x & 0 \end{bmatrix}$$

The largest possible integer value of *x*, for which at least one shortest path between some pair of vertices will contain the edge with weight *x* is _____.

- Q.39** Let *G* be a complete undirected graph on 4 vertices, having 6 edges with weights being 1, 2, 3, 4, 5, and 6. The maximum possible weight that a minimum weight spanning tree of *G* can have is _____.

Q.40 $G = (V, E)$ is an undirected simple graph in which each edge has a distinct weight, and e is a particular edge of G . Which of the following statements about the minimum spanning trees (MSTs) of G is/are **TRUE**?

- I. If e is the lightest edge of some cycle in G , then every MST of G includes e
- II. If e is the heaviest edge of some cycle in G , then every MST of G excludes e

(A) I only (B) II only (C) both I and II (D) neither I nor II

Q.41 Let Q denote a queue containing sixteen numbers and S be an empty stack. $\text{Head}(Q)$ returns the element at the head of the queue Q **without** removing it from Q . Similarly $\text{Top}(S)$ returns the element at the top of S **without** removing it from S . Consider the algorithm given below.

```

while  $Q$  is not Empty do
    if  $S$  is Empty OR  $\text{Top}(S) \leq \text{Head}(Q)$  then
         $x := \text{Dequeue}(Q);$ 
         $\text{Push}(S, x);$ 
    else
         $x := \text{Pop}(S);$ 
         $\text{Enqueue}(Q, x);$ 
    end
end

```

The maximum possible number of iterations of the **while** loop in the algorithm is _____.

Q.42 Consider the following context-free grammars:

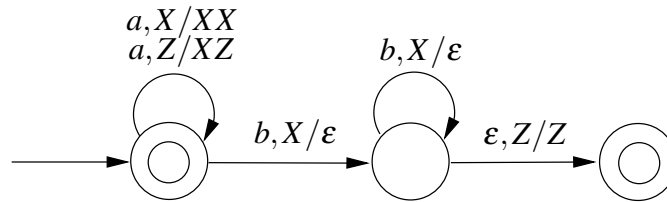
$G_1: S \rightarrow aS|B, B \rightarrow b|bB$

$G_2: S \rightarrow aA|bB, A \rightarrow aA|B|\epsilon, B \rightarrow bB|\epsilon$

Which one of the following pairs of languages is generated by G_1 and G_2 , respectively?

- (A) $\{a^m b^n | m > 0 \text{ or } n > 0\}$ and $\{a^m b^n | m > 0 \text{ and } n > 0\}$
- (B) $\{a^m b^n | m > 0 \text{ and } n > 0\}$ and $\{a^m b^n | m > 0 \text{ or } n \geq 0\}$
- (C) $\{a^m b^n | m \geq 0 \text{ or } n > 0\}$ and $\{a^m b^n | m > 0 \text{ and } n > 0\}$
- (D) $\{a^m b^n | m \geq 0 \text{ and } n > 0\}$ and $\{a^m b^n | m > 0 \text{ or } n > 0\}$

- Q.43** Consider the transition diagram of a PDA given below with input alphabet $\Sigma = \{a, b\}$ and stack alphabet $\Gamma = \{X, Z\}$. Z is the initial stack symbol. Let L denote the language accepted by the PDA.



Which one of the following is **TRUE**?

- (A) $L = \{a^n b^n | n \geq 0\}$ and is not accepted by any finite automata
- (B) $L = \{a^n | n \geq 0\} \cup \{a^n b^n | n \geq 0\}$ and is not accepted by any deterministic PDA
- (C) L is not accepted by any Turing machine that halts on every input
- (D) $L = \{a^n | n \geq 0\} \cup \{a^n b^n | n \geq 0\}$ and is deterministic context-free
- Q.44** Let X be a recursive language and Y be a recursively enumerable but not recursive language. Let W and Z be two languages such that $\bar{Y}$ reduces to W , and Z reduces to $\bar{X}$ (reduction means the standard many-one reduction). Which one of the following statements is **TRUE**?
- (A) W can be recursively enumerable and Z is recursive.
- (B) W can be recursive and Z is recursively enumerable.
- (C) W is not recursively enumerable and Z is recursive.
- (D) W is not recursively enumerable and Z is not recursive.

- Q.45** The attributes of three arithmetic operators in some programming language are given below.

Operator	Precedence	Associativity	Arity
+	High	Left	Binary
−	Medium	Right	Binary
*	Low	Left	Binary

The value of the expression $2 - 5 + 1 - 7 * 3$ in this language is _____.

Q.46 Consider the following Syntax Directed Translation Scheme (SDTS), with non-terminals $\{S, A\}$ and terminals $\{a, b\}$.

$S \rightarrow aA \quad \{ \text{print 1} \}$

$S \rightarrow a \quad \{ \text{print 2} \}$

$A \rightarrow Sb \quad \{ \text{print 3} \}$

Using the above SDTS, the output printed by a bottom-up parser, for the input **aab** is:

(A) 1 3 2

(B) 2 2 3

(C) 2 3 1

(D) syntax error

Q.47 Consider a computer system with 40-bit virtual addressing and page size of sixteen kilobytes. If the computer system has a one-level page table per process and each page table entry requires 48 bits, then the size of the per-process page table is _____ megabytes.

Q.48 Consider a disk queue with requests for I/O to blocks on cylinders 47, 38, 121, 191, 87, 11, 92, 10. The C-LOOK scheduling algorithm is used. The head is initially at cylinder number 63, moving towards larger cylinder numbers on its servicing pass. The cylinders are numbered from 0 to 199. The total head movement (in number of cylinders) incurred while servicing these requests is _____.

Q.49 Consider a computer system with ten physical page frames. The system is provided with an access sequence $(a_1, a_2, \dots, a_{20}, a_1, a_2, \dots, a_{20})$, where each a_i is a distinct virtual page number. The difference in the number of page faults between the last-in-first-out page replacement policy and the optimal page replacement policy is _____.

- Q.50** Consider the following proposed solution for the critical section problem. There are n processes: $P_0 \dots P_{n-1}$. In the code, function `pmax` returns an integer not smaller than any of its arguments. For all i , $t[i]$ is initialized to zero.

Code for P_i :

```
do {
    c[i]=1; t[i] = pmax(t[0],...,t[n-1])+1; c[i]=0;
    for every j  $\neq$  i in {0,...,n-1} {
        while (c[j]);
        while (t[j] != 0 && t[j]<=t[i]);
    }
    Critical Section;
    t[i]=0;
    Remainder Section;
} while (true);
```

Which one of the following is **TRUE** about the above solution?

- (A) At most one process can be in the critical section at any time
- (B) The bounded wait condition is satisfied
- (C) The progress condition is satisfied
- (D) It cannot cause a deadlock

- Q.51** Consider the following two phase locking protocol. Suppose a transaction T accesses (for read or write operations), a certain set of objects $\{O_1, \dots, O_k\}$. This is done in the following manner:

Step 1. T acquires exclusive locks to $O_1, \dots, O_k$ in increasing order of their addresses.

Step 2. The required operations are performed.

Step 3. All locks are released.

This protocol will

- (A) guarantee serializability and deadlock-freedom
- (B) guarantee neither serializability nor deadlock-freedom
- (C) guarantee serializability but not deadlock-freedom
- (D) guarantee deadlock-freedom but not serializability

Q.52 Consider that B wants to send a message m that is digitally signed to A. Let the pair of private and public keys for A and B be denoted by K_x^- and K_x^+ for $x = A, B$, respectively. Let $K_x(m)$ represent the operation of encrypting m with a key K_x and $H(m)$ represent the message digest. Which one of the following indicates the **CORRECT** way of sending the message m along with the digital signature to A?

- (A) $\{m, K_B^+(H(m))\}$ (B) $\{m, K_B^-(H(m))\}$ (C) $\{m, K_A^-(H(m))\}$ (D) $\{m, K_A^+(m)\}$

Q.53 An IP datagram of size 1000 bytes arrives at a router. The router has to forward this packet on a link whose MTU (maximum transmission unit) is 100 bytes. Assume that the size of the IP header is 20 bytes.

The number of fragments that the IP datagram will be divided into for transmission is _____.

Q.54 For a host machine that uses the token bucket algorithm for congestion control, the token bucket has a capacity of 1 megabyte and the maximum output rate is 20 megabytes per second. Tokens arrive at a rate to sustain output at a rate of 10 megabytes per second. The token bucket is currently full and the machine needs to send 12 megabytes of data. The minimum time required to transmit the data is _____ seconds.

Q.55 A sender uses the Stop-and-Wait ARQ protocol for reliable transmission of frames. Frames are of size 1000 bytes and the transmission rate at the sender is 80 Kbps (1Kbps = 1000 bits/second). Size of an acknowledgement is 100 bytes and the transmission rate at the receiver is 8 Kbps. The one-way propagation delay is 100 milliseconds.

Assuming no frame is lost, the sender throughput is _____ bytes/second.

GATE 2016

Set 2

65 questions | 21 pages

Paper 16 of 16

Q. 1 – Q. 5 carry one mark each.

Q.1 The man who is now Municipal Commissioner worked as _____.

- (A) the security guard at a university
- (B) a security guard at the university
- (C) a security guard at university
- (D) the security guard at the university

Q.2 Nobody knows how the Indian cricket team is going to cope with the difficult and seamer-friendly wickets in Australia.

Choose the option which is closest in meaning to the underlined phrase in the above sentence.

- (A) put up with (B) put in with (C) put down to (D) put up against

Q.3 Find the odd one in the following group of words.

mock, deride, praise, jeer

- (A) mock (B) deride (C) praise (D) jeer

Q.4 Pick the odd one from the following options.

- (A) CADBE (B) JHKIL (C) XZYWZ (D) ONPMQ

Q.5 In a quadratic function, the value of the product of the roots (α, β) is 4. Find the value of

$$\frac{\alpha^n + \beta^n}{\alpha^{-n} + \beta^{-n}}$$

- (A) n^4 (B) 4^n (C) 2^{2n-1} (D) 4^{n-1}

Q. 6 – Q. 10 carry two marks each.

Q.6 Among 150 faculty members in an institute, 55 are connected with each other through Facebook® and 85 are connected through WhatsApp®. 30 faculty members do not have Facebook® or WhatsApp® accounts. The number of faculty members connected only through Facebook® accounts is _____.

- (A) 35 (B) 45 (C) 65 (D) 90

- Q.7 Computers were invented for performing only high-end useful computations. However, it is no understatement that they have taken over our world today. The internet, for example, is ubiquitous. Many believe that the internet itself is an unintended consequence of the original invention. With the advent of mobile computing on our phones, a whole new dimension is now enabled. One is left wondering if all these developments are good or, more importantly, required.

Which of the statement(s) below is/are logically valid and can be inferred from the above paragraph?

- (i) The author believes that computers are not good for us.
- (ii) Mobile computers and the internet are both intended inventions

(A) (i) only (B) (ii) only (C) both (i) and (ii) (D) neither (i) nor (ii)

- Q.8 All hill-stations have a lake. Ooty has two lakes.

Which of the statement(s) below is/are logically valid and can be inferred from the above sentences?

- (i) Ooty is not a hill-station.
- (ii) No hill-station can have more than one lake.

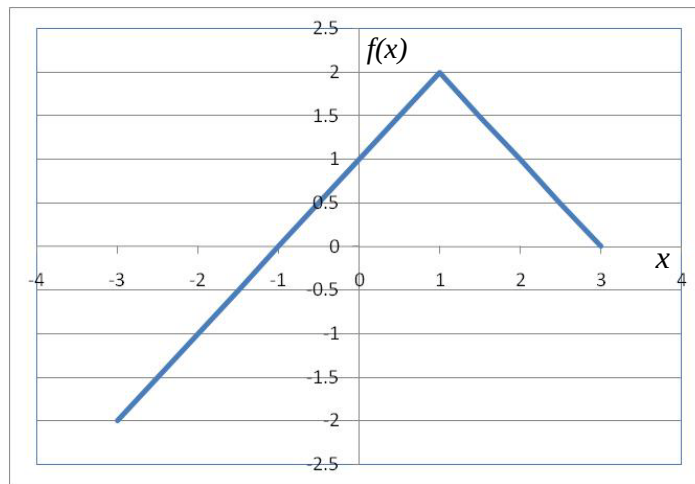
(A) (i) only (B) (ii) only
(C) both (i) and (ii) (D) neither (i) nor (ii)

- Q.9 In a 2×4 rectangle grid shown below, each cell is a rectangle. How many rectangles can be observed in the grid?



(A) 21 (B) 27 (C) 30 (D) 36

Q.10



Choose the correct expression for $f(x)$ given in the graph.

(A) $f(x) = 1 - |x - 1|$

(B) $f(x) = 1 + |x - 1|$

(C) $f(x) = 2 - |x - 1|$

(D) $f(x) = 2 + |x - 1|$

END OF THE QUESTION PAPER

Q. 1 - Q. 25 carry one mark each.

Q.1 Consider the following expressions:

- (i) *false*
- (ii) Q
- (iii) *true*
- (iv) $P \vee Q$
- (v) $\neg Q \vee P$

The number of expressions given above that are logically implied by $P \wedge (P \Rightarrow Q)$ is _____ .

Q.2 Let $f(x)$ be a polynomial and $g(x) = f'(x)$ be its derivative. If the degree of $(f(x) + f(-x))$ is 10, then the degree of $(g(x) - g(-x))$ is _____ .

Q.3 The minimum number of colours that is sufficient to vertex-colour any planar graph is _____ .

Q.4 Consider the systems, each consisting of m linear equations in n variables.

- I. If $m < n$, then all such systems have a solution
- II. If $m > n$, then none of these systems has a solution
- III. If $m = n$, then there exists a system which has a solution

Which one of the following is **CORRECT**?

- (A) I, II and III are true
- (B) Only II and III are true
- (C) Only III is true
- (D) None of them is true

- Q.5** Suppose that a shop has an equal number of LED bulbs of two different types. The probability of an LED bulb lasting more than 100 hours given that it is of Type 1 is 0.7, and given that it is of Type 2 is 0.4. The probability that an LED bulb chosen uniformly at random lasts more than 100 hours is _____ .
- Q.6** Suppose that the eigenvalues of matrix A are 1, 2, 4. The determinant of $(A^{-1})^T$ is _____ .
- Q.7** Consider an eight-bit ripple-carry adder for computing the sum of A and B , where A and B are integers represented in 2's complement form. If the decimal value of A is one, the decimal value of B that leads to the longest latency for the sum to stabilize is _____ .
- Q.8** Let, $x_1 \oplus x_2 \oplus x_3 \oplus x_4 = 0$ where x_1, x_2, x_3, x_4 are Boolean variables, and $\oplus$ is the XOR operator. Which one of the following must always be **TRUE**?
- (A) $x_1 x_2 x_3 x_4 = 0$
 - (B) $x_1 x_3 + x_2 = 0$
 - (C) $\bar{x}_1 \oplus \bar{x}_3 = \bar{x}_2 \oplus \bar{x}_4$
 - (D) $x_1 + x_2 + x_3 + x_4 = 0$
- Q.9** Let X be the number of distinct 16-bit integers in 2's complement representation. Let Y be the number of distinct 16-bit integers in sign magnitude representation. Then $X - Y$ is _____ .
- Q.10** A processor has 40 distinct instructions and 24 general purpose registers. A 32-bit instruction word has an opcode, two register operands and an immediate operand. The number of bits available for the immediate operand field is _____ .

Q.11 Breadth First Search (BFS) is started on a binary tree beginning from the root vertex. There is a vertex t at a distance four from the root. If t is the n -th vertex in this BFS traversal, then the maximum possible value of n is _____ .

Q.12 The value printed by the following program is _____ .

```
void f(int* p, int m){
    m = m + 5;
    *p = *p + m;
    return;
}

void main(){
    int i=5, j=10;

    f(&i, j);
    printf("%d", i+j);
}
```

Q.13 Assume that the algorithms considered here sort the input sequences in ascending order. If the input is already in ascending order, which of the following are **TRUE**?

- I. Quicksort runs in $\Theta(n^2)$ time
- II. Bubblesort runs in $\Theta(n^2)$ time
- III. Mergesort runs in $\Theta(n)$ time
- IV. Insertion sort runs in $\Theta(n)$ time

- (A) I and II only
- (B) I and III only
- (C) II and IV only
- (D) I and IV only

- Q.14** The Floyd-Warshall algorithm for all-pair shortest paths computation is based on
- (A) Greedy paradigm.
 - (B) Divide-and-Conquer paradigm.
 - (C) Dynamic Programming paradigm.
 - (D) neither Greedy nor Divide-and-Conquer nor Dynamic Programming paradigm.
- Q.15** N items are stored in a sorted doubly linked list. For a *delete* operation, a pointer is provided to the record to be deleted. For a *decrease-key* operation, a pointer is provided to the record on which the operation is to be performed.
- An algorithm performs the following operations on the list in this order: $\Theta(N)$ *delete*, $O(\log N)$ *insert*, $O(\log N)$ *find*, and $\Theta(N)$ *decrease-key*. What is the time complexity of all these operations put together?
- (A) $O(\log^2 N)$ (B) $O(N)$ (C) $O(N^2)$ (D) $\Theta(N^2 \log N)$
- Q.16** The number of states in the minimum sized DFA that accepts the language defined by the regular expression
- $$(0 + 1)^*(0 + 1)(0 + 1)^*$$
- is _____ .
- Q.17** Language L_1 is defined by the grammar: $S_1 \rightarrow aS_1b | \epsilon$
Language L_2 is defined by the grammar: $S_2 \rightarrow abS_2 | \epsilon$
- Consider the following statements:
- P : L_1 is regular
 Q : L_2 is regular
- Which one of the following is **TRUE**?
- (A) Both P and Q are true
 - (B) P is true and Q is false
 - (C) P is false and Q is true
 - (D) Both P and Q are false

Q.18 Consider the following types of languages: L_1 : Regular, L_2 : Context-free, L_3 : Recursive, L_4 : Recursively enumerable. Which of the following is/are **TRUE**?

- I. $\overline{L_3} \cup L_4$ is recursively enumerable
- II. $\overline{L_2} \cup L_3$ is recursive
- III. $L_1^* \cap L_2$ is context-free
- IV. $L_1 \cup \overline{L_2}$ is context-free

- (A) I only
- (B) I and III only
- (C) I and IV only
- (D) I, II and III only

Q.19 Match the following:

- | | |
|--------------------------|---------------------------|
| (P) Lexical analysis | (i) Leftmost derivation |
| (Q) Top down parsing | (ii) Type checking |
| (R) Semantic analysis | (iii) Regular expressions |
| (S) Runtime environments | (iv) Activation records |

- (A) $P \leftrightarrow i, Q \leftrightarrow ii, R \leftrightarrow iv, S \leftrightarrow iii$
- (B) $P \leftrightarrow iii, Q \leftrightarrow i, R \leftrightarrow ii, S \leftrightarrow iv$
- (C) $P \leftrightarrow ii, Q \leftrightarrow iii, R \leftrightarrow i, S \leftrightarrow iv$
- (D) $P \leftrightarrow iv, Q \leftrightarrow i, R \leftrightarrow ii, S \leftrightarrow iii$

Q.20 In which one of the following page replacement algorithms it is possible for the page fault rate to increase even when the number of allocated frames increases?

- (A) LRU (Least Recently Used)
- (B) OPT (Optimal Page Replacement)
- (C) MRU (Most Recently Used)
- (D) FIFO (First In First Out)

- Q.21** B+ Trees are considered **BALANCED** because
- (A) the lengths of the paths from the root to all leaf nodes are all equal.
 - (B) the lengths of the paths from the root to all leaf nodes differ from each other by at most 1.
 - (C) the number of children of any two non-leaf sibling nodes differ by at most 1.
 - (D) the number of records in any two leaf nodes differ by at most 1.
- Q.22** Suppose a database schedule S involves transactions $T_1, \dots, T_n$. Construct the precedence graph of S with vertices representing the transactions and edges representing the conflicts. If S is serializable, which one of the following orderings of the vertices of the precedence graph is guaranteed to yield a serial schedule?
- (A) Topological order
 - (B) Depth-first order
 - (C) Breadth-first order
 - (D) Ascending order of transaction indices
- Q.23** Anarkali digitally signs a message and sends it to Salim. Verification of the signature by Salim requires
- (A) Anarkali's public key.
 - (B) Salim's public key.
 - (C) Salim's private key.
 - (D) Anarkali's private key.
- Q.24** In an Ethernet local area network, which one of the following statements is **TRUE**?
- (A) A station stops to sense the channel once it starts transmitting a frame.
 - (B) The purpose of the jamming signal is to pad the frames that are smaller than the minimum frame size.
 - (C) A station continues to transmit the packet even after the collision is detected.
 - (D) The exponential backoff mechanism reduces the probability of collision on retransmissions.

Q.25 Identify the correct sequence in which the following packets are transmitted on the network by a host when a browser requests a webpage from a remote server, assuming that the host has just been restarted.

- (A) HTTP GET request, DNS query, TCP SYN
- (B) DNS query, HTTP GET request, TCP SYN
- (C) DNS query, TCP SYN, HTTP GET request
- (D) TCP SYN, DNS query, HTTP GET request

Q. 26 - Q. 55 carry two marks each.

Q.26 A binary relation R on $\mathbb{N} \times \mathbb{N}$ is defined as follows: $(a, b)R(c, d)$ if $a \leq c$ or $b \leq d$. Consider the following propositions:

P: R is reflexive

Q: R is transitive

Which one of the following statements is **TRUE**?

- (A) Both P and Q are true.
- (B) P is true and Q is false.
- (C) P is false and Q is true.
- (D) Both P and Q are false.

Q.27 Which one of the following well-formed formulae in predicate calculus is **NOT** valid?

- (A) $(\forall x p(x) \Rightarrow \forall x q(x)) \Rightarrow (\exists x \neg p(x) \vee \forall x q(x))$
- (B) $(\exists x p(x) \vee \exists x q(x)) \Rightarrow \exists x (p(x) \vee q(x))$
- (C) $\exists x (p(x) \wedge q(x)) \Rightarrow (\exists x p(x) \wedge \exists x q(x))$
- (D) $\forall x (p(x) \vee q(x)) \Rightarrow (\forall x p(x) \vee \forall x q(x))$

Q.28 Consider a set U of 23 different compounds in a Chemistry lab. There is a subset S of U of 9 compounds, each of which reacts with exactly 3 compounds of U . Consider the following statements:

- I. Each compound in $U \setminus S$ reacts with an odd number of compounds.
- II. At least one compound in $U \setminus S$ reacts with an odd number of compounds.
- III. Each compound in $U \setminus S$ reacts with an even number of compounds.

Which one of the above statements is **ALWAYS TRUE**?

- (A) Only I
- (B) Only II
- (C) Only III
- (D) None

Q.29 The value of the expression $13^{99} \pmod{17}$, in the range 0 to 16, is _____.

Q.30 Suppose the functions F and G can be computed in 5 and 3 nanoseconds by functional units U_F and U_G , respectively. Given two instances of U_F and two instances of U_G , it is required to implement the computation $F(G(X_i))$ for $1 \leq i \leq 10$. Ignoring all other delays, the minimum time required to complete this computation is _____ nanoseconds.

Q.31 Consider a processor with 64 registers and an instruction set of size twelve. Each instruction has five distinct fields, namely, opcode, two source register identifiers, one destination register identifier, and a twelve-bit immediate value. Each instruction must be stored in memory in a byte-aligned fashion. If a program has 100 instructions, the amount of memory (in bytes) consumed by the program text is _____.

Q.32 The width of the physical address on a machine is 40 bits. The width of the tag field in a 512 KB 8-way set associative cache is _____ bits.

- Q.33** Consider a 3 GHz (gigahertz) processor with a three-stage pipeline and stage latencies τ_1 , τ_2 , and τ_3 such that $\tau_1 = 3\tau_2/4 = 2\tau_3$. If the longest pipeline stage is split into two pipeline stages of equal latency, the new frequency is _____ GHz, ignoring delays in the pipeline registers.
- Q.34** A complete binary min-heap is made by including each integer in $[1, 1023]$ exactly once. The depth of a node in the heap is the length of the path from the root of the heap to that node. Thus, the root is at depth 0. The maximum depth at which integer 9 can appear is _____.
- Q.35** The following function computes X^Y for positive integers X and Y.

```
int exp(int X, int Y) {
    int res = 1, a = X, b = Y;

    while ( b != 0 ){
        if ( b%2 == 0 ) { a = a*a; b = b/2; }
        else           { res = res*a; b = b-1; }
    }
    return res;
}
```

Which one of the following conditions is **TRUE** before every iteration of the loop?

- (A) $X^Y = a^b$
- (B) $(res * a)^Y = (res * X)^b$
- (C) $X^Y = res * a^b$
- (D) $X^Y = (res * a)^b$

Q.36 Consider the following New-order strategy for traversing a binary tree:

- Visit the root;
- Visit the right subtree using New-order;
- Visit the left subtree using New-order;

The New-order traversal of the expression tree corresponding to the reverse polish expression $3\ 4\ *\ 5\ -\ 2\ ^\ 6\ 7\ *\ 1\ +\ -$ is given by:

- (A) $+ \ - \ 1 \ 6 \ 7 \ *\ 2 \ ^\ 5 \ - \ 3 \ 4 \ *$
 (B) $- \ + \ 1 \ *\ 6 \ 7 \ ^\ 2 \ - \ 5 \ *\ 3 \ 4$
 (C) $- \ + \ 1 \ *\ 7 \ 6 \ ^\ 2 \ - \ 5 \ *\ 4 \ 3$
 (D) $1 \ 7 \ 6 \ *\ + \ 2 \ 5 \ 4 \ 3 \ *\ - \ ^\ -$

Q.37 Consider the following program:

```
int f(int *p, int n)
{
    if (n <= 1) return 0;
    else return max(f(p+1,n-1),p[0]-p[1]);
}

int main()
{
    int a[] = {3,5,2,6,4};
    printf("%d", f(a,5));
}
```

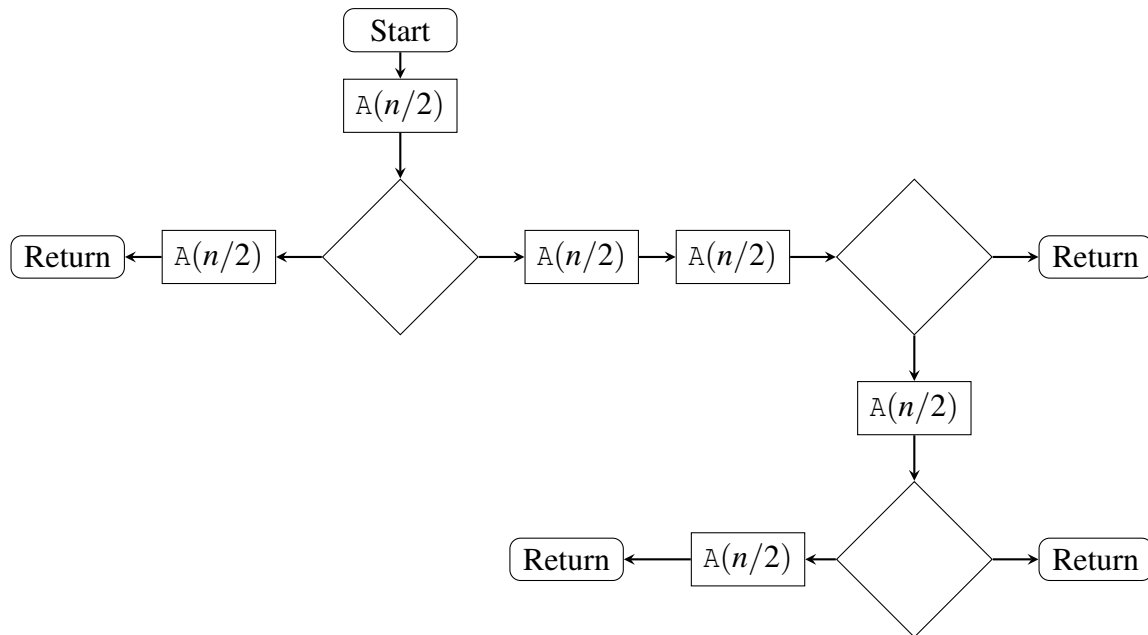
Note: $\max(x, y)$ returns the maximum of x and y .

The value printed by this program is _____ .

Q.38 Let A_1, A_2, A_3 , and A_4 be four matrices of dimensions $10 \times 5, 5 \times 20, 20 \times 10$, and 10×5 , respectively. The minimum number of scalar multiplications required to find the product $A_1 A_2 A_3 A_4$ using the basic matrix multiplication method is _____ .

- Q.39** The given diagram shows the flowchart for a recursive function $A(n)$. Assume that all statements, except for the recursive calls, have $O(1)$ time complexity. If the worst case time complexity of this function is $O(n^\alpha)$, then the least possible value (accurate up to two decimal positions) of α is _____.

Flowchart for Recursive Function $A(n)$



- Q.40** The number of ways in which the numbers 1, 2, 3, 4, 5, 6, 7 can be inserted in an empty binary search tree, such that the resulting tree has height 6, is _____.

Note: The height of a tree with a single node is 0.

Q.41 In an adjacency list representation of an undirected simple graph $G = (V, E)$, each edge (u, v) has two adjacency list entries: $[v]$ in the adjacency list of u , and $[u]$ in the adjacency list of v . These are called twins of each other. A twin pointer is a pointer from an adjacency list entry to its twin. If $|E| = m$ and $|V| = n$, and the memory size is not a constraint, what is the time complexity of the most efficient algorithm to set the twin pointer in each entry in each adjacency list?

- (A) $\Theta(n^2)$
- (B) $\Theta(n + m)$
- (C) $\Theta(m^2)$
- (D) $\Theta(n^4)$

Q.42 Consider the following two statements:

- I.** If all states of an NFA are accepting states then the language accepted by the NFA is Σ^* .
- II.** There exists a regular language A such that for all languages B , $A \cap B$ is regular.

Which one of the following is **CORRECT**?

- (A) Only **I** is true
- (B) Only **II** is true
- (C) Both **I** and **II** are true
- (D) Both **I** and **II** are false

Q.43 Consider the following languages:

$$L_1 = \{a^n b^m c^{n+m} : m, n \geq 1\}$$

$$L_2 = \{a^n b^n c^{2n} : n \geq 1\}$$

Which one of the following is **TRUE**?

- (A) Both L_1 and L_2 are context-free.
- (B) L_1 is context-free while L_2 is not context-free.
- (C) L_2 is context-free while L_1 is not context-free.
- (D) Neither L_1 nor L_2 is context-free.

Q.44 Consider the following languages.

$$\begin{aligned} L_1 &= \{ \langle M \rangle \mid M \text{ takes at least 2016 steps on some input} \}, \\ L_2 &= \{ \langle M \rangle \mid M \text{ takes at least 2016 steps on all inputs} \} \text{ and} \\ L_3 &= \{ \langle M \rangle \mid M \text{ accepts } \varepsilon \}, \end{aligned}$$

where for each Turing machine M , $\langle M \rangle$ denotes a specific encoding of M . Which one of the following is **TRUE**?

- (A) L_1 is recursive and L_2, L_3 are not recursive
- (B) L_2 is recursive and L_1, L_3 are not recursive
- (C) L_1, L_2 are recursive and L_3 is not recursive
- (D) L_1, L_2, L_3 are recursive

Q.45 Which one of the following grammars is free from *left recursion*?

- (A)
$$\begin{aligned} S &\rightarrow AB \\ A &\rightarrow Aa \mid b \\ B &\rightarrow c \end{aligned}$$
- (B)
$$\begin{aligned} S &\rightarrow Ab \mid Bb \mid c \\ A &\rightarrow Bd \mid \varepsilon \\ B &\rightarrow e \end{aligned}$$
- (C)
$$\begin{aligned} S &\rightarrow Aa \mid B \\ A &\rightarrow Bb \mid Sc \mid \varepsilon \\ B &\rightarrow d \end{aligned}$$
- (D)
$$\begin{aligned} S &\rightarrow Aa \mid Bb \mid c \\ A &\rightarrow Bd \mid \varepsilon \\ B &\rightarrow Ae \mid \varepsilon \end{aligned}$$

- Q.46** A student wrote two context-free grammars **G1** and **G2** for generating a single C-like array declaration. The dimension of the array is at least one. For example,

```
int a[10][3];
```

The grammars use D as the start symbol, and use six terminal symbols **int ; id [] num**.

Grammar **G1**

$D \rightarrow \text{int}L;$

$L \rightarrow \text{id}[E$

$E \rightarrow \text{num}]$

$E \rightarrow \text{num}][E$

Grammar **G2**

$D \rightarrow \text{int}L;$

$L \rightarrow \text{id}E$

$E \rightarrow E[\text{num}]$

$E \rightarrow [\text{num}]$

Which of the grammars correctly generate the declaration mentioned above?

- (A) Both **G1** and **G2**
- (B) Only **G1**
- (C) Only **G2**
- (D) Neither **G1** nor **G2**

- Q.47** Consider the following processes, with the arrival time and the length of the CPU burst given in milliseconds. The scheduling algorithm used is preemptive shortest remaining-time first.

Process	Arrival Time	Burst Time
P_1	0	10
P_2	3	6
P_3	7	1
P_4	8	3

The average turn around time of these processes is _____ milliseconds.

Q.48 Consider the following two-process synchronization solution.

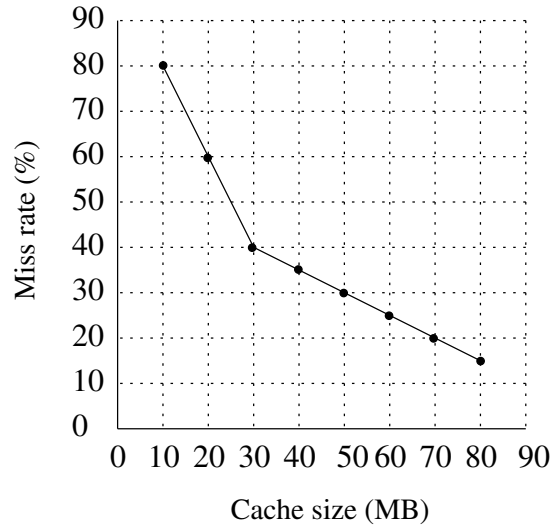
Process 0	Process 1
-----	-----
Entry: loop while (turn == 1);	Entry: loop while (turn == 0);
(critical section)	(critical section)
Exit: turn = 1;	Exit: turn = 0;

The shared variable `turn` is initialized to zero. Which one of the following is **TRUE**?

- (A) This is a correct two-process synchronization solution.
- (B) This solution violates mutual exclusion requirement.
- (C) This solution violates progress requirement.
- (D) This solution violates bounded wait requirement.

Q.49 Consider a non-negative counting semaphore S . The operation $P(S)$ decrements S , and $V(S)$ increments S . During an execution, 20 $P(S)$ operations and 12 $V(S)$ operations are issued in some order. The largest initial value of S for which at least one $P(S)$ operation will remain blocked is _____.

- Q.50** A file system uses an in-memory cache to cache disk blocks. The miss rate of the cache is shown in the figure. The latency to read a block from the cache is 1 ms and to read a block from the disk is 10 ms. Assume that the cost of checking whether a block exists in the cache is negligible. Available cache sizes are in multiples of 10 MB.



The smallest cache size required to ensure an average read latency of less than 6 ms is _____ MB.

- Q.51** Consider the following database schedule with two transactions, T_1 and T_2 .

$$S = r_2(X); r_1(X); r_2(Y); w_1(X); r_1(Y); w_2(X); a_1; a_2$$

where $r_i(Z)$ denotes a *read* operation by transaction T_i on a variable Z , $w_i(Z)$ denotes a *write* operation by T_i on a variable Z and a_i denotes an *abort* by transaction T_i .

Which one of the following statements about the above schedule is **TRUE**?

- (A) S is non-recoverable
- (B) S is recoverable, but has a cascading abort
- (C) S does not have a cascading abort
- (D) S is strict

Q.52 Consider the following database table named *water_schemes* :

<i>water_schemes</i>		
scheme_no	district_name	capacity
1	Ajmer	20
1	Bikaner	10
2	Bikaner	10
3	Bikaner	20
1	Churu	10
2	Churu	20
1	Dungargarh	10

The number of tuples returned by the following SQL query is _____ .

```
with total(name, capacity) as
  select district_name, sum(capacity)
  from water_schemes
  group by district_name
with total_avg(capacity) as
  select avg(capacity)
  from total
select name
  from total, total_avg
  where total.capacity ≥ total_avg.capacity
```

Q.53 A network has a data transmission bandwidth of 20×10^6 bits per second. It uses CSMA/CD in the MAC layer. The maximum signal propagation time from one node to another node is 40 microseconds. The minimum size of a frame in the network is _____ bytes.

Q.54 For the IEEE 802.11 MAC protocol for wireless communication, which of the following statements is/are **TRUE**?

- I. At least three non-overlapping channels are available for transmissions.
- II. The RTS-CTS mechanism is used for collision detection.
- III. Unicast frames are ACKed.

- (A) All I, II, and III
- (B) I and III only
- (C) II and III only
- (D) II only

Q.55 Consider a 128×10^3 bits/second satellite communication link with one way propagation delay of 150 milliseconds. Selective retransmission (repeat) protocol is used on this link to send data with a frame size of 1 kilobyte. Neglect the transmission time of acknowledgement. The minimum number of bits required for the sequence number field to achieve 100% utilization is _____ .